Gaussian-Process Reconstruction of the Mapping-QCLE Excess Term: A Moving-Cloud Formulation and Failure Analysis

by

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Abstract

Nonadiabatic molecular dynamics requires coupled treatment of electronic transitions and nuclear motion. The mapping-basis quantum–classical Liouville equation supplies this treatment, but its density-dependent correction requires costly mixed phase-space derivatives on multidimensional grids. This thesis tests whether a Gaussian-process model can recover the density and these derivatives from a moving trajectory cloud. A Hamiltonian scheme transports the cloud, while an explicit midpoint update applies the correction. For a one-dimensional, two-state avoided-crossing model, validation uses analytic operator tests, refinement and conservation checks, replicates, and comparison with numerically controlled reference solutions of the time-dependent Schrödinger equation and grid-based quantum–classical Liouville equation. Density values are reconstructed accurately, but derivative errors do not decrease consistently; the model violates electronic-state constraints, exhibits normalization and energy drift, and varies across replicates. Consequently, the formulation does not reliably improve the Poisson-bracket mapping equation, but establishes derivative-fidelity, structural, and conservation requirements for a viable successor.

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> "Who looks out with my eyes? What is the soul?
> I cannot stop asking.
> If I could taste one sip of an answer,
> I could break out of this prison for drunks."

—Rumi, translated by Coleman Barks

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Abbreviations

Abbreviation	Meaning
ARD	automatic relevance determination
ESS	effective sample size
GP	Gaussian process
KDE	kernel density estimation
LOO	leave-one-out
MInt	Momentum Integral
MMST	Meyer–Miller–Stock–Thoss
NMLL	negative marginal log likelihood
PBME	Poisson-bracket mapping equation
QCLE	quantum–classical Liouville equation
RBF	radial basis function
SEO	singly excited oscillator
TDSE	time-dependent Schrödinger equation

Notation and Conventions

This thesis combines fully quantum molecular notation, quantum–classical phase-space notation, MMST mapping variables, and Gaussian-process regression. The conventions summarized below are used throughout the thesis. Symbols introduced for a specialized calculation retain these meanings unless an explicit local qualification is stated.

Table 1: Canonical notation and conventions used throughout the thesis.

Mathematical object	**Canonical notation**	**Convention**
Coordinates and phase-space variables		
Electronic coordinates	$\mathbf{q} = (\mathbf{q}_1, \ldots, \mathbf{q}_{N_e})$	Used exclusively for fully quantum electronic coordinates in Chapter 1.
Nuclear or bath coordinates and momenta	$\mathbf{R}, \mathbf{P}$	The bath phase-space point is $\mathbf{X} = (\mathbf{R}, \mathbf{P})$.
Common bath mass	M	A scalar common bath mass is used in the quantum–classical and mapping models: $$T_\mathrm{b}(\mathbf{P}) = \frac{\mathbf{P}^2}{2M}, \qquad \dot{\mathbf{R}} = \frac{\mathbf{P}}{M}.$$ Individual molecular nuclear masses in the fully quantum discussion are denoted by M_A.
Mapping coordinates and momenta	$\mathbf{r}, \mathbf{p}$	The mapping phase-space point is $\mathbf{x} = (\mathbf{r}, \mathbf{p})$.
Extended bath–mapping phase-space point	$\boldsymbol{\mathcal{X}} = (\mathbf{X}, \mathbf{x})$	Equivalently, $$\boldsymbol{\mathcal{X}} = (\mathbf{R}, \mathbf{P}, \mathbf{r}, \mathbf{p}).$$ This notation is used from Chapter 3 onward.
Index conventions		

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Mathematical object	Canonical notation	Convention
Adiabatic-state indices	$\alpha, \beta, \alpha', \beta'$	Reserved for adiabatic electronic states and adiabatic-basis QCLE matrix elements.
Fixed-basis and mapping-state indices	$\lambda, \lambda', \kappa, \kappa'$	Reserved for states in the fixed subsystem basis and for the corresponding MMST mapping oscillators. For a subsystem with N_s states, $$\lambda, \lambda', \kappa, \kappa' \in \{1, \ldots, N_s\}.$$
Support-point and kernel-centre indices	i, j	Index GP support points, physical density labels, kernel centres, and kernel-expansion terms.
Local MInt eigenmode indices	a, b	Used only for the temporary local diagonalization of the fixed-basis subsystem Hamiltonian during an MInt substep.
Quantum–classical and mapping densities		
Mass-separation parameter	$\varepsilon_m = \sqrt{m/M}$	Used only in the QCLE mass-ratio expansion. The symbol μ_t is reserved for the GP posterior mean or modulation function.
Partially Wigner-transformed density operator	$\hat{\rho}_W(\mathbf{X}, t)$	Operator-valued in the finite subsystem Hilbert space and defined over the bath phase space.
Subsystem matrix elements of the Wigner density	$\rho_W^{\lambda\lambda'}(\mathbf{X}, t)$	Fixed-basis matrix elements of $\hat{\rho}_W(\mathbf{X}, t)$. Adiabatic-basis matrix elements use adiabatic indices, for example $\rho_W^{\alpha\beta}$.
Exact projected mapping density	$\widetilde{\rho}_m(\boldsymbol{\mathcal{X}}, t)$	The physical SEO-projected mapping density derived in Chapter 3.
Numerical mapping-density surrogate	$\widehat{\rho}_{m,t}(\boldsymbol{\mathcal{X}})$	A fixed-time numerical approximation to $\widetilde{\rho}_m(\boldsymbol{\mathcal{X}}, t)$.
Structured product surrogate	$\widehat{\rho}_{m,t} = g_s \mu_t$	The physical SEO profile $g_s(\mathbf{x})$ is multiplied by the smooth GP modulation function $\mu_t(\boldsymbol{\mathcal{X}})$.

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Mathematical object	Canonical notation	Convention
Physical density labels	$\mathbf{y}_t$ or $\mathbf{y}^n$	Values of the complete mapping density carried by the dynamically transported support cloud: $$y_i^n \approx \widetilde{\rho}_m\left(\boldsymbol{\mathcal{X}}_i^n, t_n\right).$$
Transformed modulation labels	$\mathbf{u}_t$ or $\mathbf{u}^n$	Safe-profile ratios used as GP regression targets: $$u_i^n = \frac{y_i^n}{g_{s,\mathrm{safe}}(\mathbf{x}_i^n)}.$$
GP kernel coefficients	$\boldsymbol{\zeta}_t$ or $\boldsymbol{\zeta}^n$	Defined by the regularized kernel solve $$\boldsymbol{\zeta}_t = K_{y,t}^{-1}\mathbf{u}_t$$ for the product representation.
Regularized kernel matrix	$K_{y,t}$	Defined by $$K_{y,t} = K_t + \sigma_n^2 I,$$ with any numerical jitter included in the diagonal regularization.
GP precision matrix	$\mathsf{P}_t = K_{y,t}^{-1}$	Used in leave-one-out formulas and distinguished from all source-generator matrices.
GP support set	$\mathcal{S}_t$ or $\mathcal{S}_n$	The time-dependent set of support points: $$\mathcal{S}_n = \{\boldsymbol{\mathcal{X}}_i^n\}_{i=1}^N.$$
Hamiltonians, Liouvillians, and propagators		
Subsystem Hamiltonian	$h(\mathbf{R})$	The finite-dimensional subsystem Hamiltonian in the fixed basis. Its matrix elements are $h^{\lambda\lambda'}(\mathbf{R})$.
Trace–traceless decomposition	$h(\mathbf{R}) = \eta(\mathbf{R})I_s + \bar{h}(\mathbf{R})$	The scalar trace contribution is incorporated into the state-independent bath potential $V_0(\mathbf{R})$, while $$\mathrm{Tr}_s\,\bar{h}(\mathbf{R}) = 0.$$
Mapping Hamiltonian	$H_m(\boldsymbol{\mathcal{X}})$	The MMST mapping Hamiltonian generating the PBME characteristic flow.

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Mathematical object	Canonical notation	Convention
Hamiltonian mapping Liouvillian	$i\mathcal{L}_m^{\circ}$	Its density generator is $$\mathcal{T} = -i\mathcal{L}_m^{\circ}.$$
Excess mapping Liouvillian	$i\mathcal{L}_m'$	Its density source is $$\mathcal{Q} = -i\mathcal{L}_m'.$$ For one bath coordinate, $$\mathcal{Q}[f] = -\frac{\hbar}{8}\sum_{\lambda,\lambda'} \frac{d\bar{h}^{\lambda\lambda'}}{dR}\left(\partial_{r_{\lambda'}}\partial_{r_\lambda} + \partial_{p_{\lambda'}}\partial_{p_\lambda}\right)\partial_P f.$$
PBME/MInt flow map	$\Psi_\tau^{\mathrm{MInt}}$	The numerical map approximating the Hamiltonian PBME flow over a time interval τ.
Transport propagator	$\mathcal{U}_\tau^{\mathrm{T}}$	Propagates the support cloud under the PBME/MInt transport leg.
Source propagator	$\mathcal{U}_\tau^{\mathrm{Q}}$	Propagates the density or transformed labels under the excess mapping-QCLE source leg.
Complete symmetric step	$\mathcal{U}_{\Delta t}^{\mathrm{TST}}$	Defined by $$\mathcal{U}_{\Delta t}^{\mathrm{TST}} = \mathcal{U}_{\Delta t/2}^{\mathrm{T}}\mathcal{U}_{\Delta t}^{\mathrm{Q}}\mathcal{U}_{\Delta t/2}^{\mathrm{T}}.$$
Discrete source matrices		
Product basis function	$\Phi_j(\boldsymbol{\mathcal{X}}) = g_s(\mathbf{x})k_j(\boldsymbol{\mathcal{X}})$	The physical basis on which the excess source is collocated.
Discrete physical source matrix	$\mathsf{L}^{(\rho)}$	Defined by $$\mathsf{L}_{ij}^{(\rho)} = \mathcal{Q}[\Phi_j]\big|_{\boldsymbol{\mathcal{X}}_i}.$$
Safe-profile diagonal matrix	D_g	Defined by $$\mathsf{D}_g = \mathrm{diag}\left[g_{s,\mathrm{safe}}(\mathbf{x}_i)\right].$$

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Mathematical object	**Canonical notation**	**Convention**
Discrete modulation source matrix	L	Defined by $$\mathsf{L} = \mathsf{D}_g^{-1}\mathsf{L}^{(\rho)}.$$
Frozen modulation generator	$\mathsf{A} = \mathsf{L}K_y^{-1}$	Generates the frozen source-leg equation $$\frac{d\mathbf{u}}{dt} = \mathsf{A}\mathbf{u}.$$
Mapping observables, invariants, and diagnostics		
Mapping observable symbol	$c_{\lambda\lambda'}(\mathbf{x})$	MMST phase-space symbol of the subsystem operator $\lvert\lambda\rangle\langle\lambda'\rvert$.
Mapped identity symbol	$I_m(\mathbf{x})$	The MMST phase-space symbol of the subsystem identity: $$I_m(\mathbf{x}) = \sum_{\lambda=1}^{N_s} c_{\lambda\lambda}(\mathbf{x}).$$
Mapping-radius invariant	$\mathcal{R}_m^2(\mathbf{x})$	Defined by $$\mathcal{R}_m^2(\mathbf{x}) = \mathbf{r}^T\mathbf{r} + \mathbf{p}^T\mathbf{p}.$$ It is conserved by the MInt Hamiltonian mapping rotation.
Raw density integral	$\mathcal{N}_{\mathrm{raw}}(t)$	The full phase-space integral of the complete mapping-density surrogate: $$\mathcal{N}_{\mathrm{raw}}(t) = \int d\boldsymbol{\mathcal{X}}\,\widehat{\rho}_{m,t}(\boldsymbol{\mathcal{X}}).$$
Subsystem trace diagnostic	$\mathcal{N}_{\mathrm{tr}}(t)$	Computed using the mapped identity or diagonal mapping observable symbols: $$\mathcal{N}_{\mathrm{tr}}(t) = \int d\boldsymbol{\mathcal{X}}\,I_m(\mathbf{x})\widehat{\rho}_{m,t}(\boldsymbol{\mathcal{X}}).$$
Physical energy	$\mathcal{E}(t)$	The density-weighted expectation of the physical mapping Hamiltonian or the corresponding physical energy observable.

Chapter 1

Introduction

This thesis tests one specific numerical construction: a product-GP density on moving PBME/MInt support, analytic evaluation of the mapping-QCLE excess source, and an explicit midpoint update of the live density labels for the one-dimensional, two-state avoided-crossing benchmark. It does not test all Gaussian-process representations or all discretizations of the continuum mapping-QCLE excess term. Reconstruction, derivative-operator fidelity, SEO projection, conservation, refinement, stochastic replication, and comparison with physical references are tested separately so that a completed calculation is not mistaken for a validated method. The resulting evidence is negative: the tested construction does not satisfy the joint conditions and does not demonstrate systematic improvement over PBME. Its contribution is therefore the formulation, the controlled failure analysis, and the design requirements that follow from it.

1.1 Motivation and scope

The mapping-basis quantum–classical Liouville equation (QCLE) provides a more complete mixed quantum–classical description than the Poisson-bracket mapping equation (PBME), but its excess term acts through mixed derivatives of a signed phase-space density. Direct evaluation on a multidimensional grid becomes prohibitively expensive as the number of environmental coordinates increases, whereas PBME achieves trajectory-level cost by discarding the term. The resulting tension between physical completeness and computational feasibility motivates this thesis: a moving trajectory cloud is used to locate dynamically relevant regions of phase space, and a Gaussian-process (GP) density representation is used to supply the differentiable field required by the excess operator.

For this benchmarked construction, usefulness would require an improvement of the QCLE approximation rather than merely a smooth interpolation. Its evaluation therefore requires four connected forms of evidence: accurate reconstruction of a manufactured excess operator, time-step sensitivity and independent-cloud enlargement, preservation of the singly-excited-oscillator (SEO) projected structure and physical moments, and smaller errors than PBME against independently resolved quantum and grid-QCLE references. These requirements provide the scientific acceptance criteria for the numerical study.

The calculations show that the tested representation does not satisfy those criteria. Its manufactured excess-operator error is non-monotone under the tested enlargement, the product surrogate develops substantial leakage from the physical SEO image, and four-seed MIDPOINT calculations show severe dispersion and raw normalization drift. No validated evidence of systematic improvement over PBME was obtained.

The general fact that on-manifold values do not determine ambient normal derivatives is not claimed as new. The distinctive contribution is instead a specific moving-cloud density-level implementation and controlled test: it formulates the excess update on PBME/MInt support, contrasts the product surrogate with the exact four-real-field SEO image, measures projection leakage, demonstrates derivative-operator failure despite small value errors, and connects that evidence to nonconservative collocation and signed-weight collapse. Previous mapping-QCLE work established the projected mapping equations and PBME approximation [1, 2], while numerical GP and meshfree work established differentiable surrogate tools [3–5]; the original contribution here is their particular moving-cloud product-GP realization and application-specific failure diagnosis, stated without an unsupported priority claim.

This conclusion is deliberately restricted to the one-dimensional, two-state Tully dual avoided-crossing benchmark studied here. The model is sufficiently controlled to separate reference, reconstruction, operator, integration, and sampling errors, but it cannot establish general accuracy or scalability for multidimensional molecular systems.

Molecular dynamics in condensed phases is often governed by events that cannot be inferred from equilibrium molecular structures alone. Many chemical and material processes involve the coupled evolution of electronic populations, nuclear coordinates, energy redistribution, and environmental fluctuations. Charge transfer, exciton migration, photochemical relaxation, proton-coupled reactions, catalytic bond rearrangement, electrochemical transformations, and energy transport in molecular materials all depend on the simultaneous evolution of electronic, nuclear, and environmental degrees of freedom [6–11].

The central theoretical difficulty is that these processes often occur outside the range of validity of a strictly adiabatic description. In the Born–Oppenheimer picture, the nuclei evolve on a single electronic potential-energy surface while the electronic state adjusts parametrically to the nuclear configuration [12, 13]. This separation is powerful when electronic states remain well separated in energy and the electronic basis varies slowly with nuclear geometry. It becomes inadequate, however, when nuclear motion carries the system through regions where two or more electronic states are close in energy or strongly coupled. Avoided crossings, conical intersections, charge-transfer resonances, bond-breaking regions, fluctuating solvent configurations, and strongly reorganizing molecular environments can all generate such conditions [14, 15]. In these regimes, nuclear motion may induce transitions between electronic states, while the evolving electronic state modifies the forces acting on the nuclei.

This is the regime of nonadiabatic molecular dynamics. Its importance is not restricted to one class of experiments or one chemical example. In electron-transfer systems, environmental reorganization can tune donor and acceptor states into resonance. In molecular aggregates and materials, nuclear fluctuations can control the migration, localization, and relaxation of electronic excitation. In photochemical reactions, passage through strongly coupled regions of configuration space can determine whether a molecule relaxes, isomerizes, dissociates, or branches into competing product channels. In proton-coupled and hydrogen-transfer processes, light-particle motion, electronic redistribution, and environmental polarization may become inseparable. These examples differ chemically, but they share a common theoretical requirement: electronic quantum dynamics and nuclear or environmental motion must be treated as coupled parts of the same dynamical process.

Fully quantum dynamics provides the reference description in principle, but it is generally infeasible for condensed-phase systems with many nuclear degrees of freedom. Mixed quantum–classical methods offer a practical compromise by treating selected electronic or subsystem degrees of freedom quantum mechanically while representing the surrounding nuclear environment with classical phase-space variables. Methods such as Ehrenfest dynamics, trajectory surface hopping, and linearized semiclassical approaches have provided important insight, but each introduces approximations whose consequences must be examined when coherence, decoherence, detailed balance, nuclear back-reaction, or long-time propagation are important [6, 8, 16].

The quantum–classical Liouville equation provides a systematic density-level starting point for this problem. It describes the coupled evolution of a quantum subsystem and a classical bath while retaining the subsystem density matrix, nuclear phase-space variables, and their mutual back-reaction within a mixed quantum–classical framework [8, 17]. In the mapping representation, discrete subsystem states are represented by continuous mapping variables, placing populations, coherences, nuclear coordinates, nuclear momenta, and mapping variables within a unified phase-space formulation [1, 18, 19].

The scope of this introductory chapter is to build the conceptual path toward that formulation. The chapter begins with the adiabatic approximation and its breakdown, then discusses adiabatic and diabatic representations, potential-energy surfaces, density matrices, subsystem–bath coupling, Landau–Zener transitions, and the Wigner phase-space representation. These ingredients provide the background needed to formulate the thesis objective: the construction and controlled assessment of one product-Gaussian-process density representation on PBME/MInt-transported support, coupled to an explicit MIDPOINT live-label update, for the mapping-basis quantum–classical Liouville equation on a one-dimensional two-state benchmark.

1.2 The adiabatic approximation and its breakdown

The microscopic starting point for molecular quantum dynamics is the nonrelativistic molecular Hamiltonian. For a system of N_e electrons and N_n nuclei, let

$$\mathbf{q} = (\mathbf{q}_1, \ldots, \mathbf{q}_{N_e})$$

denote the electronic coordinates and

$$\mathbf{R} = (\mathbf{R}_1, \ldots, \mathbf{R}_{N_n})$$

denote the nuclear coordinates. In the Born–Oppenheimer construction, the electronic problem is solved at fixed nuclear configuration. To make this conditional structure explicit, electronic quantities will be written with the notation $\mathbf{q} \mid \mathbf{R}$. Thus $\phi_\alpha(\mathbf{q} \mid \mathbf{R})$ denotes an electronic wavefunction in the electronic coordinates $\mathbf{q}$, parametrically conditioned on the nuclear geometry $\mathbf{R}$.

Neglecting relativistic corrections, spin-dependent interactions, and external fields, the molecular Hamiltonian can be written as

$$\hat{H} = \hat{T}_\mathrm{n} + \hat{H}_\mathrm{e}(\mathbf{q} \mid \mathbf{R}), \tag{1.1}$$

where

$$\hat{T}_\mathrm{n} = -\sum_{A=1}^{N_n} \frac{\hbar^2}{2M_A} \nabla_A^2 \tag{1.2}$$

is the nuclear kinetic-energy operator. Here M_A is the mass of nucleus A, and $\nabla_A \equiv \nabla_{\mathbf{R}_A}$ is the gradient with respect to the position of that nucleus.

The clamped-nuclei electronic Hamiltonian is

$$\hat{H}_\mathrm{e}(\mathbf{q} \mid \mathbf{R}) = \hat{T}_\mathrm{e} + \hat{V}_\mathrm{ee}(\mathbf{q}) + \hat{V}_\mathrm{en}(\mathbf{q} \mid \mathbf{R}) + V_\mathrm{nn}(\mathbf{R}). \tag{1.3}$$

The first term,

$$\hat{T}_\mathrm{e} = -\sum_{i=1}^{N_e} \frac{\hbar^2}{2m_e} \nabla_i^2, \tag{1.4}$$

is the electronic kinetic-energy operator, with m_e the electron mass and $\nabla_i \equiv \nabla_{\mathbf{q}_i}$. The electron–electron Coulomb repulsion is

$$\hat{V}_\mathrm{ee}(\mathbf{q}) = \frac{1}{4\pi\varepsilon_0} \sum_{i<j}^{N_e} \frac{e^2}{|\mathbf{q}_i - \mathbf{q}_j|}, \tag{1.5}$$

the electron–nuclear Coulomb attraction is

$$\hat{V}_{\mathrm{en}}(\mathbf{q} \mid \mathbf{R}) = -\frac{1}{4\pi\varepsilon_0} \sum_{i=1}^{N_e} \sum_{A=1}^{N_n} \frac{Z_A e^2}{|\mathbf{q}_i - \mathbf{R}_A|}, \tag{1.6}$$

and the nuclear–nuclear Coulomb repulsion is

$$V_{\mathrm{nn}}(\mathbf{R}) = \frac{1}{4\pi\varepsilon_0} \sum_{A<B}^{N_n} \frac{Z_A Z_B e^2}{|\mathbf{R}_A - \mathbf{R}_B|}. \tag{1.7}$$

The electronic Hamiltonian acts on electronic coordinates, while the nuclear configuration enters as a fixed parameter. The electron–nuclear attraction $\hat{V}_{\mathrm{en}}(\mathbf{q} \mid \mathbf{R})$ carries this dependence explicitly, because changing the nuclear geometry changes the Coulomb field experienced by the electrons. The nuclear–nuclear term $V_{\mathrm{nn}}(\mathbf{R})$ depends only on nuclear coordinates and therefore enters the electronic problem as a geometry-dependent scalar.

At fixed nuclear geometry, the electronic Schrödinger equation is

$$\hat{H}_{\mathrm{e}}(\mathbf{q} \mid \mathbf{R})\phi_\alpha(\mathbf{q} \mid \mathbf{R}) = E_\alpha(\mathbf{R})\phi_\alpha(\mathbf{q} \mid \mathbf{R}). \tag{1.8}$$

The eigenfunction $\phi_\alpha(\mathbf{q} \mid \mathbf{R})$ is an electronic wavefunction for a specified nuclear geometry, and the eigenvalue $E_\alpha(\mathbf{R})$ is the corresponding adiabatic energy.

The nuclear–nuclear repulsion $V_{\mathrm{nn}}(\mathbf{R})$ is independent of the electronic coordinates. Therefore, at fixed $\mathbf{R}$, it acts as a scalar shift in the electronic Hilbert space. It changes the adiabatic energy $E_\alpha(\mathbf{R})$, but it does not change the electronic eigenfunction $\phi_\alpha(\mathbf{q} \mid \mathbf{R})$. For this reason, electronic-structure calculations often solve the electronic-coordinate-dependent part first and then add $V_{\mathrm{nn}}(\mathbf{R})$ to the energy. In this thesis, however, the symbol $\hat{H}_{\mathrm{e}}(\mathbf{q} \mid \mathbf{R})$ will denote the full clamped-nuclei electronic Hamiltonian, including the geometry-dependent nuclear–nuclear term.

The electronic states satisfy the orthonormality condition

$$\langle \phi_\alpha(\mathbf{R}) | \phi_\beta(\mathbf{R}) \rangle_{\mathbf{q}} = \delta_{\alpha\beta}. \tag{1.9}$$

The inner product is taken only over electronic coordinates. In this compact notation, $\phi_\alpha(\mathbf{R})$ denotes the conditional electronic function $\phi_\alpha(\mathbf{q} \mid \mathbf{R})$. The functions $\phi_\alpha(\mathbf{q} \mid \mathbf{R})$ are the adiabatic electronic states, and the scalar functions $E_\alpha(\mathbf{R})$ are the corresponding adiabatic energies.

The exact molecular wavefunction can be expanded in this adiabatic electronic basis as [13, 15]

$$\Psi(\mathbf{q}, \mathbf{R}, t) = \sum_\alpha \chi_\alpha(\mathbf{R}, t)\phi_\alpha(\mathbf{q} \mid \mathbf{R}). \tag{1.10}$$

This is the Born–Huang expansion. Because $\hat{H}_{\mathrm{e}}(\mathbf{q} \mid \mathbf{R})$ is self-adjoint on the electronic Hilbert space at each fixed $\mathbf{R}$, its eigenfunctions $\{\phi_\alpha(\mathbf{q} \mid \mathbf{R})\}$ form a complete orthonormal set, and Eq. (1.10) is formally exact. The nuclear wavefunctions $\chi_\alpha(\mathbf{R}, t)$ are the expansion amplitudes associated with each adiabatic electronic state.

The essential mathematical point is that the adiabatic electronic basis depends on the nuclear coordinates. Therefore, the nuclear kinetic-energy operator does not act only on the nuclear wavefunctions χ_α. It also acts on the geometry-dependent electronic basis functions. For a single term in the expansion,

$$\nabla_A \left[\chi_\beta(\mathbf{R}, t)\phi_\beta(\mathbf{q} \mid \mathbf{R})\right] = \left(\nabla_A \chi_\beta\right) \phi_\beta + \chi_\beta \nabla_A \phi_\beta, \tag{1.11}$$

and

$$\nabla_A^2 \left[\chi_\beta(\mathbf{R}, t)\phi_\beta(\mathbf{q} \mid \mathbf{R})\right] = \left(\nabla_A^2 \chi_\beta\right) \phi_\beta + 2 \left(\nabla_A \chi_\beta\right) \cdot \left(\nabla_A \phi_\beta\right) + \chi_\beta \nabla_A^2 \phi_\beta. \tag{1.12}$$

These derivative terms are the origin of nonadiabatic coupling in the adiabatic representation.

Substituting Eq. (1.10) into the time-dependent Schrödinger equation,

$$i\hbar \frac{\partial}{\partial t} \Psi(\mathbf{q}, \mathbf{R}, t) = \hat{H} \Psi(\mathbf{q}, \mathbf{R}, t), \tag{1.13}$$

and projecting onto the electronic state ϕ_α gives the coupled nuclear equations

$$\begin{aligned} i\hbar \frac{\partial}{\partial t} \chi_\alpha(\mathbf{R}, t) = &\left[-\sum_A \frac{\hbar^2}{2M_A} \nabla_A^2 + E_\alpha(\mathbf{R}) \right] \chi_\alpha(\mathbf{R}, t) \\ &- \sum_\beta \sum_A \frac{\hbar^2}{2M_A} \left[2\mathbf{d}_A^{\alpha\beta}(\mathbf{R}) \cdot \nabla_A + D_A^{\alpha\beta}(\mathbf{R}) \right] \chi_\beta(\mathbf{R}, t). \end{aligned} \tag{1.14}$$

Here

$$\mathbf{d}_A^{\alpha\beta}(\mathbf{R}) = \left\langle \phi_\alpha(\mathbf{R}) \middle| \nabla_A \phi_\beta(\mathbf{R}) \right\rangle_{\mathbf{q}} \tag{1.15}$$

is the first-derivative nonadiabatic coupling vector, and

$$D_A^{\alpha\beta}(\mathbf{R}) = \left\langle \phi_\alpha(\mathbf{R}) \middle| \nabla_A^2 \phi_\beta(\mathbf{R}) \right\rangle_{\mathbf{q}} \tag{1.16}$$

is the second-derivative coupling. These quantities appear because the adiabatic electronic basis changes with nuclear geometry [15, 20].

It is useful to separate the purely diagonal nuclear motion from the coupling terms by defining

$$\hat{\tau}^{\alpha\beta} = -\sum_A \frac{\hbar^2}{2M_A} \left[2\mathbf{d}_A^{\alpha\beta}(\mathbf{R}) \cdot \nabla_A + D_A^{\alpha\beta}(\mathbf{R}) \right]. \tag{1.17}$$

The derivative operator in $\hat{\tau}^{\alpha\beta}$ acts on the nuclear wavefunction to its right. Then Eq. (1.14) becomes

$$i\hbar \frac{\partial}{\partial t} \chi_\alpha = \left[-\sum_A \frac{\hbar^2}{2M_A} \nabla_A^2 + E_\alpha(\mathbf{R}) \right] \chi_\alpha + \sum_\beta \hat{\tau}^{\alpha\beta} \chi_\beta. \tag{1.18}$$

The off-diagonal operators $\hat{\tau}^{\alpha\beta}$, with $\alpha \neq \beta$, couple different adiabatic nuclear wavefunctions and allow population and phase information to move between electronic states.

The adiabatic approximation is obtained by neglecting these couplings between different electronic states. In its simplest single-state form, the molecular wavefunction is approximated by

$$\Psi(\mathbf{q}, \mathbf{R}, t) \approx \chi_\alpha(\mathbf{R}, t)\phi_\alpha(\mathbf{q} \mid \mathbf{R}), \tag{1.19}$$

and the nuclear wavefunction is assumed to evolve independently on the adiabatic energy $E_\alpha(\mathbf{R})$:

$$i\hbar\frac{\partial}{\partial t}\chi_\alpha(\mathbf{R}, t) = \left[-\sum_A \frac{\hbar^2}{2M_A}\nabla_A^2 + E_\alpha(\mathbf{R})\right]\chi_\alpha(\mathbf{R}, t). \tag{1.20}$$

Equation (1.20) is the usual Born–Oppenheimer nuclear Schrödinger equation. In this approximation, the electronic state adjusts parametrically to the nuclear geometry, while transitions to other electronic states are neglected.

A more refined single-surface approximation may retain the diagonal part of the nonadiabatic kinetic coupling. In a real, nondegenerate adiabatic basis, or more generally after a suitable gauge choice away from degeneracies, the diagonal first-derivative coupling can be removed and the remaining scalar term is the diagonal Born–Oppenheimer correction. The essential adiabatic assumption, however, is the neglect of off-diagonal couplings $\hat{\tau}^{\alpha\beta}$ with $\alpha \neq \beta$, which are responsible for transitions among different electronic states.

The approximation breaks down when the coupling operators $\hat{\tau}^{\alpha\beta}$, with $\alpha \neq \beta$, become important. This can occur when electronic states are close in energy or when the electronic eigenfunctions vary rapidly with nuclear geometry. For nondegenerate, differentiable adiabatic states, and for $\alpha \neq \beta$, the first-derivative coupling can be related to the nuclear derivative of the electronic Hamiltonian. Differentiating the fixed-nuclei electronic eigenvalue problem in Eq. (1.8) with respect to $\mathbf{R}_A$, projecting onto $\phi_\alpha(\mathbf{q} \mid \mathbf{R})$, and taking $\alpha \neq \beta$, gives

$$\mathbf{d}_A^{\alpha\beta}(\mathbf{R}) = \frac{\left\langle \phi_\alpha(\mathbf{R}) \middle| \nabla_A \hat{H}_{\mathrm{e}}(\mathbf{q} \mid \mathbf{R}) \middle| \phi_\beta(\mathbf{R}) \right\rangle_{\mathbf{q}}}{E_\beta(\mathbf{R}) - E_\alpha(\mathbf{R})}, \qquad \alpha \neq \beta, \quad E_\alpha(\mathbf{R}) \neq E_\beta(\mathbf{R}). \tag{1.21}$$

This expression is valid away from degeneracies. When the energy gap becomes small, Eq. (1.21) shows why the adiabatic representation becomes fragile, provided the numerator is not simultaneously negligible. At an exact degeneracy, the same expression signals the singular character of the adiabatic representation rather than a regular finite coupling [14, 15, 21]. In such regions, the molecular wavefunction cannot be represented accurately by a single adiabatic product state, and the coupled equations in Eq. (1.14) must be considered.

Thus, the Born–Oppenheimer or adiabatic approximation is not merely the solution of the electronic problem at fixed nuclear geometry. The fixed-nuclei electronic problem defines an adiabatic electronic basis and the corresponding adiabatic energies $E_\alpha(\mathbf{R})$. The approximation enters when the derivative couplings generated by nuclear motion are neglected. Its breakdown is therefore a direct consequence of the same mathematical structure; the electronic state depends parametrically on nuclear geometry, and nuclear motion can couple different electronic states.

1.3 Adiabatic and diabatic representations of nonadiabatic dynamics

The preceding section introduced the adiabatic electronic states through the fixed-nuclei eigenvalue problem in Eq. (1.8). In this representation, the electronic Hamiltonian is diagonal by construction. The electronic energies $E_\alpha(\mathbf{R})$ define the adiabatic potential-energy surfaces that enter the Born–Huang expansion in Eq. (1.10). However, this diagonal electronic form does not eliminate coupling between electronic states. As shown in Eq. (1.14), the coupling appears when the nuclear kinetic-energy operator acts on the $\mathbf{R}$-dependent electronic basis, producing the nonadiabatic coupling operators $\hat{\tau}^{\alpha\beta}$ defined in Eq. (1.17).

Thus, the adiabatic representation has a precise mathematical structure. The electronic Hamiltonian is diagonal, but the nuclear kinetic-energy operator is not diagonal in the electronic-state index. The coupled nuclear equations in Eq. (1.14), or equivalently Eq. (1.18), show that transitions between adiabatic states are generated by the derivative couplings $\mathbf{d}_A^{\alpha\beta}(\mathbf{R})$ and $D_A^{\alpha\beta}(\mathbf{R})$, introduced in Eqs. (1.15) and (1.16). In this basis, nonadiabaticity appears as a kinetic coupling: the potential-energy matrix is diagonal, but nuclear motion couples the adiabatic components of the molecular wavefunction [15].

This structure explains why the adiabatic representation is most natural when the molecular state remains localized on a single electronic surface. If the off-diagonal operators $\hat{\tau}^{\alpha\beta}$, with $\alpha \neq \beta$, are small, the single-state approximation leading to Eq. (1.20) is meaningful. When these operators become important, however, the molecular wavefunction cannot be represented accurately by a single term of the Born–Huang expansion. The energy-gap relation in Eq. (1.21) gives one reason for this breakdown: small separations between adiabatic energies can enhance derivative coupling, provided the numerator in Eq. (1.21) is not simultaneously negligible.

A complementary description is obtained by introducing a different electronic representation. Let $\{\varphi_\lambda(\mathbf{q} \mid \mathbf{R})\}$ denote an electronic basis related to the adiabatic basis by an $\mathbf{R}$-dependent unitary transformation $U(\mathbf{R})$,

$$\varphi_\lambda(\mathbf{q} \mid \mathbf{R}) = \sum_\alpha \phi_\alpha(\mathbf{q} \mid \mathbf{R})\, U^{\alpha\lambda}(\mathbf{R}), \qquad U^\dagger(\mathbf{R})U(\mathbf{R}) = I. \tag{1.22}$$

Here α labels the adiabatic basis and λ labels the transformed electronic representation. The same molecular wavefunction can then be written as

$$\Psi(\mathbf{q}, \mathbf{R}, t) = \sum_\lambda \chi_\lambda^{\mathrm{d}}(\mathbf{R}, t)\varphi_\lambda(\mathbf{q} \mid \mathbf{R}), \tag{1.23}$$

where the nuclear amplitudes satisfy

$$\chi_\alpha(\mathbf{R},t) = \sum_\lambda U^{\alpha\lambda}(\mathbf{R})\chi^{\mathrm{d}}_\lambda(\mathbf{R},t). \tag{1.24}$$

This transformation is only a change of electronic representation. No approximation has been made unless terms generated by the transformation are later neglected.

In the basis $\{\varphi_\lambda\}$, the electronic Hamiltonian matrix is

$$V^{\lambda\lambda'}(\mathbf{R}) = \left\langle \varphi_\lambda(\mathbf{R}) \middle| \hat{H}_{\mathrm{e}}(\mathbf{q} \mid \mathbf{R}) \middle| \varphi_{\lambda'}(\mathbf{R}) \right\rangle_{\mathbf{q}} = \sum_\alpha \left[U^{\alpha\lambda}(\mathbf{R})\right]^* E_\alpha(\mathbf{R}) U^{\alpha\lambda'}(\mathbf{R}). \tag{1.25}$$

Unlike the adiabatic representation, this matrix is generally not diagonal. Its off-diagonal elements represent direct electronic coupling between the basis states φ_λ and $\varphi_{\lambda'}$. In this chapter, the symbol $V^{\lambda\lambda'}(\mathbf{R})$ is used for the electronic potential or Hamiltonian matrix in a molecular representation[1].

The corresponding derivative coupling in the transformed basis is

$$\mathbf{f}_A^{\lambda\lambda'}(\mathbf{R}) = \langle \varphi_\lambda(\mathbf{R}) | \nabla_A \varphi_{\lambda'}(\mathbf{R})\rangle_{\mathbf{q}}. \tag{1.26}$$

Using Eq. (1.22), this coupling is related to the adiabatic derivative coupling by

$$\mathbf{f}_A^{\lambda\lambda'} = \sum_{\alpha\beta} \left[U^{\alpha\lambda}\right]^* \mathbf{d}_A^{\alpha\beta} U^{\beta\lambda'} + \sum_\alpha \left[U^{\alpha\lambda}\right]^* \nabla_A U^{\alpha\lambda'}. \tag{1.27}$$

Equivalently, in compact matrix notation,

$$\mathbf{f}_A = U^\dagger \mathbf{d}_A U + U^\dagger \nabla_A U, \tag{1.28}$$

where the matrix U has elements $U^{\alpha\lambda}$, the matrix $\mathbf{d}_A$ has elements $\mathbf{d}_A^{\alpha\beta}$, and the matrix $\mathbf{f}_A$ has elements $\mathbf{f}_A^{\lambda\lambda'}$. Equation (1.27), or equivalently Eq. (1.28), shows that derivative couplings are representation-dependent objects. Under a change of electronic representation, coupling may be shifted between the nuclear kinetic-energy operator and the electronic Hamiltonian matrix [15, 22].

A diabatic representation is an electronic representation chosen, when possible, to remove or reduce the first-derivative couplings over a specified region of nuclear configuration space. In practice, this condition is usually local or approximate,

$$\mathbf{f}_A^{\lambda\lambda'}(\mathbf{R}) \approx 0, \tag{1.29}$$

[1] In later quantum–classical and mapping chapters, the symbol $h^{\lambda\lambda'}(\mathbf{R})$ will be used for the finite subsystem Hamiltonian matrix coupled to the bath phase-space variables. The two notations refer to closely related objects, but the distinction helps separate the introductory molecular representation from the later subsystem–bath and mapping-QCLE notation.

over the nuclear configuration space of interest. In the ideal strictly diabatic limit, one would have

$$\mathbf{f}_A^{\lambda\lambda'}(\mathbf{R}) = 0 \tag{1.30}$$

for all relevant nuclear coordinates, electronic states, and nuclear directions within the region considered. In such a basis, the nuclear kinetic-energy operator is diagonal in the electronic-state index, while the electronic Hamiltonian matrix $V^{\lambda\lambda'}(\mathbf{R})$ in Eq. (1.25) carries the nonadiabatic coupling through its off-diagonal elements [15, 22].

When the derivative couplings in this basis are neglected, or when a strictly diabatic basis exists locally in the region of interest, the nuclear equations take the schematic diabatic form

$$i\hbar\frac{\partial}{\partial t}\chi_\lambda^{\mathrm{d}}(\mathbf{R},t) = \hat{T}_{\mathrm{n}}\chi_\lambda^{\mathrm{d}}(\mathbf{R},t) + \sum_{\lambda'} V^{\lambda\lambda'}(\mathbf{R})\chi_{\lambda'}^{\mathrm{d}}(\mathbf{R},t). \tag{1.31}$$

Here $\hat{T}_{\mathrm{n}}$ is the nuclear kinetic-energy operator defined in Eq. (1.2). Equation (1.31) should be contrasted with Eq. (1.18). In the adiabatic representation, the potential-energy matrix is diagonal and nonadiabaticity is generated by derivative couplings. In a diabatic or approximately diabatic representation, the derivative couplings are removed, reduced, or neglected over the region of interest, and nonadiabaticity appears through the off-diagonal elements of $V^{\lambda\lambda'}(\mathbf{R})$.

The two descriptions are equivalent when the electronic basis is complete and all terms generated by the basis transformation are retained. They differ only in where the coupling is represented. The adiabatic basis is tied directly to the eigenvalues $E_\alpha(\mathbf{R})$ of the fixed-nuclei electronic problem in Eq. (1.8), whereas diabatic bases are often constructed to reflect chemically meaningful electronic configurations, such as localized charge states, excitonic states, or valence-bond-like structures [23, 24].

A globally exact diabatic basis does not generally exist. The condition $\mathbf{f}_A^{\lambda\lambda'}(\mathbf{R}) = 0$ for all nuclear coordinates, electronic states, and nuclear directions imposes an integrability condition on the derivative-coupling connection. In multidimensional nuclear configuration spaces, this condition can fail, and conical intersections introduce geometric phase effects that obstruct the construction of a smooth global strictly diabatic basis. Nevertheless, local and approximate diabatic representations remain extremely useful because they often provide the most chemically transparent description of nonadiabatic dynamics [14, 21, 22, 25].

1.4 Potential-energy surfaces and chemical pathways

The fixed-nuclei electronic problem in Eq. (1.8) produces the adiabatic energies $E_\alpha(\mathbf{R})$. These scalar functions over nuclear configuration space are the adiabatic potential-energy surfaces. The preceding sections established their mathematical origin. The present section emphasizes their chemical interpretation [13, 26, 27]. A potential-energy surface assigns an energy to each nuclear geometry for a specified electronic state. Regions where $E_\alpha(\mathbf{R})$ is locally low correspond to stable or metastable molecular structures on that electronic surface, while high-energy regions and saddle structures organize possible connections between different molecular arrangements. Thus, potential-energy surfaces provide the geometric and energetic background against which chemical change is described. For a given surface $E_\alpha(\mathbf{R})$, stationary geometries satisfy

$$\nabla_{\mathbf{R}} E_\alpha(\mathbf{R}_*) = 0. \tag{1.32}$$

The local character of such a geometry is determined by the Hessian. In Cartesian nuclear coordinates, the Hessian has block elements

$$K^\alpha_{AB}(\mathbf{R}_*) = \nabla_A \nabla_B E_\alpha(\mathbf{R})\big|_{\mathbf{R}=\mathbf{R}_*}, \tag{1.33}$$

where A and B label nuclei and each K^α_{AB} is a 3×3 Cartesian block of the full Hessian matrix. Equivalently, after introducing a single collective Cartesian index, the Hessian may be viewed as the matrix of second derivatives with respect to all nuclear coordinates. In molecular applications, this curvature analysis is often performed in mass-weighted coordinates and after removing translational and rotational degrees of freedom, so that the remaining curvature modes correspond to internal vibrational or reaction coordinates. Positive curvature directions correspond to locally restoring nuclear displacements, while negative curvature directions indicate unstable directions. In this way, minima, saddle regions, and connecting valleys on $E_\alpha(\mathbf{R})$ provide a geometric language for reactants, intermediates, transition regions, and products [27, 28]. A chemical pathway is conceptually a continuous curve through nuclear configuration space connecting reactant and product regions. This geometric picture should not be identified with the quantum description: in the Born–Huang formulation of Eq. (1.10), the nuclear state is not a single path but a set of amplitudes $\chi_\alpha(\mathbf{R}, t)$ distributed across configuration space [29]. The total nuclear probability density associated with the Born–Huang expansion is

$$n(\mathbf{R}, t) = \sum_\alpha |\chi_\alpha(\mathbf{R}, t)|^2, \tag{1.34}$$

where orthonormality of the electronic states in Eq. (1.9) has been used. The probability of finding the nuclei in a region Ω of configuration space is therefore

$$P_\Omega(t) = \int_\Omega n(\mathbf{R}, t)\, d\mathbf{R}. \tag{1.35}$$

From this perspective, a chemical process corresponds to the redistribution of nuclear probability density among regions of configuration space, together with possible redistribution of population and phase among electronic components. When one adiabatic component dominates and the coupling operators $\hat{\tau}^{\alpha\beta}$ in Eq. (1.17) are negligible, the pathway can often be interpreted primarily in terms of a single surface $E_\alpha(\mathbf{R})$. This is the regime in which the adiabatic approximation leading to Eq. (1.20) is chemically useful. However, many molecular processes cannot be described by a single surface. If two or more surfaces become close in energy, or if the electronic states change rapidly with nuclear geometry, the off-diagonal couplings in Eq. (1.18) can transfer population and phase information between electronic states. The separation between two adiabatic surfaces is measured by the energy gap

$$\Delta E^{\alpha\beta}(\mathbf{R}) = E_\alpha(\mathbf{R}) - E_\beta(\mathbf{R}). \tag{1.36}$$

Small values of $|\Delta E^{\alpha\beta}(\mathbf{R})|$ often identify regions where nonadiabatic effects may become important. This statement should be understood together with Eq. (1.21). The energy gap alone does not determine the dynamics, but it controls how strongly the nuclear derivative of the electronic Hamiltonian can appear as derivative coupling in the adiabatic representation [14, 15]. In a diabatic representation, the same physical situation is described in a different language. Instead of emphasizing adiabatic energy gaps and derivative couplings, one emphasizes interacting electronic configurations through the matrix $V^{\lambda\lambda'}(\mathbf{R})$ in Eq. (1.25). For a two-state diabatic model with real coupling, one often writes

$$V(\mathbf{R}) = \begin{pmatrix} V^{11}(\mathbf{R}) & V^{12}(\mathbf{R}) \\ V^{21}(\mathbf{R}) & V^{22}(\mathbf{R}) \end{pmatrix} = \begin{pmatrix} V_1(\mathbf{R}) & \Delta(\mathbf{R}) \\ \Delta(\mathbf{R}) & V_2(\mathbf{R}) \end{pmatrix}, \tag{1.37}$$

where $V^{12} = V^{21} = \Delta$ for a real symmetric two-state coupling. The corresponding adiabatic energies are the eigenvalues of this matrix,

$$E_\pm(\mathbf{R}) = \frac{V_1(\mathbf{R}) + V_2(\mathbf{R})}{2} \pm \sqrt{\left[\frac{V_1(\mathbf{R}) - V_2(\mathbf{R})}{2}\right]^2 + \Delta(\mathbf{R})^2}. \tag{1.38}$$

Equation (1.38) shows how an off-diagonal diabatic coupling opens an avoided crossing in the adiabatic surfaces. If $\Delta(\mathbf{R}) = 0$ at a geometry where $V_1(\mathbf{R}) = V_2(\mathbf{R})$, the two adiabatic energies become degenerate. In multidimensional nuclear configuration spaces, such degeneracies are associated with conical intersections when the required degeneracy conditions are satisfied [14, 21, 22]. The chemical value of the potential-energy-surface picture is therefore not limited to locating minima and barriers. It also identifies the regions where the adiabatic approximation may fail. Avoided crossings, conical intersections, charge-transfer resonances, bond-breaking regions, and strong solvent- or environment-induced energy-gap fluctuations are all examples of regions where a single-surface description may be insufficient. In such cases, the chemically observed pathway depends on both the topology of the potential-energy surfaces and the coupling structure connecting them [14, 15, 30]. Thus, potential-energy surfaces provide the energetic landscape for molecular structure and reactivity, while nonadiabatic couplings determine when motion through that landscape must be described using more than one electronic state. The adiabatic and diabatic representations introduced above give two complementary languages for the same physics. The adiabatic language uses surfaces, energy gaps, and derivative couplings. The diabatic language uses interacting electronic configurations and off-diagonal potential couplings. This connection between surface topology, electronic coupling, and nuclear quantum probability density is the chemical foundation for the nonadiabatic dynamics developed in the following chapters.

1.5 Quantum dynamics and density matrices

The preceding sections described molecular dynamics in terms of electronic states, nuclear wavefunctions, and the coupling terms generated by the nuclear kinetic-energy operator. For the developments that follow, it is useful to recast the same quantum dynamics in density-operator form. The density operator provides a representation-independent description of quantum dynamics and is especially convenient for discussing populations, coherences, statistical mixtures, expectation values, and reduced descriptions of selected degrees of freedom [31, 32].

For a pure molecular state $|\Psi(t)\rangle$, the density operator is

$$\hat{\rho}(t) = |\Psi(t)\rangle\langle\Psi(t)|. \tag{1.39}$$

More generally, if the molecular system is represented by an ensemble of states $\{|\Psi_k(t)\rangle\}$ with probabilities w_k, then

$$\hat{\rho}(t) = \sum_k w_k |\Psi_k(t)\rangle\langle\Psi_k(t)|, \qquad w_k \geq 0, \qquad \sum_k w_k = 1. \tag{1.40}$$

The density operator satisfies

$$\hat{\rho}^\dagger = \hat{\rho}, \qquad \hat{\rho} \geq 0, \qquad \mathrm{Tr}\,\hat{\rho} = 1. \tag{1.41}$$

A pure state is characterized by

$$\hat{\rho}^2 = \hat{\rho}, \qquad \mathrm{Tr}\,\hat{\rho}^2 = 1, \tag{1.42}$$

whereas a mixed state has

$$\mathrm{Tr}\,\hat{\rho}^2 < 1. \tag{1.43}$$

The quantity $\mathrm{Tr}\,\hat{\rho}^2$ is the purity of the quantum state.

For a closed molecular system evolving under the Hamiltonian introduced in Eq. (1.1), the density operator obeys the Liouville–von Neumann equation

$$\frac{\partial \hat{\rho}(t)}{\partial t} = -\frac{i}{\hbar}\left[\hat{H}, \hat{\rho}(t)\right]. \tag{1.44}$$

This equation is equivalent to the Schrödinger equation for a pure state, but it also applies directly to statistical mixtures. Under closed-system unitary evolution, Eq. (1.44) preserves the trace, Hermiticity, positivity, and purity of the density operator [31, 32].

Expectation values are obtained by taking traces. For an observable $\hat{A}$,

$$\langle \hat{A} \rangle(t) = \mathrm{Tr}\left[\hat{\rho}(t)\hat{A}\right]. \tag{1.45}$$

Thus, once $\hat{\rho}(t)$ is known, observables do not require choosing a particular wavefunction representation. A density matrix is obtained only after choosing a basis for the density operator. This distinction is important because the same physical state can be represented in an adiabatic basis, a diabatic basis, a position basis, or another convenient representation.

The concepts of electronic population and electronic coherence can also be defined directly from the density operator. Let $\mathcal{H}_{\mathrm{el}}$ denote the electronic Hilbert space and $\mathcal{H}_{\mathrm{n}}$ the nuclear Hilbert space. The molecular Hilbert space is then the tensor product[2]

$$\mathcal{H} = \mathcal{H}_{\mathrm{el}} \otimes \mathcal{H}_{\mathrm{n}}.$$

In the continuous nuclear coordinate representation, this same structure appears as a set of nuclear wavefunction components, one for each electronic basis state,

$$|\Psi(t)\rangle = \sum_{\lambda} \int d\mathbf{R}\, \Psi_{\lambda}(\mathbf{R}, t)|\lambda\rangle \otimes |\mathbf{R}\rangle. \tag{1.48}$$

The tensor-product structure therefore resolves the molecular state into electronic labels and nuclear coordinates while keeping them as parts of a single quantum state.

The reduced electronic density operator is obtained by tracing over the nuclear degrees of freedom,

$$\hat{\rho}_{\mathrm{el}}(t) = \mathrm{Tr}_{\mathrm{n}}\left[\hat{\rho}(t)\right]. \tag{1.49}$$

The partial trace removes the nuclear degrees of freedom from the operator description while retaining their influence on the electronic state through populations, coherences, and correlations already present in the full density operator. In a chosen electronic basis $\{|\lambda\rangle\}$, the matrix elements of the reduced electronic density operator are

$$\rho_{\mathrm{el}}^{\lambda\lambda'}(t) = \left\langle \lambda \middle| \hat{\rho}_{\mathrm{el}}(t) \middle| \lambda' \right\rangle. \tag{1.50}$$

The diagonal elements

$$P^{\lambda}(t) = \rho_{\mathrm{el}}^{\lambda\lambda}(t) \tag{1.51}$$

[2]The tensor product expresses the fact that a complete molecular state must specify both an electronic state and a nuclear state. If $\{|\lambda\rangle\}$ is a basis for $\mathcal{H}_{\mathrm{el}}$ and $\{|\mu\rangle\}$ is a basis for $\mathcal{H}_{\mathrm{n}}$, then the product vectors

$$|\lambda, \mu\rangle \equiv |\lambda\rangle \otimes |\mu\rangle \tag{1.46}$$

form a basis for the total molecular Hilbert space. For finite-dimensional spaces,

$$\dim(\mathcal{H}_{\mathrm{el}} \otimes \mathcal{H}_{\mathrm{n}}) = \dim(\mathcal{H}_{\mathrm{el}})\dim(\mathcal{H}_{\mathrm{n}}). \tag{1.47}$$

Thus, if the electronic subspace contains N_s states and the nuclear space is represented by N_R basis functions or grid points, the total molecular state has $N_s N_R$ amplitudes.

are the electronic populations, while the off-diagonal elements

$$\rho_{\rm el}^{\lambda\lambda'}(t), \qquad \lambda \neq \lambda', \tag{1.52}$$

are electronic coherences.

Equivalently, these quantities may be written as expectation values of electronic projection and transition operators. The population of state λ is

$$P^{\lambda}(t) = \mathrm{Tr}\left[\hat{\rho}(t)\left(|\lambda\rangle\langle\lambda| \otimes \hat{I}_{\rm n}\right)\right], \tag{1.53}$$

and the electronic coherence between states λ and λ' is

$$\rho_{\rm el}^{\lambda\lambda'}(t) = \mathrm{Tr}\left[\hat{\rho}(t)\left(|\lambda'\rangle\langle\lambda| \otimes \hat{I}_{\rm n}\right)\right], \qquad \lambda \neq \lambda'. \tag{1.54}$$

The reversed order in the transition operator follows from the identity $\rho_{\rm el}^{\lambda\lambda'} = \langle\lambda|\hat{\rho}_{\rm el}|\lambda'\rangle$. The diagonal quantities $P^{\lambda}(t)$ measure occupation of electronic basis states, while the off-diagonal quantities $\rho_{\rm el}^{\lambda\lambda'}(t)$ measure phase correlations between different electronic components. These phase correlations are essential in nonadiabatic dynamics because the coupling operators introduced in Eq. (1.17) can transfer amplitude between electronic components and generate coherence.

The density-operator formulation remains fully quantum. No phase-space representation has been introduced, and no degree of freedom has been replaced by a classical variable. Its role is to provide the operator language needed to describe closed molecular evolution, electronic populations, electronic coherences, expectation values, and reduced electronic states. It also prepares the ground for treating condensed-phase systems, where one is often interested in a subsystem whose state is correlated with a larger molecular environment [32, 33].

1.6 Subsystem–bath coupling in condensed phases

Many chemical processes of interest occur not in isolated molecules, but in condensed-phase environments. Solvents, proteins, molecular crystals, interfaces, and other surrounding media can shift electronic energies, modulate couplings, reshape reaction pathways, and suppress or modify electronic coherence. To describe such situations, it is useful to partition the full quantum system into a subsystem and an environment [33–35].

Let the total Hilbert space be organized as

$$\mathcal{H} = \mathcal{H}_{\rm s} \otimes \mathcal{H}_{\rm b}, \tag{1.55}$$

where $\mathcal{H}_{\rm s}$ denotes the subsystem Hilbert space and $\mathcal{H}_{\rm b}$ denotes the bath or environmental Hilbert space. The subsystem contains the degrees of freedom most directly involved in the chemical process, such as selected electronic states, a reactive molecular unit, a chromophore, a proton-transfer coordinate, or a charge-transfer complex. The bath contains the remaining molecular and environmental degrees of freedom, such as solvent coordinates, protein modes, lattice vibrations, or other surrounding nuclear coordinates.

With this partition, the Hamiltonian may be organized as

$$\hat{H} = \hat{H}_{\mathrm{s}} + \hat{H}_{\mathrm{b}} + \hat{H}_{\mathrm{sb}}. \tag{1.56}$$

Here $\hat{H}_{\mathrm{s}}$ acts on the subsystem, $\hat{H}_{\mathrm{b}}$ acts on the bath, and $\hat{H}_{\mathrm{sb}}$ contains interactions between them. Equation (1.56) is only a partition of the full quantum Hamiltonian. It does not by itself introduce a classical, semiclassical, weak-coupling, or Markovian approximation.

The density operator introduced in Sec. 1.5 acts on the full Hilbert space in Eq. (1.55). The reduced subsystem density operator is obtained by tracing over the bath,

$$\hat{\rho}_{\mathrm{s}}(t) = \mathrm{Tr}_{\mathrm{b}}\left[\hat{\rho}(t)\right], \tag{1.57}$$

and the reduced bath density operator is

$$\hat{\rho}_{\mathrm{b}}(t) = \mathrm{Tr}_{\mathrm{s}}\left[\hat{\rho}(t)\right]. \tag{1.58}$$

For an observable $\hat{A}_{\mathrm{s}}$ acting only on the subsystem, the corresponding full-space observable is $\hat{A}_{\mathrm{s}} \otimes \hat{I}_{\mathrm{b}}$, and its expectation value may be written as

$$\langle \hat{A}_{\mathrm{s}} \rangle(t) = \mathrm{Tr}_{\mathrm{s}}\left[\hat{\rho}_{\mathrm{s}}(t)\hat{A}_{\mathrm{s}}\right]. \tag{1.59}$$

Thus, the reduced subsystem density operator contains all information required to compute subsystem observables, even though the full density operator may contain correlations between subsystem and bath degrees of freedom [32, 33].

Using the matrix-element convention introduced in Sec. 1.5, the reduced subsystem density matrix in a chosen finite subsystem basis is

$$\rho_{\mathrm{s}}^{\lambda\lambda'}(t) = \left\langle \lambda \middle| \hat{\rho}_{\mathrm{s}}(t) \middle| \lambda' \right\rangle. \tag{1.60}$$

The diagonal and off-diagonal components have the population and coherence interpretation already described in Eqs. (1.51)– (1.52). In a condensed-phase system, their evolution is affected by correlations generated through $\hat{H}_{\mathrm{sb}}$. Even when the total density operator evolves unitarily according to Eq. (1.44), the reduced subsystem dynamics is generally not unitary, because tracing over the bath removes information about subsystem–environment correlations. This is the microscopic origin of relaxation, dephasing, and decoherence in reduced descriptions [33, 36].

For nonadiabatic molecular systems, the subsystem is often represented by a finite set of electronic, protonic, excitonic, or charge-transfer states. The remaining nuclear and environmental coordinates enter the Hamiltonian matrix through their influence on subsystem energies and couplings. Writing $\mathbf{R} = (\mathbf{R}_{\mathrm{s}}, \mathbf{R}_{\mathrm{b}})$, one may write the subsystem Hamiltonian matrix elements as

$h^{\lambda\lambda'}(\mathbf{R}) = h^{\lambda\lambda'}(\mathbf{R}_{\mathrm{s}}, \mathbf{R}_{\mathrm{b}})$. The bath coordinates can therefore shift diagonal subsystem energies and modulate off-diagonal couplings.

In an adiabatic description, environmental motion changes the energy gaps between subsystem states. For two states, this dependence may be written as

$$\Delta E^{\alpha\beta}(\mathbf{R}_{\mathrm{s}}, \mathbf{R}_{\mathrm{b}}) = E_{\alpha}(\mathbf{R}_{\mathrm{s}}, \mathbf{R}_{\mathrm{b}}) - E_{\beta}(\mathbf{R}_{\mathrm{s}}, \mathbf{R}_{\mathrm{b}}). \tag{1.61}$$

Since the derivative coupling in Eq. (1.21) depends on the electronic energy separation, environmental fluctuations can move the system into or out of regions where nonadiabatic effects are important.

In a diabatic or approximately diabatic representation, the same environmental influence appears through bath-dependent diabatic potentials and couplings. For two diabatic states, one may write

$$V(\mathbf{R}_{\mathrm{s}}, \mathbf{R}_{\mathrm{b}}) = \begin{pmatrix} V^{11}(\mathbf{R}_{\mathrm{s}}, \mathbf{R}_{\mathrm{b}}) & V^{12}(\mathbf{R}_{\mathrm{s}}, \mathbf{R}_{\mathrm{b}}) \\ V^{21}(\mathbf{R}_{\mathrm{s}}, \mathbf{R}_{\mathrm{b}}) & V^{22}(\mathbf{R}_{\mathrm{s}}, \mathbf{R}_{\mathrm{b}}) \end{pmatrix} = \begin{pmatrix} V_1(\mathbf{R}_{\mathrm{s}}, \mathbf{R}_{\mathrm{b}}) & \Delta(\mathbf{R}_{\mathrm{s}}, \mathbf{R}_{\mathrm{b}}) \\ \Delta^*(\mathbf{R}_{\mathrm{s}}, \mathbf{R}_{\mathrm{b}}) & V_2(\mathbf{R}_{\mathrm{s}}, \mathbf{R}_{\mathrm{b}}) \end{pmatrix}. \tag{1.62}$$

The diagonal terms describe how the environment stabilizes or destabilizes the diabatic configurations, while the off-diagonal terms describe coupling between them. Thus, the bath can influence nonadiabatic dynamics both by changing relative state energies and by changing the coupling between states [34, 35].

A useful condensed-phase quantity is the energy-gap coordinate. For two states in a diabatic representation, it may be written as

$$\mathcal{E}^{\lambda\lambda'}(\mathbf{R}_{\mathrm{s}}, \mathbf{R}_{\mathrm{b}}) = h^{\lambda\lambda}(\mathbf{R}_{\mathrm{s}}, \mathbf{R}_{\mathrm{b}}) - h^{\lambda'\lambda'}(\mathbf{R}_{\mathrm{s}}, \mathbf{R}_{\mathrm{b}}). \tag{1.63}$$

Solvent rearrangement, protein motion, lattice distortion, or other bath fluctuations can change this gap in time. Consequently, the environment can tune the subsystem toward resonance, shift avoided-crossing regions, modify the importance of electronic coupling, and influence branching between chemical pathways [30, 34, 35].

The subsystem–bath formulation introduced here remains fully quantum. It organizes the molecular problem into chemically relevant degrees of freedom and their environment, but it does not yet introduce a phase-space representation or a quantum–classical approximation. Its purpose is to identify the structure that later approximations must respect: subsystem populations and coherences are not isolated variables, but are coupled to environmental motion through the Hamiltonian and through the reduced density operator. This provides the natural bridge from the fully quantum description developed above to simple nonadiabatic transition models and, subsequently, to phase-space formulations of quantum dynamics.

1.7 Landau–Zener transitions as a minimal nonadiabatic model

The preceding sections introduced nonadiabatic dynamics as the breakdown of a single-surface adiabatic description and showed that the same physics can be represented either through derivative couplings in an adiabatic basis or through off-diagonal couplings in a diabatic basis. A minimal model that captures this mechanism is the Landau–Zener model. Its purpose here is not to provide a complete theory of molecular dynamics, but to isolate the local ingredients controlling two-state nonadiabatic transfer: an energy gap, an electronic coupling, and a finite time scale for passing through the coupling region [37–42]. Consider two diabatic states $|1\rangle$ and $|2\rangle$. Near a crossing region, the trace-subtracted two-state Hamiltonian may be written as

$$\bar{H}_{\mathrm{LZ}}(t) = \begin{pmatrix} \delta(t)/2 & \Delta \\ \Delta^* & -\delta(t)/2 \end{pmatrix}, \tag{1.64}$$

where

$$\delta(t) = E_1(t) - E_2(t) \tag{1.65}$$

is the diabatic energy gap and Δ is the electronic coupling. The standard Landau–Zener model assumes that, near the crossing,

$$\delta(t) = \nu t, \qquad \nu = \left.\frac{d\delta}{dt}\right|_{t=0}, \tag{1.66}$$

while Δ is approximately constant. The instantaneous adiabatic energies are the eigenvalues of Eq. (1.64),

$$E_{\pm}(t) = \pm\sqrt{\left[\frac{\delta(t)}{2}\right]^2 + |\Delta|^2}. \tag{1.67}$$

Thus, a diabatic crossing is converted into an avoided crossing with minimum gap

$$E_+(0) - E_-(0) = 2|\Delta|. \tag{1.68}$$

This illustrates the central physical idea: electronic coupling mixes the two diabatic states and creates a local region where population transfer can occur. For the Hamiltonian convention in Eq. (1.64), the Landau–Zener probability of remaining in the same diabatic state after passing through the crossing region is

$$P_{\mathrm{D}} = \exp\left[-\frac{2\pi|\Delta|^2}{\hbar|\nu|}\right]. \tag{1.69}$$

Here P_{D} denotes diabatic survival. Different conventions in the literature define the Landau–Zener probability with respect to either diabatic labels or adiabatic labels, so this convention must be stated explicitly. With the present convention, the probability of changing diabatic character, or equivalently following the adiabatic branch through the avoided crossing, is

$$P_{\text{follow}} = 1 - P_{\text{D}}. \tag{1.70}$$

The dimensionless parameter

$$\Gamma_{\text{LZ}} = \frac{|\Delta|^2}{\hbar|\nu|} \tag{1.71}$$

therefore measures the competition between electronic coupling and the rate of motion through the crossing. Large Γ_{LZ} corresponds to strong coupling or slow passage and favors adiabatic following. Small Γ_{LZ} corresponds to weak coupling or rapid passage and favors diabatic survival. In molecular applications, the time dependence of the gap is often interpreted as the local change of an energy gap along a nuclear or collective coordinate. If Q is such a coordinate and the system passes through $Q = Q_c$, then locally

$$\nu = \left.\frac{d\delta}{dQ}\right|_{Q_c} \left.\frac{dQ}{dt}\right|_{Q_c}. \tag{1.72}$$

This connects the Landau–Zener picture to the energy-gap coordinate and to the motion of the nuclear environment. Environmental fluctuations can change the instantaneous gap, the electronic coupling, or the effective passage time, and can therefore influence the probability of nonadiabatic transfer. The next step is to move from this local two-state picture to a phase-space description of quantum dynamics, where nuclear position and momentum information are represented through the Wigner transform without yet making a classical approximation.

1.8 Phase-space representation of quantum dynamics

The density-operator formulation introduced in Sec. 1.5 gives an exact description of quantum dynamics, but it does not place nuclear positions and momenta on equal footing. A phase-space representation provides a way to rewrite the nuclear coordinate dependence of quantum operators in terms of variables $(\mathbf{R}, \mathbf{P})$. This is the purpose of the Wigner transform [43–49].

In molecular nonadiabatic dynamics, one often Wigner transforms only the nuclear or bath coordinates while retaining a finite subsystem description for the remaining degrees of freedom. This partial Wigner transform is an exact change of representation, not a classical approximation. The subsystem matrix-element convention introduced in Sec. 1.5 will be used whenever component equations are needed.

1.8.1 The Wigner quasi-density distribution function

Let $\mathbf{R} \in \mathbb{R}^N$ denote the nuclear coordinates to be transformed, and let $\mathbf{P} \in \mathbb{R}^N$ denote their conjugate momenta. The nuclear coordinate eigenstates satisfy

$$\hat{\mathbf{R}}|\mathbf{R}\rangle = \mathbf{R}|\mathbf{R}\rangle, \qquad \int d\mathbf{R}\,|\mathbf{R}\rangle\langle\mathbf{R}| = \hat{I}_{\mathrm{n}}. \tag{1.73}$$

The Wigner transform rewrites the nuclear coordinate-space kernel of an operator using the center coordinate $\mathbf{R}$ and the relative coordinate $\mathbf{Z}$. The center coordinate becomes the phase-space position, while the Fourier conjugate of $\mathbf{Z}$ becomes the momentum.

For the density operator, the partial Wigner transform is defined as

$$\hat{\rho}_W(\mathbf{R},\mathbf{P},t) = \frac{1}{(2\pi\hbar)^N}\int d\mathbf{Z}\,\exp\left(\frac{i}{\hbar}\mathbf{P}\cdot\mathbf{Z}\right)\left\langle \mathbf{R}-\frac{\mathbf{Z}}{2}\middle|\hat{\rho}(t)\middle|\mathbf{R}+\frac{\mathbf{Z}}{2}\right\rangle. \tag{1.74}$$

For a general operator $\hat{A}$, we use the convention

$$\hat{A}_W(\mathbf{R},\mathbf{P}) = \int d\mathbf{Z}\,\exp\left(\frac{i}{\hbar}\mathbf{P}\cdot\mathbf{Z}\right)\left\langle \mathbf{R}-\frac{\mathbf{Z}}{2}\middle|\hat{A}\middle|\mathbf{R}+\frac{\mathbf{Z}}{2}\right\rangle. \tag{1.75}$$

With this convention, the factor $(2\pi\hbar)^{-N}$ is included in the density transform but not in the transform of a general operator. The Hamiltonian is transformed in the same way:

$$\hat{H}_W(\mathbf{R},\mathbf{P}) = \int d\mathbf{Z}\,\exp\left(\frac{i}{\hbar}\mathbf{P}\cdot\mathbf{Z}\right)\left\langle \mathbf{R}-\frac{\mathbf{Z}}{2}\middle|\hat{H}\middle|\mathbf{R}+\frac{\mathbf{Z}}{2}\right\rangle. \tag{1.76}$$

Expectation values then take the phase-space form

$$\langle\hat{A}\rangle(t) = \mathrm{Tr}_{\mathrm{s}}\int d\mathbf{R}\,d\mathbf{P}\,\hat{\rho}_W(\mathbf{R},\mathbf{P},t)\hat{A}_W(\mathbf{R},\mathbf{P}), \tag{1.77}$$

where Tr_{s} denotes the trace over the retained subsystem space. Equation (1.77) is the Wigner representation of $\mathrm{Tr}[\hat{\rho}(t)\hat{A}]$.

Using the matrix-element convention introduced earlier, the corresponding components are

$$\rho_W^{\lambda\lambda'}(\mathbf{R},\mathbf{P},t) = \left\langle\lambda\middle|\hat{\rho}_W(\mathbf{R},\mathbf{P},t)\middle|\lambda'\right\rangle. \tag{1.78}$$

The component form of Eq. (1.77) is

$$\langle\hat{A}\rangle(t) = \sum_{\lambda\lambda'}\int d\mathbf{R}\,d\mathbf{P}\,\rho_W^{\lambda\lambda'}(\mathbf{R},\mathbf{P},t)A_W^{\lambda'\lambda}(\mathbf{R},\mathbf{P}). \tag{1.79}$$

The reduced subsystem density matrix is recovered by integrating over the Wigner variables,

$$\rho_{\mathrm{s}}^{\lambda\lambda'}(t) = \int d\mathbf{R}\,d\mathbf{P}\,\rho_W^{\lambda\lambda'}(\mathbf{R},\mathbf{P},t). \tag{1.80}$$

Its population and coherence interpretation is the same as in Eqs. (1.51)– (1.52).

1.8.2 Properties of the Wigner density distribution function

The partial Wigner transform preserves the complete information contained in the nuclear coordinate-space kernel of the density operator. Because the transform is applied only to the nuclear or bath degrees of freedom, $\hat{\rho}_W(\mathbf{R},\mathbf{P},t)$ remains an operator in the retained subsystem Hilbert space.

The coordinate marginal follows by integrating Eq. (1.74) over momentum. Using

$$\int d\mathbf{P}\, \exp\left(\frac{i}{\hbar}\mathbf{P}\cdot\mathbf{Z}\right) = (2\pi\hbar)^N \delta(\mathbf{Z}), \tag{1.81}$$

one obtains

$$\int d\mathbf{P}\, \hat{\rho}_W(\mathbf{R},\mathbf{P},t) = \langle \mathbf{R}|\hat{\rho}(t)|\mathbf{R}\rangle\,. \tag{1.82}$$

The right-hand side is the diagonal nuclear coordinate-space kernel of the density operator and remains an operator in the subsystem space. In subsystem matrix elements,

$$\int d\mathbf{P}\, \rho_W^{\lambda\lambda'}(\mathbf{R},\mathbf{P},t) = \left\langle \lambda,\mathbf{R}\middle|\hat{\rho}(t)\middle|\lambda',\mathbf{R}\right\rangle. \tag{1.83}$$

Integration over the nuclear coordinate gives the corresponding momentum-space marginal,

$$\int d\mathbf{R}\, \hat{\rho}_W(\mathbf{R},\mathbf{P},t) = \langle \mathbf{P}|\hat{\rho}(t)|\mathbf{P}\rangle\,. \tag{1.84}$$

In subsystem matrix elements,

$$\int d\mathbf{R}\, \rho_W^{\lambda\lambda'}(\mathbf{R},\mathbf{P},t) = \left\langle \lambda,\mathbf{P}\middle|\hat{\rho}(t)\middle|\lambda',\mathbf{P}\right\rangle. \tag{1.85}$$

Equations (1.82) and (1.84) show that the partial Wigner transform reproduces the exact coordinate and momentum distributions of the transformed degrees of freedom.

The normalization of the density operator becomes

$$\mathrm{Tr}\,\hat{\rho}(t) = \mathrm{Tr_s}\int d\mathbf{R}\, d\mathbf{P}\, \hat{\rho}_W(\mathbf{R},\mathbf{P},t) = 1, \tag{1.86}$$

where $\mathrm{Tr_s}$ denotes the trace over the retained subsystem Hilbert space. In component form,

$$\sum_\lambda \int d\mathbf{R}\, d\mathbf{P}\, \rho_W^{\lambda\lambda}(\mathbf{R},\mathbf{P},t) = 1. \tag{1.87}$$

The reduced subsystem density matrix is recovered by integrating the matrix-valued Wigner density over the nuclear phase space, as stated in Eq. (1.80).

Hermiticity of the density operator implies

$$\hat{\rho}_W^{\dagger}(\mathbf{R}, \mathbf{P}, t) = \hat{\rho}_W(\mathbf{R}, \mathbf{P}, t). \tag{1.88}$$

Consequently,

$$\rho_W^{\lambda'\lambda}(\mathbf{R}, \mathbf{P}, t) = \left[\rho_W^{\lambda\lambda'}(\mathbf{R}, \mathbf{P}, t)\right]^*. \tag{1.89}$$

The diagonal components $\rho_W^{\lambda\lambda}(\mathbf{R}, \mathbf{P}, t)$ are therefore real, whereas the off-diagonal components may be complex and occur in complex-conjugate pairs. The off-diagonal components describe phase-space-resolved subsystem coherences.

Positivity of the density operator does not imply pointwise positivity of its Wigner transform. Although

$$\hat{\rho}(t) \geq 0 \tag{1.90}$$

as an operator on the full Hilbert space, the matrix $\hat{\rho}_W(\mathbf{R}, \mathbf{P}, t)$ need not be positive semidefinite at each phase-space point. In particular, its real diagonal components may take negative values. More generally, for any normalized subsystem state $|u\rangle$, the scalar projection

$$W_u(\mathbf{R}, \mathbf{P}, t) = \langle u|\hat{\rho}_W(\mathbf{R}, \mathbf{P}, t)|u\rangle \tag{1.91}$$

is real but need not be nonnegative. The off-diagonal components are generally complex and therefore do not have a positivity interpretation.

The Wigner density is consequently a quasi-probability representation rather than a classical joint probability density over simultaneously specified positions and momenta. Its possible negativity does not conflict with the positivity of physical probabilities. Observable coordinate distributions, momentum distributions, and subsystem populations are obtained from the appropriate marginals or expectation values and retain the positivity properties of the underlying density operator.

The partial Wigner transform is invertible. The nuclear coordinate-space kernel is recovered from

$$\left\langle \mathbf{R} - \frac{\mathbf{Z}}{2} \middle| \hat{\rho}(t) \middle| \mathbf{R} + \frac{\mathbf{Z}}{2} \right\rangle = \int d\mathbf{P}\, \exp\left(-\frac{i}{\hbar}\mathbf{P}\cdot\mathbf{Z}\right) \hat{\rho}_W(\mathbf{R}, \mathbf{P}, t). \tag{1.92}$$

In subsystem components,

$$\begin{aligned}
&\left\langle \lambda, \mathbf{R} - \frac{\mathbf{Z}}{2} \middle| \hat{\rho}(t) \middle| \lambda', \mathbf{R} + \frac{\mathbf{Z}}{2} \right\rangle \\
&\qquad = \int d\mathbf{P}\, \exp\left(-\frac{i}{\hbar}\mathbf{P}\cdot\mathbf{Z}\right) \rho_W^{\lambda\lambda'}(\mathbf{R}, \mathbf{P}, t).
\end{aligned} \tag{1.93}$$

The partial Wigner transform is therefore an exact change of representation. It places the transformed nuclear coordinates and momenta on equal footing while retaining the subsystem degrees of freedom in operator or matrix form.

1.8.3 Products of Wigner-transformed operators and the Moyal bracket

The Wigner transform is linear, but it does not convert the product of two operators into the ordinary pointwise product of their Wigner transforms. Let $\hat{C} = \hat{A}\hat{B}$. The Wigner transform of $\hat{C}$ is

$$\hat{C}_W(\mathbf{R}, \mathbf{P}) = (\hat{A}\hat{B})_W(\mathbf{R}, \mathbf{P}) = \hat{A}_W(\mathbf{R}, \mathbf{P}) \star \hat{B}_W(\mathbf{R}, \mathbf{P}), \tag{1.94}$$

where $\star$ denotes the Moyal product, or star product [44, 47, 48].

For the Wigner-transform convention introduced in Eq. (1.75), the star product is defined by

$$\hat{A}_W \star \hat{B}_W = \hat{A}_W \exp\left(\frac{i\hbar}{2}\Lambda\right) \hat{B}_W, \tag{1.95}$$

where

$$\Lambda = \overleftarrow{\nabla}_{\mathbf{R}} \cdot \overrightarrow{\nabla}_{\mathbf{P}} - \overleftarrow{\nabla}_{\mathbf{P}} \cdot \overrightarrow{\nabla}_{\mathbf{R}} \tag{1.96}$$

is the Janus operator. It is a bidifferential operator acting on the phase-space dependence of the Wigner symbols. The left-pointing derivatives act on the symbol to the left of Λ, whereas the right-pointing derivatives act on the symbol to its right. The derivatives in Eq. (1.96) act only on the transformed nuclear or bath variables $(\mathbf{R}, \mathbf{P})$.

The first-order action of the Janus operator is

$$\hat{A}_W \Lambda \hat{B}_W = \nabla_{\mathbf{R}} \hat{A}_W \cdot \nabla_{\mathbf{P}} \hat{B}_W - \nabla_{\mathbf{P}} \hat{A}_W \cdot \nabla_{\mathbf{R}} \hat{B}_W. \tag{1.97}$$

When A_W and B_W are scalar-valued phase-space functions, Eq. (1.97) is the canonical Poisson bracket,

$$A_W \Lambda B_W = \{A_W, B_W\}_{\mathrm{PB}}. \tag{1.98}$$

For a partial Wigner transform, however, the Wigner symbols generally remain operators in the retained subsystem Hilbert space. Their phase-space derivatives are evaluated elementwise, but the order of multiplication in the subsystem space must be preserved.

Expanding the exponential in Eq. (1.95) gives

$$\hat{A}_W \star \hat{B}_W = \sum_{n=0}^{\infty} \frac{1}{n!} \left(\frac{i\hbar}{2}\right)^n \hat{A}_W \Lambda^n \hat{B}_W. \tag{1.99}$$

The first terms of this expansion are

$$\begin{aligned} \hat{A}_W \star \hat{B}_W = \hat{A}_W \hat{B}_W &+ \frac{i\hbar}{2} \hat{A}_W \Lambda \hat{B}_W - \frac{\hbar^2}{8} \hat{A}_W \Lambda^2 \hat{B}_W \\ &- \frac{i\hbar^3}{48} \hat{A}_W \Lambda^3 \hat{B}_W + \cdots . \end{aligned} \tag{1.100}$$

Here, $\hat{A}_W \Lambda^n \hat{B}_W$ denotes the n-fold bidifferential action generated by Λ, with all left derivatives acting on $\hat{A}_W$ and all right derivatives acting on $\hat{B}_W$.

The Wigner transform of the commutator of two operators is therefore

$$[\hat{A}, \hat{B}]_W = \hat{A}_W \star \hat{B}_W - \hat{B}_W \star \hat{A}_W. \tag{1.101}$$

The corresponding Moyal bracket is defined as

$$\{\hat{A}_W, \hat{B}_W\}_{\mathrm{M}} = \frac{1}{i\hbar}\left(\hat{A}_W \star \hat{B}_W - \hat{B}_W \star \hat{A}_W\right). \tag{1.102}$$

Thus, the Moyal bracket is the Wigner representation of the quantum commutator divided by $i\hbar$.

For scalar-valued Wigner symbols, ordinary multiplication is commutative. Under interchange of the two symbols, the even powers of Λ cancel from the star commutator, while the odd powers remain. The Moyal bracket can then be written as

$$\{A_W, B_W\}_{\mathrm{M}} = \frac{2}{\hbar} A_W \sin\left(\frac{\hbar}{2}\Lambda\right) B_W. \tag{1.103}$$

Its expansion is

$$\{A_W, B_W\}_{\mathrm{M}} = A_W \Lambda B_W - \frac{\hbar^2}{24} A_W \Lambda^3 B_W + \frac{\hbar^4}{1920} A_W \Lambda^5 B_W - \cdots. \tag{1.104}$$

The leading term is the canonical Poisson bracket, whereas the higher odd powers of Λ contain quantum corrections to classical phase-space evolution.

The sine representation in Eq. (1.103) is specific to scalar-valued symbols. For subsystem-operator-valued Wigner symbols, ordinary multiplication is not generally commutative. Consequently, the cancellation of the even powers of Λ does not occur in general, and the ordered star commutator in Eq. (1.102) must be retained.

For two operators transformed according to Eq. (1.75), the trace of their product is

$$\mathrm{Tr}(\hat{A}\hat{B}) = \frac{1}{(2\pi\hbar)^N} \mathrm{Tr_s} \int d\mathbf{R}\, d\mathbf{P}\, \left(\hat{A}_W \star \hat{B}_W\right). \tag{1.105}$$

If the Wigner symbols and their relevant derivatives vanish sufficiently rapidly at the phase-space boundaries, integration by parts gives

$$\int d\mathbf{R}\, d\mathbf{P}\, \mathrm{Tr_s}\left(\hat{A}_W \star \hat{B}_W\right) = \int d\mathbf{R}\, d\mathbf{P}\, \mathrm{Tr_s}\left(\hat{A}_W \hat{B}_W\right). \tag{1.106}$$

It follows that

$$\mathrm{Tr}(\hat{A}\hat{B}) = \frac{1}{(2\pi\hbar)^N} \mathrm{Tr_s} \int d\mathbf{R}\, d\mathbf{P}\, \hat{A}_W(\mathbf{R}, \mathbf{P}) \hat{B}_W(\mathbf{R}, \mathbf{P}). \tag{1.107}$$

In subsystem matrix elements, the Wigner transform of an operator product is

$$(\hat{A}\hat{B})_W^{\lambda\lambda'} = \sum_\kappa A_W^{\lambda\kappa} \star B_W^{\kappa\lambda'}. \tag{1.108}$$

The subsystem index κ is contracted according to matrix multiplication, whereas the star product acts only on the Wigner variables $(\mathbf{R}, \mathbf{P})$.

1.8.4 The Wigner-transformed quantum evolution equation

The density operator satisfies the Liouville–von Neumann equation,

$$\frac{\partial \hat{\rho}(t)}{\partial t} = -\frac{i}{\hbar}[\hat{H}, \hat{\rho}(t)]. \tag{1.109}$$

Applying the partial Wigner transform and using Eq. (1.101) gives the exact phase-space evolution

$$\frac{\partial \hat{\rho}_W}{\partial t} = -\frac{i}{\hbar}\left(\hat{H}_W \star \hat{\rho}_W - \hat{\rho}_W \star \hat{H}_W\right) = \{\hat{H}_W, \hat{\rho}_W\}_{\mathrm{M}}. \tag{1.110}$$

No quantum–classical approximation has been introduced in Eq. (1.110). The star product acts only on the Wigner-transformed nuclear or bath variables, while the order of multiplication in the retained subsystem Hilbert space remains unchanged.

For the subsystem–bath partition used in the following chapters, the partially Wigner-transformed Hamiltonian is written as

$$\hat{H}_W(\mathbf{X}) = H_b(\mathbf{X})\hat{I}_{\mathrm{s}} + \hat{h}(\mathbf{R}), \qquad H_b(\mathbf{X}) = \frac{\mathbf{P}^2}{2M} + V_b(\mathbf{R}), \tag{1.111}$$

where M is the common bath mass used in the quantum–classical model, V_b is the scalar bath potential, and $\hat{h}(\mathbf{R})$ is the finite subsystem Hamiltonian, including its coupling to the bath coordinates. This notation is retained in Chapters 2–5.

In subsystem matrix elements, Eq. (1.110) becomes

$$\frac{\partial \rho_W^{\lambda\lambda'}(\mathbf{X}, t)}{\partial t} = -\frac{i}{\hbar}\sum_{\kappa}\left[H_W^{\lambda\kappa} \star \rho_W^{\kappa\lambda'} - \rho_W^{\lambda\kappa} \star H_W^{\kappa\lambda'}\right]. \tag{1.112}$$

The partial Wigner transform therefore introduces a phase-space representation of the bath without replacing it by classical dynamics. The controlled mass-ratio expansion that converts Eq. (1.110) into the quantum–classical Liouville equation is derived once, in Sec. 2.3.1.

1.9 Thesis objective and organization

The objective is to determine whether the single product-GP/moving-cloud/ explicit-MIDPOINT construction tested here can make the excess mapping-QCLE operator computationally accessible on a one-dimensional, two-state avoided-crossing benchmark. The Meyer–Miller–Stock–Thoss representation places the finite electronic subsystem and classical environment in a common extended phase space [1, 18, 19, 50]. Hamiltonian PBME propagation transports the support, while the GP posterior mean represents the density between support points and supplies analytic derivatives. An explicit midpoint update then applies the excess density operator.

The thesis distinguishes formulation from validation. Constructing the finite-dimensional operator verifies that the continuum equations have been translated into an executable numerical method; it does not establish that the chosen density representation determines the required derivatives or improves the physical solution. Manufactured operators, projection residuals, like-for-like reconstruction controls, refinement studies, replicated support clouds, and raw conservation diagnostics are therefore treated as part of the method rather than as optional post-processing [51, 52].

Chapter 2 derives the QCLE and defines the reference hierarchy. Chapter 3 develops the mapping representation and its SEO projection. Chapter 4 introduces and tests the fixed-time GP density model, and Chapter 5 constructs the density-level time integrator. Chapter 6 presents the controlled validation sequence before comparing the physical dynamics. Chapter 7 separates demonstrated conclusions from limitations and future reformulations.

Taken together, the introduction specifies the exact excess-operator problem, its accuracy–scalability motivation, the moving-cloud product-GP angle, the application-specific contribution, and the negative outcome. The remaining chapters derive and test that single construction.

Chapter 2

Mixed Quantum–Classical Approaches to Nonadiabatic Dynamics

This background chapter asks which approximations connect exact partial-Wigner dynamics to the QCLE and PBME hierarchy used later. That hierarchy is essential because it fixes the continuum equation that the numerical prototype attempts to discretize and prevents a numerical grid-QCLE reference from being described as exact. The derivation proceeds through the mass-ratio truncation and then reviews the semiclassical and mean-field reductions needed for the mapping formulation. The outcome is the generator and approximation hierarchy on which the later construction rests; this chapter provides background and makes no claim of original methodological contribution.

Chapter 1 introduced the exact partial-Wigner representation of quantum dynamics, including the Wigner transform, the Moyal product, the Janus operator, and the corresponding Wigner-transformed Liouville–von Neumann equation. The present chapter develops the approximate dynamical descriptions needed for mixed quantum–classical nonadiabatic systems. It first reviews semiclassical initial value representations and then derives the quantum–classical Liouville equation (QCLE) from the exact partial-Wigner dynamics by a mass-ratio expansion. The QCLE is subsequently written in adiabatic and coordinate-independent subsystem bases, followed by its Ehrenfest mean-field reduction. These formulations provide the starting point for the mapping representation developed in Chapter 3.

Throughout this chapter, $\mathbf{X} = (\mathbf{R}, \mathbf{P})$ denotes a phase-space point associated with the Wigner-transformed nuclear variables. Beginning with Sec. 2.3, these variables refer specifically to the bath degrees of freedom. A common scalar bath mass M is used, so that the bath kinetic energy and velocity are $\mathbf{P}^2/(2M)$ and $\mathbf{P}/M$, respectively.

2.1 From exact Wigner dynamics to approximate phase-space methods

The exact Wigner-transformed equation derived in Sec. 1.8 is equivalent to the Liouville–von Neumann equation. Its phase-space form does not reduce the amount of information required to represent an exact quantum density. Rather, it exposes the differential structure from which semiclassical and quantum–classical approximations can be organized.

Let N_R and N_P denote the numbers of grid points used for the coordinate and momentum representations of the transformed nuclear degrees of freedom. A scalar Wigner density requires

$$N_W = N_R N_P \tag{2.1}$$

values. If the coordinate and momentum resolutions are comparable, then

$$N_W \sim N_R^2. \tag{2.2}$$

For a subsystem containing N_s states, the matrix-valued Wigner density contains one phase-space field for each subsystem matrix element and therefore requires

$$N_W^{\mathrm{s}} = N_s^2 N_R N_P \sim N_s^2 N_R^2 \tag{2.3}$$

values. In more than one nuclear dimension, both N_R and N_P already represent multidimensional grids, so the storage and propagation costs grow rapidly with the number of transformed degrees of freedom.

The methods considered below reduce this exact density-level problem in different ways. Semiclassical initial value representations approximate a quantum propagator by classical trajectories carrying action phases and stability information. The QCLE instead retains a quantum subsystem in operator form and applies a controlled mass-ratio expansion only to the partially Wigner-transformed bath dynamics [8, 17, 53–55].

2.2 Semiclassical initial value representations

Semiclassical initial value representations approximate quantum propagation using classical trajectories augmented by phase and stability information. The classical action supplies the oscillatory phase, while derivatives of the Hamiltonian flow determine the semiclassical amplitude. These methods avoid direct propagation of a full density matrix but retain more dynamical information than a purely classical trajectory ensemble.

2.2.1 Van Vleck semiclassical propagator

Let D denote the number of nuclear coordinates. The exact coordinate-space propagator is

$$K(\mathbf{R}_t, t; \mathbf{R}_0, 0) = \left\langle \mathbf{R}_t \middle| e^{-i\hat{H}t/\hbar} \middle| \mathbf{R}_0 \right\rangle . \tag{2.4}$$

The Van Vleck approximation expresses this propagator as a sum over classical trajectories j that connect $\mathbf{R}_0$ to $\mathbf{R}_t$ in time t:

$$\begin{aligned} K_{\mathrm{vV}}(\mathbf{R}_t, t; \mathbf{R}_0, 0) = \sum_j \left(\frac{1}{2\pi i \hbar} \right)^{D/2} & \left| \det \left[- \frac{\partial^2 S_j(\mathbf{R}_t, \mathbf{R}_0; t)}{\partial \mathbf{R}_t \, \partial \mathbf{R}_0} \right] \right|^{1/2} \\ & \times \exp \left[\frac{i}{\hbar} S_j(\mathbf{R}_t, \mathbf{R}_0; t) - i \frac{\pi}{2} \nu_j \right] , \end{aligned} \tag{2.5}$$

where

$$S_j = \int_0^t \Big[\mathbf{P}_j(\tau) \cdot \dot{\mathbf{R}}_j(\tau) - H(\mathbf{R}_j(\tau), \mathbf{P}_j(\tau)) \Big] \, d\tau \tag{2.6}$$

is the classical action and ν_j is the Maslov index [53, 55]. The determinant in Eq. (2.5) measures the focusing or defocusing of nearby classical trajectories.

The Van Vleck expression is a boundary-value representation: all classical paths joining the prescribed endpoints must be found. Initial value representations replace this boundary-value search by integration over initial phase-space conditions.

2.2.2 Herman–Kluk initial value representation

The Herman–Kluk propagator is a coherent-state initial value representation in which each initial point $(\mathbf{R}_0, \mathbf{P}_0)$ is propagated by classical Hamiltonian dynamics [54, 55]. A frozen Gaussian coherent state centered at $(\mathbf{R}_c, \mathbf{P}_c)$ is written as

$$\begin{aligned} \langle \mathbf{R} | g_{\mathbf{R}_c \mathbf{P}_c} \rangle = \left(\frac{\det \Gamma}{\pi^D} \right)^{1/4} \exp \Bigg[& -\frac{1}{2} (\mathbf{R} - \mathbf{R}_c)^{\mathrm{T}} \Gamma (\mathbf{R} - \mathbf{R}_c) \\ & + \frac{i}{\hbar} \mathbf{P}_c \cdot (\mathbf{R} - \mathbf{R}_c) \Bigg] , \end{aligned} \tag{2.7}$$

where Γ is a positive width matrix. With this convention, the Herman–Kluk propagator is

$$\begin{aligned} \hat{U}_{\mathrm{HK}}(t) = \frac{1}{(2\pi\hbar)^D} & \int d\mathbf{R}_0 \, d\mathbf{P}_0 \, C_t(\mathbf{R}_0, \mathbf{P}_0) \exp \left[\frac{i}{\hbar} S_t(\mathbf{R}_0, \mathbf{P}_0) \right] \\ & \times | g_{\mathbf{R}_t \mathbf{P}_t} \rangle \langle g_{\mathbf{R}_0 \mathbf{P}_0} | , \end{aligned} \tag{2.8}$$

where $(\mathbf{R}_t, \mathbf{P}_t)$ and S_t are obtained from the classical trajectory initiated at $(\mathbf{R}_0, \mathbf{P}_0)$.

The prefactor depends on the monodromy matrix

$$\mathsf{M}_t = \begin{pmatrix} \mathsf{M}_{RR} & \mathsf{M}_{RP} \\ \mathsf{M}_{PR} & \mathsf{M}_{PP} \end{pmatrix} = \begin{pmatrix} \partial \mathbf{R}_t/\partial \mathbf{R}_0 & \partial \mathbf{R}_t/\partial \mathbf{P}_0 \\ \partial \mathbf{P}_t/\partial \mathbf{R}_0 & \partial \mathbf{P}_t/\partial \mathbf{P}_0 \end{pmatrix}. \tag{2.9}$$

For the coherent-state convention in Eq. (2.7), one standard form is

$$C_t = \left\{ \det \left[\frac{1}{2} \left(\mathsf{M}_{RR} + \mathsf{M}_{PP} - i\hbar\Gamma \mathsf{M}_{RP} + \frac{i}{\hbar}\Gamma^{-1}\mathsf{M}_{PR} \right) \right] \right\}^{1/2}. \tag{2.10}$$

The trajectory is classical, while the action and monodromy matrix carry the semiclassical phase and stability information. The oscillatory phase and the prefactor make direct Herman–Kluk calculations increasingly difficult as the number of degrees of freedom grows.

2.2.3 Linearized semiclassical initial value representation

The linearized semiclassical initial value representation (LSC-IVR) is obtained by linearizing the phase difference between the forward and backward semiclassical paths entering a quantum time-correlation function [55, 56]. Consider

$$C_{AB}(t) = \mathrm{Tr}\left[\hat{\rho}\, \hat{A}\, e^{i\hat{H}t/\hbar} \hat{B} e^{-i\hat{H}t/\hbar} \right]. \tag{2.11}$$

Using the Wigner-transform conventions and star product introduced in Sec. 1.8, the initial operator product is represented by

$$(\hat{\rho}\hat{A})_W = \rho_W \star A_W. \tag{2.12}$$

LSC-IVR replaces the exact evolution of the Wigner symbol of $\hat{B}$ by classical transport along

$$\mathbf{X}_t = \Phi^t(\mathbf{X}), \tag{2.13}$$

where Φ^t is the classical Hamiltonian flow. The resulting approximation is

$$C_{AB}^{\mathrm{LSC}}(t) = \int d\mathbf{X}\, \left[\rho_W \star A_W \right](\mathbf{X}) B_W(\mathbf{X}_t). \tag{2.14}$$

No additional normalization factor appears because the density transform was normalized in Eq. (1.74).

Replacing the initial star product by an ordinary product,

$$\rho_W \star A_W \approx \rho_W A_W, \tag{2.15}$$

produces the classical-Wigner expression

$$C_{AB}^{\mathrm{CW}}(t) = \int d\mathbf{X}\, \rho_W(\mathbf{X}) A_W(\mathbf{X}) B_W(\mathbf{X}_t). \tag{2.16}$$

The approximation in Eq. (2.15) is separate from the linearization of the time evolution. For Hamiltonians that are at most quadratic in the phase-space variables, the Moyal evolution of a Wigner symbol reduces to classical Liouville transport. For anharmonic systems, LSC-IVR does not retain the full forward–backward interference structure of the underlying semiclassical propagator.

2.3 The quantum–classical Liouville equation

The QCLE is obtained by partitioning the molecular system into a quantum subsystem and a heavier bath. Only the bath variables are Wigner transformed, and the exact bath Moyal product is expanded in the ratio of characteristic subsystem and bath masses. The subsystem remains an operator throughout the derivation [8, 17].

2.3.1 Mass-ratio expansion

Let m denote a characteristic light-subsystem mass and M the common bath mass. The expansion parameter is

$$\varepsilon_m = \left(\frac{m}{M}\right)^{1/2}, \qquad \varepsilon_m \ll 1. \tag{2.17}$$

Choose a positive characteristic energy E_{ref} and introduce the reference scales

$$\lambda_m = \left(\frac{\hbar^2}{m E_{\mathrm{ref}}}\right)^{1/2}, \qquad P_M = (M E_{\mathrm{ref}})^{1/2}. \tag{2.18}$$

With

$$\mathbf{R} = \lambda_m \widetilde{\mathbf{R}}, \qquad \mathbf{P} = P_M \widetilde{\mathbf{P}}, \tag{2.19}$$

the differential operator in the star-product exponent satisfies

$$\hbar\Lambda = \varepsilon_m \widetilde{\Lambda}, \tag{2.20}$$

where Λ is the Janus operator introduced in Sec. 1.8.3 and $\widetilde{\Lambda}$ has the same form in the scaled variables. Hence, the bath star product defined in Eq. (1.95) becomes

$$\hat{A}_W \star \hat{B}_W = \hat{A}_W \exp\left(\frac{i\varepsilon_m}{2}\widetilde{\Lambda}\right) \hat{B}_W. \tag{2.21}$$

Expansion through first order in ε_m gives

$$\hat{A}_W \star \hat{B}_W = \hat{A}_W \hat{B}_W + \frac{i\hbar}{2} \hat{A}_W \Lambda \hat{B}_W + \mathcal{O}(\varepsilon_m^2). \tag{2.22}$$

Substituting Eq. (2.22) into the exact partially Wigner-transformed Liouville–von Neumann equation in Eq. (1.110) retains the subsystem commutator and truncates only the bath Moyal structure.

2.3.2 Quantum–classical Liouville equation

For two subsystem operators that depend on the bath phase-space point, define

$$\{\hat{A}_W, \hat{B}_W\}_{\mathbf{X}} = \nabla_{\mathbf{R}} \hat{A}_W \cdot \nabla_{\mathbf{P}} \hat{B}_W - \nabla_{\mathbf{P}} \hat{A}_W \cdot \nabla_{\mathbf{R}} \hat{B}_W. \tag{2.23}$$

The derivatives act only on the bath variables, while the order of subsystem operator multiplication is retained. The first-order mass-ratio truncation gives

$$\frac{\partial \hat{\rho}_W}{\partial t} = -\frac{i}{\hbar}[\hat{H}_W, \hat{\rho}_W] + \frac{1}{2}\left(\{\hat{H}_W, \hat{\rho}_W\}_{\mathbf{X}} - \{\hat{\rho}_W, \hat{H}_W\}_{\mathbf{X}}\right). \tag{2.24}$$

The second term is an antisymmetrized combination of operator-valued phase-space brackets. Equation (2.24) is written in Liouville form as

$$\frac{\partial \hat{\rho}_W}{\partial t} = -i\hat{\mathcal{L}}\hat{\rho}_W. \tag{2.25}$$

For the subsystem–bath Hamiltonian

$$\hat{H}_W(\mathbf{X}) = \left[\frac{\mathbf{P}^2}{2M} + V_b(\mathbf{R})\right] \hat{I}_{\mathrm{s}} + \hat{h}(\mathbf{R}), \tag{2.26}$$

Eq. (2.24) becomes

$$\begin{aligned}\frac{\partial \hat{\rho}_W}{\partial t} = &-\frac{i}{\hbar}[\hat{h}(\mathbf{R}), \hat{\rho}_W] - \frac{\mathbf{P}}{M} \cdot \nabla_{\mathbf{R}} \hat{\rho}_W + \nabla_{\mathbf{R}} V_b(\mathbf{R}) \cdot \nabla_{\mathbf{P}} \hat{\rho}_W \\ &+ \frac{1}{2}\left[\nabla_{\mathbf{R}} \hat{h}(\mathbf{R}) \cdot \nabla_{\mathbf{P}} \hat{\rho}_W + \nabla_{\mathbf{P}} \hat{\rho}_W \cdot \nabla_{\mathbf{R}} \hat{h}(\mathbf{R})\right].\end{aligned} \tag{2.27}$$

The first term generates subsystem quantum evolution. The second and third terms generate bath streaming under the bath-only Hamiltonian. The final line contains the subsystem-dependent bath back-reaction.

In the one-state limit, the subsystem commutator vanishes and Eq. (2.24) reduces to

$$\frac{\partial \rho_W}{\partial t} = \{H_W, \rho_W\}_{\mathbf{X}}, \tag{2.28}$$

which is the classical Liouville equation for the bath phase-space density.

2.3.3 Density-level computational structure

After a finite subsystem basis has been selected, $\hat{\rho}_W(\mathbf{X}, t)$ becomes a matrix of phase-space fields. The scaling discussed in Sec. 2.1 therefore remains relevant: the QCLE propagates subsystem–bath correlations over the bath phase space rather than a single bath trajectory. Practical methods consequently seek trajectory or basis representations that avoid a direct multidimensional grid while preserving the operator-valued density structure of Eq. (2.24).

2.4 Adiabatic-basis QCLE and trajectory branching

The QCLE may be represented in the adiabatic eigenbasis of $\hat{h}(\mathbf{R})$,

$$\hat{h}(\mathbf{R})|\alpha; \mathbf{R}\rangle = E_\alpha(\mathbf{R})|\alpha; \mathbf{R}\rangle. \tag{2.29}$$

The total adiabatic potential governing bath motion is

$$U_\alpha(\mathbf{R}) = V_b(\mathbf{R}) + E_\alpha(\mathbf{R}), \tag{2.30}$$

and the corresponding force is

$$\mathbf{F}^\alpha(\mathbf{R}) = -\nabla_{\mathbf{R}} U_\alpha(\mathbf{R}). \tag{2.31}$$

The derivative-coupling vectors and adiabatic energy gaps are those defined in Eqs. (1.15) and (1.36). The phase-space-resolved adiabatic density components are

$$\rho_W^{\alpha\alpha'}(\mathbf{X}, t) = \langle \alpha; \mathbf{R}|\hat{\rho}_W(\mathbf{X}, t)|\alpha'; \mathbf{R}\rangle. \tag{2.32}$$

2.4.1 QCLE in the adiabatic representation

The adiabatic-basis QCLE can be written as

$$\frac{\partial}{\partial t}\rho_W^{\alpha\alpha'}(\mathbf{X}, t) = -\left[i\omega^{\alpha\alpha'}(\mathbf{R}) + iL^{\alpha\alpha'}\right]\rho_W^{\alpha\alpha'}(\mathbf{X}, t) + \sum_{\beta\beta'} J^{\alpha\alpha',\beta\beta'}(\mathbf{X})\rho_W^{\beta\beta'}(\mathbf{X}, t), \tag{2.33}$$

where

$$\omega^{\alpha\alpha'}(\mathbf{R}) = \frac{E_\alpha(\mathbf{R}) - E_{\alpha'}(\mathbf{R})}{\hbar} \tag{2.34}$$

is the Bohr frequency and

$$iL^{\alpha\alpha'} = \frac{\mathbf{P}}{M}\cdot\nabla_{\mathbf{R}} + \frac{1}{2}\left[\mathbf{F}^\alpha(\mathbf{R}) + \mathbf{F}^{\alpha'}(\mathbf{R})\right]\cdot\nabla_{\mathbf{P}} \tag{2.35}$$

is the phase-space transport operator [17, 57, 58].

For $\alpha = \alpha'$, Eq. (2.35) reduces to classical transport on $U_\alpha(\mathbf{R})$. For $\alpha \neq \alpha'$, the coherence is transported by the average force

$$\mathbf{F}^{\alpha\alpha'}_{\text{mean}}(\mathbf{R}) = \frac{1}{2}\left[\mathbf{F}^{\alpha}(\mathbf{R}) + \mathbf{F}^{\alpha'}(\mathbf{R})\right]. \tag{2.36}$$

This average defines the transport field of the coherence component; it is not an additional adiabatic electronic surface. The coherence also accumulates the phase generated by $\omega^{\alpha\alpha'}$ and remains coupled to other density components through J.

2.4.2 Nonadiabatic transition operator

With the sign convention in Eq. (2.33), the transition operator is

$$\begin{aligned} J^{\alpha\alpha',\beta\beta'}(\mathbf{X}) = &-\mathbf{d}^{\alpha\beta}(\mathbf{R}) \cdot \left[\frac{\mathbf{P}}{M} + \frac{1}{2}\Delta E^{\alpha\beta}(\mathbf{R})\nabla_{\mathbf{P}}\right]\delta_{\alpha'\beta'} \\ &- \left[\mathbf{d}^{\alpha'\beta'}(\mathbf{R})\right]^{*} \cdot \left[\frac{\mathbf{P}}{M} + \frac{1}{2}\Delta E^{\alpha'\beta'}(\mathbf{R})\nabla_{\mathbf{P}}\right]\delta_{\alpha\beta}. \end{aligned} \tag{2.37}$$

The momentum derivatives act on the density components to the right. The first line changes the left adiabatic index, while the second changes the right index. Population and coherence components are therefore coupled within the same density-matrix evolution.

The factors $\mathbf{P} \cdot \mathbf{d}^{\alpha\beta}/M$ describe transitions induced by motion through the coordinate-dependent adiabatic basis. The momentum derivatives account for the associated bath back-reaction. Their magnitude is governed by the combined local behavior of the derivative coupling, energy gap, and bath velocity, rather than by the energy gap alone.

2.4.3 Momentum-jump approximation

The momentum-jump approximation replaces the momentum-derivative action in Eq. (2.37) by a finite translation along the derivative-coupling direction [57, 58]. In a region where a real adiabatic gauge is used and $\mathbf{d}^{\alpha\beta}(\mathbf{R}) \neq \mathbf{0}$, define

$$\widehat{\mathbf{d}}^{\alpha\beta}(\mathbf{R}) = \frac{\mathbf{d}^{\alpha\beta}(\mathbf{R})}{|\mathbf{d}^{\alpha\beta}(\mathbf{R})|}. \tag{2.38}$$

The translation operator acts as

$$\hat{\jmath}^{\alpha\beta} f(\mathbf{P}) = f\left(\mathbf{P} + \Delta\mathbf{P}^{\alpha\beta}\right), \tag{2.39}$$

where, with the energy-gap convention of Eq. (1.36),

$$\begin{aligned} \Delta\mathbf{P}^{\alpha\beta} = \widehat{\mathbf{d}}^{\alpha\beta}\Bigg[&\operatorname{sgn}\left(\mathbf{P}\cdot\widehat{\mathbf{d}}^{\alpha\beta}\right)\sqrt{\left(\mathbf{P}\cdot\widehat{\mathbf{d}}^{\alpha\beta}\right)^2 + M\Delta E^{\alpha\beta}} \\ &-\mathbf{P}\cdot\widehat{\mathbf{d}}^{\alpha\beta}\Bigg]. \end{aligned} \tag{2.40}$$

Only the momentum component parallel to $\widehat{\mathbf{d}}^{\alpha\beta}$ changes. If the square-root argument is negative, the attempted transition has no real momentum translation.

The momentum-jump transition operator is

$$
\begin{aligned}
J_{\mathrm{MJ}}^{\alpha\alpha',\beta\beta'}(\mathbf{X}) = &- \left[\frac{\mathbf{P}}{M} \cdot \mathbf{d}^{\alpha\beta}(\mathbf{R})\right] \hat{j}^{\alpha\beta} \delta_{\alpha'\beta'} \\
&- \left[\frac{\mathbf{P}}{M} \cdot \mathbf{d}^{\alpha'\beta'}(\mathbf{R})\right]^* \hat{j}^{\alpha'\beta'} \delta_{\alpha\beta}.
\end{aligned} \tag{2.41}
$$

A single QCLE transition changes one density-matrix index. This is why Eq. (2.40) contains $M\Delta E^{\alpha\beta}$, whereas a direct population-to-population surface-hopping momentum adjustment contains the full factor $2M\Delta E^{\alpha\beta}$.

2.4.4 Surface-hopping-like trajectory representation

A short-time propagator may be factorized into Bohr-phase, continuous transport, and transition contributions. To first order in Δt,

$$
\left(e^{-i\mathcal{L}\Delta t}\right)^{\alpha\alpha',\beta\beta'} \approx W^{\alpha\alpha'}(t+\Delta t, t) e^{-iL^{\alpha\alpha'}\Delta t} \left[\delta_{\alpha\beta}\delta_{\alpha'\beta'} + \Delta t\, J_{\mathrm{MJ}}^{\alpha\alpha',\beta\beta'}(\mathbf{X})\right], \tag{2.42}
$$

where

$$
W^{\alpha\alpha'}(t+\Delta t, t) = \exp\left[-i\int_t^{t+\Delta t} \omega^{\alpha\alpha'}(\mathbf{R}_s)\, ds\right]. \tag{2.43}
$$

Repeated application generates a branched ensemble containing both population and coherence segments. This density-matrix trajectory construction is surface-hopping-like, but it is not identical to the fewest-switches surface-hopping algorithm. The two approaches propagate different dynamical objects and use different transition prescriptions; their connection requires additional approximations to the QCLE branching structure.

2.5 QCLE in a fixed subsystem basis

Let $\{|\lambda\rangle\}_{\lambda=1}^{N_s}$ be an orthonormal subsystem basis independent of $\mathbf{R}$. Coordinate independence removes the derivative couplings generated by an $\mathbf{R}$-dependent basis. It does not require the subsystem Hamiltonian to be diagonal, and it should not by itself be interpreted as the existence of a globally diabatic molecular basis.

The matrix elements of the subsystem Hamiltonian and density are

$$
h^{\lambda\lambda'}(\mathbf{R}) = \langle\lambda|\hat{h}(\mathbf{R})|\lambda'\rangle, \qquad \rho_W^{\lambda\lambda'}(\mathbf{X}, t) = \langle\lambda|\hat{\rho}_W(\mathbf{X}, t)|\lambda'\rangle. \tag{2.44}
$$

These definitions will be used in the mapping representation of Chapter 3.

2.5.1 Fixed-basis QCLE

Projecting Eq. (2.27) onto the fixed basis gives

$$\begin{aligned}\frac{\partial}{\partial t}\rho_W^{\lambda\lambda'}(\mathbf{X},t) = &-\frac{i}{\hbar}\sum_{\lambda''}\left[h^{\lambda\lambda''}(\mathbf{R})\rho_W^{\lambda''\lambda'}(\mathbf{X},t) - \rho_W^{\lambda\lambda''}(\mathbf{X},t)h^{\lambda''\lambda'}(\mathbf{R})\right] \\ &- \frac{\mathbf{P}}{M}\cdot\nabla_{\mathbf{R}}\rho_W^{\lambda\lambda'}(\mathbf{X},t) + \nabla_{\mathbf{R}}V_b(\mathbf{R})\cdot\nabla_{\mathbf{P}}\rho_W^{\lambda\lambda'}(\mathbf{X},t) \\ &+ \frac{1}{2}\sum_{\lambda''}\Big[\nabla_{\mathbf{R}}h^{\lambda\lambda''}(\mathbf{R})\cdot\nabla_{\mathbf{P}}\rho_W^{\lambda''\lambda'}(\mathbf{X},t) \\ &\qquad + \nabla_{\mathbf{P}}\rho_W^{\lambda\lambda''}(\mathbf{X},t)\cdot\nabla_{\mathbf{R}}h^{\lambda''\lambda'}(\mathbf{R})\Big]. \end{aligned} \tag{2.45}$$

The first line contains the subsystem commutator, the second line contains bath streaming, and the final two lines contain the subsystem-dependent bath back-reaction. Off-diagonal elements of $h^{\lambda\lambda'}(\mathbf{R})$ generate population transfer and coherence in the fixed basis.

For a one-state subsystem, Eq. (2.45) reduces to the classical Liouville equation on the scalar potential $V_b(\mathbf{R}) + h(\mathbf{R})$.

2.5.2 Bath back-reaction

The bath momentum expectation is

$$\langle\mathbf{P}\rangle(t) = \mathrm{Tr_s}\int d\mathbf{X}\,\mathbf{P}\,\hat{\rho}_W(\mathbf{X},t). \tag{2.46}$$

Multiplying Eq. (2.27) by $\mathbf{P}$, integrating over phase space, and assuming that the density and its required derivatives vanish at the phase-space boundaries gives

$$\frac{d}{dt}\langle\mathbf{P}\rangle = -\mathrm{Tr_s}\int d\mathbf{X}\,\hat{\rho}_W(\mathbf{X},t)\nabla_{\mathbf{R}}\left[V_b(\mathbf{R})\hat{I}_s + \hat{h}(\mathbf{R})\right]. \tag{2.47}$$

In the fixed basis,

$$\begin{aligned}\frac{d}{dt}\langle\mathbf{P}\rangle = &-\int d\mathbf{X}\left[\sum_{\lambda}\rho_W^{\lambda\lambda}(\mathbf{X},t)\right]\nabla_{\mathbf{R}}V_b(\mathbf{R}) \\ &-\int d\mathbf{X}\sum_{\lambda\lambda'}\rho_W^{\lambda'\lambda}(\mathbf{X},t)\nabla_{\mathbf{R}}h^{\lambda\lambda'}(\mathbf{R}).\end{aligned} \tag{2.48}$$

Both populations and coherences contribute to the subsystem-dependent force. This fixed-basis structure is mapped to continuous variables in Chapter 3; it need not be rederived there.

2.6 Ehrenfest mean-field dynamics

Ehrenfest dynamics is a mean-field reduction of the subsystem–bath dynamics. Each bath trajectory carries one subsystem density matrix, and the bath feels the expectation value of the subsystem-dependent force. An ensemble of such trajectories may represent a distributed initial bath state, but an individual trajectory does not branch into distinct bath paths conditioned on different subsystem states [6, 8].

For trajectory a, the subsystem density matrix evolves according to

$$\frac{d}{dt}\hat{\rho}_{\rm s}^{(a)}(t) = -\frac{i}{\hbar}\left[\hat{h}(\mathbf{R}_a(t)), \hat{\rho}_{\rm s}^{(a)}(t)\right], \tag{2.49}$$

while the bath obeys

$$\dot{\mathbf{R}}_a = \frac{\mathbf{P}_a}{M}, \tag{2.50}$$

$$\dot{\mathbf{P}}_a = -\nabla_{\mathbf{R}} V_b(\mathbf{R}_a) - \mathrm{Tr}_{\rm s}\left[\hat{\rho}_{\rm s}^{(a)}(t)\nabla_{\mathbf{R}}\hat{h}(\mathbf{R}_a)\right]. \tag{2.51}$$

In the fixed subsystem basis,

$$\dot{\mathbf{P}}_a = -\nabla_{\mathbf{R}} V_b(\mathbf{R}_a) - \sum_{\lambda\lambda'} \rho_{\rm s}^{(a),\lambda'\lambda}(t)\nabla_{\mathbf{R}} h^{\lambda\lambda'}(\mathbf{R}_a). \tag{2.52}$$

For a pure subsystem state

$$|\psi_a(t)\rangle = \sum_{\lambda} c_{\lambda}^{(a)}(t)|\lambda\rangle, \tag{2.53}$$

the coefficients satisfy

$$i\hbar\dot{c}_{\lambda}^{(a)}(t) = \sum_{\lambda'} h^{\lambda\lambda'}(\mathbf{R}_a(t))c_{\lambda'}^{(a)}(t). \tag{2.54}$$

The mean-field closure replaces the full phase-space-resolved subsystem density by one subsystem density matrix attached to each bath trajectory. It therefore cannot generate separate electronic-state-conditioned bath histories from a common initial bath point. When such histories separate dynamically, Ehrenfest propagation commonly retains coherence that should be reduced by subsystem–bath correlation. The method also does not generally reproduce the correct quantum–classical detailed-balance relation.

2.7 Summary and transition to the mapping representation

Semiclassical initial value representations approximate quantum propagation by classical trajectories carrying action and stability information. LSC-IVR further replaces exact Wigner-symbol evolution by classical transport. The QCLE follows a different reduction: the subsystem remains quantum, while the bath Moyal product is truncated through first order in the mass-ratio parameter.

In the adiabatic representation, the QCLE separates Bohr-phase evolution, phase-space transport, and transitions among density-matrix indices. In a coordinate-independent subsystem basis, derivative couplings are absent and nonadiabatic coupling is represented by off-diagonal elements of $h^{\lambda\lambda'}(\mathbf{R})$. Ehrenfest dynamics further replaces the phase-space-resolved subsystem–bath correlations by an unbranched mean-field evolution.

Chapter 3 applies the mapping construction to the fixed-basis Hamiltonian and QCLE defined in Eqs. (2.26) and (2.45). The next chapter therefore introduces only the mapping transformation and the structures that are new to the mapping representation, rather than repeating the quantum–classical derivation given here.

Thus, the QCLE is the first-order mass-ratio truncation of the partial-Wigner dynamics, while PBME is a further Hamiltonian-only mapping approximation. This hierarchy supplies background, not an original solver result.

Chapter 3

The Mapping Representation of Quantum–Classical Dynamics

This background chapter asks which physical subspace and density structure a mapping-QCLE numerical representation must preserve. The question matters because MMST variables enlarge the state space, so an arbitrary six-dimensional function need not represent a physical electronic density. The chapter derives the SEO embedding and projection, the mapping symbols, the PBME generator, and the formal excess density operator. For a two-state system, the derivation shows that the exact SEO image contains only four real bath-dependent fields. That finite structure becomes the design constraint against which the product surrogate is tested; the derivation itself is background and is not presented as an original methodological contribution.

The fixed subsystem-basis quantum–classical Liouville equation (QCLE) derived in Chapter 2 represents the quantum subsystem through a finite matrix algebra. The Meyer–Miller–Stock–Thoss (MMST) mapping replaces that discrete algebra by operators acting in an auxiliary harmonic-oscillator space [18, 19, 50, 59–61]. After a Wigner transform over the mapping oscillators, the subsystem is represented by continuous mapping coordinates and momenta, while the bath variables retain the phase-space representation introduced in the preceding chapters.

The mapping is an embedding, not a classicalization of the subsystem. The physical subsystem Hilbert space is identified with the singly-excited oscillator (SEO) subspace of the auxiliary oscillator space. Consequently, physical operators and densities must be distinguished from arbitrary functions on the enlarged mapping phase space. This distinction is central to the mapping-basis QCLE and to the approximation obtained by retaining only its Poisson-bracket contribution [1, 2, 62].

This chapter constructs the MMST representation required in the remainder of the thesis. It introduces the SEO embedding, derives the mapping symbols of subsystem operators, defines the physical phase-space projection and the projected mapping density, and constructs the trace–traceless mapping Hamiltonian. The fixed-basis QCLE is then transformed into mapping phase space and separated into a Hamiltonian Liouville operator and an excess density-level operator.

The Poisson-bracket mapping equation (PBME) is obtained by omitting the latter contribution.

3.1 MMST mapping of discrete subsystem states

Consider the fixed orthonormal subsystem basis $\{|\lambda\rangle\}_{\lambda=1}^{N_s}$ introduced in Chapter 2. The MMST construction associates each subsystem basis state with an SEO state of N_s harmonic oscillators:

$$|\lambda\rangle \longmapsto |m_\lambda\rangle = |0_1, \ldots, 1_\lambda, \ldots, 0_{N_s}\rangle. \tag{3.1}$$

Let $\hat{a}_\lambda^\dagger$ and $\hat{a}_\lambda$ be the creation and annihilation operators of oscillator λ, with

$$[\hat{a}_\lambda, \hat{a}_{\lambda'}^\dagger] = \delta_{\lambda\lambda'}. \tag{3.2}$$

Then

$$|m_\lambda\rangle = \hat{a}_\lambda^\dagger|0\rangle, \tag{3.3}$$

where $|0\rangle$ is the joint oscillator vacuum. The projector onto the physical SEO subspace is

$$\hat{\mathcal{P}}_{\text{SEO}} = \sum_{\lambda=1}^{N_s} |m_\lambda\rangle\langle m_\lambda|. \tag{3.4}$$

The subsystem transition operator is mapped according to

$$|\lambda\rangle\langle\lambda'| \longmapsto \hat{a}_\lambda^\dagger \hat{a}_{\lambda'}. \tag{3.5}$$

Within the SEO subspace,

$$\langle m_\alpha|\hat{a}_\lambda^\dagger \hat{a}_{\lambda'}|m_\beta\rangle = \delta_{\alpha\lambda}\delta_{\lambda'\beta}, \tag{3.6}$$

which reproduces the corresponding subsystem matrix element. Thus, for any subsystem operator

$$\hat{A}_W(\mathbf{X}) = \sum_{\lambda\lambda'} A_W^{\lambda\lambda'}(\mathbf{X})|\lambda\rangle\langle\lambda'|, \tag{3.7}$$

its mapped operator is

$$\hat{A}_m(\mathbf{X}) = \sum_{\lambda\lambda'} A_W^{\lambda\lambda'}(\mathbf{X})\hat{a}_\lambda^\dagger \hat{a}_{\lambda'}. \tag{3.8}$$

The equivalence with the original finite subsystem algebra holds after restriction to the SEO subspace. States with zero, two, or more oscillator excitations do not represent physical subsystem states.

3.2 Mapping-space Wigner symbols of subsystem operators

For each mapping oscillator, introduce canonical operators $\hat{r}_\lambda$ and $\hat{p}_\lambda$, with

$$[\hat{r}_\lambda, \hat{p}_{\lambda'}] = i\hbar\delta_{\lambda\lambda'}. \tag{3.9}$$

The oscillator ladder operators are

$$\hat{a}_\lambda = \frac{1}{\sqrt{2\hbar}}(\hat{r}_\lambda + i\hat{p}_\lambda), \qquad \hat{a}_\lambda^\dagger = \frac{1}{\sqrt{2\hbar}}(\hat{r}_\lambda - i\hat{p}_\lambda). \tag{3.10}$$

The mapping phase-space point is

$$\mathbf{x} = (\mathbf{r}, \mathbf{p}), \qquad \mathbf{r} = (r_1, \ldots, r_{N_s}), \qquad \mathbf{p} = (p_1, \ldots, p_{N_s}), \tag{3.11}$$

and the extended bath–mapping phase-space point is

$$\boldsymbol{\mathcal{X}} = (\mathbf{X}, \mathbf{x}) = (\mathbf{R}, \mathbf{P}, \mathbf{r}, \mathbf{p}). \tag{3.12}$$

Here $\mathbf{X} = (\mathbf{R}, \mathbf{P})$ is the bath phase-space point already defined in Chapter 2.

Applying the Wigner transform of Chapter 1 to the mapping oscillators gives

$$A_m(\mathbf{x}, \mathbf{X}) = \sum_{\lambda\lambda'} A_W^{\lambda\lambda'}(\mathbf{X}) c_{\lambda\lambda'}(\mathbf{x}), \tag{3.13}$$

where

$$c_{\lambda\lambda'}(\mathbf{x}) = \left(\hat{a}_\lambda^\dagger \hat{a}_{\lambda'}\right)_W \tag{3.14}$$

is the mapping symbol of a subsystem transition operator. Direct evaluation gives

$$c_{\lambda\lambda'}(\mathbf{x}) = \frac{1}{2\hbar}\left[r_\lambda r_{\lambda'} + p_\lambda p_{\lambda'} + i\left(r_\lambda p_{\lambda'} - r_{\lambda'} p_\lambda\right) - \hbar\delta_{\lambda\lambda'}\right]. \tag{3.15}$$

In particular,

$$c_{\lambda\lambda}(\mathbf{x}) = \frac{1}{2\hbar}\left(r_\lambda^2 + p_\lambda^2 - \hbar\right). \tag{3.16}$$

For a Hermitian subsystem operator, the relation $A_W^{\lambda'\lambda} = (A_W^{\lambda\lambda'})^*$ ensures that the complete mapping symbol in Eq. (3.13) is real.

The subsystem identity has the mapping symbol

$$I_m(\mathbf{x}) = \sum_\lambda c_{\lambda\lambda}(\mathbf{x}) = \frac{1}{2\hbar}\left(\mathbf{r}^2 + \mathbf{p}^2 - N_s\hbar\right). \tag{3.17}$$

It is not equal to unity pointwise on the enlarged mapping phase space. Its identity action is recovered only within the SEO-projected representation defined below.

3.3 Projected mapping density and the SEO projection kernel

The Wigner representation of the SEO matrix operator $|m_\lambda\rangle\langle m_{\lambda'}|$ defines the kernel

$$g_{\lambda\lambda'}(\mathbf{x}) = \frac{2}{\hbar}\mathcal{G}_{\rm SEO}(\mathbf{x})\left[r_\lambda r_{\lambda'} + p_\lambda p_{\lambda'} - i\left(r_\lambda p_{\lambda'} - r_{\lambda'} p_\lambda\right) - \frac{\hbar}{2}\delta_{\lambda\lambda'}\right], \tag{3.18}$$

where

$$\mathcal{G}_{\rm SEO}(\mathbf{x}) = (\pi\hbar)^{-N_s}\exp\left[-\frac{\mathbf{r}^2+\mathbf{p}^2}{\hbar}\right]. \tag{3.19}$$

The c- and g-families satisfy the duality relation

$$\int d\mathbf{x}\, g_{\lambda\lambda'}(\mathbf{x}) c_{\kappa\kappa'}(\mathbf{x}) = \delta_{\lambda\kappa}\delta_{\lambda'\kappa'}. \tag{3.20}$$

The g-kernels also obey

$$(2\pi\hbar)^{N_s}\int d\mathbf{x}\, g_{\lambda\lambda'}(\mathbf{x}) g_{\kappa'\kappa}(\mathbf{x}) = \delta_{\lambda\kappa}\delta_{\lambda'\kappa'}. \tag{3.21}$$

The phase-space projection onto the physical mapping image is the linear operator

$$\mathcal{P}F(\mathbf{x}) = (2\pi\hbar)^{N_s}\sum_{\lambda\lambda'} g_{\lambda'\lambda}(\mathbf{x})\int d\mathbf{x}'\, g_{\lambda\lambda'}(\mathbf{x}')F(\mathbf{x}'). \tag{3.22}$$

Equivalently,

$$\mathcal{P}F(\mathbf{x}) = \int d\mathbf{x}'\, K_{\mathcal{P}}(\mathbf{x},\mathbf{x}')F(\mathbf{x}'), \tag{3.23}$$

with

$$K_{\mathcal{P}}(\mathbf{x},\mathbf{x}') = (2\pi\hbar)^{N_s}\sum_{\lambda\lambda'} g_{\lambda'\lambda}(\mathbf{x}) g_{\lambda\lambda'}(\mathbf{x}'). \tag{3.24}$$

Equation (3.21) implies

$$\mathcal{P}^2 = \mathcal{P}. \tag{3.25}$$

The physical mapping representative of the fixed-basis density matrix is

$$\tilde{\rho}_m(\mathbf{x},\mathbf{X},t) = \sum_{\lambda\lambda'} \rho_W^{\lambda\lambda'}(\mathbf{X},t) g_{\lambda'\lambda}(\mathbf{x}). \tag{3.26}$$

It satisfies

$$\mathcal{P}\tilde{\rho}_m = \tilde{\rho}_m. \tag{3.27}$$

Conversely, the subsystem density components are reconstructed from the projected mapping density by either of the equivalent relations

$$\rho_W^{\lambda\lambda'}(\mathbf{X},t) = \int d\mathbf{x}\, c_{\lambda'\lambda}(\mathbf{x})\tilde{\rho}_m(\mathbf{x},\mathbf{X},t) \tag{3.28}$$

$$= (2\pi\hbar)^{N_s}\int d\mathbf{x}\, g_{\lambda\lambda'}(\mathbf{x})\tilde{\rho}_m(\mathbf{x},\mathbf{X},t). \tag{3.29}$$

The first relation follows from Eq. (3.20); the second follows from Eq. (3.21). These formulas distinguish the physical projected density from an arbitrary function on mapping phase space.

3.4 Expectation values in the mapping basis

For the transform conventions established in Chapter 1, the fixed-basis expectation value is

$$\langle A\rangle(t) = \int d\mathbf{X} \sum_{\lambda\lambda'} A_W^{\lambda\lambda'}(\mathbf{X})\rho_W^{\lambda'\lambda}(\mathbf{X},t). \tag{3.30}$$

Using Eqs. (3.13) and (3.26), this becomes

$$\langle A\rangle(t) = \int d\mathbf{X}\, d\mathbf{x}\, A_m(\mathbf{x},\mathbf{X})\tilde{\rho}_m(\mathbf{x},\mathbf{X},t). \tag{3.31}$$

No additional mapping-space normalization factor appears because it is already contained in the definition of the g-kernels.

The reduced subsystem density matrix follows from

$$\rho_s^{\lambda\lambda'}(t) = \int d\mathbf{X}\, d\mathbf{x}\, c_{\lambda'\lambda}(\mathbf{x})\tilde{\rho}_m(\mathbf{x},\mathbf{X},t), \tag{3.32}$$

and the population of state λ is

$$P_\lambda(t) = \int d\mathbf{X}\, d\mathbf{x}\, c_{\lambda\lambda}(\mathbf{x})\tilde{\rho}_m(\mathbf{x},\mathbf{X},t). \tag{3.33}$$

For a normalized physical density,

$$\int d\mathbf{X}\, d\mathbf{x}\, \tilde{\rho}_m(\mathbf{x},\mathbf{X},t) = 1, \tag{3.34}$$

and, equivalently,

$$\int d\mathbf{X}\, d\mathbf{x}\, I_m(\mathbf{x})\tilde{\rho}_m(\mathbf{x},\mathbf{X},t) = 1. \tag{3.35}$$

The equality of Eqs. (3.34) and (3.35) is a projected-space identity; it does not imply that $I_m(\mathbf{x}) = 1$ pointwise.

3.5 The trace–traceless mapping Hamiltonian

The fixed-basis subsystem Hamiltonian matrix $h^{\lambda\lambda'}(\mathbf{R})$ was defined in Sec. 2.5. Decompose it into scalar and traceless parts:

$$\eta(\mathbf{R}) = \frac{1}{N_s}\,\mathrm{Tr}_s\, h(\mathbf{R}), \tag{3.36}$$

$$h(\mathbf{R}) = \eta(\mathbf{R})I_s + \bar{h}(\mathbf{R}), \qquad \mathrm{Tr}_s\, \bar{h}(\mathbf{R}) = 0. \tag{3.37}$$

The scalar contribution is combined with the bath potential:

$$V_0(\mathbf{R}) = V_b(\mathbf{R}) + \eta(\mathbf{R}), \tag{3.38}$$

so that

$$H_0(\mathbf{X}) = \frac{\mathbf{P}^2}{2M} + V_0(\mathbf{R}). \tag{3.39}$$

From this point onward, the fixed-basis model Hamiltonians considered in this thesis are taken to be real and symmetric. The mapping symbol of the traceless matrix is then

$$\bar{h}_m(\mathbf{x}, \mathbf{R}) = \frac{1}{2\hbar} \sum_{\lambda\lambda'} \bar{h}^{\lambda\lambda'}(\mathbf{R}) \left(r_\lambda r_{\lambda'} + p_\lambda p_{\lambda'} \right). \tag{3.40}$$

The imaginary antisymmetric part of Eq. (3.15) cancels because $\bar{h}^{\lambda\lambda'} = \bar{h}^{\lambda'\lambda}$, and the term proportional to $\delta_{\lambda\lambda'}$ vanishes because $\bar{h}$ is traceless. For a general Hermitian matrix, the full symbol in Eq. (3.15) must be retained.

The trace–traceless mapping Hamiltonian is

$$H_m(\boldsymbol{\mathcal{X}}) = H_0(\mathbf{X}) + \bar{h}_m(\mathbf{x}, \mathbf{R}). \tag{3.41}$$

Within the SEO-projected representation, this Hamiltonian is equivalent to the mapping of the original subsystem Hamiltonian because the mapped identity acts as unity inside projected matrix elements. The role of this restriction under full mapping-QCLE and PBME propagation is addressed below.

3.6 The mapping-basis QCLE

The starting point is the fixed-basis density-form QCLE in Eq. (2.24). The bath-only scalar part of the Hamiltonian maps directly to the bath Poisson bracket generated by H_0. Products of subsystem operators are transformed by applying the star-product rule of Sec. 1.8.3 to the mapping variables. The corresponding Janus operator is

$$\Lambda_{\mathbf{x}} = \overleftarrow{\nabla}_{\mathbf{r}} \cdot \overrightarrow{\nabla}_{\mathbf{p}} - \overleftarrow{\nabla}_{\mathbf{p}} \cdot \overrightarrow{\nabla}_{\mathbf{r}}. \tag{3.42}$$

Thus, for two mapped subsystem operators,

$$(\hat{A}\hat{B})_m = A_m \exp\left(\frac{i\hbar}{2}\Lambda_{\mathbf{x}}\right) B_m. \tag{3.43}$$

Define the mapping-space Poisson bracket by

$$\{A, B\}_{\mathbf{x}} = \nabla_{\mathbf{r}} A \cdot \nabla_{\mathbf{p}} B - \nabla_{\mathbf{p}} A \cdot \nabla_{\mathbf{r}} B, \tag{3.44}$$

and the extended bracket by

$$\{A, B\}_{\boldsymbol{\mathcal{X}}} = \{A, B\}_{\mathbf{X}} + \{A, B\}_{\mathbf{x}}, \tag{3.45}$$

where the bath bracket $\{\cdot, \cdot\}_{\mathbf{X}}$ is defined in Eq. (2.23).

Because $\bar{h}_m$ is quadratic in the mapping variables, the mapping-space star commutator terminates at first order. Therefore,

$$-\frac{i}{\hbar}[\bar{h}, \hat{\rho}_W]_m = \{\bar{h}_m, \tilde{\rho}_m\}_{\mathbf{x}}. \tag{3.46}$$

The symmetrized bath-force term in the fixed-basis QCLE becomes the even part of the mapping star product:

$$\begin{aligned}\frac{1}{2}\left[(\nabla_{\mathbf{R}}\bar{h}) \cdot \nabla_{\mathbf{P}}\hat{\rho}_W + \nabla_{\mathbf{P}}\hat{\rho}_W \cdot (\nabla_{\mathbf{R}}\bar{h})\right]_m \\ = (\nabla_{\mathbf{R}}\bar{h}_m) \cos\left(\frac{\hbar}{2}\Lambda_{\mathbf{x}}\right) \cdot \nabla_{\mathbf{P}}\tilde{\rho}_m.\end{aligned} \tag{3.47}$$

Since $\nabla_{\mathbf{R}}\bar{h}_m$ is also quadratic in $(\mathbf{r}, \mathbf{p})$, the cosine expansion terminates after the second-order term. Define the mapping-space operator

$$\mathcal{D}_{\lambda\lambda'} = \frac{\partial^2}{\partial r_{\lambda'}\partial r_\lambda} + \frac{\partial^2}{\partial p_{\lambda'}\partial p_\lambda}. \tag{3.48}$$

Then

$$\begin{aligned}(\nabla_{\mathbf{R}}\bar{h}_m) \cos\left(\frac{\hbar}{2}\Lambda_{\mathbf{x}}\right) \cdot \nabla_{\mathbf{P}}\tilde{\rho}_m = (\nabla_{\mathbf{R}}\bar{h}_m) \cdot \nabla_{\mathbf{P}}\tilde{\rho}_m \\ - \frac{\hbar}{8}\sum_{\lambda\lambda'}\left(\nabla_{\mathbf{R}}\bar{h}^{\lambda\lambda'}(\mathbf{R})\right) \cdot \mathcal{D}_{\lambda\lambda'}\nabla_{\mathbf{P}}\tilde{\rho}_m.\end{aligned} \tag{3.49}$$

Combining the bath scalar term, the mapping commutator, and the force term gives the mapping-basis QCLE:

$$\begin{aligned}\frac{\partial\tilde{\rho}_m(\boldsymbol{\mathcal{X}}, t)}{\partial t} = \{H_m(\boldsymbol{\mathcal{X}}), \tilde{\rho}_m(\boldsymbol{\mathcal{X}}, t)\}_{\boldsymbol{\mathcal{X}}} \\ - \frac{\hbar}{8}\sum_{\lambda\lambda'}\left(\nabla_{\mathbf{R}}\bar{h}^{\lambda\lambda'}(\mathbf{R})\right) \cdot \mathcal{D}_{\lambda\lambda'}\nabla_{\mathbf{P}}\tilde{\rho}_m(\boldsymbol{\mathcal{X}}, t).\end{aligned} \tag{3.50}$$

The first term is Hamiltonian transport in the extended phase space. The second is the excess mapping-QCLE contribution, containing two derivatives in mapping phase space and one bath-momentum derivative.

For later use, write

$$i\mathcal{L}_m = i\mathcal{L}_m^\circ + i\mathcal{L}_m', \tag{3.51}$$

with the density equation

$$\frac{\partial \tilde{\rho}_m}{\partial t} = -i\mathcal{L}_m \tilde{\rho}_m. \tag{3.52}$$

The Hamiltonian and excess actions on a sufficiently differentiable function $f(\boldsymbol{\mathcal{X}})$ are

$$-i\mathcal{L}_m^{\circ} f = \{H_m, f\}_{\boldsymbol{\mathcal{X}}}, \tag{3.53}$$

$$-i\mathcal{L}_m' f = -\frac{\hbar}{8} \sum_{\lambda\lambda'} \left(\nabla_{\mathbf{R}} \bar{h}^{\lambda\lambda'}(\mathbf{R})\right) \cdot \left[\left(\frac{\partial^2}{\partial r_{\lambda'} \partial r_{\lambda}} + \frac{\partial^2}{\partial p_{\lambda'} \partial p_{\lambda}}\right) \nabla_{\mathbf{P}} f\right]. \tag{3.54}$$

The full mapping-QCLE generator preserves the physical mapping image:

$$[\mathcal{P}, i\mathcal{L}_m] = 0. \tag{3.55}$$

Hence projected initial data remain projected under the full mapping-QCLE,

$$\mathcal{P}\tilde{\rho}_m(0) = \tilde{\rho}_m(0) \quad \Longrightarrow \quad \mathcal{P}\tilde{\rho}_m(t) = \tilde{\rho}_m(t). \tag{3.56}$$

This property connects the continuous mapping equation to the original finite subsystem dynamics and is essential to the equivalence of operator representations inside the physical mapping subspace [2].

3.7 PBME as Hamiltonian mapping-QCLE dynamics

PBME is obtained by omitting the excess operator $i\mathcal{L}_m'$ from Eq. (3.51). Its density equation is

$$\frac{\partial \rho_m^{\mathrm{PB}}}{\partial t} = \{H_m, \rho_m^{\mathrm{PB}}\}_{\boldsymbol{\mathcal{X}}}, \tag{3.57}$$

where the initial condition may be chosen as the projected physical density,

$$\rho_m^{\mathrm{PB}}(\boldsymbol{\mathcal{X}}, 0) = \tilde{\rho}_m(\boldsymbol{\mathcal{X}}, 0). \tag{3.58}$$

The notation ρ_m^{PB} is used because the PBME evolution does not generally remain in the image of $\mathcal{P}$.

The Hamiltonian characteristics generated by Eq. (3.41) satisfy

$$\dot{\mathbf{R}} = \frac{\mathbf{P}}{M}, \tag{3.59}$$

$$\dot{\mathbf{P}} = -\nabla_{\mathbf{R}} H_m(\boldsymbol{\mathcal{X}}), \tag{3.60}$$

$$\dot{\mathbf{r}} = \frac{1}{\hbar}\bar{h}(\mathbf{R})\mathbf{p}, \tag{3.61}$$

$$\dot{\mathbf{p}} = -\frac{1}{\hbar}\bar{h}(\mathbf{R})\mathbf{r}. \tag{3.62}$$

The bath force is

$$\dot{\mathbf{P}} = -\nabla_{\mathbf{R}} V_0(\mathbf{R}) - \frac{1}{2\hbar}\sum_{\lambda\lambda'} \nabla_{\mathbf{R}} \bar{h}^{\lambda\lambda'}(\mathbf{R}) \left(r_\lambda r_{\lambda'} + p_\lambda p_{\lambda'}\right). \tag{3.63}$$

Along an exact PBME characteristic,

$$\frac{d}{dt}\rho_m^{\mathrm{PB}}\left(\boldsymbol{\mathcal{X}}(t), t\right) = 0, \tag{3.64}$$

and the Hamiltonian flow preserves the extended phase-space volume,

$$\det\left(\frac{\partial \boldsymbol{\mathcal{X}}(t)}{\partial \boldsymbol{\mathcal{X}}(0)}\right) = 1. \tag{3.65}$$

Unlike the full generator, the Hamiltonian mapping generator does not commute with the SEO phase-space projector in general:

$$[\mathcal{P}, i\mathcal{L}_m^\circ] \neq 0. \tag{3.66}$$

Thus, PBME may carry initially projected data outside the physical mapping image. This also limits the equivalence of different mapping Hamiltonians: forms that agree inside projected integrals need not generate identical PBME trajectories after the approximate evolution has developed components outside that subspace. The trace–traceless Hamiltonian in Eq. (3.41) is therefore adopted consistently throughout this thesis.

PBME retains the Hamiltonian subsystem–bath back-reaction encoded in H_m, but it omits the density-level action in Eq. (3.54). Omitting that term is an additional approximation to the mapping-QCLE; it is not a consequence of the mass-ratio truncation already used to derive the QCLE [2, 62].

3.8 Summary and transition to density reconstruction

The MMST construction embeds the finite subsystem into the SEO subspace of an auxiliary oscillator Hilbert space. The operator kernels $c_{\lambda\lambda'}$ and the SEO kernels $g_{\lambda\lambda'}$ provide the dual phase-space representation required to evaluate observables and reconstruct subsystem density-matrix elements. The projector $\mathcal{P}$ distinguishes physical mapping densities from arbitrary functions on the enlarged mapping phase space.

For the two-state model, Eq. (3.26) shows that the complete physical mapping dependence is determined by four real bath-dependent fields: ρ_W^{11}, ρ_W^{22}, $\mathrm{Re}\,\rho_W^{12}$, and $\mathrm{Im}\,\rho_W^{12}$, subject to Hermiticity and trace constraints. Consequently, a surrogate used inside the excess operator must either remain in this four-function SEO image or be projected back into it before differentiation.

After separating the scalar and traceless parts of the fixed-basis subsystem Hamiltonian, the mapping-basis QCLE takes the operator form $i\mathcal{L}_m = i\mathcal{L}_m^{\circ} + i\mathcal{L}_m'$. The first part generates Hamiltonian transport in the extended phase space; the second acts on the density through mixed mapping and bath-momentum derivatives. The full generator preserves the projected physical subspace, whereas PBME, obtained by retaining only $i\mathcal{L}_m^{\circ}$, generally does not.

The next chapter addresses the representation of the time-dependent projected mapping density. The operators in Eqs. (3.53) and (3.54) are already fixed by the mapping-QCLE; the remaining computational requirement is a differentiable approximation to $\tilde{\rho}_m(\boldsymbol{\mathcal{X}}, t)$ on which those operators can act. Projection and approximate propagation cannot be interchanged because the full generator commutes with $\mathcal{P}$, whereas the PBME generator generally does not [2]. The next chapter tests a representation that does not enforce this design requirement and therefore reports the resulting projection residual explicitly.

Consequently, the physical mapping density lies in the SEO image; for two states its mapping dependence is fixed by four real bath-dependent fields. Any surrogate for the excess operator must remain in, or be projected into, that image.

Chapter 4

Gaussian-Process Representation of the Mapping-Basis Density

This chapter asks whether a fixed SEO profile multiplied by a six-dimensional GP modulation can provide the density derivatives required by the excess operator. This distinction matters because small value residuals cannot answer that question if the surrogate leaves the SEO image or supplies inaccurate derivatives away from its training support. The original construction tested here is a differentiable product-GP density architecture already coupled to the moving-cloud formulation, and it is examined through manufactured on-support and off-support operator tests together with a diagnostic SEO projection. The evidence shows that the ansatz is not projection preserving and that its manufactured excess-operator error does not decrease consistently with the tested enlargement. The construction therefore exposes the representation requirements of a successor rather than providing a validated projected density model.

Chapter 3 established the mapping-basis QCLE for the SEO-projected mapping density $\tilde{\rho}_m(\boldsymbol{\mathcal{X}}, t)$ and identified the excess operator that acts on its mapping-variable and bath-momentum derivatives. The present chapter addresses only the fixed-time representation problem. Given values of this projected density at a finite set of phase-space support points, the objective is to construct a smooth differentiable surrogate for $\tilde{\rho}_m(\boldsymbol{\mathcal{X}}, t)$. The update of the support values between successive time slices is developed in Chapter 5.

The mathematical target throughout this chapter is the projected density itself. Its fixed-time surrogate is parameterized through a product model: the diagonal SEO profile of the initially occupied state is used as an analytic reference factor, while a Gaussian process represents a multiplicative modulation over the full extended phase space. The covariance model for the modulation is a stationary anisotropic squared-exponential kernel. The SEO factor is exact for the specified initial electronic state, but it is a model choice at later times: a general projected mapping density need not retain the same nodal set. The Gaussian process represents only the density modulation; it does not replace the mapping Hamiltonian, the PBME flow, or either part of the mapping-QCLE generator [63–67].

4.1 Fixed-time product representation

4.1.1 Density target and multiplicative SEO profile

At a fixed time t, the regression target is the projected mapping density $\tilde{\rho}_m(\boldsymbol{\mathcal{X}}, t)$ defined in Chapter 3. Its surrogate is written as

$$\widehat{\rho}_{m,t}(\boldsymbol{\mathcal{X}}) = g_s(\mathbf{x})\, \mu_t(\boldsymbol{\mathcal{X}}), \tag{4.1}$$

where s denotes the initially occupied subsystem state and μ_t is the smooth modulation represented by the inner Gaussian process. The subscript t labels a fixed reconstruction time; it does not define a separate Gaussian-process time-evolution equation.

The reference factor is the diagonal SEO Wigner kernel $g_{ss}(\mathbf{x})$ introduced in Eq. (3.18). For the occupied state s, it is

$$g_s(\mathbf{x}) = (\pi\hbar)^{-N_s} \exp\left[-\frac{\mathbf{r}^2 + \mathbf{p}^2}{\hbar}\right] \left[\frac{2}{\hbar}\left(r_s^2 + p_s^2\right) - 1\right]. \tag{4.2}$$

Thus, the Gaussian envelope and the occupied-state polynomial factor are known analytically and are held fixed. The inner GP learns only the modulation required to match the projected density data.

The factorization in Eq. (4.1) is the structured representation used for the projected-density target. The fixed function g_s supplies the exact initial SEO reference structure, while the inner Gaussian process supplies the modulation. All density values, derivatives, moments, and observables are evaluated from the complete product $g_s\mu_t$. The factorization is exact at the initial time whenever the initial projected density is the occupied-state SEO profile multiplied by a smooth bath-dependent factor. At later times, its adequacy is assessed directly in density space. In particular, g_s vanishes on the nodal set

$$r_s^2 + p_s^2 = \frac{\hbar}{2}. \tag{4.3}$$

The fitted ratio must therefore be regularized near this set. The regularized label transform stabilizes the inner regression, but it does not remove the structural zero of the final product in Eq. (4.1). A general later-time projected density is a linear combination of SEO kernels and need not vanish on this ring. The product ansatz can therefore be structurally biased even when its support-point residual is small. Density-space residuals near the nodal set and comparison with a direct-density representation must consequently be treated as tests of model adequacy, not only as conditioning diagnostics.

The density may take either sign. The Gaussian process is therefore a real-valued regression model with no positivity transformation. Physical traces, populations, coherences, and energy moments are evaluated using the SEO-consistent observable formulas of Chapter 3.

4.1.2 SEO compatibility of the product ansatz

Equation (3.26) fixes the mapping dependence of a physical two-state density through four real, bath-dependent SEO coefficient fields. By contrast, Eq. (4.1) learns a generic six-dimensional modulation and does not enforce $\mathcal{P}\widehat{\rho}_{m,t} = \widehat{\rho}_{m,t}$. The product ansatz is therefore a modeling assumption to be validated, not a consequence of the exact projector.

The projection test fits the product surrogate on full-scattering PBME snapshots and projects its mapping dependence onto the exact rank-four SEO basis. The mean relative L^2 leakage is 0.2737 at $P_{\text{init}} = 20$ and 0.9532 at $P_{\text{init}} = 100$, with maxima 0.3735 and 0.9958, respectively. Thus, the approximation is not projection preserving, and its high-momentum agreement cannot validate the excess correction. This thesis adopts the diagnostic route: it retains the ansatz, reports the residual, and does not claim a projected solver. The structure-preserving successor is to regress the four real bath-dependent fields ρ_W^{11}, ρ_W^{22}, $\operatorname{Re}\rho_W^{12}$, and $\operatorname{Im}\rho_W^{12}$, with Hermiticity and trace constraints, so that mapping derivatives are analytic and projection leakage vanishes by construction [2, 60, 61]. That projected alternative is a proposed reformulation rather than part of the reported calculations.

4.1.3 Support data and transformed labels

At time t, let

$$\mathcal{S}_t = \{\boldsymbol{\mathcal{X}}_i(t)\}_{i=1}^{N} \tag{4.4}$$

be the phase-space support set, and let

$$y_i(t) \approx \tilde{\rho}_m(\boldsymbol{\mathcal{X}}_i(t), t) \tag{4.5}$$

be the density value associated with the i-th support point. This chapter assumes that the pairs

$$\mathcal{D}_t = \{(\boldsymbol{\mathcal{X}}_i(t), y_i(t))\}_{i=1}^{N} \tag{4.6}$$

are available. The procedure that updates the labels during propagation is not part of the fixed-time regression model and is deferred to Chapter 5.

The projected density remains the mathematical target of the fit. The ratio used below is only an auxiliary parameterization that removes the known reference profile from the regression labels. The inner GP is trained on the ratio of the density label to the occupied-state SEO profile. To regularize division near the nodal ring, define

$$g_{s,\text{safe}}(\mathbf{x}_i) = \operatorname{sgn}_0[g_s(\mathbf{x}_i)] \max\left\{|g_s(\mathbf{x}_i)|, \varepsilon_g g_{\text{ref},t}\right\}, \tag{4.7}$$

where $\varepsilon_g > 0$ is a relative floor and

$$g_{\text{ref},t} = \max_{1\le i\le N} |g_s(\mathbf{x}_i(t))| \tag{4.8}$$

is the maximum magnitude of the analytic profile on the current support set. Furthermore, $\mathrm{sgn}_0(u) = 1$ for $u \geq 0$ and $\mathrm{sgn}_0(u) = -1$ for $u < 0$. The transformed labels are

$$u_i(t) = \frac{y_i(t)}{g_{s,\mathrm{safe}}(\mathbf{x}_i(t))}. \tag{4.9}$$

The vector of transformed labels is denoted by $\mathbf{u}_t = (u_1(t), \ldots, u_N(t))^T$.

For focused initial mapping data, the mapping radii are fixed by construction. The value of g_s is therefore constant with respect to the angular variables on each focus torus. If the initial density contains the same SEO factor, Eq. (4.9) removes that fixed mapping-space shape exactly, leaving a modulation that is constant along the angular mapping directions at fixed bath phase-space position.

4.2 Gaussian-process model for the modulation

4.2.1 Zero-mean prior and observation model

Let $f_t(\boldsymbol{\mathcal{X}})$ denote the latent modulation function. It is assigned the zero-mean Gaussian-process prior

$$f_t(\boldsymbol{\mathcal{X}}) \sim \mathcal{GP}\big(0, k_{\boldsymbol{\theta}_t}\left(\boldsymbol{\mathcal{X}}, \boldsymbol{\mathcal{X}}'\right)\big), \tag{4.10}$$

where $\boldsymbol{\theta}_t$ denotes the kernel hyperparameters used at the current refit. A zero prior mean is adopted throughout this chapter, so no prior-mean derivatives or moment contributions are omitted in the formulas below.

The transformed labels satisfy

$$u_i(t) = f_t(\boldsymbol{\mathcal{X}}_i(t)) + \epsilon_i(t), \qquad \boldsymbol{\epsilon}_t \sim \mathcal{N}\big(\mathbf{0}, \sigma_n^2 I\big). \tag{4.11}$$

Here σ_n^2 models imperfect labels and regularizes the kernel system.

Define

$$(K_t)_{ij} = k_{\boldsymbol{\theta}_t}\left(\boldsymbol{\mathcal{X}}_i(t), \boldsymbol{\mathcal{X}}_j(t)\right), \qquad K_{y,t} = K_t + \sigma_n^2 I. \tag{4.12}$$

The coefficient vector is obtained from

$$\boldsymbol{\zeta}_t = K_{y,t}^{-1}\mathbf{u}_t. \tag{4.13}$$

In computation, Eq. (4.13) is evaluated by a Cholesky solve rather than by forming the inverse explicitly.

4.2.2 Posterior mean and covariance

The posterior mean of the latent modulation is denoted by $\mu_t(\boldsymbol{\mathcal{X}})$ and is

$$\mu_t(\boldsymbol{\mathcal{X}}) = \sum_{j=1}^{N} \zeta_j(t) k_{\boldsymbol{\theta}_t} \left(\boldsymbol{\mathcal{X}}, \boldsymbol{\mathcal{X}}_j(t)\right). \tag{4.14}$$

Substitution into Eq. (4.1) gives the product surrogate for the projected-density target,

$$\widehat{\rho}_{m,t}(\boldsymbol{\mathcal{X}}) = g_s(\mathbf{x}) \sum_{j=1}^{N} \zeta_j(t) k_{\boldsymbol{\theta}_t} \left(\boldsymbol{\mathcal{X}}, \boldsymbol{\mathcal{X}}_j(t)\right). \tag{4.15}$$

For an evaluation point $\boldsymbol{\mathcal{X}}_*$, let $\mathbf{k}_{*,t}$ be the vector whose j-th component is $k_{\boldsymbol{\theta}_t}(\boldsymbol{\mathcal{X}}_*, \boldsymbol{\mathcal{X}}_j(t))$. The posterior covariance of the latent modulation is

$$\Sigma_{f,t} \left(\boldsymbol{\mathcal{X}}_*, \boldsymbol{\mathcal{X}}'_*\right) = k_{\boldsymbol{\theta}_t} \left(\boldsymbol{\mathcal{X}}_*, \boldsymbol{\mathcal{X}}'_*\right) - \mathbf{k}_{*,t}^T K_{y,t}^{-1} \mathbf{k}'_{*,t}. \tag{4.16}$$

Because the SEO profile is fixed, the covariance of the product density is

$$\Sigma_{\rho,t} \left(\boldsymbol{\mathcal{X}}, \boldsymbol{\mathcal{X}}'\right) = g_s(\mathbf{x}) g_s(\mathbf{x}') \Sigma_{f,t} \left(\boldsymbol{\mathcal{X}}, \boldsymbol{\mathcal{X}}'\right). \tag{4.17}$$

These covariances quantify uncertainty conditional on the selected kernel family, hyperparameters, regularization, and support geometry. They are not physical stochastic terms in the mapping-QCLE, and they do not by themselves quantify discretization or model-form error [5, 68].

4.3 Stationary ARD-RBF kernel

4.3.1 Six-dimensional kernel

The numerical model considered in this thesis contains one bath coordinate, one bath momentum, and two mapping oscillators. The extended input therefore has six components. Let z_d denote the d-th component of $\boldsymbol{\mathcal{X}}$ in the fixed ordering used by the numerical implementation. The inner GP uses stationary automatic relevance determination (ARD) radial-basis-function (RBF), or squared-exponential, kernel

$$k_{\boldsymbol{\theta}} \left(\boldsymbol{\mathcal{X}}, \boldsymbol{\mathcal{X}}'\right) = \sigma_f^2 \exp\left[-\frac{1}{2} \sum_{d=1}^{6} \frac{(z_d - z'_d)^2}{\ell_d^2}\right], \tag{4.18}$$

with

$$\boldsymbol{\theta} = (\sigma_f, \sigma_n, \ell_1, \ldots, \ell_6). \tag{4.19}$$

The SEO factor in Eq. (4.2) does not enter Eq. (4.18). The kernel represents correlations in the smooth modulation μ_t across all six coordinates.

Different length scales are retained because bath and mapping directions have different numerical ranges and may carry different correlation lengths. The squared-exponential kernel is infinitely differentiable, which permits the analytic third derivatives required after the product rule is applied to the mapping-QCLE excess operator. Smoothness of the kernel makes these derivatives available; it does not establish their accuracy from finite support data [3, 4, 69–71].

4.3.2 Input standardization

When standardized inputs are used, write

$$\widehat{z}_d = \frac{z_d - b_d}{s_d}, \qquad s_d > 0, \tag{4.20}$$

where b_d and s_d are fixed centering and scaling constants during a refit. The regression may be performed in the standardized coordinates, but physical derivatives must be restored through

$$\frac{\partial}{\partial z_d} = \frac{1}{s_d} \frac{\partial}{\partial \widehat{z}_d}. \tag{4.21}$$

The corresponding product of scale factors must be included for every higher derivative. Phase-space integrals in standardized coordinates also require the Jacobian $\prod_d s_d$.

4.4 Analytic derivatives of the product surrogate

The excess mapping-QCLE action was defined in Eq. (3.54). In the general vector bath notation, the required derivative is

$$\left(\frac{\partial^2}{\partial r_{\lambda'} \partial r_\lambda} + \frac{\partial^2}{\partial p_{\lambda'} \partial p_\lambda} \right) \nabla_{\mathbf{P}} \widehat{\rho}_{m,t}. \tag{4.22}$$

For the one-dimensional bath used in the numerical calculations, $\nabla_{\mathbf{P}}$ reduces to $\partial/\partial P$. The explicit kernel derivatives below are written in this one-dimensional form. Since the reconstructed density is the product $g_s \mu_t$, both factors must be differentiated. Differentiating the inner GP alone would omit terms that are of the same differential order as the desired operator.

4.4.1 Derivatives of the stationary kernel

For the j-th kernel function, define

$$k_j(\boldsymbol{\mathcal{X}}) = k_{\boldsymbol{\theta}_t}\left(\boldsymbol{\mathcal{X}}, \boldsymbol{\mathcal{X}}_j(t)\right), \qquad q_{j,d} = \frac{z_{j,d}(t) - z_d}{\ell_d^2}. \tag{4.23}$$

Then

$$\frac{\partial k_j}{\partial z_d} = q_{j,d} k_j, \tag{4.24}$$

$$\frac{\partial^2 k_j}{\partial z_e \partial z_d} = \left(q_{j,e} q_{j,d} - \frac{\delta_{de}}{\ell_d^2} \right) k_j. \tag{4.25}$$

Let $q_{j,P}$ denote the component associated with the bath momentum. For a mapping coordinate x_a,

$$\frac{\partial^2 k_j}{\partial x_a \partial P} = q_{j,a} q_{j,P} k_j, \tag{4.26}$$

and

$$\frac{\partial^3 k_j}{\partial x_b \partial x_a \partial P} = q_{j,P} \left(q_{j,b} q_{j,a} - \frac{\delta_{ab}}{\ell_a^2} \right) k_j. \tag{4.27}$$

The absence of a Kronecker term coupling P to a mapping coordinate follows from their independence as separate input components.

The modulation derivatives are obtained by summing these kernel derivatives with coefficients $\zeta_j(t)$. In particular,

$$\frac{\partial \mu_t}{\partial P} = \sum_{j=1}^{N} \zeta_j(t) q_{j,P} k_j, \tag{4.28}$$

$$\frac{\partial^2 \mu_t}{\partial x_a \partial P} = \sum_{j=1}^{N} \zeta_j(t) q_{j,a} q_{j,P} k_j, \tag{4.29}$$

and

$$\frac{\partial^3 \mu_t}{\partial x_b \partial x_a \partial P} = \sum_{j=1}^{N} \zeta_j(t) q_{j,P} \left(q_{j,b} q_{j,a} - \frac{\delta_{ab}}{\ell_a^2} \right) k_j. \tag{4.30}$$

4.4.2 Closed-form derivatives of the SEO profile

For compactness, define

$$\mathcal{G}_{\text{SEO}}(\mathbf{x}) = (\pi\hbar)^{-N_s} \exp\left[-\frac{\mathbf{r}^2 + \mathbf{p}^2}{\hbar} \right], \qquad B_s(\mathbf{x}) = \frac{2}{\hbar}\left(r_s^2 + p_s^2 \right) - 1, \tag{4.31}$$

so that $g_s = \mathcal{G}_{\text{SEO}} B_s$. The first derivatives are

$$\frac{\partial g_s}{\partial r_a} = \mathcal{G}_{\text{SEO}} \left[-\frac{2r_a}{\hbar} B_s + \frac{4r_s}{\hbar} \delta_{as} \right], \tag{4.32}$$

$$\frac{\partial g_s}{\partial p_a} = \mathcal{G}_{\text{SEO}} \left[-\frac{2p_a}{\hbar} B_s + \frac{4p_s}{\hbar} \delta_{as} \right]. \tag{4.33}$$

The second derivatives required by the excess operator are

$$\frac{\partial^2 g_s}{\partial r_b \partial r_a} = \mathcal{G}_{\text{SEO}} \left[\left(\frac{4 r_a r_b}{\hbar^2} - \frac{2\delta_{ab}}{\hbar} \right) B_s - \frac{8}{\hbar^2} \left(\delta_{as} r_s r_b + \delta_{bs} r_s r_a \right) + \frac{4}{\hbar} \delta_{as} \delta_{bs} \right], \tag{4.34}$$

$$\frac{\partial^2 g_s}{\partial p_b \partial p_a} = \mathcal{G}_{\text{SEO}} \left[\left(\frac{4p_a p_b}{\hbar^2} - \frac{2\delta_{ab}}{\hbar} \right) B_s - \frac{8}{\hbar^2} \left(\delta_{as} p_s p_b + \delta_{bs} p_s p_a \right) + \frac{4}{\hbar} \delta_{as} \delta_{bs} \right]. \tag{4.35}$$

These expressions are evaluated analytically. No finite-difference approximation is introduced for the reference profile.

4.4.3 Product-rule form of the excess derivative

Because g_s is independent of the bath momentum,

$$\frac{\partial}{\partial P} \widehat{\rho}_{m,t} = g_s \frac{\partial \mu_t}{\partial P}. \tag{4.36}$$

For mapping coordinates x_a and x_b, the required third derivative is

$$\begin{aligned} \frac{\partial^3}{\partial x_b \partial x_a \partial P} \widehat{\rho}_{m,t} = & \frac{\partial^2 g_s}{\partial x_b \partial x_a} \frac{\partial \mu_t}{\partial P} + \frac{\partial g_s}{\partial x_a} \frac{\partial^2 \mu_t}{\partial x_b \partial P} \\ & + \frac{\partial g_s}{\partial x_b} \frac{\partial^2 \mu_t}{\partial x_a \partial P} + g_s \frac{\partial^3 \mu_t}{\partial x_b \partial x_a \partial P}. \end{aligned} \tag{4.37}$$

Applying Eq. (4.37) separately to the r- and p-mapping coordinates gives the derivative combination in Eq. (4.22). The excess action on the surrogate is then obtained by substituting this analytic result into Eq. (3.54). Thus, all dependence on the mapping Hamiltonian remains in the known coefficients $\nabla_{\mathbf{R}} \bar{h}^{\lambda\lambda'}$, while all density derivatives are supplied by the product surrogate.

If standardized coordinates are used, each term in Eq. (4.37) is multiplied by the appropriate inverse scale factors from Eq. (4.21). The analytic SEO profile is differentiated with respect to the physical mapping variables, not the standardized variables, unless the same chain rule is applied explicitly.

4.5 Initial hyperparameter identification

4.5.1 Joint optimization at the initial time

The correlation scales are identified once at $t = 0$ by joint optimization of

$$\boldsymbol{\theta}_0 = \left(\sigma_f, \sigma_n, \ell_1, \ldots, \ell_6 \right). \tag{4.38}$$

The optimization uses either the negative marginal log likelihood (NMLL) or a leave-one-out cross-validation (LOOCV) objective. The negative marginal log likelihood is

$$\mathcal{J}_{\text{NMLL}}(\boldsymbol{\theta}) = \frac{1}{2} \mathbf{u}_0^T K_{y,0}^{-1} \mathbf{u}_0 + \frac{1}{2} \log \det K_{y,0} + \frac{N}{2} \log(2\pi). \tag{4.39}$$

For leave-one-out optimization, define

$$\mathsf{P}_0 = K_{y,0}^{-1}, \qquad \boldsymbol{\zeta}_0 = \mathsf{P}_0 \mathbf{u}_0. \tag{4.40}$$

The leave-one-out residual in modulation space is

$$e_i^{\mathrm{LOO}} = \frac{\zeta_i(0)}{(\mathsf{P}_0)_{ii}}, \tag{4.41}$$

and a corresponding objective is

$$\mathcal{J}_{\mathrm{LOO}} = \frac{1}{N} \sum_{i=1}^{N} \left(e_i^{\mathrm{LOO}}\right)^2. \tag{4.42}$$

The initial optimization is performed with an L-BFGS method using a strong Wolfe line search. A validation criterion may terminate the optimization before the training objective begins to improve at the expense of held-out performance. Failed Cholesky factorizations, nonfinite objectives, and failed line searches are treated as rejected trial points rather than terminating the entire fit. The regularization scale σ_n participates in the joint fit by default; an optional fixed-σ_n mode restores a pinned regularization value.

The converged initial hyperparameters are stored as the anchor values

$$\boldsymbol{\theta}_{\mathrm{anchor}} = \boldsymbol{\theta}_0. \tag{4.43}$$

Subsequent refits are restricted relative to these anchors rather than treated as independent model-selection problems.

4.5.2 Focused-mode geometric anchors and LOOCV scale gauge

For the focused mapping initialization used in this thesis, the mapping length-scale anchors may be fixed from the focus-circle geometry. With zero-point parameter $\gamma = 1/2$, electronic state s, and independent phases $\phi_\lambda \sim \mathrm{Unif}[0, 2\pi)$, the focused radii are

$$r_s^2 + p_s^2 = 2\hbar(1+\gamma), \qquad (r_s, p_s) = \sqrt{2\hbar(1+\gamma)}(\cos\phi_s, \sin\phi_s), \tag{4.44}$$

$$r_{\lambda\neq s}^2 + p_{\lambda\neq s}^2 = 2\hbar\gamma, \qquad (r_\lambda, p_\lambda) = R_{\mathrm{focus}}(\cos\phi_\lambda, \sin\phi_\lambda), \tag{4.45}$$

where

$$R_{\mathrm{focus}} \equiv \sqrt{2\hbar\gamma}. \tag{4.46}$$

The occupied and unoccupied length-scale anchors in the two-state model are

$$\ell_{r_s}^{(0)} = \ell_{p_s}^{(0)} = \frac{\sqrt{3}}{2}, \qquad \ell_{r_{\bar{s}}}^{(0)} = \ell_{p_{\bar{s}}}^{(0)} = \frac{R_{\mathrm{focus}}}{2}, \tag{4.47}$$

where $\bar{s}$ denotes the other subsystem state. For $\hbar = 1$ and $\gamma = 1/2$, $R_{\text{focus}} = 1$, so the occupied and unoccupied anchors are $\sqrt{3}/2$ and $1/2$, respectively. These are geometric anchors of the focused sampling layout, not independent estimates of dynamical smoothness.

When the hyperparameters are selected only through the leave-one-out residuals, a common rescaling of $K_{y,0}$ leaves Eq. (4.41) invariant. Indeed, under $K_{y,0} \mapsto c^2 K_{y,0}$, both $\boldsymbol{\zeta}_0$ and the diagonal of P_0 scale as c^{-2}. Consequently, LOOCV identifies the ratio of signal and regularization scales, together with the length scales, but not the overall covariance amplitude. In this mode, $\sigma_f \equiv 1$ is used as a gauge choice. This gauge argument applies to the LOOCV objective; the NMLL retains explicit dependence on the overall covariance scale through its log-determinant term.

4.6 Restricted hyperparameter policies after the initial fit

The modeling assumption to be validated is that the principal physical correlation scales are identified at the initial time: the bath-coordinate scale is set by the initial bath distribution, while the mapping-sector scale is set by the $\hbar$-dependent SEO envelope and the focused mapping geometry. Later changes arise primarily from deformation and spreading of the support cloud. Unrestricted reoptimization at every time slice is therefore not used as the production strategy.

Let $\boldsymbol{\theta}_{\text{anchor}}$ denote the stored initial values. Four refit policies are defined.

Frozen policy. Before every refit, all hyperparameters are copied from $\boldsymbol{\theta}_{\text{anchor}}$. The kernel matrix is rebuilt on the current support locations and only the coefficient solve is repeated. Copying the anchors back, rather than merely skipping optimization, prevents drift from a previous nonfrozen update from accumulating.

Restricted breathing policy. The signal scale remains fixed at its anchor value. A short limited-memory BFGS optimization updates selected length scales using the configured LOO/NMLL objective augmented by the shrinkage penalty

$$\mathcal{R}_\ell = \gamma_\ell \sum_d \left[\log \ell_d - \log \ell_{d,\text{anchor}}\right]^2 , \tag{4.48}$$

with bounds on each update. Depending on the selected statistical model, the same optimization may update σ_n; the noise scale must therefore be described as estimated rather than anchored whenever it is allowed to vary. Mapping length scales fixed by the focused geometry are excluded from the optimization, so the remaining update is restricted to informative bath directions and any permitted noise scale. This policy is an available diagnostic branch, not the policy used by the focused production calculations reported below.

Trigger-based adaptive policy. No optimization is performed unless the support cloud becomes broad relative to the current kernel bandwidth on an informative coordinate. Define

$$a_d(t) = \frac{\mathrm{Var}_t(z_d)}{\ell_d^2}. \tag{4.49}$$

An update is triggered only when a configured informative-axis ratio exceeds

$$a_{\mathrm{trigger}} = 4. \tag{4.50}$$

For the present model, the bath-coordinate and bath-momentum ratios are reported separately. When the trigger is active, the restricted limited-memory BFGS update is applied; otherwise the anchored values are retained. Trigger activity is a run-specific diagnostic and is not inferred from the quality of the final fit. The trigger is geometric rather than residual based because a stationary kernel can fit values accurately at the support points while providing unreliable derivatives between separated lobes of the cloud.

Unrestricted policy. All hyperparameters are reoptimized at every refit. This policy is retained as a diagnostic comparison rather than as the production default. Repeated value-based optimization may trade short-scale response variation for a larger noise scale and longer length scales, thereby flattening the reconstructed density and its derivatives.

A legacy hard-freeze option overrides all policies and restores the anchor hyperparameters before every coefficient solve. In the time integrator of Chapter 5, hyperparameters are held fixed within each propagation leg and may change only at refit boundaries. The operator used within a leg is therefore constructed in one fixed regression representation; an inter-leg hyperparameter update changes the representation used by the next leg rather than the internal definition of the current leg.

4.7 Reconstruction and operator diagnostics

The product surrogate is judged in two different spaces. Value diagnostics measure the accuracy of $\widehat{\rho}_{m,t}$ as a density function. Operator diagnostics measure the accuracy and stability of the derivatives used by the excess mapping-QCLE generator. A small interpolation residual is not, by itself, evidence that the third derivatives between support points are reliable.

4.7.1 Training and leave-one-out diagnostics

At the support points, define the density-space prediction

$$\widehat{y}_i(t) = g_s(\mathbf{x}_i(t))\mu_t(\boldsymbol{\mathcal{X}}_i(t)) \,. \tag{4.51}$$

The density-space fit error is summarized by

$$\mathrm{RMS}_{\mathrm{fit}}(t) = \left[\frac{1}{N}\sum_{i=1}^{N}\left(\widehat{y}_i(t) - y_i(t)\right)^2\right]^{1/2}, \tag{4.52}$$

and by the coefficient of determination

$$R^2(t) = 1 - \frac{\sum_i \left(\widehat{y}_i - y_i\right)^2}{\sum_i \left(y_i - \bar{y}_t\right)^2}, \qquad \bar{y}_t = \frac{1}{N}\sum_i y_i. \tag{4.53}$$

For the inner GP, let

$$\mathsf{P}_t = K_{y,t}^{-1}. \tag{4.54}$$

The algebraic leave-one-out residual in modulation space is

$$e_i^{\mathrm{LOO}}(t) = \frac{\zeta_i(t)}{(\mathsf{P}_t)_{ii}}, \tag{4.55}$$

with standardized score

$$z_i^{\mathrm{LOO}}(t) = e_i^{\mathrm{LOO}}(t)\sqrt{(\mathsf{P}_t)_{ii}}. \tag{4.56}$$

The reported summaries include

$$\mathrm{RMS}_{\mathrm{LOO}}(t) = \left[\frac{1}{N}\sum_i \left(e_i^{\mathrm{LOO}}(t)\right)^2\right]^{1/2}, \tag{4.57}$$

$$e_{\max}^{\mathrm{LOO}}(t) = \max_i \left|e_i^{\mathrm{LOO}}(t)\right|, \tag{4.58}$$

and the standardized-outlier count

$$N_{3\sigma}(t) = \sum_{i=1}^{N} \mathbf{1}\left(\left|z_i^{\mathrm{LOO}}(t)\right| > 3\right), \tag{4.59}$$

where $\mathbf{1}(\cdot)$ is the indicator function. The optional fraction is $N_{3\sigma}/N$. This inverse-matrix identity is standard in kriging and radial-basis cross-validation [72, 73]. A density-space cross-validation would reconstruct each held-out modulation and multiply it by the unfloored profile value $g_s(\mathbf{x}_i)$. That transformation is important near the profile floor, where a small modulation-space error need not correspond to a small density-space error. The production files report only the algebraic modulation-space LOO diagnostic above; the density-space LOO procedure was not computed and is not presented as performed validation.

The regularized training residual of the inner GP is

$$\mathbf{r}_t^{\mathrm{train}} = K_t\boldsymbol{\zeta}_t - \mathbf{u}_t = -\sigma_n^2\boldsymbol{\zeta}_t. \tag{4.60}$$

It is therefore not expected to vanish when $\sigma_n > 0$.

4.7.2 Conditioning and coefficient concentration

The kernel-system condition number is

$$\kappa(K_{y,t}) = \|K_{y,t}\| \|K_{y,t}^{-1}\|. \tag{4.61}$$

The coefficient norm $\|\boldsymbol{\zeta}_t\|_2$ and its change between refits measure sensitivity of the representation to the moving support cloud.

To quantify whether the kernel expansion is distributed across many support points or concentrated on a small subset, define

$$w_i^{(\zeta)}(t) = \frac{|\zeta_i(t)|}{\sum_j |\zeta_j(t)|}, \tag{4.62}$$

$$N_{\text{eff}}^{(\zeta)}(t) = \frac{1}{\sum_i \left(w_i^{(\zeta)}(t)\right)^2}. \tag{4.63}$$

The ratio $N_{\text{eff}}^{(\zeta)}/N$ is a coefficient-concentration diagnostic; it is not a Monte Carlo effective sample size.

4.7.3 Physical linear functionals

For any mapping-space observable symbol $F(\boldsymbol{\mathcal{X}})$, define

$$\mathcal{I}_F(t) = \int d\boldsymbol{\mathcal{X}}\, F(\boldsymbol{\mathcal{X}}) \widehat{\rho}_{m,t}(\boldsymbol{\mathcal{X}}). \tag{4.64}$$

The SEO-consistent trace, populations, coherences, and energy are obtained by using the symbols already defined in Chapter 3; they are not redefined here. The raw density integral is

$$\mathcal{N}_{\text{raw}}(t) = \mathcal{I}_1(t), \tag{4.65}$$

while the subsystem trace is

$$\mathcal{N}_{\text{tr}}(t) = \sum_{\lambda=1}^{N_s} \mathcal{I}_{c_{\lambda\lambda}}(t). \tag{4.66}$$

For the exact projected density, both equal one by Eqs. (3.34) and (3.35). For the finite product surrogate, their residuals and the discrepancy

$$\varepsilon_{\text{raw}-\text{tr}}(t) = \mathcal{N}_{\text{raw}}(t) - \mathcal{N}_{\text{tr}}(t) \tag{4.67}$$

measure different aspects of the reconstruction error. The raw integral tests the global integral of the represented density, whereas $\mathcal{N}_{\text{tr}}(t)$ tests the subsystem trace obtained from the mapping observable symbols.

Analytic Gaussian moments are used whenever they are available. Otherwise, a quadrature rule is used:

$$\mathcal{I}_F(t) \approx \sum_{q=1}^{N_q} w_q F(\boldsymbol{\mathcal{Z}}_q)\widehat{\rho}_{m,t}(\boldsymbol{\mathcal{Z}}_q). \tag{4.68}$$

The same reconstructed density is used for all physical functionals and for the derivative operator.

4.7.4 Posterior and operator uncertainty

For a real linear functional $\mathcal{I}_F$, the GP model assigns the conditional variance

$$\mathrm{Var}_{\mathrm{GP}}[\mathcal{I}_F] = \int d\boldsymbol{\mathcal{X}} \int d\boldsymbol{\mathcal{X}}'\, F(\boldsymbol{\mathcal{X}})\Sigma_{\rho,t}\left(\boldsymbol{\mathcal{X}}, \boldsymbol{\mathcal{X}}'\right) F(\boldsymbol{\mathcal{X}}'). \tag{4.69}$$

For complex coherences, the real and imaginary parts are treated separately or with the corresponding complex covariance convention.

Let $\mathcal{Q}$ denote the linear excess action $-i\mathcal{L}'_m$ defined in Eq. (3.54). Its pointwise posterior variance is formally

$$\mathrm{Var}_{\mathrm{GP}}\left[\mathcal{Q}\tilde{\rho}_m(\boldsymbol{\mathcal{X}})\right] = \mathcal{Q}_{\boldsymbol{\mathcal{X}}}\mathcal{Q}_{\boldsymbol{\mathcal{X}}'}\Sigma_{\rho,t}\left(\boldsymbol{\mathcal{X}}, \boldsymbol{\mathcal{X}}'\right)\big|_{\boldsymbol{\mathcal{X}}'=\boldsymbol{\mathcal{X}}}. \tag{4.70}$$

This diagnostic is more directly related to the quantity used by the dynamics than the posterior variance of the density value alone. It was not evaluated for the production calculations. Equation (4.70) therefore defines a proposed model-conditional diagnostic and must not be read as a reported uncertainty band. In particular, convergence and calibration guarantees depend on kernel smoothness, likelihood specification, and experimental design; they do not survive arbitrary model misspecification automatically [66, 74].

A second operator-fidelity diagnostic compares the analytic excess action with a held-out numerical derivative estimate on a validation set,

$$\varepsilon_{\mathcal{Q}} = \left[\frac{1}{N_{\mathrm{val}}}\sum_{q=1}^{N_{\mathrm{val}}} \left|\mathcal{Q}_{\mathrm{analytic}}(\boldsymbol{\mathcal{Z}}_q) - \mathcal{Q}_{\mathrm{reference}}(\boldsymbol{\mathcal{Z}}_q)\right|^2\right]^{1/2}. \tag{4.71}$$

The reference may be supplied by a controlled finite-difference calculation or another independent derivative evaluation. The current hyperparameter policies are selected using value-fit and cloud-geometry criteria; they do not directly minimize Eqs. (4.70) or (4.71). This distinction is retained when the numerical results are interpreted.

4.7.5 Manufactured-density test of derivative fidelity

The decisive reconstruction test uses a manufactured smooth density for which the density, its gradient, and the complete excess action $\mathcal{Q}[\rho]$ are known analytically. The final validation campaign, reported in Chapter 6 and in complete per-seed form in Tables G.1–G.3, evaluates seeds

123, 124, and 125 at every support size $N = 300, 600, 1200, 2400$ under all three regularization policies $\ell_2 \in \{10^{-6}, 0.01, 0.05\}$, with paired evaluations on the training cloud and on an independent query cloud that are identical across the ℓ_2 values. The following relative L^2 errors illustrate that campaign with its seed-123, $\ell_2 = 10^{-6}$ on-support subset:

N_{train}	density	gradient	$\mathcal{Q}[\rho]$
300	3.37×10^{-3}	5.89×10^{-3}	1.277×10^{-2}
600	4.95×10^{-3}	8.84×10^{-3}	1.684×10^{-2}
1200	9.93×10^{-3}	1.254×10^{-2}	2.275×10^{-2}

At $N = 1200$, the off-support operator error is 2.471×10^{-2}. Repeating the on-support test at that resolution with seeds 123 and 124 gives 2.275×10^{-2} and 2.433×10^{-2}, respectively. Seed 123 alone increases over the three tabulated levels, but that sequence must not be interpreted as a deterministic power law. Across the complete three-seed campaign of Tables G.1–G.3, the operator error is approximately flat at $2\text{–}3 \times 10^{-2}$ over the factor-eight range $N = 300\text{–}2400$; seed scatter at the two smaller supports spans nearly an order of magnitude. Off-support error is at least as large as on-support error at each recorded level. Thus, adding independently sampled support does not reduce the operator error. No evidence of decreasing operator error was obtained under the tested independent-cloud enlargement; the clouds at different N are independently sampled, so this is an enlargement sensitivity result, not a deterministic nested-refinement study, and it is also not monotonic growth of a seed-averaged curve.

The inline subset above uses a frozen hyperparameter policy at the manufactured-test regularization $\ell_2 = 10^{-6}$. The complete campaign repeats every seed and support size at the pilot-selected value 0.01 and the production value 0.05 on identical training and query clouds (Fig. 6.1 and Tables 6.2–6.4); the same $2\text{–}3 \times 10^{-2}$ operator-error plateau is observed under all three tested policies. Changing regularization within the tested set therefore does not produce decreasing error under independent-cloud enlargement. Degradation of Gram-matrix conditioning relative to coverage is a plausible mechanism, but it was not separately measured and is not asserted as the cause. More generally, interpolation results on an embedded manifold control the restriction of a function under stated smoothness and sampling assumptions; they do not uniquely select its ambient normal derivatives [75, 76]. The result establishes only that no decrease in manufactured excess-operator error was demonstrated for the tested representation and policy under the tested independent-cloud enlargement; because the support clouds were independently sampled rather than nested or coupled, deterministic support convergence was not assessed.

4.8 Summary and transition to density-level integration

This chapter introduced a fixed-time product representation of the projected mapping density. The occupied-state SEO profile $g_s(\mathbf{x})$ supplies an analytic mapping-space reference structure, while a stationary six-dimensional ARD-RBF Gaussian process represents the smooth modulation

μ_t. The GP is trained on sign-preserving, floor-regularized ratios of the density labels to the SEO profile, and the reported density is always reconstructed as $g_s\mu_t$.

The analytic derivatives required by the excess mapping-QCLE operator contain product-rule contributions from both the SEO profile and the inner GP. Closed forms were given for the first and second derivatives of g_s and for the first through third derivatives of the ARD-RBF expansion. The construction uses analytic differentiation of the fixed SEO profile and the stationary covariance kernel.

Hyperparameters are identified jointly at the initial time and stored as anchors. Later refits use deliberately restricted policies: hard freezing, length-scale breathing with anchor shrinkage, geometry-triggered adaptation, or unrestricted refitting as a diagnostic comparison. The support-cloud ratio $\mathrm{Var}(z_d)/\ell_d^2$ provides a geometric warning that value interpolation may no longer control derivatives between separated regions of phase space.

The chapter also separated value-fit diagnostics from operator-fidelity diagnostics. Trace and observable moments, leave-one-out residuals, kernel conditioning, coefficient concentration, posterior density uncertainty, and uncertainty in the excess action all probe different aspects of the same reconstruction. None of these diagnostics removes the fixed nodal set imposed by the product ansatz; that restriction is assessed explicitly in the Results and Discussion chapters. Chapter 5 uses this fixed-time representation inside the split density-propagation scheme and specifies when support values, coefficients, and hyperparameters are updated.

The chapter therefore establishes that the product ansatz supplies analytic derivatives of its finite posterior mean, but it does not guarantee the normal derivatives or SEO structure of the unknown physical density. The manufactured operator test, not training or LOO residuals, is the decisive derivative-fidelity check.

Chapter 5

Density-Level Integration of the Mapping-QCLE

This chapter asks how the formal excess source can be coupled to PBME/MInt support motion at the density-label level. A defensible integrator must keep Hamiltonian transport distinct from density redistribution and must expose raw conservation failures rather than conceal them through normalization. The original numerical construction is an explicit midpoint predictor–corrector for labels on PBME/MInt-transported support, with the GP source evaluated intrinsically on the midpoint cloud. Although the update is formally second order when its semi-discrete source is smooth, accurate, and stable, the production MIDPOINT calculations do not demonstrate positive temporal order because representation, operator, and sampling errors dominate the refinement signal.

Chapter 3 expressed the mapping-basis quantum–classical Liouville equation as the sum of a Hamiltonian mapping Liouvillian and an excess density-level Liouvillian. Chapter 4 then introduced the fixed-time product surrogate

$$\widehat{\rho}_{m,t}(\boldsymbol{\mathcal{X}}) = g_s(\mathbf{x})\,\mu_t(\boldsymbol{\mathcal{X}}) \tag{5.1}$$

for the projected mapping density. The analytic factor g_s contains the known singly-excited-oscillator mapping profile, while the inner Gaussian process represents the evolving modulation μ_t. The present chapter combines these ingredients into a density-level propagation scheme.

The central physical distinction is between transport and redistribution. The Hamiltonian contribution moves phase-space support points along PBME characteristics. The excess contribution is not an additional force and does not generate a second trajectory field. It acts on the density through mixed mapping-variable and bath-momentum derivatives. In characteristic form, it is therefore a source that changes the signed density carried by the transported support cloud.

The calculations reported in Chapter 6 use a centred predictor–corrector. The support is advanced to the temporal midpoint with the Momentum Integral (MInt) algorithm, a symmetric symplectic integrator for the nonseparable MMST mapping Hamiltonian [77, 78]. An initial source evaluation predicts the midpoint labels; a temporary midpoint GP is then fitted, and its source evaluation advances the physical labels over the complete step. The support subsequently completes the second MInt half-step. This is the explicit midpoint branch documented in the implementation appendix and used by every figure labelled MIDPOINT.

For fixed support geometry and fixed GP hyperparameters, the excess action is also linear in the modulation labels and defines a constant finite-dimensional matrix generator. Crank–Nicolson and matrix-exponential updates of that frozen system are useful alternative branches, but they were not used to generate the reported MIDPOINT results. The distinction is maintained below so that formal properties of an optional source integrator are not attributed to the production data.

5.1 Characteristic form of the mapping-QCLE

Using the Liouvillian decomposition defined in Eq. (3.51), define

$$\mathcal{T}[f] \equiv -i\mathcal{L}_m^{\circ} f, \qquad \mathcal{Q}[f] \equiv -i\mathcal{L}_m' f. \tag{5.2}$$

The operator $\mathcal{T}$ generates PBME Hamiltonian transport, whereas $\mathcal{Q}$ is the excess density-level source. Its differential form is the operator given in Eq. (3.54); it is not rederived here. The mapping-QCLE may be written as

$$\frac{\partial \tilde{\rho}_m}{\partial t} = -i\mathcal{L}_m^{\circ}\tilde{\rho}_m + \mathcal{Q}[\tilde{\rho}_m]. \tag{5.3}$$

Let $\mathcal{F}_t^{\circ}$ denote the exact PBME Hamiltonian flow. With the Poisson-bracket convention used in Chapter 3, the Hamiltonian term is equivalent to advection by the PBME vector field $\dot{\boldsymbol{\mathcal{X}}}^{\circ}$. Hence

$$\frac{\partial \tilde{\rho}_m}{\partial t} + \dot{\boldsymbol{\mathcal{X}}}^{\circ} \cdot \nabla_{\boldsymbol{\mathcal{X}}} \tilde{\rho}_m = \mathcal{Q}[\tilde{\rho}_m]. \tag{5.4}$$

Along a PBME characteristic,

$$\boldsymbol{\mathcal{X}}(t) = \mathcal{F}_{t-t_0}^{\circ}\left(\boldsymbol{\mathcal{X}}(t_0)\right), \tag{5.5}$$

the density satisfies

$$\frac{d}{dt}\tilde{\rho}_m\left(\boldsymbol{\mathcal{X}}(t), t\right) = \mathcal{Q}[\tilde{\rho}_m]\left(\boldsymbol{\mathcal{X}}(t), t\right). \tag{5.6}$$

When $\mathcal{Q}$ is omitted, the density is constant along the Hamiltonian characteristic, as established in Eq. (3.64). In the full equation, the excess source changes the density value attached to the same moving phase-space point.

Equation (5.6) gives the physical interpretation used throughout this chapter. PBME determines where a support point moves. The excess operator determines how much projected density is carried by that point. Since the projected mapping density is signed, the source update must remain well defined when the density passes through zero. For this reason, density values are updated additively or through a linear source propagator; no logarithmic density rate is introduced.

5.1.1 Transported support and live density labels

At time t_n, let

$$\mathcal{S}_n = \{\boldsymbol{\mathcal{X}}_i^n\}_{i=1}^N \tag{5.7}$$

be the support cloud and let

$$y_i^n \approx \tilde{\rho}_m\left(\boldsymbol{\mathcal{X}}_i^n, t_n\right) \tag{5.8}$$

be the physical density label carried by its i-th point. The live label vector is

$$\mathbf{y}^n = \left(y_1^n, \ldots, y_N^n\right)^T. \tag{5.9}$$

These labels are distinct from the transformed modulation labels used by the inner GP. At any refit boundary, the transformed labels are obtained using the sign-preserving safe SEO profile defined in Eq. (4.7):

$$u_i^n = \frac{y_i^n}{g_{s,\mathrm{safe}}(\mathbf{x}_i^n)}. \tag{5.10}$$

The floor regularizes the regression near the nodal ring. The reconstructed field and the excess operator nevertheless use the exact analytic factor g_s, not the floored profile.

The support points evolve independently during a PBME transport leg, but the labels are coupled through the reconstructed field. Changing one label changes the GP coefficient vector and therefore changes the excess source evaluated at all support points. The method is consequently not a set of independent weighted trajectories. It is a moving-cloud discretization of a continuous density equation.

5.2 Finite-dimensional excess operator in the product-GP representation

Consider a source leg during which the support cloud $\mathcal{S} = \{\boldsymbol{\mathcal{X}}_i\}_{i=1}^N$, the input scaling, the SEO profile, and the kernel hyperparameters are held fixed. Let

$$\mathsf{D}_g = \mathrm{diag}\left[g_{s,\mathrm{safe}}(\mathbf{x}_1), \ldots, g_{s,\mathrm{safe}}(\mathbf{x}_N)\right]. \tag{5.11}$$

Then

$$\mathbf{u} = \mathsf{D}_g^{-1}\mathbf{y}. \tag{5.12}$$

For the stationary ARD–RBF kernel and observation regularization introduced in Chapter 4, define

$$K_y = K + \sigma_n^2 I, \qquad \boldsymbol{\zeta} = K_y^{-1}\mathbf{u}. \tag{5.13}$$

The inverse in Eq. (5.13) denotes a linear solve; it is not formed explicitly.

For each support centre $\boldsymbol{\mathcal{X}}_j$, define the physical product basis function

$$\Phi_j(\boldsymbol{\mathcal{X}}) = g_s(\mathbf{x}) k_{\boldsymbol{\theta}}\left(\boldsymbol{\mathcal{X}}, \boldsymbol{\mathcal{X}}_j\right). \tag{5.14}$$

The reconstructed density is

$$\widehat{\rho}_m(\boldsymbol{\mathcal{X}}) = \sum_{j=1}^{N} \zeta_j \Phi_j(\boldsymbol{\mathcal{X}}). \tag{5.15}$$

Since $\mathcal{Q}$ is linear, its action at the support points is

$$\mathcal{Q}[\widehat{\rho}_m]\left(\boldsymbol{\mathcal{X}}_i\right) = \sum_{j=1}^{N} \mathsf{L}_{ij}^{(\rho)} \zeta_j, \tag{5.16}$$

where the physical-density operator matrix is

$$\mathsf{L}_{ij}^{(\rho)} = \left. \mathcal{Q}\left[\Phi_j\right]\right|_{\boldsymbol{\mathcal{X}}=\boldsymbol{\mathcal{X}}_i}. \tag{5.17}$$

Using Eq. (3.54), this entry is

$$\mathsf{L}_{ij}^{(\rho)} = -\frac{\hbar}{8} \sum_{\lambda\lambda'} \Big(\nabla_{\mathbf{R}} \bar{h}^{\lambda\lambda'}(\mathbf{R}_i)\Big) \cdot \left[\left(\frac{\partial^2}{\partial r_{\lambda'} \partial r_\lambda} + \frac{\partial^2}{\partial p_{\lambda'} \partial p_\lambda}\right) \nabla_{\mathbf{P}} \Phi_j(\boldsymbol{\mathcal{X}})\right]_{\boldsymbol{\mathcal{X}}=\boldsymbol{\mathcal{X}}_i}. \tag{5.18}$$

Every entry of $\mathsf{L}^{(\rho)}$ is evaluated analytically from the product-rule derivatives derived in Sec. 4.4. Thus the fixed SEO profile contributes to the operator matrix, but it does not alter the stationary kernel itself.

During a fixed-support source leg, the safe profile values are constants. The modulation-label rate is therefore

$$\frac{d\mathbf{u}}{dt} = \mathsf{D}_g^{-1} \mathsf{L}^{(\rho)} \boldsymbol{\zeta}. \tag{5.19}$$

Define

$$\mathsf{L} = \mathsf{D}_g^{-1} \mathsf{L}^{(\rho)} \tag{5.20}$$

and

$$\mathsf{A} = \mathsf{L} K_y^{-1}. \tag{5.21}$$

Equivalently, the fixed-support evolution of the physical density labels is

$$\frac{d\mathbf{y}}{dt} = \mathsf{A}_\rho \mathbf{y}, \qquad \mathsf{A}_\rho = \mathsf{D}_g \mathsf{A} \mathsf{D}_g^{-1} = \mathsf{L}^{(\rho)} K_y^{-1} \mathsf{D}_g^{-1}. \tag{5.22}$$

The transformed-label and physical-label systems are similar and therefore have the same eigenvalues. The transformed-label form is used in the implementation because it is the coordinate system in which the inner GP is fitted. Equations (5.13) and (5.19) then give the closed linear system

$$\frac{d\mathbf{u}}{dt} = \mathsf{A}\mathbf{u}. \tag{5.23}$$

If the support locations, scaling factors, SEO profile values, kernel hyperparameters, and noise regularization are frozen, the matrix A is constant. The corresponding finite-dimensional source system has the formal propagator

$$\mathbf{u}(t+\tau) = \exp(\tau\mathsf{A})\mathbf{u}(t). \tag{5.24}$$

Equation (5.24) is exact for the frozen collocation system in Eq. (5.23); it is not exact for the continuum density, and it is not the update used for the MIDPOINT results. The reported predictor–corrector instead rebuilds the surrogate at a predicted midpoint state and uses the midpoint source slope.

For moderate N, the exponential may be formed by a stable Schur–Padé algorithm. For larger support sets, only its action on $\mathbf{u}$ is required and a Krylov or scaling-and-truncated-Taylor method may be used [79, 80]. The numerical tolerance of the exponential action must be chosen below the regression and operator errors so that it does not control the observed dynamics.

5.2.1 Exact profile versus safe profile

The matrices D_g and L contain the safe profile only where division by g_s is required. The product basis in Eq. (5.14) always contains the exact analytic profile. Hence the physical field propagated by the differential operator is

$$\widehat{\rho}_m = g_s\mu, \tag{5.25}$$

whereas the safe profile is a conditioning device for the finite label coordinates. Away from the nodal ring, $g_{s,\mathrm{safe}} = g_s$, and the distinction disappears. Near the nodal ring, the discrepancy is a controlled regularization error and must be reported through the floor-activity and nodal-region residuals defined later in this chapter.

5.3 MInt propagation of the Hamiltonian characteristics

The PBME characteristics and the trace–traceless mapping Hamiltonian were derived in Chapter 3. The common bath mass is the scalar M. To construct the numerical flow, split the mapping Hamiltonian as

$$H_m = H_A + H_B, \qquad H_A = \frac{\mathbf{P}^2}{2M}, \qquad H_B = V_0(\mathbf{R}) + \bar{h}_m(\mathbf{x}, \mathbf{R}). \tag{5.26}$$

Let $\mathcal{A}_\tau$ and $\mathcal{B}_\tau$ denote the exact Hamiltonian subflows generated by H_A and H_B. The MInt map is the symmetric composition

$$\Psi_\tau^{\text{MInt}} = \mathcal{A}_{\tau/2} \circ \mathcal{B}_\tau \circ \mathcal{A}_{\tau/2}. \tag{5.27}$$

Because both factors are exact Hamiltonian subflows, the composition is symplectic. The palindromic ordering makes it time reversible, and its local Hamiltonian-flow error is $\mathcal{O}(\tau^3)$ [77, 81].

5.3.1 Free bath drift

The H_A subflow is

$$\mathbf{R}(\tau) = \mathbf{R}(0) + \frac{\tau}{M}\mathbf{P}(0), \tag{5.28}$$

$$\mathbf{P}(\tau) = \mathbf{P}(0), \qquad \mathbf{r}(\tau) = \mathbf{r}(0), \qquad \mathbf{p}(\tau) = \mathbf{p}(0). \tag{5.29}$$

The bath tangent block is

$$D\mathcal{A}_\tau = \begin{pmatrix} I & (\tau/M)I \\ 0 & I \end{pmatrix}, \tag{5.30}$$

with identity action on the mapping variables.

5.3.2 Exact mapping rotation at fixed bath coordinate

During the H_B subflow, $\mathbf{R}$ is fixed and the mapping equations are linear with constant coefficients. For the real symmetric subsystem Hamiltonian used in this thesis, define

$$\mathsf{C}_\tau(\mathbf{R}) = \cos\left(\frac{\tau}{\hbar}\bar{h}(\mathbf{R})\right), \qquad \mathsf{S}_\tau(\mathbf{R}) = \sin\left(\frac{\tau}{\hbar}\bar{h}(\mathbf{R})\right). \tag{5.31}$$

The exact mapping update is

$$\mathbf{r}(\tau) = \mathsf{C}_\tau\mathbf{r}(0) + \mathsf{S}_\tau\mathbf{p}(0), \tag{5.32}$$

$$\mathbf{p}(\tau) = -\mathsf{S}_\tau\mathbf{r}(0) + \mathsf{C}_\tau\mathbf{p}(0). \tag{5.33}$$

The combined mapping transformation is orthogonal. It therefore preserves

$$\mathcal{R}_m^2 = \mathbf{r}^T\mathbf{r} + \mathbf{p}^T\mathbf{p} \tag{5.34}$$

exactly. Since the free-drift subflow does not change the mapping variables, MInt preserves $\mathcal{R}_m^2$ at every time step. This invariant is a numerical property of the Hamiltonian backbone; the excess source changes density labels but does not alter the phase-space coordinates.

For numerical evaluation, one may diagonalize

$$\bar{h}(\mathbf{R}) = U(\mathbf{R})\,\varepsilon(\mathbf{R})\,U(\mathbf{R})^T, \tag{5.35}$$

where U is orthogonal. This is a local computational diagonalization of the fixed-basis matrix, not a second derivation of the adiabatic-basis QCLE. In the local eigenmode coordinates $\widetilde{\mathbf{r}} = U^T\mathbf{r}$ and $\widetilde{\mathbf{p}} = U^T\mathbf{p}$, each pair rotates with frequency $\varepsilon_a/\hbar$.

5.3.3 Exact bath-momentum integral

During the same H_B subflow, the bath momentum satisfies

$$\dot{\mathbf{P}} = -\nabla_{\mathbf{R}}V_0(\mathbf{R}) - \nabla_{\mathbf{R}}\bar{h}_m(\mathbf{x}(\tau), \mathbf{R}). \tag{5.36}$$

Since $\mathbf{R}$ is fixed and the mapping rotation is known analytically, the force can be integrated exactly:

$$\mathbf{P}(\tau) = \mathbf{P}(0) - \tau\nabla_{\mathbf{R}}V_0(\mathbf{R}) - \mathbf{I}_\tau, \tag{5.37}$$

where

$$\mathbf{I}_\tau = \int_0^\tau \nabla_{\mathbf{R}}\bar{h}_m\left(\mathbf{x}(s), \mathbf{R}\right)\, ds. \tag{5.38}$$

This exact integral gives the method its name.

For completeness, define the bath-vector-valued matrix

$$\mathbf{G}(\mathbf{R}) = U(\mathbf{R})^T\left[\nabla_{\mathbf{R}}\bar{h}(\mathbf{R})\right]U(\mathbf{R}) \tag{5.39}$$

and the frequency differences

$$\omega_{ab} = \frac{\varepsilon_a - \varepsilon_b}{\hbar}. \tag{5.40}$$

Introduce

$$\boldsymbol{\Gamma}_{ab}(\tau) = \begin{cases} \mathbf{G}_{aa}\tau, & a = b, \\ \mathbf{G}_{ab}\dfrac{\sin(\omega_{ab}\tau)}{\omega_{ab}}, & a \neq b, \end{cases} \tag{5.41}$$

$$\boldsymbol{\Xi}_{ab}(\tau) = \begin{cases} \mathbf{0}, & a = b, \\ \mathbf{G}_{ab}\dfrac{1-\cos(\omega_{ab}\tau)}{\omega_{ab}}, & a \neq b. \end{cases} \tag{5.42}$$

The quotients are evaluated by their analytic zero-gap limits. With the convention that a quadratic form containing a bath-vector-valued matrix returns a bath vector,

$$\mathbf{I}_\tau = \frac{1}{2\hbar}\Big[\widetilde{\mathbf{r}}(0)^T\boldsymbol{\Gamma}(\tau)\widetilde{\mathbf{r}}(0) + \widetilde{\mathbf{p}}(0)^T\boldsymbol{\Gamma}(\tau)\widetilde{\mathbf{p}}(0) - 2\widetilde{\mathbf{r}}(0)^T\boldsymbol{\Xi}(\tau)\widetilde{\mathbf{p}}(0)\Big]. \tag{5.43}$$

For stable numerical evaluation, the scalar factors may be written as

$$\frac{\sin(\omega\tau)}{\omega} = \tau\ \mathrm{sinc}(\omega\tau), \qquad \frac{1-\cos(\omega\tau)}{\omega} = \tau\ \mathrm{cosc}(\omega\tau), \tag{5.44}$$

with $\operatorname{sinc}(0) = 1$ and $\operatorname{cosc}(0) = 0$.

5.3.4 Geometric properties of MInt

Let $\mathbb{J}$ be the canonical symplectic matrix in the extended ordering used in Chapter 3. The MInt tangent map satisfies

$$\left(D\Psi_\tau^{\mathrm{MInt}}\right)^T \mathbb{J} D\Psi_\tau^{\mathrm{MInt}} = \mathbb{J}, \qquad \det D\Psi_\tau^{\mathrm{MInt}} = 1. \tag{5.45}$$

The map is reversible:

$$\left(\Psi_\tau^{\mathrm{MInt}}\right)^{-1} = \Psi_{-\tau}^{\mathrm{MInt}}. \tag{5.46}$$

MInt does not conserve the exact Hamiltonian algebraically at finite step size. Its symplectic structure instead implies evolution under a nearby modified Hamiltonian, so the energy error is expected to remain bounded and oscillatory in a resolved time-step regime rather than exhibit systematic secular drift.

The tangent map may be propagated for geometric diagnostics, but it is not required to evaluate the excess source in the present algorithm. The source operator is collocated directly on the fixed midpoint support. Consequently, no differentiation of a pulled-back density through the inverse MInt map, and no second- or third-order derivative tensor of that map, enters the production scheme.

5.4 Explicit midpoint propagation used in the reported calculations

The reported MIDPOINT calculations advance the support and the live physical labels with a centred predictor–corrector. This branch must be distinguished from the optional frozen-support matrix exponential in Eq. (5.24). The support motion is independent of the live labels because it is generated by the PBME Hamiltonian, whereas the source slope depends on the GP reconstructed from the current labels.

5.4.1 Initial source slope and first Hamiltonian half-step

At the beginning of the step, evaluate

$$k_i^{(1)} = \mathcal{Q}[\widehat{\rho}_{m,n}]\left(\boldsymbol{\mathcal{X}}_i^n\right). \tag{5.47}$$

The support is then advanced to the temporal midpoint,

$$\boldsymbol{\mathcal{X}}_i^{n+1/2} = \Psi_{\Delta t/2}^{\mathrm{MInt}}\left(\boldsymbol{\mathcal{X}}_i^n\right), \tag{5.48}$$

while the PBME transport itself leaves the physical density label unchanged along the characteristic. The first source slope predicts a midpoint label,

$$y_i^{n+1/2,*} = y_i^n + \frac{\Delta t}{2} k_i^{(1)}. \tag{5.49}$$

The asterisk identifies a predictor, not an accepted physical state.

5.4.2 Midpoint refit and corrector slope

At the midpoint support, form the transformed predictor labels

$$\mathbf{u}^{n+1/2,*} = \left(\mathsf{D}_g^{n+1/2}\right)^{-1} \mathbf{y}^{n+1/2,*}, \tag{5.50}$$

rebuild the kernel matrix, and solve

$$\boldsymbol{\zeta}^{n+1/2,*} = K_{y,n+1/2}^{-1} \mathbf{u}^{n+1/2,*}. \tag{5.51}$$

The temporary midpoint surrogate is

$$\widehat{\rho}_{m,n+1/2}^{*} = g_s \mu_{n+1/2}^{*}. \tag{5.52}$$

Its excess action supplies the corrector slope,

$$k_i^{(2)} = \mathcal{Q}[\widehat{\rho}_{m,n+1/2}^{*}] \left(\boldsymbol{\mathcal{X}}_i^{n+1/2}\right). \tag{5.53}$$

Equivalently, for the product representation,

$$\mathbf{k}^{(2)} = \mathsf{L}_{n+1/2}^{(\rho)} \boldsymbol{\zeta}^{n+1/2,*}. \tag{5.54}$$

The accepted physical labels are advanced additively:

$$\mathbf{y}^{n+1} = \mathbf{y}^n + \Delta t\, \mathbf{k}^{(2)}. \tag{5.55}$$

This update remains well defined when a label crosses zero. The fitted midpoint values need not coincide exactly with the predictor labels when kernel regularization or the safe SEO profile is active; the corresponding density-space residual is recorded.

For diagnostic storage, the implementation may use the correction multiplier

$$w_i^n = \frac{y_i^n}{y_i^0}, \tag{5.56}$$

where the ratio is numerically well defined. Equation (5.55) then implies

$$\Delta w_i^n = \frac{\Delta t\, k_i^{(2)}}{y_i^0}. \tag{5.57}$$

The multiplier is an internal coordinate and a statistical diagnostic; the physical propagated quantity remains y_i^n. Because Eqs. (5.56) and (5.57) become ill-conditioned in the initial-density tails, the ratio is accompanied by an explicit post-processing threshold study. For a relative threshold η, the retained index set is

$$\mathcal{M}_\eta = \left\{ i : \ |y_i^0| > \eta \max_j |y_j^0| \right\}. \tag{5.58}$$

The minimum and empirical quantiles of $|y_i^0|$, the retained and excluded point fractions, the excluded absolute physical mass, and the response of the raw observables are reported in Section 6.6 and Appendix G. This threshold study does not alter the propagation. Density-label updates in Eq. (5.55), which do not divide by y_i^0, remain the primary dynamical variables.

5.4.3 Second Hamiltonian half-step and endpoint refit

The support completes the Hamiltonian step,

$$\boldsymbol{\mathcal{X}}_i^{n+1} = \Psi_{\Delta t/2}^{\mathrm{MInt}} \left(\boldsymbol{\mathcal{X}}_i^{n+1/2} \right). \tag{5.59}$$

The source-updated live density labels are carried unchanged during this second PBME transport leg. At the endpoint,

$$u_i^{n+1} = \frac{y_i^{n+1}}{g_{s,\mathrm{safe}}(\mathbf{x}_i^{n+1})}, \tag{5.60}$$

and the endpoint surrogate is refitted for the next step and for diagnostic evaluation.

The complete reported update is therefore

$$\mathbf{k}^{(1)} = \mathcal{Q}[\widehat{\rho}_{m,n}](\boldsymbol{\mathcal{X}}^n), \tag{5.61}$$

$$\boldsymbol{\mathcal{X}}^{n+1/2} = \Psi_{\Delta t/2}^{\mathrm{MInt}}(\boldsymbol{\mathcal{X}}^n), \tag{5.62}$$

$$\mathbf{y}^{n+1/2,*} = \mathbf{y}^n + \frac{\Delta t}{2}\mathbf{k}^{(1)}, \tag{5.63}$$

$$\mathbf{k}^{(2)} = \mathcal{Q}[\widehat{\rho}_{m,n+1/2}^{*}](\boldsymbol{\mathcal{X}}^{n+1/2}), \tag{5.64}$$

$$\mathbf{y}^{n+1} = \mathbf{y}^n + \Delta t\, \mathbf{k}^{(2)}, \tag{5.65}$$

$$\boldsymbol{\mathcal{X}}^{n+1} = \Psi_{\Delta t/2}^{\mathrm{MInt}}(\boldsymbol{\mathcal{X}}^{n+1/2}). \tag{5.66}$$

5.4.4 Hyperparameter and representation policy within a step

The kernel hyperparameters and input scaling are fixed over an accepted predictor–corrector step. The kernel matrix is rebuilt at the midpoint and endpoint because the support geometry changes, but the hyperparameter values are not reoptimized inside the step. Any permitted breathing or adaptive update is applied only after the endpoint diagnostics have been evaluated. This rule prevents a hyperparameter change from being conflated with the midpoint source correction.

5.5 Temporal order and approximation structure

For a fixed, sufficiently smooth semi-discrete characteristic system, the explicit midpoint rule is a second-order Runge–Kutta method. Its one-step local truncation error is $\mathcal{O}(\Delta t^3)$, and the symmetric MInt map has the same local order for the Hamiltonian support motion [77, 78, 82]. The combined reported update is therefore formally second order in time for the semi-discrete equations, provided that the GP reconstruction, safe-profile map, and midpoint source evaluation vary smoothly and that the step remains in the stable regime. A discontinuous floor activation or an unresolved source can invalidate the observed asymptotic order.

This temporal statement does not establish convergence to the continuum mapping-QCLE. Let $e_n = \widehat{\rho}_{m,n} - \tilde{\rho}_m(\cdot, t_n)$ be the field- reconstruction error. Since $\mathcal{Q}$ contains mixed third derivatives,

$$|\mathcal{Q}[\widehat{\rho}_{m,n}] - \mathcal{Q}[\tilde{\rho}_m]| \leq \frac{\hbar}{8} \sum_{\lambda\lambda'} \left\| \nabla_{\mathbf{R}} \bar{h}^{\lambda\lambda'} \right\| \\ \times \left(\left\| \partial_{r_{\lambda'}} \partial_{r_\lambda} \nabla_{\mathbf{P}} e_n \right\| + \left\| \partial_{p_{\lambda'}} \partial_{p_\lambda} \nabla_{\mathbf{P}} e_n \right\| \right). \tag{5.67}$$

A small value-fit residual does not therefore imply a small excess-operator error. The relevant approximation norm includes the mixed derivatives that enter $\mathcal{Q}$.

A schematic one-step error decomposition is

$$\varepsilon_{\text{step}}^{n+1} = \mathcal{O}(\Delta t^3) + \Delta t\, \varepsilon_{\mathcal{Q}}^{n+1/2} + \varepsilon_{\text{fit}}^{n+1} + \varepsilon_{\text{floor}}^{n+1} + \varepsilon_{\text{cloud}}^{n+1} + \varepsilon_{\text{round}}^{n+1}, \tag{5.68}$$

where $\varepsilon_{\mathcal{Q}}$ is the operator error of the product surrogate, ε_{fit} is the endpoint reconstruction error, $\varepsilon_{\text{floor}}$ is the safe-profile regularization error, $\varepsilon_{\text{cloud}}$ contains finite-support and cloud- estimator errors, and $\varepsilon_{\text{round}}$ denotes linear-algebra and floating-point error. A time-step scan isolates temporal error only when the support set and regression policy are held fixed; a separate support and model refinement is required to assess continuum convergence.

5.5.1 Relation to fixed-support source integrators

The first-order frozen-source update

$$\mathbf{u}^+ = \left(I + \Delta t\, \mathsf{A}_{n+1/2}\right) \mathbf{u}^- \tag{5.69}$$

retains only the linear term of the matrix exponential and is not the predictor–corrector in Eq. (5.55). The latter evaluates a second source slope after refitting a predicted midpoint density.

The Crank–Nicolson and matrix-exponential branches instead solve alternative frozen-midpoint systems based on $\mathsf{A}_{n+1/2}$. They are documented for reproducibility, but no temporal-order or stability property of those branches is used to interpret the MIDPOINT figures.

5.6 Physical moments and discrete consistency

Physical observables are evaluated from the endpoint product surrogate using the mapping formulas established in Chapter 3. The regression and propagation layers must not replace these observable definitions by an unweighted average of support labels.

The cloud coefficient depends on the sampling branch. For a full-dimensional positive proposal q_0, ordinary importance sampling gives $c_i^n = y_i^n/[Nq_0(\boldsymbol{\mathcal{X}}_i^0)]$. The focused calculations reported in this thesis are different: their mapping support lies on fixed- action circles and is singular with respect to mapping-space Lebesgue measure. They therefore use the calibrated focused estimator

$$c_i^n = \frac{1}{N}\frac{y_i^n}{y_i^0} = \frac{w_i^n}{N}, \tag{5.70}$$

for nonzero initial labels. This calibration gives equal base coefficients at $t = 0$ and the standard focused MMST observable averages [2, 19, 59]. It must not be identified with full-dimensional importance sampling of the smooth SEO density.

A cloud estimate of an observable with mapping symbol A_m is

$$\langle A\rangle_n^{\text{cloud}} = \sum_{i=1}^{N} A_m(\boldsymbol{\mathcal{X}}_i^n)c_i^n. \tag{5.71}$$

This is a signed quadrature sum. It is not normalized by $\sum_i |c_i^n|$. The corresponding functional of the continuous surrogate is evaluated as described in Chapter 4. For full-dimensional sampling, comparison of the two estimators tests two representations of the same density integral. For focused sampling, it is a cross-representation diagnostic between a calibrated lower-dimensional mapping ensemble and a smooth off-manifold continuation; equality is not an automatic quadrature identity.

For any selected moment functional, absorb the fixed sampling calibration and observable values into a row vector $\mathbf{m}^T$. Under the reported midpoint update, its one-step change is

$$\mathbf{m}^T\mathbf{y}^{n+1} - \mathbf{m}^T\mathbf{y}^n = \Delta t\,\mathbf{m}^T\mathbf{k}^{(2)}. \tag{5.72}$$

For the optional frozen linear source system, the left-null residual

$$\varepsilon_{\text{null}}(\mathbf{m}) = \left\|\mathbf{m}^T\mathsf{D}_g\mathsf{A}\mathsf{D}_g^{-1}\right\|_2 \tag{5.73}$$

measures violation caused by collocation, regularization, and finite support. For the reported predictor–corrector, the directly relevant conservation diagnostic is $|\mathbf{m}^T\mathbf{k}^{(2)}|$, together with accumulated raw moment drift.

The raw integral, SEO-consistent trace, subsystem populations and coherences, and total energy are computed from $\widehat{\rho}_{m,n} = g_s\mu_n$, never from μ_n alone. Their definitions are not repeated here. The raw integral and the subsystem trace coincide for an exact normalized projected density; their difference for the finite surrogate is a diagnostic of representation error.

5.7 Physical interpretation of the split evolution

The numerical method separates three mechanisms that have different physical and numerical meanings. First, the MInt legs provide Hamiltonian transport of the bath and mapping coordinates under the trace–traceless mapping Hamiltonian. They preserve the canonical geometry, the extended phase-space volume, and the mapping radius. These properties concern the PBME backbone and do not imply that the full mapping density is constant once the excess source is included.

Second, the explicit midpoint label update supplies the density-level back-reaction associated with the part of the mapping-QCLE that is absent from PBME and cannot be written as an additional force on an independent trajectory [2, 62]. It changes the signed density evaluated on the midpoint cloud. Through the SEO-consistent observable kernels, this redistribution can alter subsystem populations and coherences without changing the midpoint coordinates during the source leg.

Third, function reconstruction converts finite support data through the product GP into the smooth field required by the excess operator. The analytic SEO factor supplies the known mapping-space envelope and nodal structure, while the stationary inner kernel supplies the evolving modulation. Since the source depends on mixed third derivatives, accurate interpolation at the training points is not sufficient. The between-point length scales, support geometry, and coefficient stability control the corrector slope. The fixed SEO factor also imposes a nodal set that the exact later-time projected density need not share.

These mechanisms lead to distinguishable failure modes. Loss of symplecticity, reversibility, or mapping-radius conservation indicates an error in the Hamiltonian propagation. A large training or leave-one-out residual indicates a value-reconstruction problem. A small value residual but a large operator-fidelity residual indicates that the density values are interpolated while the derivatives entering L are unreliable. A rapidly growing corrector slope or strong time-step sensitivity indicates an unresolved or unstable source update.

5.8 Algorithmic specification

One complete production step from t_n to $t_{n+1} = t_n + \Delta t$ is:

1. Start from the support cloud and physical labels $\{\boldsymbol{\mathcal{X}}_i^n, y_i^n\}_{i=1}^N$, together with the endpoint product surrogate $\widehat{\rho}_{m,n} = g_s\mu_n$.

2. Select the kernel hyperparameters for the current step according to the policy in Sec. 4.6. Hold these hyperparameters and all input-scaling parameters fixed until the complete symmetric step has finished.

3. Evaluate the initial source slope $\mathbf{k}^{(1)}$ from the endpoint surrogate using Eq. (5.47).

4. Propagate every support point through the first MInt half-step and predict the midpoint labels using Eq. (5.49).

5. Form the midpoint transformed predictor labels, rebuild $K_{y,n+1/2}$ by a Cholesky factorization, and fit the temporary midpoint surrogate.

6. Evaluate the corrector slope $\mathbf{k}^{(2)}$ at the midpoint support and advance the physical labels additively with Eq. (5.55).

7. Propagate the midpoint support through the second MInt half-step. Carry the source-updated live density labels unchanged during this transport leg.

8. Transform the endpoint labels by Eq. (5.60), solve for $\boldsymbol{\zeta}^{n+1}$, and construct the endpoint product surrogate.

9. Evaluate the physical, geometric, regression, source, conservation, and operator diagnostics. If the step is accepted, any permitted breathing or adaptive hyperparameter update is applied only when preparing the next step.

The primary propagated density state is the vector of physical labels $\mathbf{y}^n$. Ratios to the SEO profile are internal regression coordinates. They are never treated as physical densities and are not used without the sign-preserving nodal regularization.

5.9 Diagnostics for the results chapter

The results analysis must distinguish errors in the Hamiltonian map, the GP representation, and the discrete source operator. The following diagnostics are therefore recorded separately.

Whenever a source contribution is summarized as a ratio of global RMS norms, that scalar is only a global diagnostic. It is not a pointwise bound on the source update, and no statement such as “less than one percent per step” is inferred for every support point.

5.9.1 Hamiltonian-map diagnostics

For a computed MInt tangent map, define the symplectic residual

$$\varepsilon_{\text{symp}} = \frac{\left\|(D\Psi)^T \mathbb{J}(D\Psi) - \mathbb{J}\right\|_F}{\|\mathbb{J}\|_F}. \tag{5.74}$$

Time reversibility is tested by

$$\varepsilon_{\mathrm{rt}} = \frac{\left\| \Psi_{-\tau}^{\mathrm{MInt}} \left(\Psi_{\tau}^{\mathrm{MInt}} (\boldsymbol{\mathcal{X}}) \right) - \boldsymbol{\mathcal{X}} \right\|}{1 + \|\boldsymbol{\mathcal{X}}\|}. \tag{5.75}$$

The mapping-radius residual is

$$\varepsilon_{\mathcal{R},i}^{n} = \left| \mathcal{R}_m^2(\mathbf{x}_i^n) - \mathcal{R}_m^2(\mathbf{x}_i^0) \right|, \tag{5.76}$$

and the trajectory energy error is

$$\Delta H_i^n = H_m(\boldsymbol{\mathcal{X}}_i^n) - H_m(\boldsymbol{\mathcal{X}}_i^0). \tag{5.77}$$

Bounded oscillation of ΔH_i^n is consistent with a resolved symplectic integration; systematic drift signals an implementation problem or an unresolved time step.

5.9.2 Regression and floor diagnostics

The training, leave-one-out, conditioning, coefficient-concentration, posterior-variance, cloud-spread, and operator-fidelity diagnostics are those defined in Chapter 4 and are not redefined here. They are evaluated at both midpoint and endpoint refits.

The fraction of points on which the profile floor is active is

$$f_{\mathrm{floor}}^{n+1/2} = \frac{1}{N} \# \left\{ i : |g_s(\mathbf{x}_i^{n+1/2})| < \varepsilon_g g_{\mathrm{ref},n+1/2} \right\}. \tag{5.78}$$

A corresponding density-space residual is evaluated on the active subset. The floor may stabilize the coefficient solve while still biasing the density near the nodal ring; these two effects must be reported separately.

5.9.3 Optional spectral diagnostics of a frozen source matrix

The spectral abscissa

$$a_{\mathrm{spec}}(\mathsf{A}) = \max_j \operatorname{Re} \xi_j(\mathsf{A}) \tag{5.79}$$

indicates asymptotic growth or decay in the frozen source system. Since A need not be normal, its eigenvalues do not completely determine finite-time amplification. The quantities

$$\|\mathsf{A}\|_2, \qquad \|\exp(\Delta t \mathsf{A})\|_2, \qquad \frac{\|\exp(\Delta t \mathsf{A})\mathbf{u}\|_2}{\|\mathbf{u}\|_2} \tag{5.80}$$

may therefore be monitored together with $a_{\mathrm{spec}}(\mathsf{A})$. Large transient amplification with a modest spectral abscissa is evidence of non-normality and can magnify small coefficient or support perturbations.

The linear-Dyson comparator is measured by

$$\varepsilon_{\text{Dyson}} = \frac{\|\exp(\Delta t \mathsf{A})\mathbf{u} - (I + \Delta t \mathsf{A})\mathbf{u}\|_2}{1 + \|\exp(\Delta t \mathsf{A})\mathbf{u}\|_2}. \tag{5.81}$$

This diagnostic directly measures the significance of quadratic and higher powers of a frozen discrete excess generator over one source leg. These quantities diagnose the optional matrix branches; they were not recorded for, and are not used to support claims about, the reported explicit MIDPOINT calculations.

5.9.4 Moment and signed-cloud diagnostics

The physical raw integral, subsystem trace, populations, coherences, and energy are evaluated from the complete product density. Their residuals are those defined in Chapter 4. The left-null residual in Eq. (5.73) is additionally reported at the operator level.

For signed cloud contributions

$$c_i^n = \frac{1}{N} \frac{y_i^n}{y_i^0} \tag{5.82}$$

in the focused branch, define the cancellation ratio

$$\chi^n_{\text{cancel}} = \frac{|\sum_i c_i^n|}{\sum_i |c_i^n|} \tag{5.83}$$

and the signed effective sample size

$$\text{ESS}^n_{\pm} = \frac{(\sum_i c_i^n)^2}{\sum_i (c_i^n)^2}. \tag{5.84}$$

Small values indicate that an observable is obtained through cancellation of large positive and negative contributions. In that regime, support quality and quadrature resolution may dominate the temporal integration error.

5.9.5 Time-step and operator-fidelity tests

A controlled time-step study compares at least two successively reduced time steps while keeping the support and GP policy fixed. It reports observables, moment residuals, MInt energy errors, and both source-slope evaluations. A separate independent-cloud enlargement in support size or placement is required to distinguish temporal sensitivity from stochastic representation sensitivity [51, 52].

For every three-level order diagnostic, numerical resolvability is declared before the order is calculated. With level values u_1, u_2, u_3,

$$\tau_{\text{noise}} = \tau_{\text{abs}} + \tau_{\text{rel}} \max_{k=1,2,3} |u_k|, \qquad \tau_{\text{abs}} = 10^{-12}, \quad \tau_{\text{rel}} = 10^{-12}. \tag{5.85}$$

Both successive differences must exceed τ_{noise}; otherwise the row is labelled "roundoff- or saturation-limited; order not interpreted." For aligned time-series differences, the maximum absolute level in Eq. (5.85) is replaced by the maximum RMS scale of the three aligned series. A resolvable row must then contract monotonically and must not show implausibly rapid non-asymptotic contraction. For both stochastic moving-cloud methods, PBME and MIDPOINT, both differences must also exceed the pooled independent-seed variation. Thus a ratio of two roundoff-level quantities is never promoted to an observed order.

The validation campaign completed the intended run inventory through full scattering. The time-step campaign contains 24 paired momentum–seed–step configurations: $P_{\text{init}} \in \{20, 100\}$, seeds $11, 29, 47, 73$, and $\Delta t \in \{0.5, 0.25, 0.125\}$ with $N = 1000$; each paired configuration contains matched PBME and MIDPOINT calculations initialized from the same cloud, so the campaign comprises 48 individual method executions. The independent-cloud campaign contains 18 paired configurations at $P_{\text{init}} \in \{20, 100\}$, seeds $11, 29, 47$, and $N \in \{500, 1000, 2000\}$ with $\Delta t = 0.25$. The initially proposed representative low/intermediate pair 20/40 was replaced by the wider low/high pair 20/100; no $P_{\text{init}} = 40$ refinement run is claimed. For MIDPOINT at $P_{\text{init}} = 20$, the population-L^2 differences are approximately 0.0936 from coarse to intermediate step and 0.1068 from intermediate to fine step, whereas the independent-seed standard deviation is 0.7949. The refinement difference therefore does not shrink and is an order of magnitude smaller than realization scatter. Across the 128 method–momentum–seed–observable rows of the four-seed run-by-run analysis (Table G.4), 102 rows are rejected because the refinement signal is not resolvable against independent-seed dispersion, 15 are roundoff- or saturation-limited under Eq. (5.85), 3 rows give zero or negative observed order, and the remaining 8 rows give small positive values that never exceed 0.64. No positive temporal order approaching a formal integrator order is demonstrated for MIDPOINT.

The reference post-processing recovers the expected temporal behaviour: approximately second order for the symmetric TDSE split-operator scheme and approximately fourth order for the grid-QCLE Runge–Kutta integrator, so failure to recover order for MIDPOINT is not an artefact of the post-processing. Spatial or phase-space-grid refinement is assessed separately and is not assigned the temporal order of either integrator. The support clouds are independent between resolutions rather than nested. Support-size differences therefore combine resolution and sampling effects and cannot be assigned a deterministic support order.

The held-out operator-fidelity residual defined in Eq. (4.71) is evaluated at midpoint support and independent validation points. Agreement of populations under time-step refinement while this residual remains large is interpreted as error cancellation rather than as evidence that the density-level operator has been resolved.

5.10 Summary

This chapter formulated the reported numerical mapping-QCLE evolution as an explicit midpoint predictor–corrector along Hamiltonian support characteristics. The PBME support cloud is propagated by MInt, whose exact subflows consist of free bath drift, mapping rotation at fixed bath coordinate, and the analytic bath-momentum integral. The resulting map is symmetric, symplectic, volume preserving, and exactly conserves the mapping radius.

An initial excess slope predicts the live labels at the transported midpoint. A temporary midpoint GP is fitted to those labels, and its analytic excess action supplies the corrector slope used in the full label update. For fixed support, the same product representation also defines the matrix $\mathsf{A} = \mathsf{L}K_y^{-1}$; Crank–Nicolson and matrix-exponential updates of this frozen system are optional branches, not the source of the results reported in this thesis.

The reported semi-discrete update is formally second order in time under the smoothness and stability assumptions of the explicit midpoint rule, but its total error also contains finite-support, product-model, nodal-floor, derivative, and focused-cloud estimator contributions. The diagnostics defined here separate these mechanisms. Geometric residuals test MInt; value and leave-one-out diagnostics test the regression; corrector and operator-fidelity diagnostics test the excess action; and signed-cloud diagnostics quantify cancellation and sampling quality.

In summary, the explicit midpoint rule is a formal second-order integrator for its semi-discrete source, not evidence of convergence to the continuum mapping-QCLE. The source is evaluated intrinsically on transported midpoint support; applied source equals raw source in production because clipping controls are inactive.

Chapter 6

Numerical Results and Physical Interpretation

The central question is whether the product-GP excess source supplies a physically reliable correction to PBME for the one-dimensional two-state scattering model. Reliability requires more than accurate interpolation of the density values used for training. The reconstructed derivatives must remain accurate away from the sampled cloud, the fitted density must remain close to the physical singly-excited-oscillator (SEO) image, the propagation must become insensitive to time step and cloud size, raw invariants must be preserved, and physical errors relative to controlled TDSE or grid-QCLE references must decrease. These questions are tested separately below so that a favourable value-reconstruction result cannot conceal a failure of the derivative operator or the dynamics. The combined evidence gives a controlled negative result: the tested MIDPOINT construction does not satisfy these joint requirements and does not show a systematic improvement over PBME.

6.1 Numerical design and hierarchy of evidence

Table 6.1 summarizes the independent numerical questions. Raw physical observables and raw invariants are primary. Errors relative to analytic manufactured functions or controlled reference solutions are the next level. GP training residuals, leave-one-out residuals, posterior variance, and R^2 diagnose the interpolant internally but are not treated as physical-error estimates. Likewise, unit-mass density shapes are useful only after the corresponding raw integral has been examined.

Table 6.1: Numerical studies used to distinguish interpolation accuracy, operator fidelity, dynamical stability, estimator structure, and physical predictive accuracy. Here and below, N denotes the number of independently sampled phase-space trajectories in a cloud.

Question	Quantity examined	Controlled variation	Evidence reported
Manufactured operator	ρ, $\nabla\rho$, and $Q[\rho]$ on and away from the training cloud	$\ell_2 = 10^{-6}, 0.01, 0.05$; $N = 300, 600, 1200, 2400$; seeds 123–125	Relative L_1, L_2, and L_∞ errors
Time-step refinement	Dynamical observables at three common physical-time grids	$\Delta t = 0.5, 0.25, 0.125$; $P_{\text{init}} = 20, 100$; four independent seeds	Successive differences compared with seed dispersion
Cloud-size sensitivity	Effect of enlarging independently sampled phase-space clouds	$N = 500, 1000, 2000$; three independent seeds	Mean, sample standard deviation, and change relative to seed dispersion
Replication	Sensitivity of endpoint observables to independently sampled initial clouds	Seeds 11, 29, 47, and 73; $N = 1000$; $\Delta t = 0.25$	Mean, sample standard deviation, and Student-t interval
Estimator structure	SEO leakage, raw invariants, and signed-label conditioning	Three collision-time snapshots and threshold sweep	Projection residual, raw drift, effective sample sizes, and excluded physical mass
Reference calculations	Numerical control of TDSE and grid-QCLE baselines	Separate three-level time and grid refinements	Values, successive differences, observed order, edge mass, and CFL ratio
Physical comparison	PBME and MIDPOINT against the same TDSE or grid-QCLE reference	Paired four-seed comparisons	Density and observable error differences with paired Student-t intervals

6.2 Manufactured density, gradient, and excess operator

The manufactured calculation isolates the reconstruction problem from dynamical feedback. An analytic positive density is sampled in the same six-dimensional phase space used by the product surrogate. Its density, gradient, and excess action $Q[\rho]$ are then compared with their analytic counterparts both on the training cloud and on 1000 independently drawn query points. Tables 6.2–6.4 report the relative L_1, L_2, and L_∞ errors as means and sample standard deviations over three independent cloud pairs for every regularization and cloud size.

Table 6.2: Manufactured-function errors for the reconstructed density ρ. Entries are mean $\pm$ sample standard deviation over three independently generated training/query-cloud pairs. The relative norms use the corresponding analytic quantity in the denominator.

ℓ_2	N	Evaluation domain	E_1	E_2	E_∞
1.00×10^{-6}	300	training cloud	0.001 ± 0.00143	0.00128 ± 0.00182	0.00209 ± 0.00313
1.00×10^{-6}	300	independent query cloud	0.00451 ± 0.00349	0.00699 ± 0.00679	0.0152 ± 0.0191
1.00×10^{-6}	600	training cloud	0.00523 ± 0.00595	0.00619 ± 0.00669	0.00748 ± 0.00676
1.00×10^{-6}	600	independent query cloud	0.0075 ± 0.00558	0.00946 ± 0.00541	0.0154 ± 0.00413
1.00×10^{-6}	1200	training cloud	0.00679 ± 0.0022	0.00764 ± 0.00217	$0.0091 \pm 2.41 \times 10^{-4}$
1.00×10^{-6}	1200	independent query cloud	0.00766 ± 0.00214	0.00839 ± 0.00219	0.0122 ± 0.00115
1.00×10^{-6}	2400	training cloud	0.00649 ± 0.00163	0.00684 ± 0.00124	0.00876 ± 0.00111

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Table 6.2 (continued)

ℓ_2	N	Evaluation domain	E_1	E_2	E_∞
1.00×10^{-6}	2400	independent query cloud	0.00747 ± 0.0012	$0.0083 \pm 4.15 \times 10^{-4}$	0.0158 ± 0.015
0.01	300	training cloud	0.00123 ± 0.00128	0.00159 ± 0.00161	0.0028 ± 0.00272
0.01	300	independent query cloud	0.00467 ± 0.00381	0.00773 ± 0.00822	0.019 ± 0.0262
0.01	600	training cloud	0.00521 ± 0.00594	0.00617 ± 0.00665	0.00744 ± 0.00666
0.01	600	independent query cloud	0.00734 ± 0.00558	0.00928 ± 0.00547	0.0151 ± 0.00486
0.01	1200	training cloud	0.00679 ± 0.00227	0.00765 ± 0.00222	$0.00908 \pm 4.61 \times 10^{-4}$
0.01	1200	independent query cloud	0.00762 ± 0.00215	0.00831 ± 0.00222	0.0117 ± 0.00116
0.01	2400	training cloud	0.00715 ± 0.00251	0.00755 ± 0.0022	$0.00978 \pm 2.94 \times 10^{-4}$
0.01	2400	independent query cloud	0.00799 ± 0.00213	0.00843 ± 0.00148	0.0135 ± 0.0069
0.05	300	training cloud	0.00101 ± 0.00143	0.0013 ± 0.00181	0.00215 ± 0.00311
0.05	300	independent query cloud	0.00459 ± 0.00367	0.00711 ± 0.00717	0.0158 ± 0.0207
0.05	600	training cloud	0.0052 ± 0.00589	0.00614 ± 0.0066	0.00743 ± 0.00668
0.05	600	independent query cloud	0.00746 ± 0.00552	0.0094 ± 0.00533	0.0152 ± 0.00414
0.05	1200	training cloud	0.00681 ± 0.00219	0.00764 ± 0.00216	$0.00893 \pm 3.06 \times 10^{-4}$
0.05	1200	independent query cloud	0.00768 ± 0.00212	0.00834 ± 0.00222	0.0122 ± 0.00103
0.05	2400	training cloud	0.00719 ± 0.00251	0.00758 ± 0.00223	$0.00955 \pm 4.83 \times 10^{-4}$
0.05	2400	independent query cloud	0.00805 ± 0.00208	$0.0089 \pm 9.81 \times 10^{-4}$	0.017 ± 0.0132

Table 6.3: Manufactured-function errors for the reconstructed density gradient $\nabla\rho$. Entries are mean $\pm$ sample standard deviation over three independently generated training/query-cloud pairs. The relative norms use the corresponding analytic quantity in the denominator.

ℓ_2	N	Evaluation domain	E_1	E_2	E_∞
1.00×10^{-6}	300	training cloud	0.00606 ± 0.00497	0.00792 ± 0.00765	0.0132 ± 0.0126
1.00×10^{-6}	300	independent query cloud	0.00753 ± 0.0062	0.0099 ± 0.00913	0.0274 ± 0.0203
1.00×10^{-6}	600	training cloud	0.0104 ± 0.0077	0.0117 ± 0.00609	0.0199 ± 0.00625
1.00×10^{-6}	600	independent query cloud	0.0115 ± 0.00724	0.0125 ± 0.00528	0.0279 ± 0.0159
1.00×10^{-6}	1200	training cloud	0.0111 ± 0.00296	0.0102 ± 0.00264	$0.0123 \pm 5.89 \times 10^{-4}$
1.00×10^{-6}	1200	independent query cloud	0.0119 ± 0.00291	0.0111 ± 0.00265	0.0161 ± 0.00726
1.00×10^{-6}	2400	training cloud	0.0118 ± 0.00149	$0.0105 \pm 8.74 \times 10^{-4}$	0.0136 ± 0.00444
1.00×10^{-6}	2400	independent query cloud	0.0126 ± 0.00109	$0.0113 \pm 6.03 \times 10^{-4}$	0.0158 ± 0.00374
0.01	300	training cloud	0.00638 ± 0.0055	0.00825 ± 0.00831	0.0128 ± 0.0126
0.01	300	independent query cloud	0.00775 ± 0.00669	0.0109 ± 0.0113	0.0331 ± 0.0337
0.01	600	training cloud	0.0103 ± 0.00768	0.0115 ± 0.00608	0.0199 ± 0.00586
0.01	600	independent query cloud	0.0112 ± 0.00724	0.0121 ± 0.00527	0.0234 ± 0.0133
0.01	1200	training cloud	0.011 ± 0.00295	0.0101 ± 0.00264	$0.0119 \pm 4.28 \times 10^{-4}$
0.01	1200	independent query cloud	0.0118 ± 0.00287	0.0109 ± 0.00264	0.0138 ± 0.00402
0.01	2400	training cloud	0.0124 ± 0.00234	0.0113 ± 0.00169	0.0147 ± 0.00383
0.01	2400	independent query cloud	0.0133 ± 0.00205	0.012 ± 0.00137	$0.014 \pm 4.58 \times 10^{-4}$
0.05	300	training cloud	0.00605 ± 0.00498	0.00792 ± 0.00754	0.0133 ± 0.0124
0.05	300	independent query cloud	0.00761 ± 0.00649	0.00983 ± 0.00969	0.0265 ± 0.024
0.05	600	training cloud	0.0104 ± 0.00761	0.0116 ± 0.00599	0.0197 ± 0.00625
0.05	600	independent query cloud	0.0114 ± 0.00717	0.0125 ± 0.00522	0.0287 ± 0.0159
0.05	1200	training cloud	0.0111 ± 0.00291	0.0102 ± 0.00263	$0.0121 \pm 7.71 \times 10^{-4}$
0.05	1200	independent query cloud	0.0119 ± 0.00286	0.011 ± 0.00263	0.0127 ± 0.00182
0.05	2400	training cloud	0.0125 ± 0.00236	0.0113 ± 0.00176	0.0144 ± 0.00342
0.05	2400	independent query cloud	0.0133 ± 0.00204	0.0121 ± 0.00131	0.0163 ± 0.00339

Table 6.4: Manufactured-function errors for the reconstructed excess action $Q[\rho]$. Entries are mean $\pm$ sample standard deviation over three independently generated training/query-cloud pairs. The relative norms use the corresponding analytic quantity in the denominator.

ℓ_2	N	Evaluation domain	E_1	E_2	E_∞
1.00×10^{-6}	300	training cloud	0.017 ± 0.0155	0.0189 ± 0.0185	0.0304 ± 0.0364
1.00×10^{-6}	300	independent query cloud	0.0206 ± 0.019	0.0256 ± 0.0246	0.0643 ± 0.0702
1.00×10^{-6}	600	training cloud	0.0269 ± 0.0131	0.0306 ± 0.0131	0.0456 ± 0.029
1.00×10^{-6}	600	independent query cloud	0.0291 ± 0.0134	0.0307 ± 0.0139	0.0405 ± 0.00411
1.00×10^{-6}	1200	training cloud	0.0225 ± 0.00407	0.0218 ± 0.00351	0.0281 ± 0.00512
1.00×10^{-6}	1200	independent query cloud	0.0232 ± 0.00324	0.0236 ± 0.00222	$0.0267 \pm 6.70 \times 10^{-4}$
1.00×10^{-6}	2400	training cloud	0.0241 ± 0.00199	0.022 ± 0.00223	0.0255 ± 0.0046
1.00×10^{-6}	2400	independent query cloud	0.0234 ± 0.00216	0.0213 ± 0.00272	0.027 ± 0.00334
0.01	300	training cloud	0.0182 ± 0.0178	0.0225 ± 0.0251	0.0304 ± 0.0368
0.01	300	independent query cloud	0.0199 ± 0.0183	0.0328 ± 0.0381	0.115 ± 0.163
0.01	600	training cloud	0.0268 ± 0.0125	0.0304 ± 0.0126	0.0456 ± 0.0303
0.01	600	independent query cloud	0.0285 ± 0.0129	0.0302 ± 0.0137	0.0399 ± 0.00491
0.01	1200	training cloud	0.0222 ± 0.00392	0.0214 ± 0.00342	0.0274 ± 0.00362
0.01	1200	independent query cloud	0.0228 ± 0.003	0.0233 ± 0.00225	0.0269 ± 0.0011
0.01	2400	training cloud	0.0239 ± 0.00161	0.022 ± 0.00195	0.0279 ± 0.00533
0.01	2400	independent query cloud	0.0231 ± 0.0014	0.0215 ± 0.00247	0.0286 ± 0.00244
0.05	300	training cloud	0.0171 ± 0.0159	0.0189 ± 0.0188	0.0288 ± 0.0341
0.05	300	independent query cloud	0.0208 ± 0.0197	0.0256 ± 0.0261	0.0654 ± 0.0779
0.05	600	training cloud	0.027 ± 0.0133	0.0307 ± 0.0132	0.0458 ± 0.0285
0.05	600	independent query cloud	0.0292 ± 0.0136	0.0307 ± 0.0139	0.0399 ± 0.00367
0.05	1200	training cloud	0.0225 ± 0.00363	0.0219 ± 0.003	0.0293 ± 0.00615
0.05	1200	independent query cloud	0.0232 ± 0.00291	0.0238 ± 0.00182	0.0272 ± 0.00129
0.05	2400	training cloud	0.0239 ± 0.00196	0.0219 ± 0.00224	0.0275 ± 0.00565
0.05	2400	independent query cloud	0.0233 ± 0.00193	0.0216 ± 0.00288	0.0285 ± 0.00333

The value reconstruction is generally the least demanding part of this test; the gradient and $Q[\rho]$ amplify local errors through differentiation. Most importantly, increasing N does not produce a monotone reduction of the operator error. For the regularization used in the dynamical calculations, $\ell_2 = 0.05$, the off-cloud mean relative L_1 error in $Q[\rho]$ is 2.08×10^{-2}, 2.92×10^{-2}, 2.32×10^{-2}, and 2.33×10^{-2} for $N = 300, 600, 1200, 2400$, respectively. The smaller regularizations 10^{-6} and 0.01 change the error level but do not restore a systematic refinement trend. The data therefore support neither convergence of the ambient derivative reconstruction nor an attribution of the non-monotonicity to one numerical cause.

Figure 6.1 condenses the decisive operator comparison.

6.3 Physical SEO image and identical-cloud reconstruction

The two-state physical SEO image has four real bath-dependent coefficient fields. Table 6.5 measures the relative distance of the fitted product density from this image using 20 bath anchors and 400 mapping probes. The projection is diagnosed but not imposed. Across the reported snapshots and initial momenta, the mean leakage spans approximately 0.221–0.977, which is too large to regard the propagated surrogate as a small perturbation within the physical SEO

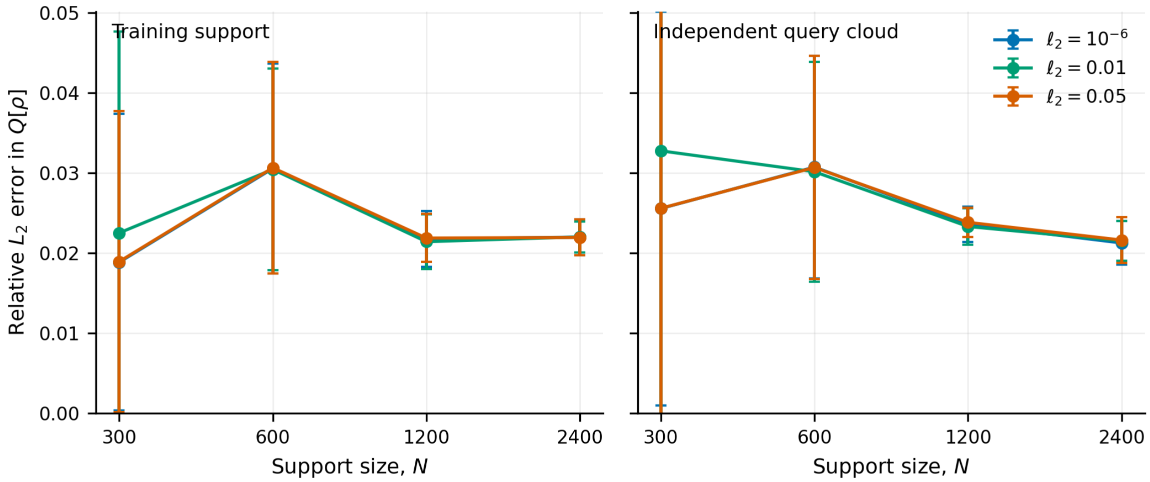

Figure 6.1: Manufactured relative L_2 error in the excess action $Q[\rho]$ on the training cloud and on an independent query cloud. Points are means over seeds 123, 124, and 125; bars are sample standard deviations. The three regularization policies use paired training and query clouds at each cloud size. The error remains nonzero and nonmonotone, so neither cloud enlargement nor changing ℓ_2 from 10^{-6} to the pilot 0.01 or production 0.05 policy establishes a monotone error decrease under independent-cloud enlargement.

subspace. This is a representation diagnostic, not a TDSE or QCLE error.

The identical-cloud comparison in Table 6.6 addresses a narrower question: can the GP reproduce a KDE density when both estimators use the same PBME snapshot, weights, bandwidth, normalization, and evaluation grid? The maximum E_1 values are well below the declared 0.02 criterion. This positive control confirms accurate same-cloud value reconstruction, but it does not test derivatives, the SEO constraint, or time propagation.

Table 6.5: Relative L_2 distance from the four-field physical SEO image at three collision-time snapshots. The entries are mean $\pm$ sample standard deviation over four independently propagated clouds; projection is diagnosed by least squares but is not imposed during propagation.

Method	P_{init}	t/t_c	Mean leakage	Largest local leakage
PBME	20	0	0.222 ± 0.00155	0.256
PBME	20	1	0.803 ± 0.0769	0.993
PBME	20	2	0.32 ± 0.0339	0.777
PBME	100	0	0.222 ± 0.00155	0.256
PBME	100	1	0.962 ± 0.00312	0.998
PBME	100	2	0.959 ± 0.0109	0.997
MIDPOINT	20	0	0.222 ± 0.00155	0.256
MIDPOINT	20	1	0.864 ± 0.0338	0.998
MIDPOINT	20	2	0.626 ± 0.163	0.99
MIDPOINT	100	0	0.222 ± 0.00155	0.256
MIDPOINT	100	1	0.968 ± 0.00653	0.999
MIDPOINT	100	2	0.967 ± 0.00671	0.998

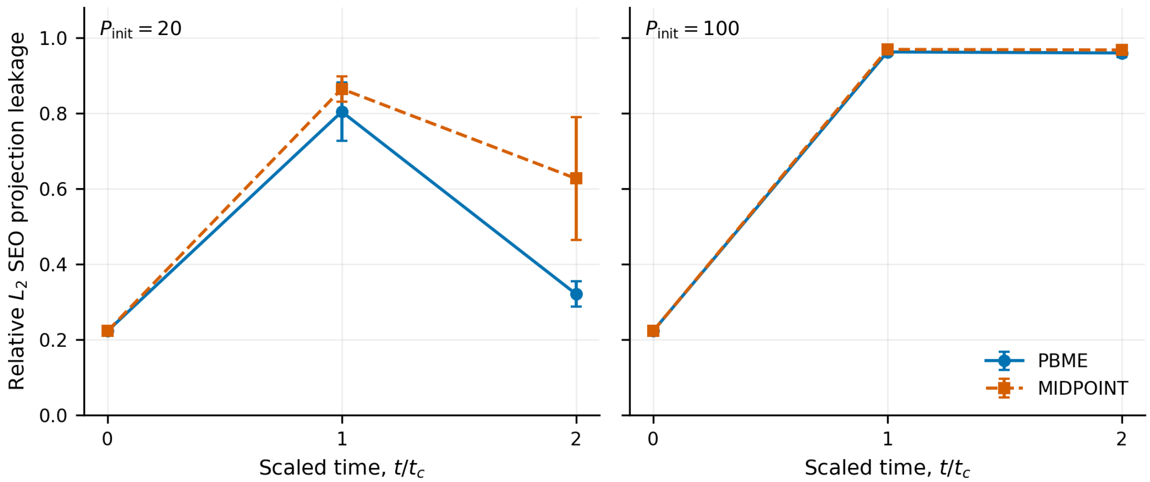

Figure 6.2: Diagnostic SEO projection leakage at $t/t_c = 0, 1, 2$. Each point is the mean over four independent propagation seeds and each bar is the sample standard deviation. The plotted quantity is the relative L_2 residual $\|y - Bc\|_2 / \max(\|y\|_2, 10^{-30})$ from a four-field least-squares SEO projection using 20 bath anchors and 400 mapping probes. Projection is diagnosed but not enforced, and this quantity is not a physical TDSE or QCLE error.

Table 6.6: Identical-cloud KDE/GP reconstruction control. Both estimators use the same PBME snapshot, weights, bandwidth convention, normalization, and evaluation grid. The declared criterion concerns E_1 only and does not test derivatives or propagation.

P_{init}	Cases	$\max E_1$	$\max E_2$	$\max E_\infty$	Declared reconstruction criterion
20	12	0.00118	4.24×10^{-4}	5.88×10^{-4}	$E_1 \leq 0.02$ (criterion met)
100	12	0.00155	5.16×10^{-4}	9.28×10^{-4}	$E_1 \leq 0.02$ (criterion met)

The nonzero initial value in Table 6.5 requires explicit interpretation. At $t = 0$ the analytic initial density is the occupied-state SEO profile multiplied by a smooth bath factor, so it lies exactly in the four-field image and its residual under the same projector vanishes identically in exact arithmetic. The reported initial leakage of approximately 0.222 is therefore a property of the fitted finite surrogate as evaluated by this diagnostic, not of the physical initial target: the 20-anchor/400-probe evaluation samples mapping directions that the focused initial cloud does not constrain, where the fitted product density is determined by the Gaussian-process extension and the safe-profile floor rather than by data. PBME and MIDPOINT report identical initial values because both branches share the same initial fit before any excess source is applied. The present calculations do not apportion this initial residual among the Gaussian-process extension, the floor regularization, and the finite anchor/probe sampling; the subsequent growth of the leakage along the propagation, not the initial offset, carries the propagation-relevant conclusion.

The four-seed snapshot summary in Figure 6.2 shows that the leakage is not a small local residual confined to one calculation.

6.4 Time-step refinement

Table 6.7 compares $\Delta t = 0.5, 0.25, 0.125$ on common physical times using linear interpolation without extrapolation. Both successive time-normalized differences must first exceed $\tau_{\text{noise}} = 10^{-12} + 10^{-12} \max_k \text{RMS}(O_k)$. Rows below that floor are labelled roundoff- or saturation-limited; rows with nondecreasing successive differences are labelled nonmonotone. For MIDPOINT, both differences must additionally exceed the pooled dispersion over four independently sampled clouds. The 128 run/observable rows comprise 15 noise-limited rejections, 102 seed-variability rejections, three nonpositive orders, and eight positive orders. The latter do not repair the simultaneous failures in replication and raw conservation, so the formal order of the midpoint formula is not promoted to a demonstrated order of the full moving-cloud dynamics.

Table 6.7: Time-step sensitivity of endpoint observables. Values and time-normalized successive differences are averages over seeds 11, 29, 47, and 73 after interpolation to common physical times without extrapolation. Both differences must exceed the declared absolute-plus-relative floor $\tau_{\text{noise}} = 10^{-12} + 10^{-12} \max_k \mathrm{RMS}(O_k)$ before an order is interpreted. For both stochastic moving-cloud methods, PBME and MIDPOINT, both differences must also exceed pooled independent-seed dispersion.

Method	P_{init}	Observable	$\langle y_{0.5} \rangle$	$\langle y_{0.25} \rangle$	$\langle y_{0.125} \rangle$	$\langle D_{12} \rangle$	$\langle D_{23} \rangle$	Interpretation
PBME	20	ρ_{11}^{SN}	0.964	0.964	0.964	7.02×10^{-7}	1.75×10^{-7}	step-size signal is not resolved above seed spread
PBME	20	ρ_{22}^{SN}	0.0357	0.0357	0.0357	7.02×10^{-7}	1.75×10^{-7}	step-size signal is not resolved above seed spread
PBME	20	electronic trace	1	1	1	2.90×10^{-13}	5.91×10^{-13}	roundoff- or saturation-limited; order not interpreted
PBME	20	energy	0.1	0.1	0.1	4.93×10^{-9}	1.23×10^{-9}	step-size signal is not resolved above seed spread
PBME	20	$\langle R \rangle$	15.3	15.3	15.3	1.48×10^{-6}	3.69×10^{-7}	step-size signal is not resolved above seed spread
PBME	20	$\langle P \rangle$	19.8	19.8	19.8	3.50×10^{-6}	8.74×10^{-7}	step-size signal is not resolved above seed spread
PBME	20	$\mathrm{var}(R)$	2.31	2.31	2.31	4.89×10^{-7}	1.22×10^{-7}	step-size signal is not resolved above seed spread
PBME	20	$\mathrm{var}(P)$	1.66	1.66	1.66	3.00×10^{-6}	7.49×10^{-7}	step-size signal is not resolved above seed spread
PBME	100	ρ_{11}^{SN}	0.137	0.137	0.137	6.74×10^{-6}	1.69×10^{-6}	step-size signal is not resolved above seed spread
PBME	100	ρ_{22}^{SN}	0.863	0.863	0.863	6.74×10^{-6}	1.69×10^{-6}	step-size signal is not resolved above seed spread

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Table 6.7 (continued)

Method	P_{init}	Observable	$\langle y_{0.5} \rangle$	$\langle y_{0.25} \rangle$	$\langle y_{0.125} \rangle$	$\langle D_{12} \rangle$	$\langle D_{23} \rangle$	Interpretation
PBME	100	electronic trace	1	1	1	5.64×10^{-14}	1.16×10^{-13}	roundoff- or saturation-limited; order not interpreted
PBME	100	energy	2.5	2.5	2.5	5.98×10^{-8}	1.50×10^{-8}	step-size signal is not resolved above seed spread
PBME	100	$\langle R \rangle$	14.9	14.9	14.9	3.94×10^{-7}	9.85×10^{-8}	step-size signal is not resolved above seed spread
PBME	100	$\langle P \rangle$	99.1	99.1	99.1	6.82×10^{-6}	1.70×10^{-6}	step-size signal is not resolved above seed spread
PBME	100	$\mathrm{var}(R)$	0.985	0.985	0.985	1.81×10^{-7}	4.51×10^{-8}	step-size signal is not resolved above seed spread
PBME	100	$\mathrm{var}(P)$	0.444	0.444	0.444	7.48×10^{-6}	1.87×10^{-6}	step-size signal is not resolved above seed spread
MIDPOINT	20	ρ_{11}^{SN}	1.18	1.31	0.818	0.525	0.422	contracting signal resolved above noise and seed spread
MIDPOINT	20	ρ_{22}^{SN}	−0.18	−0.312	0.182	0.525	0.422	contracting signal resolved above noise and seed spread
MIDPOINT	20	electronic trace	1	1	1	1.45×10^{-12}	1.68×10^{-12}	roundoff- or saturation-limited; order not interpreted
MIDPOINT	20	energy	0.107	0.126	0.0937	0.0317	0.0232	step-size signal is not resolved above seed spread
MIDPOINT	20	$\langle R \rangle$	16.6	14.3	13.9	2.38	2.21	step-size signal is not resolved above seed spread
MIDPOINT	20	$\langle P \rangle$	21.5	24.2	18.5	4.91	3.96	contracting signal resolved above noise and seed spread

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Table 6.7 (continued)

Method	P_{init}	Observable	$\langle y_{0.5} \rangle$	$\langle y_{0.25} \rangle$	$\langle y_{0.125} \rangle$	$\langle D_{12} \rangle$	$\langle D_{23} \rangle$	Interpretation
MIDPOINT	20	var(R)	-1.52×10^{3}	-5.07	-13.1	503	8.38	step-size signal is not resolved above seed spread
MIDPOINT	20	var(P)	-2.67×10^{3}	11.4	-13.4	1.94×10^{3}	82.5	step-size signal is not resolved above seed spread
MIDPOINT	100	ρ_{11}^{SN}	-0.201	-0.198	-0.18	5.74	5.82	nonmonotone step-size response
MIDPOINT	100	ρ_{22}^{SN}	1.2	1.2	1.18	5.74	5.82	nonmonotone step-size response
MIDPOINT	100	electronic trace	1	1	1	7.61×10^{-13}	1.44×10^{-12}	roundoff- or saturation-limited; order not interpreted
MIDPOINT	100	energy	2.5	2.53	2.52	0.704	0.722	nonmonotone step-size response
MIDPOINT	100	$\langle R \rangle$	12.8	12.5	12.6	14.5	12.4	step-size signal is not resolved above seed spread
MIDPOINT	100	$\langle P \rangle$	98.8	99.4	99.2	18.2	18	step-size signal is not resolved above seed spread
MIDPOINT	100	var(R)	-1.02×10^{44}	-1.27×10^{43}	-6.88×10^{45}	2.95×10^{43}	2.01×10^{45}	nonmonotone step-size response
MIDPOINT	100	var(P)	-5.96×10^{45}	-9.50×10^{44}	-4.79×10^{47}	3.06×10^{45}	2.38×10^{47}	nonmonotone step-size response

6.5 Independent-cloud enlargement and four-seed replication

The cloud-size study in Table 6.8 compares $N = 500, 1000, 2000$ using three independently sampled clouds at each size. These clouds are not nested; the results therefore measure sensitivity to enlarging a stochastic representation rather than deterministic convergence. Several MIDPOINT changes remain comparable to, or larger than, the seed dispersion, while PBME is generally much more stable.

Table 6.9 makes that contrast explicit at $N = 1000$ and $\Delta t = 0.25$. For the endpoint ρ_{11}^{SN} estimator, the PBME sample standard deviation is 0.00788 at initial momentum 20 and 0.0154 at initial momentum 100; the corresponding MIDPOINT values are 0.795 and 0.956. The MIDPOINT mean and interval can also extend outside the physical population range $[0, 1]$. Four seeds provide a sensitive instability diagnostic but not a precise uncertainty distribution, so the intervals are interpreted descriptively.

Table 6.8: Sensitivity to enlarging independently sampled trajectory clouds. Each entry is mean ± sample standard deviation over seeds 11, 29, and 47. Because the clouds are independently sampled rather than nested, this is a stochastic cloud-size test, not a deterministic convergence order.

Method	P_{init}	Observable	$N = 500$	$N = 1000$	$N = 2000$	Interpretation
PBME	20	ρ_{11}^{SN}	0.963 ± 0.0109	0.964 ± 0.00965	0.953 ± 0.00151	cloud-size change exceeds seed spread
PBME	20	ρ_{22}^{SN}	0.0369 ± 0.0109	0.0356 ± 0.00965	0.0472 ± 0.00151	cloud-size change exceeds seed spread
PBME	20	electronic trace	$1 \pm 7.72 \times 10^{-15}$	$1 \pm 3.67 \times 10^{-15}$	$1 \pm 1.12 \times 10^{-15}$	cloud-size change exceeds seed spread
PBME	20	energy	$0.0999 \pm 1.94 \times 10^{-4}$	$0.1 \pm 9.32 \times 10^{-5}$	$0.1 \pm 9.57 \times 10^{-5}$	change remains within seed spread
PBME	20	$\langle R \rangle$	15.3 ± 0.033	15.3 ± 0.0236	15.3 ± 0.0264	cloud-size change exceeds seed spread
PBME	20	$\langle P \rangle$	19.8 ± 0.0466	19.8 ± 0.0583	19.7 ± 0.013	cloud-size change exceeds seed spread
PBME	20	$\text{var}(R)$	2.18 ± 0.124	2.37 ± 0.133	2.31 ± 0.0335	change remains within seed spread
PBME	20	$\text{var}(P)$	1.46 ± 0.0285	1.7 ± 0.097	1.7 ± 0.0634	cloud-size change exceeds seed spread
PBME	100	ρ_{11}^{SN}	0.137 ± 0.00679	0.135 ± 0.0178	0.143 ± 0.006	change remains within seed spread
PBME	100	ρ_{22}^{SN}	0.863 ± 0.00679	0.865 ± 0.0178	0.857 ± 0.006	change remains within seed spread
PBME	100	electronic trace	$1 \pm 9.02 \times 10^{-16}$	$1 \pm 1.78 \times 10^{-15}$	$1 \pm 3.51 \times 10^{-16}$	change remains within seed spread
PBME	100	energy	$2.5 \pm 9.52 \times 10^{-4}$	$2.5 \pm 4.71 \times 10^{-4}$	$2.5 \pm 4.84 \times 10^{-4}$	change remains within seed spread
PBME	100	$\langle R \rangle$	14.9 ± 0.0205	14.9 ± 0.0249	14.9 ± 0.0124	change remains within seed spread
PBME	100	$\langle P \rangle$	99.1 ± 0.0226	99.1 ± 0.0247	99.1 ± 0.00766	cloud-size change exceeds seed spread
PBME	100	$\text{var}(R)$	0.979 ± 0.0751	0.986 ± 0.054	1 ± 0.0299	change remains within seed spread
PBME	100	$\text{var}(P)$	0.426 ± 0.0187	0.447 ± 0.0117	0.46 ± 0.00392	cloud-size change exceeds seed spread

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Table 6.8 (continued)

Method	P_{init}	Observable	$N = 500$	$N = 1000$	$N = 2000$	Interpretation
MIDPOINT	20	ρ_{11}^{SN}	1.08 ± 0.0635	0.916 ± 0.0672	0.951 ± 0.0744	cloud-size change exceeds seed spread
MIDPOINT	20	ρ_{22}^{SN}	-0.079 ± 0.0635	0.0841 ± 0.0672	0.0485 ± 0.0744	cloud-size change exceeds seed spread
MIDPOINT	20	electronic trace	$1 \pm 6.46 \times 10^{-13}$	$1 \pm 6.09 \times 10^{-13}$	$1 \pm 7.07 \times 10^{-13}$	change remains within seed spread
MIDPOINT	20	energy	0.101 ± 0.0214	0.0994 ± 0.0167	0.0934 ± 0.0207	change remains within seed spread
MIDPOINT	20	$\langle R \rangle$	16.9 ± 0.989	14.9 ± 2	14.5 ± 2.24	change remains within seed spread
MIDPOINT	20	$\langle P \rangle$	20.5 ± 1.93	19.5 ± 1.73	19 ± 1.75	change remains within seed spread
MIDPOINT	20	$\text{var}(R)$	$-1.31 \times 10^{3} \pm 262$	-9.5 ± 52.5	$-1.94 \times 10^{8} \pm 3.31 \times 10^{8}$	change remains within seed spread
MIDPOINT	20	$\text{var}(P)$	$-1.94 \times 10^{3} \pm 571$	-6.33 ± 85.3	$-3.87 \times 10^{8} \pm 6.63 \times 10^{8}$	change remains within seed spread
MIDPOINT	100	ρ_{11}^{SN}	0.923 ± 1.01	-0.46 ± 0.98	-0.297 ± 0.842	change remains within seed spread
MIDPOINT	100	ρ_{22}^{SN}	0.0775 ± 1.01	1.46 ± 0.98	1.3 ± 0.842	change remains within seed spread
MIDPOINT	100	electronic trace	$1 \pm 1.39 \times 10^{-13}$	$1 \pm 7.76 \times 10^{-14}$	$1 \pm 3.65 \times 10^{-14}$	change remains within seed spread
MIDPOINT	100	energy	2.41 ± 0.0692	2.53 ± 0.107	2.5 ± 0.05	change remains within seed spread
MIDPOINT	100	$\langle R \rangle$	12.4 ± 3.43	12.5 ± 0.978	12.6 ± 1.27	change remains within seed spread
MIDPOINT	100	$\langle P \rangle$	98.2 ± 0.412	99.2 ± 1.28	98.7 ± 1.29	change remains within seed spread
MIDPOINT	100	$\text{var}(R)$	$-1.47 \times 10^{19} \pm 2.54 \times 10^{19}$	$-1.69 \times 10^{43} \pm 2.56 \times 10^{43}$	$-4.76 \times 10^{94} \pm 8.25 \times 10^{94}$	change remains within seed spread
MIDPOINT	100	$\text{var}(P)$	$-8.89 \times 10^{20} \pm 1.53 \times 10^{21}$	$-1.27 \times 10^{45} \pm 1.96 \times 10^{45}$	$-2.43 \times 10^{96} \pm 4.20 \times 10^{96}$	change remains within seed spread

Table 6.9: Four-seed endpoint replication at $N = 1000$ and $\Delta t = 0.25$. The interval is a two-sided Student-t interval with three degrees of freedom and is interpreted as a small-sample sensitivity measure, not as strong uncertainty calibration.

Method	P_{init}	Observable	Mean	Sample SD	95% interval	Max. spread
PBME	20	ρ_{11}^{SN}	0.964	0.00788	$[0.952, 0.977]$	0.019
PBME	20	ρ_{22}^{SN}	0.0357	0.00788	$[0.0231, 0.0482]$	0.019
PBME	20	electronic trace	1	3.08×10^{-15}	$[1, 1]$	7.33×10^{-15}
PBME	20	energy	0.1	8.18×10^{-5}	$[0.0999, 0.1]$	1.79×10^{-4}
PBME	20	$\langle R \rangle$	15.3	0.0227	$[15.3, 15.4]$	0.0458
PBME	20	$\langle P \rangle$	19.8	0.0476	$[19.7, 19.9]$	0.111
PBME	20	$\mathrm{var}(R)$	2.31	0.163	$[2.05, 2.57]$	0.357
PBME	20	$\mathrm{var}(P)$	1.66	0.118	$[1.47, 1.85]$	0.281
PBME	100	ρ_{11}^{SN}	0.137	0.0154	$[0.113, 0.162]$	0.0333
PBME	100	ρ_{22}^{SN}	0.863	0.0154	$[0.838, 0.887]$	0.0333
PBME	100	electronic trace	1	1.46×10^{-15}	$[1, 1]$	3.33×10^{-15}
PBME	100	energy	2.5	4.13×10^{-4}	$[2.5, 2.5]$	9.07×10^{-4}
PBME	100	$\langle R \rangle$	14.9	0.0271	$[14.9, 14.9]$	0.0543
PBME	100	$\langle P \rangle$	99.1	0.0203	$[99.1, 99.2]$	0.0461
PBME	100	$\mathrm{var}(R)$	0.985	0.0441	$[0.915, 1.06]$	0.0944
PBME	100	$\mathrm{var}(P)$	0.444	0.0116	$[0.426, 0.463]$	0.021
MIDPOINT	20	ρ_{11}^{SN}	1.31	0.795	$[0.0475, 2.58]$	1.65
MIDPOINT	20	ρ_{22}^{SN}	-0.312	0.795	$[-1.58, 0.952]$	1.65
MIDPOINT	20	electronic trace	1	2.52×10^{-12}	$[1, 1]$	5.34×10^{-12}
MIDPOINT	20	energy	0.126	0.0557	$[0.0378, 0.215]$	0.122
MIDPOINT	20	$\langle R \rangle$	14.3	1.98	$[11.2, 17.5]$	4.08
MIDPOINT	20	$\langle P \rangle$	24.2	9.49	$[9.11, 39.3]$	20.4
MIDPOINT	20	$\mathrm{var}(R)$	-5.07	43.8	$[-74.8, 64.6]$	104
MIDPOINT	20	$\mathrm{var}(P)$	11.4	78.1	$[-113, 136]$	170
MIDPOINT	100	ρ_{11}^{SN}	-0.198	0.956	$[-1.72, 1.32]$	2.07
MIDPOINT	100	ρ_{22}^{SN}	1.2	0.956	$[-0.324, 2.72]$	2.07
MIDPOINT	100	electronic trace	1	6.41×10^{-14}	$[1, 1]$	1.41×10^{-13}
MIDPOINT	100	energy	2.53	0.0879	$[2.39, 2.67]$	0.207
MIDPOINT	100	$\langle R \rangle$	12.5	0.799	$[11.2, 13.8]$	1.92
MIDPOINT	100	$\langle P \rangle$	99.4	1.1	$[97.6, 101]$	2.22
MIDPOINT	100	$\mathrm{var}(R)$	-1.27×10^{43}	2.25×10^{43}	$[-4.85 \times 10^{43}, 2.31 \times 10^{43}]$	4.63×10^{43}
MIDPOINT	100	$\mathrm{var}(P)$	-9.50×10^{44}	1.72×10^{45}	$[-3.69 \times 10^{45}, 1.79 \times 10^{45}]$	3.52×10^{45}

Figure 6.3 places the seed instability and the associated raw conservation failure on the same footing.

6.6 Raw conservation and signed-label tail sensitivity

The population and density curves shown elsewhere can be normalized for visual comparison, but that operation removes the very integral whose conservation must be tested. Table 6.10 therefore reports the normalization, electronic trace, and energy before any such rescaling. PBME keeps the raw normalization constant in these calculations and its trace and energy drifts remain small. MIDPOINT shows large and strongly seed-dependent drifts: even in the canonical four-seed set the largest normalization drift reaches 8.69×10^{20}, and across the tested refinement/cloud-size calculations it reaches 7.35×10^{46}. A unit-normalized display curve cannot be used as evidence against this failure.

The signed-label diagnostic in Table 6.11 quantifies the conditioning of the ratio $y_i(t)/y_i^0$. At zero threshold, the largest absolute ratio reaches 7.61×10^{23} and the signed effective sample size falls as low as 0.0247. The first nonzero threshold, however, already removes about 10^{-3} of the absolute physical mass, which is larger than the predeclared negligible-mass allowance of 10^{-6}. Thus there is no nontrivial negligible-mass plateau. The signed estimator is demonstrably ill-conditioned, but these calculations do not isolate a vanishing-initial-label tail as its unique cause.

Table 6.10: Raw pre-normalization conservation diagnostics for the canonical four-seed calculations ($N = 1000$, $\Delta t = 0.25$). The largest drift is the maximum over both time and seeds; the RMS column is averaged over seeds. These raw quantities are evaluated before any display normalization.

Method	P_{init}	Quantity	Largest $\lvert\Delta\rvert$	Mean RMS drift	Endpoint-drift range
PBME	20	normalization	0	0	$[0, 0]$
PBME	20	trace	1.17×10^{-12}	6.08×10^{-13}	$[1.16 \times 10^{-12}, 1.17 \times 10^{-12}]$
PBME	20	energy	6.38×10^{-9}	1.64×10^{-9}	$[1.45 \times 10^{-13}, 4.38 \times 10^{-13}]$
PBME	100	normalization	0	0	$[0, 0]$
PBME	100	trace	2.28×10^{-13}	1.18×10^{-13}	$[2.25 \times 10^{-13}, 2.28 \times 10^{-13}]$
PBME	100	energy	8.02×10^{-8}	1.99×10^{-8}	$[-9.10 \times 10^{-13}, -3.81 \times 10^{-13}]$
MIDPOINT	20	normalization	2.07	0.353	$[-0.938, 0.174]$
MIDPOINT	20	trace	2.07	0.353	$[-0.938, 0.174]$
MIDPOINT	20	energy	0.244	0.0387	$[-0.0871, 0.0383]$
MIDPOINT	100	normalization	8.69×10^{20}	9.95×10^{19}	$[-1.66 \times 10^{20}, 5.94 \times 10^{20}]$
MIDPOINT	100	trace	8.69×10^{20}	9.95×10^{19}	$[-1.66 \times 10^{20}, 5.94 \times 10^{20}]$
MIDPOINT	100	energy	2.22×10^{21}	2.56×10^{20}	$[-4.36 \times 10^{20}, 1.52 \times 10^{21}]$

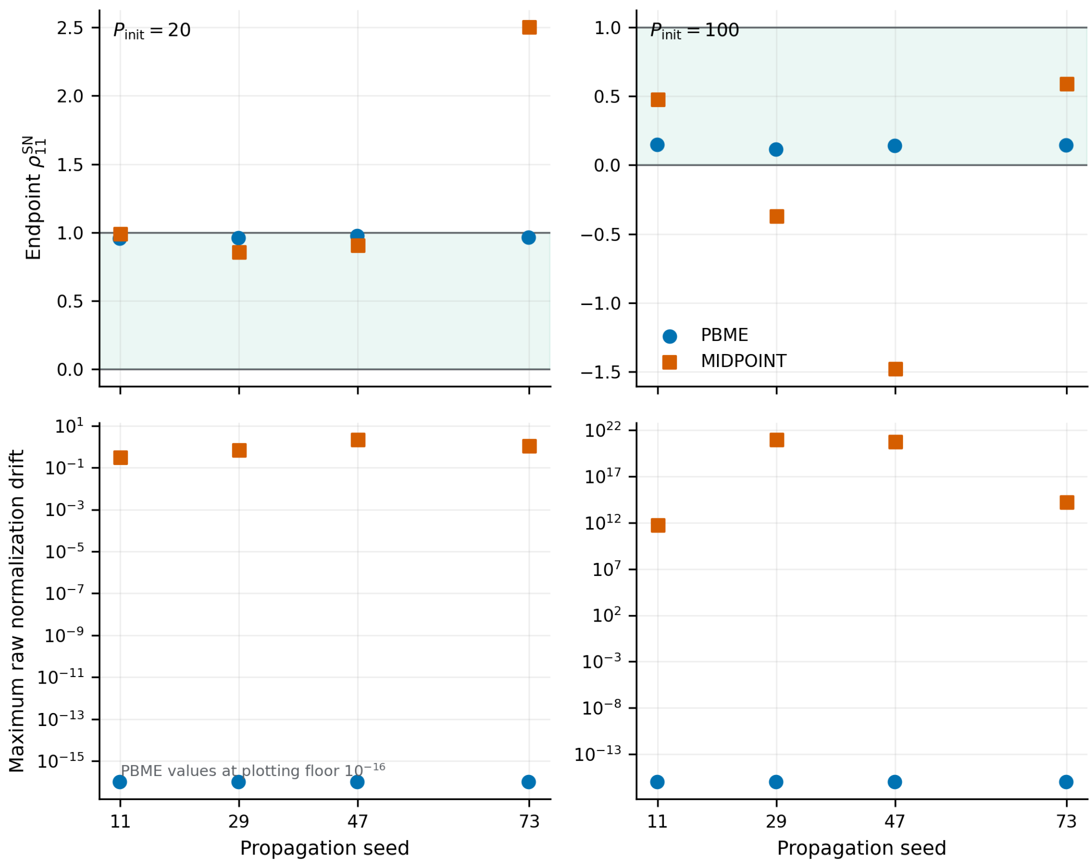

Figure 6.3: Four-seed replication and raw pre-normalization conservation for the canonical $N = 1000$, $\Delta t = 0.25$ calculations. The upper panels show the endpoint ρ_{11}^{SN} estimator; the shaded band marks the physical interval $[0, 1]$. The lower panels show the maximum absolute drift of the raw normalization. Exact PBME zeros are displayed at 10^{-16} only to make them visible on the logarithmic axis. The seed count is four, not the number of trajectories, and the plot supplies a descriptive sensitivity comparison rather than strong uncertainty calibration.

Table 6.11: Signed-label conditioning and tail sensitivity. The first nonzero threshold excludes about 10^{-3} of the absolute physical mass, already above the predeclared negligible-mass allowance 10^{-6}. Consequently no nontrivial negligible-mass plateau exists from which to attribute the instability uniquely to a vanishing-$|y_i^0|$ tail.

Method	P_{init}	Seed	$\max \lvert y/y^0 \rvert$	$N_{\mathrm{eff}}^{\mathrm{signed}}$	$N_{\mathrm{eff}}^{\mathrm{abs}}$	Raw norm	First η	Excluded mass
MIDPOINT	20	11	333	4.07	46	1	0.001	1.00×10^{-3}
MIDPOINT	20	29	176	3.97	92.6	0.439	0.001	1.00×10^{-3}
MIDPOINT	20	47	999	0.669	32.3	1.17	0.001	1.00×10^{-3}
MIDPOINT	20	73	256	0.0247	73.5	0.0621	0.01	0.008
MIDPOINT	100	11	2.55×10^{14}	0.961	27.6	-4.75×10^{11}	0.001	1.00×10^{-3}
MIDPOINT	100	29	7.61×10^{23}	0.5	16.4	5.94×10^{20}	0.001	1.00×10^{-3}
MIDPOINT	100	47	3.38×10^{23}	0.204	12.2	-1.66×10^{20}	0.001	1.00×10^{-3}
MIDPOINT	100	73	4.83×10^{16}	0.478	70.4	-9.29×10^{13}	0.01	0.008
PBME	20	11	1	1.00×10^{3}	1.00×10^{3}	1	0.001	1.00×10^{-3}
PBME	20	29	1	1.00×10^{3}	1.00×10^{3}	1	0.001	1.00×10^{-3}
PBME	20	47	1	1.00×10^{3}	1.00×10^{3}	1	0.001	1.00×10^{-3}
PBME	20	73	1	1.00×10^{3}	1.00×10^{3}	1	0.01	0.008
PBME	100	11	1	1.00×10^{3}	1.00×10^{3}	1	0.001	1.00×10^{-3}
PBME	100	29	1	1.00×10^{3}	1.00×10^{3}	1	0.001	1.00×10^{-3}
PBME	100	47	1	1.00×10^{3}	1.00×10^{3}	1	0.001	1.00×10^{-3}
PBME	100	73	1	1.00×10^{3}	1.00×10^{3}	1	0.01	0.008

Figure 6.4 shows the paired initial-label distributions and the threshold sweep behind Table 6.11.

6.7 TDSE and grid-QCLE numerical controls

TDSE is model-exact only within the displayed domain, time step, periodic FFT convention, edge-mass, normalization, and reflected-momentum controls. Grid QCLE is a numerical solution of the approximate QCLE equation. Its separate temporal and phase-space-grid studies report three levels and both successive differences for each observable. The exact domains and discretizations are collected in Table 6.12, while Tables 6.13 and 6.14 give the observable-level comparisons.

The maximum TDSE boundary probability in the outer spatial bands is 1.22×10^{-24}, and the maximum negative-momentum probability for the high-momentum case is 6.09×10^{-27}, both negligible on the declared scale. For grid QCLE, the largest physical marginal fractions in the outer 5% bands are 2.51×10^{-4} in R and 5.92×10^{-4} in P, below the 10^{-3} criterion. An auxiliary phase-space $|W|$-weighted edge diagnostic can reach 7.94×10^{-2} because the Wigner density is sign-indefinite and exhibits numerical ringing; it is retained as a ringing diagnostic and is not substituted for the physical marginal boundary mass. The CFL ratios also remain below unity. For $P_{\text{init}} = 100$, the nominal TDSE grid levels all resolve to the required 8192-point grid; that case therefore establishes boundary adequacy and numerical repeatability on the chosen grid, but not an independent spatial convergence rate.

The expected temporal scaling is approximately second order for the TDSE split-operator calculation and approximately fourth order for the grid-QCLE classical RK4 calculation. Spatial or phase-space-grid behaviour is assessed separately and is not assigned either temporal order. An observed order is reported only when both successive differences exceed the same declared absolute-plus-relative noise floor, decrease monotonically, and give $0 < p \leq 6$; more rapid contractions are retained as nonasymptotic rather than promoted as convergence evidence.

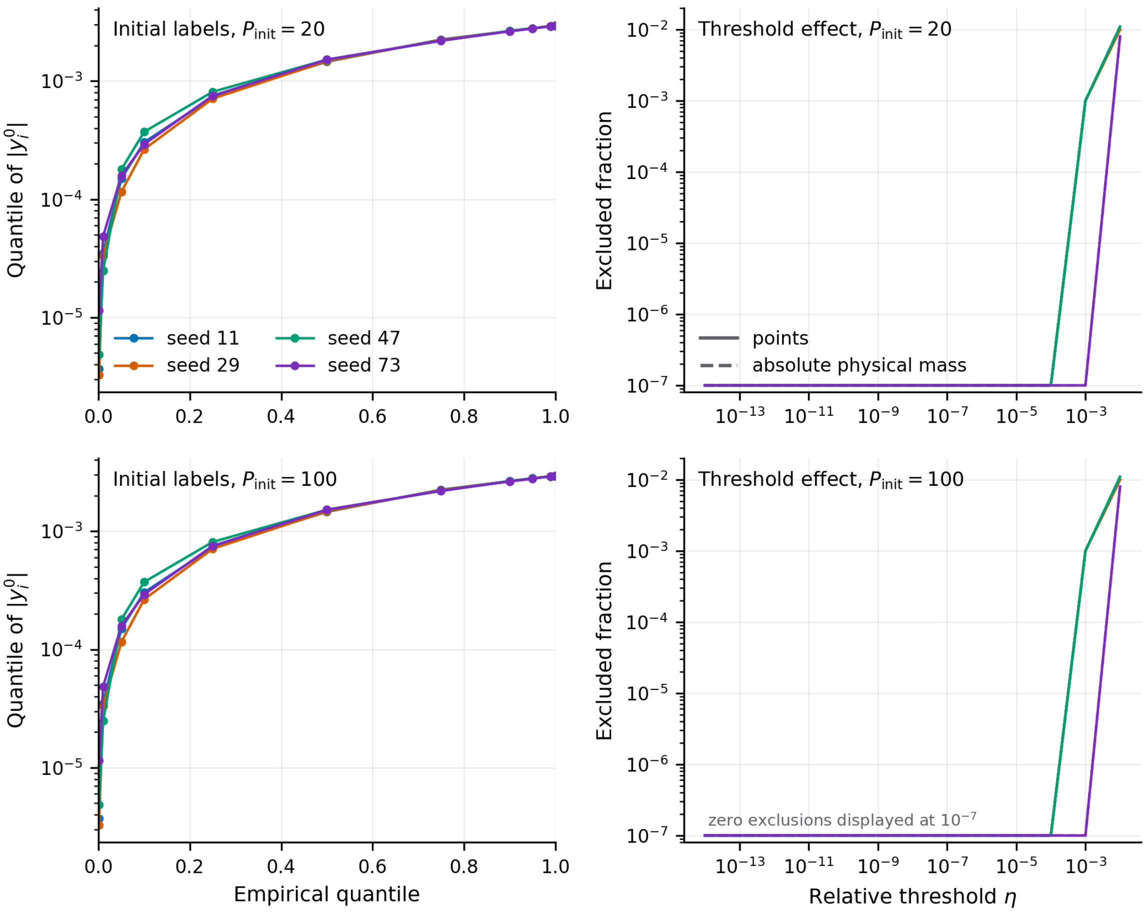

Figure 6.4: Initial-label tail sensitivity for all eight paired P_{init}/seed cases. The quantile panel reports the $|y_i^0|$ distribution on the grid $q = 0.001, \ldots, 0.999$; the threshold panels use the PBME copy of the method-identical inclusion masks at every positive threshold η. Solid lines are excluded point fractions and dashed lines are excluded absolute physical-mass fractions; exact zeros are plotted at 10^{-7} only for logarithmic display and do not alter any reported error. The first nonzero threshold already excludes about 10^{-3} of the absolute physical mass, exceeding the predeclared 10^{-6} allowance, so no negligible-mass plateau exists.

Table 6.12: Reference settings split by equation, initial momentum, and refinement mode.

Method	P_{init}	Mode	t_{final}	Δt ladder	Grid ladder	R domains	P domains	Finest edge fraction	Finest CFL	Interpretation
TDSE	20	grid	3000	0.05/0.05/0.05	2048/4096/8192	[-40,45.958]/[-40,45.979]/[-40,45.9895]	—	6.164956e-25	—	three-level refinement
TDSE	20	time	3000	0.2/0.1/0.05	8192/8192/8192	[-40,45.9895]/[-40,45.9895]/[-40,45.9895]	—	1.223884e-24	—	three-level refinement
TDSE	100	grid	600	0.05/0.05/0.05	8192/8192/8192	[-40,45.9895]/[-40,45.9895]/[-40,45.9895]	—	4.825683e-26	—	boundary adequacy/repeatability; no independent grid order
TDSE	100	time	600	0.2/0.1/0.05	8192/8192/8192	[-40,45.9895]/[-40,45.9895]/[-40,45.9895]	—	4.825683e-26	—	three-level refinement
grid QCLE	20	grid	3000	0.5/0.5/0.5	48x64/96x128/192x256	[-30,30]/[-30,30]/[-30,30]	[-35,35]/[-35,35]/[-35,35]	5.915682e-04	0.09195161	three-level refinement
grid QCLE	20	time	3000	2/1/0.5	192x256/192x256/192x256	[-30,30]/[-30,30]/[-30,30]	[-35,35]/[-35,35]/[-35,35]	5.915682e-04	0.09195161	three-level refinement
grid QCLE	100	grid	600	0.5/0.5/0.5	48x32/96x64/192x128	[-30,30]/[-30,30]/[-30,30]	[80,120]/[80,120]/[80,120]	1.697013e-07	0.1066453	three-level refinement
grid QCLE	100	time	600	2/1/0.5	192x128/192x128/192x128	[-30,30]/[-30,30]/[-30,30]	[80,120]/[80,120]/[80,120]	2.152345e-07	0.1066453	three-level refinement

Slash-separated entries are coarse/fine/finer. For grid QCLE, the reported edge value is the larger of the finest-grid physical R and P marginal edge fractions. The sign-indefinite phase-space absolute-W ringing diagnostic is not substituted for this boundary-mass control. The dimensionless CFL entry is Delta t/Delta t__max, where Delta t__max is the minimum of the RK4 imaginary-axis bound 2.828 divided by the spectral advection, force, and electronic-frequency scales; values below one satisfy this declared linear-stability check.

Table 6.13: Three-level TDSE reference study. Time and spatial/grid refinement are performed separately. δ_{12} and δ_{23} are successive absolute differences. Both must exceed $\tau_{\text{noise}} = 10^{-12} + 10^{-12}\max_k |O_k|$; nonmonotone rows and values above the plausibility bound $p \le 6$ are retained but not interpreted as convergence orders.

Refinement	P_{init}	Observable	Level 1	Level 2	Level 3	δ_{12}	δ_{23}	Interpretation
time	20	ρ_{11}^{SN}	0.952	0.952	0.952	1.20×10^{-7}	3.00×10^{-8}	$p = 2$
time	20	ρ_{22}^{SN}	0.0481	0.0481	0.0481	1.20×10^{-7}	3.00×10^{-8}	$p = 2$
time	20	electronic trace	1	1	1	4.14×10^{-12}	6.22×10^{-12}	non-monotone or unresolved
time	20	energy	0.1	0.1	0.1	4.15×10^{-13}	6.20×10^{-13}	roundoff- or saturation-limited; order not interpreted
time	20	$\langle R \rangle$	15.3	15.3	15.3	4.91×10^{-7}	1.23×10^{-7}	$p = 2$
time	20	$\langle P \rangle$	19.7	19.7	19.7	7.10×10^{-7}	1.78×10^{-7}	$p = 2$
time	20	$\text{var}(R)$	2.41	2.41	2.41	1.85×10^{-6}	4.64×10^{-7}	$p = 2$
time	20	$\text{var}(P)$	2.07	2.07	2.07	4.00×10^{-6}	1.00×10^{-6}	$p = 2$
time	100	ρ_{11}^{SN}	0.143	0.143	0.143	1.57×10^{-6}	3.92×10^{-7}	$p = 2$
time	100	ρ_{22}^{SN}	0.857	0.857	0.857	1.57×10^{-6}	3.92×10^{-7}	$p = 2$
time	100	electronic trace	1	1	1	1.22×10^{-12}	2.20×10^{-12}	roundoff- or saturation-limited; order not interpreted
time	100	energy	2.5	2.5	2.5	2.86×10^{-12}	5.44×10^{-12}	roundoff- or saturation-limited; order not interpreted
time	100	$\langle R \rangle$	14.9	14.9	14.9	1.81×10^{-7}	4.54×10^{-8}	$p = 2$
time	100	$\langle P \rangle$	99.1	99.1	99.1	1.58×10^{-6}	3.94×10^{-7}	$p = 2$
time	100	$\text{var}(R)$	1.01	1.01	1.01	4.65×10^{-8}	1.16×10^{-8}	$p = 2$
time	100	$\text{var}(P)$	0.376	0.376	0.376	1.12×10^{-6}	2.79×10^{-7}	$p = 2$
grid	20	ρ_{11}^{SN}	0.952	0.952	0.952	4.59×10^{-12}	1.80×10^{-12}	roundoff- or saturation-limited; order not interpreted
grid	20	ρ_{22}^{SN}	0.0481	0.0481	0.0481	4.48×10^{-14}	2.12×10^{-14}	roundoff- or saturation-limited; order not interpreted
grid	20	electronic trace	1	1	1	4.64×10^{-12}	1.77×10^{-12}	roundoff- or saturation-limited; order not interpreted

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Table 6.13 (continued)

Refinement	P_{init}	Observable	Level 1	Level 2	Level 3	δ_{12}	δ_{23}	Interpretation
grid	20	energy	0.1	0.1	0.1	4.68×10^{-13}	1.72×10^{-13}	roundoff- or saturation-limited; order not interpreted
grid	20	$\langle R \rangle$	15.3	15.3	15.3	1.79×10^{-12}	3.77×10^{-13}	roundoff- or saturation-limited; order not interpreted
grid	20	$\langle P \rangle$	19.7	19.7	19.7	1.43×10^{-12}	1.17×10^{-13}	roundoff- or saturation-limited; order not interpreted
grid	20	$\mathrm{var}(R)$	2.41	2.41	2.41	2.10×10^{-12}	3.15×10^{-12}	roundoff- or saturation-limited; order not interpreted
grid	20	$\mathrm{var}(P)$	2.07	2.07	2.07	5.91×10^{-12}	3.58×10^{-12}	$p = 0.723$
grid	100	ρ_{11}^{SN}	0.143	0.143	0.143	0	0	roundoff- or saturation-limited; order not interpreted
grid	100	ρ_{22}^{SN}	0.857	0.857	0.857	0	0	roundoff- or saturation-limited; order not interpreted
grid	100	electronic trace	1	1	1	0	0	roundoff- or saturation-limited; order not interpreted
grid	100	energy	2.5	2.5	2.5	0	0	roundoff- or saturation-limited; order not interpreted
grid	100	$\langle R \rangle$	14.9	14.9	14.9	0	0	roundoff- or saturation-limited; order not interpreted
grid	100	$\langle P \rangle$	99.1	99.1	99.1	0	0	roundoff- or saturation-limited; order not interpreted
grid	100	$\mathrm{var}(R)$	1.01	1.01	1.01	0	0	roundoff- or saturation-limited; order not interpreted

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Table 6.13 (continued)

Refinement	P_{init}	Observable	Level 1	Level 2	Level 3	δ_{12}	δ_{23}	Interpretation
grid	100	var(P)	0.376	0.376	0.376	0	0	roundoff- or saturation-limited; order not interpreted

Table 6.14: Three-level grid QCLE reference study. Time and spatial/grid refinement are performed separately. δ_{12} and δ_{23} are successive absolute differences. Both must exceed $\tau_{\text{noise}} = 10^{-12} + 10^{-12}\max_k |O_k|$; nonmonotone rows and values above the plausibility bound $p \leq 6$ are retained but not interpreted as convergence orders.

Refinement	P_{init}	Observable	Level 1	Level 2	Level 3	δ_{12}	δ_{23}	Interpretation
time	20	ρ_{11}^{SN}	0.948	0.948	0.948	1.27×10^{-10}	7.86×10^{-12}	$p = 4.01$
time	20	ρ_{22}^{SN}	0.0518	0.0518	0.0518	1.27×10^{-10}	7.86×10^{-12}	$p = 4.01$
time	20	electronic trace	1	1	1	6.66×10^{-16}	4.44×10^{-16}	roundoff- or saturation-limited; order not interpreted
time	20	energy	0.1	0.1	0.1	2.08×10^{-16}	1.39×10^{-17}	roundoff- or saturation-limited; order not interpreted
time	20	$\langle R\rangle$	15.3	15.3	15.3	1.43×10^{-10}	9.07×10^{-12}	roundoff- or saturation-limited; order not interpreted
time	20	$\langle P\rangle$	19.7	19.7	19.7	1.21×10^{-11}	1.13×10^{-12}	roundoff- or saturation-limited; order not interpreted
time	20	$\mathrm{var}(R)$	2.14	2.14	2.14	4.36×10^{-9}	2.76×10^{-10}	$p = 3.98$
time	20	$\mathrm{var}(P)$	1.05	1.05	1.05	1.46×10^{-9}	1.09×10^{-10}	$p = 3.75$
time	100	ρ_{11}^{SN}	0.143	0.143	0.143	1.68×10^{-10}	9.50×10^{-12}	$p = 4.14$
time	100	ρ_{22}^{SN}	0.857	0.857	0.857	1.68×10^{-10}	9.50×10^{-12}	$p = 4.14$
time	100	electronic trace	1	1	1	0	2.22×10^{-16}	roundoff- or saturation-limited; order not interpreted
time	100	energy	2.5	2.5	2.5	2.22×10^{-15}	4.44×10^{-16}	roundoff- or saturation-limited; order not interpreted
time	100	$\langle R\rangle$	14.9	14.9	14.9	1.78×10^{-12}	1.05×10^{-13}	roundoff- or saturation-limited; order not interpreted
time	100	$\langle P\rangle$	99.1	99.1	99.1	1.96×10^{-10}	1.22×10^{-11}	roundoff- or saturation-limited; order not interpreted

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Table 6.14 (continued)

Refinement	P_{init}	Observable	Level 1	Level 2	Level 3	δ_{12}	δ_{23}	Interpretation
time	100	var(R)	1.01	1.01	1.01	1.42×10^{-11}	7.67×10^{-13}	roundoff- or saturation-limited; order not interpreted
time	100	var(P)	0.382	0.382	0.382	1.62×10^{-10}	7.28×10^{-12}	$p = 4.48$
grid	20	ρ_{11}^{SN}	0.412	1.05	0.948	0.638	0.102	$p = 2.64$
grid	20	ρ_{22}^{SN}	0.595	-0.0502	0.0518	0.645	0.102	$p = 2.66$
grid	20	electronic trace	1.01	1	1	0.00719	1.23×10^{-7}	rapid contraction; order not interpreted (not demonstrably asymptotic)
grid	20	energy	0.101	0.1	0.1	0.00129	3.76×10^{-5}	$p = 5.1$
grid	20	$\langle R \rangle$	14.2	15.5	15.3	1.37	0.242	$p = 2.5$
grid	20	$\langle P \rangle$	14.7	20.1	19.7	5.33	0.37	$p = 3.85$
grid	20	var(R)	10.7	1.26	2.14	9.47	0.874	$p = 3.44$
grid	20	var(P)	37	0.451	1.05	36.6	0.595	$p = 5.94$
grid	100	ρ_{11}^{SN}	0.107	0.143	0.143	0.0361	6.54×10^{-5}	rapid contraction; order not interpreted (not demonstrably asymptotic)
grid	100	ρ_{22}^{SN}	0.808	0.857	0.857	0.0488	5.89×10^{-5}	rapid contraction; order not interpreted (not demonstrably asymptotic)
grid	100	electronic trace	0.915	1	1	0.085	6.52×10^{-6}	rapid contraction; order not interpreted (not demonstrably asymptotic)
grid	100	energy	2.29	2.5	2.5	0.213	1.56×10^{-5}	rapid contraction; order not interpreted (not demonstrably asymptotic)
grid	100	$\langle R \rangle$	14.9	14.9	14.9	0.00136	1.71×10^{-4}	$p = 2.98$
grid	100	$\langle P \rangle$	99.1	99.1	99.1	0.0337	2.61×10^{-5}	rapid contraction; order not interpreted (not demonstrably asymptotic)

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Table 6.14 (continued)

Refinement	P_{init}	Observable	Level 1	Level 2	Level 3	δ_{12}	δ_{23}	Interpretation
grid	100	var(R)	0.954	1.01	1.01	0.0529	5.98×10^{-4}	rapid contraction; order not interpreted (not demonstrably asymptotic)
grid	100	var(P)	0.51	0.381	0.382	0.129	0.00106	rapid contraction; order not interpreted (not demonstrably asymptotic)

6.8 PBME versus MIDPOINT against common references

Each paired difference is $E_{\text{MIDPOINT,ref}} - E_{\text{PBME,ref}}$, so a negative value favours MIDPOINT for that particular metric. Isolated negative differences do not establish systematic improvement. Table 6.15 reports the paired mean and Student-t interval for raw and unit-mass density errors and for the principal time-dependent observables. Several intervals include zero, and resolved comparisons are not consistently in favour of MIDPOINT. More fundamentally, a physical success claim also requires seed-stable dynamics, raw conservation, and controlled discretization sensitivity. Those conditions are not met, so systematic improvement over PBME is not demonstrated.

Table 6.15: Paired physical-error comparison against common references. The reported difference is $\Delta E = E_{\mathrm{MIDPOINT}} - E_{\mathrm{PBME}}$ over the same four independent seeds; negative values favour MIDPOINT for that metric. Intervals are paired Student-t 95% intervals with three degrees of freedom. Raw and unit-mass density errors are distinguished so that shape normalization cannot hide normalization error.

Reference	P_{init}	Quantity	Error measure	$\langle \Delta E \rangle$	95% interval	Interpretation
QCLE	20	R–P density	raw E_1	0.506	$[-0.297, 1.31]$	interval includes zero
QCLE	20	R–P density	unit-mass shape E_1	2.61	$[-3.87, 9.09]$	interval includes zero
QCLE	20	ρ_{11}^{SN}	time-series RMS error	0.378	$[-0.435, 1.19]$	interval includes zero
QCLE	20	ρ_{22}^{SN}	time-series RMS error	0.378	$[-0.435, 1.19]$	interval includes zero
QCLE	20	$\langle P \rangle$	time-series RMS error	3.69	$[-5.32, 12.7]$	interval includes zero
QCLE	20	$\langle R \rangle$	time-series RMS error	1.97	$[-2.22, 6.16]$	interval includes zero
QCLE	100	R–P density	raw E_1	4.89×10^{20}	$[-5.69 \times 10^{20}, 1.55 \times 10^{21}]$	interval includes zero
QCLE	100	R–P density	unit-mass shape E_1	3.07	$[2.62, 3.52]$	PBME smaller for this metric
QCLE	100	ρ_{11}^{SN}	time-series RMS error	51.4	$[-103, 206]$	interval includes zero
QCLE	100	ρ_{22}^{SN}	time-series RMS error	51.4	$[-103, 206]$	interval includes zero
QCLE	100	$\langle P \rangle$	time-series RMS error	144	$[-296, 584]$	interval includes zero
QCLE	100	$\langle R \rangle$	time-series RMS error	108	$[-225, 441]$	interval includes zero
TDSE	20	ρ_{11}^{SN}	time-series RMS error	0.38	$[-0.431, 1.19]$	interval includes zero
TDSE	20	ρ_{22}^{SN}	time-series RMS error	0.38	$[-0.431, 1.19]$	interval includes zero
TDSE	20	$\langle P \rangle$	time-series RMS error	3.7	$[-5.3, 12.7]$	interval includes zero
TDSE	20	$\langle R \rangle$	time-series RMS error	1.98	$[-2.22, 6.17]$	interval includes zero
TDSE	100	ρ_{11}^{SN}	time-series RMS error	51.4	$[-103, 206]$	interval includes zero
TDSE	100	ρ_{22}^{SN}	time-series RMS error	51.4	$[-103, 206]$	interval includes zero
TDSE	100	$\langle P \rangle$	time-series RMS error	144	$[-296, 584]$	interval includes zero
TDSE	100	$\langle R \rangle$	time-series RMS error	108	$[-225, 441]$	interval includes zero

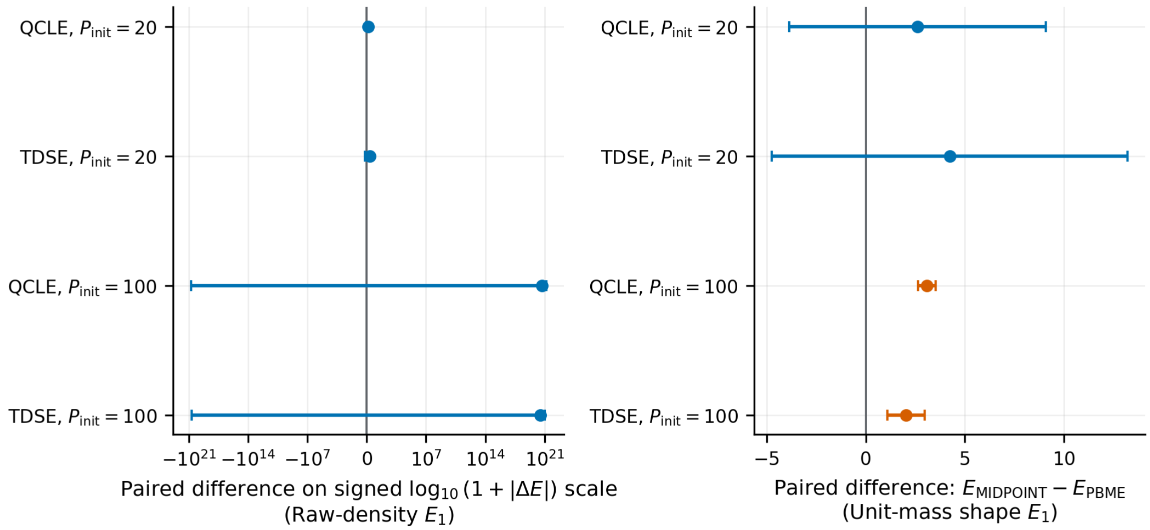

Figure 6.5: Four-seed paired MIDPOINT-minus-PBME differences for density E_1 errors against the matched TDSE and grid-QCLE references. Points are paired means and bars are two-sided Student-t 95% intervals with three degrees of freedom. Positive values mean that MIDPOINT has the larger error. The raw-density panel uses the signed display transform $\text{sgn}(\Delta E)\log_{10}(1+|\Delta E|)$, whose tick labels show the corresponding untransformed magnitude; the shape panel uses unit-mass normalization and a linear axis. Shape agreement is therefore not used as conservation evidence.

This conclusion is visible directly in Figure 6.5. The raw-density panel is kept separate from the independently normalized shape panel so that shape normalization cannot conceal a failed raw integral.

6.9 Synthesis of the numerical evidence

The positive identical-cloud reconstruction control establishes that the GP can interpolate density values accurately under matched conditions. That result does not carry through the sequence required by the physical calculation. The manufactured derivative operator is non-monotone with cloud size; the fitted density has substantial SEO-image leakage; the dynamical response is dominated by cloud-to-cloud variation; and the raw MIDPOINT normalization, trace, and energy can drift catastrophically. Controlled TDSE and grid-QCLE references do not reveal a consistent physical-error reduction relative to PBME. The failure is therefore not inferred from one diagnostic: it is the common conclusion of independent operator, representation, refinement, replication, conservation, and reference tests.

The defensible result is correspondingly narrow. The tested product-GP, moving-cloud MIDPOINT discretization is not a reliable dynamical method for this benchmark. This statement concerns the discretization studied here; it does not constitute a failure of the continuum mapping-QCLE excess term.

Chapter 7

Discussion and Conclusions

This thesis set out to test whether a differentiable product-GP representation could make the mapping-QCLE excess term usable on a moving trajectory cloud. The answer for the construction studied here is no. Accurate interpolation of density values on a fixed cloud does not imply accurate ambient derivatives, preservation of the physical SEO image, stable signed estimators, or conservative dynamics. The contribution is therefore a precisely defined formulation, a sequence of physically interpretable tests that expose its limitations, and concrete requirements for a successor. It is not a validated GP-QCLE solver and it is not evidence that MIDPOINT improves PBME.

7.1 Application-specific contribution

Values on a lower-dimensional manifold do not generally identify ambient normal derivatives; that fact is not original to this thesis. The original application-specific contribution is narrower: constructing a differentiable product-GP excess-source prototype on moving PBME/MInt clouds, testing it against an analytic manufactured operator and the exact two-state SEO image, and connecting its observed derivative, projection, conservation, and signed-weight failures to the propagated dynamics without claiming a unique cause [1–5].

7.2 Physical interpretation of the failure

The results in Chapter 6 reveal a connected, but not uniquely causal, failure pathway. A product kernel can reconstruct the sampled density values while leaving ambient derivatives underdetermined away from the cloud. The excess operator depends precisely on those derivatives. Once its source is applied without enforcing the four-field SEO structure, the fitted density can develop large components outside the physical image. The focused-sampling ratio then combines sign cancellation with very large individual label ratios, which magnifies cloud-to-cloud variation. The resulting raw normalization, trace, and energy drift show that this is not merely a cosmetic change in density shape.

These observations establish compatibility among the failure modes, not a proof that one is the sole cause of the others. The manufactured operator study, SEO projection diagnostic, signed-label analysis, and conservation test deliberately isolate different links in the chain. Their agreement supports the method-level conclusion without overstating causal identification. The controlled TDSE and grid-QCLE studies then ensure that the absence of a systematic physical improvement is not an artefact of an uncontrolled reference calculation.

7.3 Claim boundary

This thesis does not establish a validated discretization of the complete mapping-basis QCLE, deterministic cloud convergence, strong uncertainty calibration from four seeds, chemical accuracy outside the one-dimensional two-state benchmark, multidimensional scalability, or systematic improvement over PBME. TDSE and grid-QCLE have different epistemic roles: TDSE is model-exact within numerical controls, whereas grid QCLE solves an approximate quantum–classical equation numerically.

The production regularization 0.05, pilot 0.01, and manufactured-test 10^{-6} policies are all tested and reported rather than conflated. The tail study demonstrates extreme signed-weight instability, but its negligible-mass rule does not permit a nontrivial point-removal plateau, so the instability is not assigned uniquely to small initial labels.

7.4 Requirements for a successor

A credible successor should regress the four real bath-dependent fields of the two-state SEO image so that Hermiticity and projection are enforced by construction. Its discrete excess source should preserve normalization and the relevant trace and energy identities before any renormalization. The regression should incorporate derivative information or structural constraints capable of controlling directions normal to the sampled manifold. A future success claim requires decreasing manufactured-operator errors under a controlled refinement, time-step changes resolvable above seed variation, independent-cloud uncertainty bands, raw conservation, and reproducibly smaller PBME-matched errors against controlled references where the excess source is appreciable [83–85].

7.5 Numerical scope and transferability

The numerical conclusions are tied to the parameter ranges, independent seed sets, domains, and discretizations stated in Chapter 6 and Appendix F. Repeating those calculations tests this particular construction on the same benchmark; it does not by itself establish transferability to additional electronic states, nuclear dimensions, coupling regimes, or sampling measures. Such transfer would require renewed manufactured-operator, projection, conservation, and reference comparisons rather than extrapolation from the present negative result.

7.6 Conclusion

The combined numerical evidence supports a controlled negative result. Accurate same-cloud value reconstruction does not guarantee an accurate derivative-dependent excess operator, preservation of the SEO image, raw conservation, or seed-stable dynamics. The tested MIDPOINT branch fails the joint reliability criteria and does not demonstrate systematic improvement over PBME. This conclusion applies to the tested discretization, not to the continuum QCLE excess term.

Accordingly, the thesis contributes a precisely scoped formulation, a physics-based failure analysis, and evidence-backed redesign criteria. It does not claim that the tested construction is reliable, deterministically cloud-converged, or improved.

Appendix A

Initial Conditions, Sampling, and Cloud Quadrature

This appendix defines the numerical state carried by the moving support cloud and documents the initial nuclear and mapping distributions used in the production calculations. It also distinguishes full-dimensional importance sampling from the singular focused mapping construction and states the quadrature measure used for physical observables.

A.1 State carried by the moving support cloud

At time t_n, the numerical state consists of the support cloud

$$\mathcal{S}_n = \{\boldsymbol{\mathcal{X}}_i^n\}_{i=1}^N \,, \tag{A.1}$$

the physical density labels

$$y_i^n \approx \tilde{\rho}_m\left(\boldsymbol{\mathcal{X}}_i^n, t_n\right), \tag{A.2}$$

and a GP surrogate fitted to these data.

For the correction-weight implementation, the code stores the initial sign-bearing labels y_i^0 separately from the dimensionless array named `correction_weight`. No additional mathematical label is introduced: at step n, the physical density label remains y_i^n, and the stored array represents the ratio y_i^n/y_i^0 wherever that ratio is numerically well defined. All derivations below are therefore written directly in terms of y_i^n, as in the main text.

For the product surrogate of Chapter 4, the transformed modulation labels are

$$u_i^n = \frac{y_i^n}{g_{s,\mathrm{safe}}(\mathbf{x}_i^n)}, \tag{A.3}$$

with $g_{s,\mathrm{safe}}$ defined in Eq. (4.7). The vectors of physical and transformed labels are

$$\mathbf{y}^n = (y_1^n, \ldots, y_N^n)^T, \qquad \mathbf{u}^n = (u_1^n, \ldots, u_N^n)^T.$$

A.2 Initial nuclear and mapping distributions

A.2.1 Gaussian nuclear Wigner density

For one bath degree of freedom, the initial nuclear wavepacket is

$$\psi(R) \propto \exp\left[-\frac{(R-R_0)^2}{4\sigma_R^2} + \frac{i}{\hbar}P_{\mathrm{init}}(R-R_0)\right]. \tag{A.4}$$

Its Wigner transform is

$$W_{\mathrm{cl}}(R,P) = \frac{1}{\pi\hbar}\exp\left[-\frac{(R-R_0)^2}{2\sigma_R^2} - \frac{2\sigma_R^2(P-P_{\mathrm{init}})^2}{\hbar^2}\right]. \tag{A.5}$$

Consequently,

$$R \sim \mathcal{N}(R_0, \sigma_R^2), \qquad P \sim \mathcal{N}\left(P_{\mathrm{init}}, \frac{\hbar^2}{4\sigma_R^2}\right), \tag{A.6}$$

so that $\sigma_P = \hbar/(2\sigma_R)$.

A.2.2 Signed SEO sampling

For an initially occupied subsystem state s, the diagonal SEO kernel is the analytic function

$$g_s(\mathbf{x}) = \mathcal{G}_{\mathrm{SEO}}(\mathbf{x})B_s(\mathbf{x}), \tag{A.7}$$

where

$$\mathcal{G}_{\mathrm{SEO}}(\mathbf{x}) = (\pi\hbar)^{-N_s}\exp\left[-\frac{\mathbf{r}^2+\mathbf{p}^2}{\hbar}\right] \tag{A.8}$$

and

$$B_s(\mathbf{x}) = \frac{2}{\hbar}\left(r_s^2+p_s^2\right) - 1. \tag{A.9}$$

The initial projected mapping density used by the sampler is

$$\tilde{\rho}_m(\boldsymbol{\mathcal{X}}, 0) = W_{\mathrm{cl}}(R,P)g_s(\mathbf{x}). \tag{A.10}$$

The standard signed-SEO proposal samples the bath variables from W_{cl} and the mapping variables from the positive Gaussian envelope:

$$\pi_0(\boldsymbol{\mathcal{X}}) = W_{\mathrm{cl}}(R,P)\mathcal{G}_{\mathrm{SEO}}(\mathbf{x}). \tag{A.11}$$

The associated importance ratio is

$$\frac{\tilde{\rho}_m(\boldsymbol{\mathcal{X}}, 0)}{\pi_0(\boldsymbol{\mathcal{X}})} = B_s(\mathbf{x}), \tag{A.12}$$

which carries the sign of the SEO Wigner density.

A.2.3 Absolute-target sampling

The optional absolute-target sampler uses

$$\pi_{\text{abs}}(\boldsymbol{\mathcal{X}}) = \frac{|\tilde{\rho}_m(\boldsymbol{\mathcal{X}}, 0)|}{C_{\text{abs}}}, \qquad C_{\text{abs}} = \int d\boldsymbol{\mathcal{X}}\, |\tilde{\rho}_m(\boldsymbol{\mathcal{X}}, 0)|. \tag{A.13}$$

The importance ratio is then

$$\frac{\tilde{\rho}_m(\boldsymbol{\mathcal{X}}, 0)}{\pi_{\text{abs}}(\boldsymbol{\mathcal{X}})} = C_{\text{abs}}\, \text{sgn}\left[\tilde{\rho}_m(\boldsymbol{\mathcal{X}}, 0)\right]. \tag{A.14}$$

Hence the magnitude of the importance factor is constant. The implemented rejection sampler uses a high quantile of $|B_s|$ as a practical acceptance ceiling; this is a variance-reduction device with an explicitly controlled truncation bias. This branch is inactive in the focused production and validation runs. For the optional absolute-target branch, the campaign guide does not report the selected quantile, realized ceiling, rejection rate, or induced truncation bias. No numerical claim in this thesis relies on that inactive sampler.

A.2.4 Focused MMST sampling

The focused occupation convention [2, 19, 59] is

$$n_\lambda = \frac{r_\lambda^2 + p_\lambda^2}{2\hbar} - \gamma. \tag{A.15}$$

For a trajectory focused on state s,

$$n_s = 1, \qquad n_{\lambda \neq s} = 0,$$

and therefore

$$r_s^2 + p_s^2 = 2\hbar(1+\gamma), \qquad r_{\lambda\neq s}^2 + p_{\lambda\neq s}^2 = 2\hbar\gamma. \tag{A.16}$$

The implementation uses $\gamma = 1/2$. For the two-state problem,

$$r_s^2 + p_s^2 = 3\hbar, \qquad r_{\bar{s}}^2 + p_{\bar{s}}^2 = \hbar,$$

where $\bar{s}$ denotes the state different from s. Independent angles $\theta_\lambda \sim U(0, 2\pi)$ determine

$$r_\lambda = \sqrt{2\hbar(n_\lambda + \gamma)}\cos\theta_\lambda, \qquad p_\lambda = \sqrt{2\hbar(n_\lambda + \gamma)}\sin\theta_\lambda. \tag{A.17}$$

On the focused manifold, the mapping part of the SEO target has the constant value

$$K_{\text{focus}} = (\pi\hbar)^{-N_s} \exp\left[-2(1 + N_s\gamma)\right](4\gamma + 3). \tag{A.18}$$

The initial labels are therefore

$$y_i^0 = W_{\mathrm{cl}}(R_i^0, P_i^0) K_{\mathrm{focus}}. \tag{A.19}$$

Focused sampling is supported on a product of circles in mapping space and is therefore singular with respect to the full mapping-space Lebesgue measure. This geometric fact is important when moment constraints and mapping-sector length scales are interpreted.

A.2.5 Frozen cloud quadrature measure

For a full-dimensional positive proposal density q_0, the ordinary importance measure is defined at initialization by

$$\omega_i = \frac{1}{N q_0(\boldsymbol{\mathcal{X}}_i^0)}. \tag{A.20}$$

It is then held fixed. For a phase-space observable $A_m(\boldsymbol{\mathcal{X}})$,

$$\int d\boldsymbol{\mathcal{X}}\, A_m(\boldsymbol{\mathcal{X}}) \tilde{\rho}_m(\boldsymbol{\mathcal{X}}, t_n) \approx \sum_{i=1}^{N} \omega_i A_m(\boldsymbol{\mathcal{X}}_i^n) y_i^n. \tag{A.21}$$

The estimator is not self-normalized. An optional quantile cap on ω_i is implemented as a diagnostic variance-reduction measure and must be reported whenever it is used.

Equation (A.20) does not apply to focused mapping sampling, which is singular in the full mapping space. For the focused production branch, the implemented frozen calibration may be written as $\omega_i^{\mathrm{focus}} = 1/(N y_i^0)$, so that

$$c_i^n = \omega_i^{\mathrm{focus}} y_i^n = \frac{1}{N} \frac{y_i^n}{y_i^0}, \qquad c_i^0 = \frac{1}{N}. \tag{A.22}$$

Here $\omega_i^{\mathrm{focus}}$ is a conventional estimator calibration, not the reciprocal of a full-dimensional proposal density.

Appendix B

Hamiltonian Mapping Propagation and the MInt Map

This appendix gives the trace–traceless mapping Hamiltonian and the symmetric MInt factorization used to propagate the PBME characteristics. These formulas concern Hamiltonian support motion only; the excess density source is developed separately in Appendix D.

B.1 PBME Hamiltonian and the MInt map

B.1.1 Trace–traceless mapping Hamiltonian

The mapping Hamiltonian is

$$H_m(\boldsymbol{\mathcal{X}}) = \frac{\mathbf{P}^2}{2M} + V_0(\mathbf{R}) + \frac{1}{2\hbar}\left[\mathbf{r}^T\bar{h}(\mathbf{R})\mathbf{r} + \mathbf{p}^T\bar{h}(\mathbf{R})\mathbf{p}\right]. \tag{B.1}$$

For the one-dimensional numerical model,

$$\dot{R} = \frac{P}{M}, \tag{B.2}$$

$$\dot{P} = -\frac{dV_0}{dR} - \frac{1}{2\hbar}\left[\mathbf{r}^T\frac{d\bar{h}}{dR}\mathbf{r} + \mathbf{p}^T\frac{d\bar{h}}{dR}\mathbf{p}\right], \tag{B.3}$$

$$\dot{\mathbf{r}} = \frac{1}{\hbar}\bar{h}(R)\mathbf{p}, \tag{B.4}$$

$$\dot{\mathbf{p}} = -\frac{1}{\hbar}\bar{h}(R)\mathbf{r}. \tag{B.5}$$

For the dual avoided-crossing model,

$$h(R) = \begin{pmatrix} V^{11}(R) & V^{12}(R) \\ V^{12}(R) & V^{22}(R) \end{pmatrix}, \tag{B.6}$$

with

$$V^{11}(R) = 0, \tag{B.7}$$
$$V^{22}(R) = -Ae^{-BR^2} + E_0, \tag{B.8}$$
$$V^{12}(R) = Ce^{-DR^2}, \tag{B.9}$$

and

$$\frac{dV^{11}}{dR} = 0, \tag{B.10}$$
$$\frac{dV^{22}}{dR} = 2ABR\, e^{-BR^2}, \tag{B.11}$$
$$\frac{dV^{12}}{dR} = -2CDR\, e^{-DR^2}. \tag{B.12}$$

The trace–traceless decomposition is

$$V_0(R) = \frac{1}{2}\,\mathrm{Tr_s}\, h(R), \qquad \bar{h}(R) = h(R) - V_0(R) I_\mathrm{s}. \tag{B.13}$$

B.1.2 Symmetric splitting

Write

$$H_m = H_A + H_B, \qquad H_A = \frac{P^2}{2M}, \qquad H_B = V_0(R) + \frac{1}{2\hbar}\left[\mathbf{r}^T \bar{h}(R)\mathbf{r} + \mathbf{p}^T \bar{h}(R)\mathbf{p}\right].$$

The MInt step is

$$\Psi_\tau^{\mathrm{MInt}} = \mathcal{A}_{\tau/2} \circ \mathcal{B}_\tau \circ \mathcal{A}_{\tau/2}. \tag{B.14}$$

The first drift gives

$$R_{1/2} = R_0 + \frac{\tau}{2M} P_0. \tag{B.15}$$

At fixed $R_{1/2}$, diagonalize

$$\bar{h}(R_{1/2}) = \mathsf{U}_\mathrm{loc} \boldsymbol{\varepsilon} \mathsf{U}_\mathrm{loc}^T, \tag{B.16}$$

where $\boldsymbol{\varepsilon} = \mathrm{diag}(\varepsilon_1, \ldots, \varepsilon_{N_s})$. Define

$$\widetilde{\mathbf{r}} = \mathsf{U}_\mathrm{loc}^T \mathbf{r}, \qquad \widetilde{\mathbf{p}} = \mathsf{U}_\mathrm{loc}^T \mathbf{p}.$$

The fixed-R mapping equations are decoupled rotations:

$$\widetilde{\mathbf{r}}(\tau) = \cos\left(\frac{\tau\boldsymbol{\varepsilon}}{\hbar}\right) \widetilde{\mathbf{r}}(0) + \sin\left(\frac{\tau\boldsymbol{\varepsilon}}{\hbar}\right) \widetilde{\mathbf{p}}(0), \tag{B.17}$$
$$\widetilde{\mathbf{p}}(\tau) = \cos\left(\frac{\tau\boldsymbol{\varepsilon}}{\hbar}\right) \widetilde{\mathbf{p}}(0) - \sin\left(\frac{\tau\boldsymbol{\varepsilon}}{\hbar}\right) \widetilde{\mathbf{r}}(0). \tag{B.18}$$

B.1.3 Analytic bath-momentum integral

Let

$$\mathsf{G} = \mathsf{U}_{\text{loc}}^{T} \frac{d\bar{h}}{dR} \mathsf{U}_{\text{loc}}, \qquad \omega_{ab} = \frac{\varepsilon_a - \varepsilon_b}{\hbar}. \tag{B.19}$$

Introduce

$$\mathsf{C}_{ab} = \widetilde{r}_a \widetilde{r}_b + \widetilde{p}_a \widetilde{p}_b, \tag{B.20}$$

$$\mathsf{S}_{ab} = \widetilde{p}_a \widetilde{r}_b - \widetilde{r}_a \widetilde{p}_b. \tag{B.21}$$

The oscillatory integrals are

$$I_c(\omega, \tau) = \int_0^{\tau} \cos(\omega t)\, dt = \tau \cos\left(\frac{\omega\tau}{2}\right) \operatorname{sinc}\left(\frac{\omega\tau}{2}\right), \tag{B.22}$$

$$I_s(\omega, \tau) = \int_0^{\tau} \sin(\omega t)\, dt = \tau \sin\left(\frac{\omega\tau}{2}\right) \operatorname{sinc}\left(\frac{\omega\tau}{2}\right), \tag{B.23}$$

where $\operatorname{sinc}(x) = \sin(x)/x$. These expressions remain regular as $\omega \to 0$.

The exact fixed-R mapping-force integral is

$$I_{\text{map}}(\tau) = \sum_{a,b} \mathsf{G}_{ab} \left[\mathsf{C}_{ab} I_c(\omega_{ab}, \tau) + \mathsf{S}_{ab} I_s(\omega_{ab}, \tau)\right]. \tag{B.24}$$

Hence

$$P_1 = P_0 - \tau \frac{dV_0}{dR}(R_{1/2}) - \frac{1}{2\hbar} I_{\text{map}}(\tau), \tag{B.25}$$

and the closing drift is

$$R_1 = R_{1/2} + \frac{\tau}{2M} P_1. \tag{B.26}$$

The mapping rotation preserves

$$\mathcal{R}_m^2 = \mathbf{r}^T \mathbf{r} + \mathbf{p}^T \mathbf{p}, \tag{B.27}$$

and MInt is symmetric and symplectic. The numerical checks use the round-trip residual

$$\varepsilon_{\text{rt}} = \frac{\left\| \Psi_{-\tau}^{\text{MInt}} \left[\Psi_{\tau}^{\text{MInt}}(\boldsymbol{\mathcal{X}})\right] - \boldsymbol{\mathcal{X}} \right\|}{1 + \|\boldsymbol{\mathcal{X}}\|} \tag{B.28}$$

and the symplectic residual

$$\varepsilon_{\text{symp}} = \frac{\left\| \left(D\Psi_{\tau}^{\text{MInt}}\right)^T \mathbb{J} D\Psi_{\tau}^{\text{MInt}} - \mathbb{J} \right\|_F}{\|\mathbb{J}\|_F}. \tag{B.29}$$

Appendix C

Gaussian-Process Density Representation and Analytic Derivatives

This appendix collects the finite Gaussian-process representations, analytic kernel and SEO-profile derivatives, hyperparameter objectives, and moment constraints. The product representation is the production architecture used for the reported calculations. Direct-density and density-difference forms are retained as explicitly identified alternatives and should not be inferred to have generated the MIDPOINT results unless stated otherwise.

C.1 Gaussian-process density representations

C.1.1 Product representation used in the thesis

The structured density representation is

$$\widehat{\rho}_{m,t}(\boldsymbol{\mathcal{X}}) = g_s(\mathbf{x})\mu_t(\boldsymbol{\mathcal{X}}), \tag{C.1}$$

where

$$\mu_t(\boldsymbol{\mathcal{X}}) = \sum_{j=1}^{N} \zeta_j(t) k_{\boldsymbol{\theta}_t}\left(\boldsymbol{\mathcal{X}}, \boldsymbol{\mathcal{X}}_j(t)\right). \tag{C.2}$$

The coefficients satisfy

$$\boldsymbol{\zeta}_t = K_{y,t}^{-1}\mathbf{u}_t, \qquad K_{y,t} = K_t + \sigma_n^2 I. \tag{C.3}$$

The exact g_s is used in prediction and differentiation. The safe profile appears only in the label transformation Eq. (A.3).

C.1.2 Optional direct-density GP

The formulation also admits a direct-density branch,

$$\widehat{\rho}_{m,t}^{\mathrm{dir}}(\boldsymbol{\mathcal{X}}) = \sum_{j=1}^{N} \zeta_j^{\mathrm{dir}}(t) k_{\boldsymbol{\theta}_t}\left(\boldsymbol{\mathcal{X}}, \boldsymbol{\mathcal{X}}_j(t)\right), \tag{C.4}$$

with

$$\boldsymbol{\zeta}_t^{\mathrm{dir}} = K_{y,t}^{-1}\mathbf{y}_t.$$

This branch approximates the full density directly and must not be confused with the product representation of Eq. (C.1). In particular, its kernel derivatives supply the complete density derivatives, whereas the product branch requires the derivatives of both g_s and μ_t.

C.1.3 Density-difference representation

The density-difference branch decomposes the density as

$$\begin{aligned} \widehat{\rho}_{m,t}^{\mathrm{diff}}(\boldsymbol{\mathcal{X}}) = \widehat{\rho}_{m,0}&\left[\mathcal{F}_{-t}^{\circ}(\boldsymbol{\mathcal{X}})\right] \\ &+ \sum_{j=1}^{N} \zeta_j^{\Delta}(t) k_{\boldsymbol{\theta}_t}\left(\boldsymbol{\mathcal{X}}, \boldsymbol{\mathcal{X}}_j(t)\right). \end{aligned} \tag{C.5}$$

The superscript Δ is a branch qualifier on the GP coefficients; it does not denote an additional physical density. The PBME Hamiltonian flow $\mathcal{F}_t^{\circ}$ is already defined in Chapter 5. At the transported support points,

$$\mathcal{F}_{-t}^{\circ}\left(\boldsymbol{\mathcal{X}}_i(t)\right) = \boldsymbol{\mathcal{X}}_i(0).$$

The baseline GP is frozen after its initial fit, and the correction coefficients $\boldsymbol{\zeta}_t^{\Delta}$ are obtained from the residual physical labels

$$y_i(t) - y_i(0). \tag{C.6}$$

In the present implementation, the off-support transported baseline is approximated by moving the baseline kernel centres with the support cloud. This identity is exact at the support points but is not an exact pullback of each kernel under a nonlinear MInt map. The density-difference architecture is therefore a distinct approximation and is reported separately from the product representation. The current software does not combine the density-difference and product branches.

C.2 ARD squared-exponential kernel derivatives

For a kernel section centred at $\boldsymbol{\mathcal{X}}_j(t)$, write

$$k_j(\boldsymbol{\mathcal{X}}) = \sigma_f^2 \exp\left[-\frac{1}{2}\sum_d \frac{(\mathcal{X}_d - \mathcal{X}_{j,d})^2}{\ell_d^2}\right], \tag{C.7}$$

and define

$$q_{j,d} = \frac{\mathcal{X}_{j,d} - \mathcal{X}_d}{\ell_d^2}. \tag{C.8}$$

Then

$$\partial_d k_j = q_{j,d} k_j. \tag{C.9}$$

Differentiating once more gives

$$\partial_e \partial_d k_j = \left(q_{j,e} q_{j,d} - \frac{\delta_{de}}{\ell_d^2}\right) k_j. \tag{C.10}$$

A third derivative is

$$\partial_f \partial_e \partial_d k_j = \left[q_{j,f} q_{j,e} q_{j,d} - \frac{\delta_{de}}{\ell_d^2} q_{j,f} - \frac{\delta_{df}}{\ell_d^2} q_{j,e} - \frac{\delta_{ef}}{\ell_e^2} q_{j,d}\right] k_j. \tag{C.11}$$

For one bath momentum component P and mapping coordinates x_d, x_e, the bath and mapping components are independent, so

$$\frac{\partial^3 k_j}{\partial x_e \partial x_d \partial P} = q_{j,P}\left(q_{j,e} q_{j,d} - \frac{\delta_{de}}{\ell_d^2}\right) k_j. \tag{C.12}$$

The derivatives of μ_t are obtained by summing Eqs. (C.9)–(C.12) with coefficients $\zeta_j(t)$.

C.3 Derivatives of the SEO profile and the product density

Write

$$g_s(\mathbf{x}) = \mathcal{G}_{\mathrm{SEO}}(\mathbf{x}) B_s(\mathbf{x}),$$

with $\mathcal{G}_{\mathrm{SEO}}$ and B_s defined in Eqs. (A.8) and (A.9). For a mapping coordinate x_d, where x_d denotes either an r_λ or a p_λ component,

$$\partial_d \mathcal{G}_{\mathrm{SEO}} = -\frac{2x_d}{\hbar} \mathcal{G}_{\mathrm{SEO}}. \tag{C.13}$$

The polynomial derivatives are

$$\frac{\partial B_s}{\partial r_\lambda} = \frac{4}{\hbar}\delta_{\lambda s} r_s, \qquad \frac{\partial B_s}{\partial p_\lambda} = \frac{4}{\hbar}\delta_{\lambda s} p_s, \tag{C.14}$$

and

$$\frac{\partial^2 B_s}{\partial r_{\lambda'}\partial r_\lambda} = \frac{4}{\hbar}\delta_{\lambda s}\delta_{\lambda' s}, \qquad \frac{\partial^2 B_s}{\partial p_{\lambda'}\partial p_\lambda} = \frac{4}{\hbar}\delta_{\lambda s}\delta_{\lambda' s}. \tag{C.15}$$

The first derivative of the profile is

$$\partial_d g_s = \mathcal{G}_{\text{SEO}}\left[\partial_d B_s - \frac{2x_d}{\hbar}B_s\right], \tag{C.16}$$

and the second derivative is

$$\begin{aligned}\partial_e\partial_d g_s = \mathcal{G}_{\text{SEO}}\Bigg[&\partial_e\partial_d B_s - \frac{2x_e}{\hbar}\partial_d B_s - \frac{2x_d}{\hbar}\partial_e B_s \\ &+ \left(\frac{4x_d x_e}{\hbar^2} - \frac{2\delta_{ab}}{\hbar}\right)B_s\Bigg].\end{aligned} \tag{C.17}$$

For the static product profile, g_s is independent of $\mathbf{R}$ and $\mathbf{P}$. The mixed third derivative required by the excess operator is therefore

$$\begin{aligned}\frac{\partial^3\widehat{\rho}_{m,t}}{\partial x_e\partial x_d\partial P} = &\frac{\partial^2 g_s}{\partial x_e\partial x_d}\frac{\partial\mu_t}{\partial P} + \frac{\partial g_s}{\partial x_d}\frac{\partial^2\mu_t}{\partial x_e\partial P} \\ &+ \frac{\partial g_s}{\partial x_e}\frac{\partial^2\mu_t}{\partial x_d\partial P} + g_s\frac{\partial^3\mu_t}{\partial x_e\partial x_d\partial P}.\end{aligned} \tag{C.18}$$

This is the exact product rule used for the static product surrogate.

C.3.1 Transported SEO profile

The transported-profile branch stores a mapping footpoint $\mathbf{x}_i^0$ for each support trajectory and represents

$$g_{s,t}^{\text{tr}}(\boldsymbol{\mathcal{X}}_i^t) = g_s(\mathbf{x}_i^0). \tag{C.19}$$

Let $D\mathbf{x}_i^0 = D_{\boldsymbol{\mathcal{X}}_i^t}\mathbf{x}_i^0$ denote the Jacobian of the mapping footpoint with respect to the current extended phase-space point. The implemented gradient is

$$\nabla_{\boldsymbol{\mathcal{X}}^t} g_{s,t}^{\text{tr}} = \left(D\mathbf{x}^0\right)^T \nabla_{\mathbf{x}^0} g_s. \tag{C.20}$$

The implemented Hessian retains only the product of first footpoint derivatives,

$$\nabla^2_{\boldsymbol{\mathcal{X}}^t} g_{s,t}^{\text{tr}} \approx \left(D\mathbf{x}^0\right)^T \nabla^2_{\mathbf{x}^0} g_s \left(D\mathbf{x}^0\right). \tag{C.21}$$

The omitted term is the contraction of $\nabla_{\mathbf{x}^0} g_s$ with the Hessian $D^2\mathbf{x}^0$ of the footpoint map. Accordingly, Eq. (C.21) is an approximation rather than the exact Hessian of the pulled-back profile.

For a profile that depends on P through the transported footpoint, the implemented Leibniz expansion is

$$\begin{aligned}\partial_d \partial_e \partial_P (g\mu) \approx g_{de}\mu_P + g_{dP}\mu_e + g_d\mu_{eP} + g_{eP}\mu_d \\ + g_e\mu_{dP} + g_P\mu_{de} + g\mu_{deP}.\end{aligned} \tag{C.22}$$

The pure profile term $g_{deP}\mu$ is omitted because the current transported profile stores only first-order footpoint geometry. This seven-term closure must therefore be distinguished from the exact static-profile formula Eq. (C.18).

C.4 GP hyperparameters and physical moment constraints

C.4.1 Kernel system and optimization objectives

The regularized kernel matrix is

$$K_y = K + \sigma_n^2 I + \epsilon_{\text{jit}} I. \tag{C.23}$$

The negative log marginal likelihood for a label vector $\mathbf{d}$ is

$$\mathcal{J}_{\text{NMLL}} = \frac{1}{2}\mathbf{d}^T K_y^{-1}\mathbf{d} + \frac{1}{2}\log\det K_y + \frac{N}{2}\log(2\pi), \tag{C.24}$$

where $\mathbf{d} = \mathbf{u}$ for the product GP and $\mathbf{d} = \mathbf{y}$ for the direct-density GP.

For the LOO objective, define the precision matrix

$$\mathsf{P} = K_y^{-1}, \qquad \boldsymbol{\zeta} = \mathsf{P}\mathbf{d}. \tag{C.25}$$

The exact LOO residual is

$$e_i^{\text{LOO}} = \frac{\zeta_i}{\mathsf{P}_{ii}}, \tag{C.26}$$

and the corresponding mean-square objective is

$$\mathcal{J}_{\text{LOO}} = \frac{1}{N}\sum_{i=1}^{N}\left(e_i^{\text{LOO}}\right)^2.$$

The numerical implementation permits frozen, breathing, adaptive, and free refit policies described in Sec. 4.6. The hyperparameters used to construct a source operator are held fixed during that operator evaluation or source leg.

C.4.2 Karush–Kuhn–Tucker/Schur-complement projection

The Karush–Kuhn–Tucker (KKT) formulation is used for the linear moment constraints. Let $\psi_c(\boldsymbol{\mathcal{X}})$ denote a constrained physical moment, and define

$$(\mathsf{C}_{\text{mom}})_{cj} = \int d\boldsymbol{\mathcal{X}}\, \psi_c(\boldsymbol{\mathcal{X}}) k_j(\boldsymbol{\mathcal{X}}). \tag{C.27}$$

For the direct-density branch, the constrained coefficient vector minimizes

$$\frac{1}{2}\left(\boldsymbol{\zeta}-\boldsymbol{\zeta}_0\right)^T K_y \left(\boldsymbol{\zeta}-\boldsymbol{\zeta}_0\right) \tag{C.28}$$

subject to

$$\mathsf{C}_{\mathrm{mom}}\boldsymbol{\zeta}=\mathbf{m}, \tag{C.29}$$

where $\boldsymbol{\zeta}_0 = K_y^{-1}\mathbf{y}$. The Lagrangian stationarity conditions give

$$K_y\left(\boldsymbol{\zeta}-\boldsymbol{\zeta}_0\right)+\mathsf{C}_{\mathrm{mom}}^T\boldsymbol{\xi}_{\mathrm{K}}=0,$$

and therefore

$$\boldsymbol{\zeta}=\boldsymbol{\zeta}_0-K_y^{-1}\mathsf{C}_{\mathrm{mom}}^T\boldsymbol{\xi}_{\mathrm{K}}.$$

Substitution into Eq. (C.29) yields

$$\left(\mathsf{C}_{\mathrm{mom}}K_y^{-1}\mathsf{C}_{\mathrm{mom}}^T\right)\boldsymbol{\xi}_{\mathrm{K}}=\mathsf{C}_{\mathrm{mom}}\boldsymbol{\zeta}_0-\mathbf{m}.$$

Hence

$$\boldsymbol{\zeta}=\boldsymbol{\zeta}_0-K_y^{-1}\mathsf{C}_{\mathrm{mom}}^T\left(\mathsf{C}_{\mathrm{mom}}K_y^{-1}\mathsf{C}_{\mathrm{mom}}^T\right)^{+}\left(\mathsf{C}_{\mathrm{mom}}\boldsymbol{\zeta}_0-\mathbf{m}\right) \tag{C.30}$$

where $+$ denotes the rank-revealing pseudoinverse. The implementation adds a small ridge to the Schur matrix and uses a singular-value decomposition (SVD) to remove numerically null constraint directions.

The constraint contract depends on the density representation and sampling geometry. Signed-SEO direct-density fits may enforce normalization, subsystem trace, and energy. Focused labels do not define a regular six-dimensional density fit and therefore disable this projection. The current product wrapper also disables the direct-GP KKT rows because those rows integrate the inner modulation μ_t, whereas the physical moments involve the product $g_s\mu_t$. Applying the direct-GP moment matrix to the inner GP would impose the wrong physical constraints.

C.5 Support locality and derivative identifiability

For a zero-mean ARD squared-exponential posterior mean with a finite set of centres, define the kernel-scaled distance to the support by

$$d_{\boldsymbol{\theta}}\left(\boldsymbol{\mathcal{X}},\mathcal{S}_t\right)=\min_j\left[\sum_{a=1}^{D}\frac{\left(\mathcal{X}_a-\mathcal{X}_{j,a}(t)\right)^2}{\ell_a^2}\right]^{1/2}. \tag{C.31}$$

The finite expansion satisfies

$$\left|\mu_t^{(N)}(\boldsymbol{\mathcal{X}})\right|\leq\sigma_f^2\left\|\boldsymbol{\zeta}_t\right\|_1\exp\left[-\frac{1}{2}d_{\boldsymbol{\theta}}\left(\boldsymbol{\mathcal{X}},\mathcal{S}_t\right)^2\right]. \tag{C.32}$$

This bound describes the locality of the chosen finite representation. It does not imply that the continuum density vanishes away from the support. It states that the posterior mean has no data-driven control there and reverts toward its prior according to the kernel geometry.

On a target region Ω, the Euclidean fill distance is

$$h_{\Omega,\mathcal{S}_t} = \sup_{\boldsymbol{\mathcal{X}}\in\Omega} \min_i \|\boldsymbol{\mathcal{X}} - \boldsymbol{\mathcal{X}}_i(t)\| . \tag{C.33}$$

Approximation bounds on a compact region depend on this quantity, the kernel smoothness, and the regularity of the target [70, 71]. A finite empirical cloud and a smooth finite-bandwidth kernel expansion are mathematically different numerical objects; agreement of cloud quadrature and analytic GP integration must therefore be tested rather than assumed [86].

Focused MMST sampling further places the mapping variables on a constrained shell whose total radius is preserved by the Hamiltonian map. Values of a multivariate function on this lower-dimensional manifold do not, in general, determine derivatives normal to it. The kernel prior selects one smooth continuation of the modulation μ_t, but that continuation is a modeling assumption rather than information independently supplied by the focused data. This is especially important for the mixed mapping derivatives entering the excess operator. The analytic SEO factor reduces the ambiguity by supplying known mapping-space structure, but it does not eliminate the need for off-manifold validation of the evolving modulation.

Appendix D

Excess Mapping Liouvillian and Semi-Discrete Source Operator

This appendix derives the differential action of the excess mapping Liouvillian for the supported density representations. It includes the conservative flux form, the backward-map derivative tensors needed by the pulled-back path, and the matrices used to collocate the source on the moving support.

D.1 Excess mapping-QCLE operator

D.1.1 General operator and one-dimensional specialization

The thesis defines

$$\mathcal{Q} = -i\mathcal{L}'_m.$$

For a general bath,

$$\mathcal{Q}[f] = -\frac{\hbar}{8}\sum_{\lambda,\lambda'=1}^{N_s} \nabla_{\mathbf{R}}\bar{h}^{\lambda\lambda'}(\mathbf{R}) \cdot \left[\left(\frac{\partial^2}{\partial r_{\lambda'}\partial r_\lambda} + \frac{\partial^2}{\partial p_{\lambda'}\partial p_\lambda}\right)\nabla_{\mathbf{P}} f\right]. \tag{D.1}$$

For the one-dimensional Tully implementation,

$$\mathcal{Q}[f] = -\frac{\hbar}{8}\sum_{\lambda,\lambda'=1}^{2} \frac{d\bar{h}^{\lambda\lambda'}(R)}{dR}\left[\frac{\partial^3 f}{\partial P\partial r_{\lambda'}\partial r_\lambda} + \frac{\partial^3 f}{\partial P\partial p_{\lambda'}\partial p_\lambda}\right]. \tag{D.2}$$

The sign in Eq. (D.2) is included in $\mathcal{Q}$. The characteristic equation is therefore

$$\frac{d}{dt}\tilde{\rho}_m\left(\boldsymbol{\mathcal{X}}(t),t\right) = \mathcal{Q}[\tilde{\rho}_m]\left(\boldsymbol{\mathcal{X}}(t),t\right), \tag{D.3}$$

and a midpoint label update contains $+\Delta t\,\mathcal{Q}$, not a second minus sign.

D.1.2 Direct-density GP contraction

For the direct-density GP of Eq. (C.4), substitution of Eq. (C.12) into Eq. (D.2) gives

$$\begin{aligned}\mathcal{Q}\left[\widehat{\rho}^{\mathrm{dir}}_{m,t}\right](\boldsymbol{\mathcal{X}}) = -\frac{\hbar}{8}\sum_{j=1}^{N}\zeta^{\mathrm{dir}}_{j}(t)q_{j,P}k_j \\ \times \sum_{\lambda,\lambda'=1}^{2}\frac{d\bar{h}^{\lambda\lambda'}}{dR}\Bigg[q_{j,r_{\lambda'}}q_{j,r_\lambda} - \frac{\delta_{\lambda\lambda'}}{\ell^2_{r_\lambda}} \\ + q_{j,p_{\lambda'}}q_{j,p_\lambda} - \frac{\delta_{\lambda\lambda'}}{\ell^2_{p_\lambda}}\Bigg].\end{aligned} \tag{D.4}$$

D.1.3 Product-density contraction

For the static product surrogate, Eq. (C.18) gives

$$\begin{aligned}\mathcal{Q}[g_s\mu] = -\frac{\hbar}{8}\sum_{\lambda,\lambda'=1}^{2}\frac{d\bar{h}^{\lambda\lambda'}}{dR}\Bigg[\frac{\partial^2 g_s}{\partial r_{\lambda'}\partial r_\lambda}\frac{\partial\mu}{\partial P} \\ + \frac{\partial g_s}{\partial r_\lambda}\frac{\partial^2\mu}{\partial r_{\lambda'}\partial P} + \frac{\partial g_s}{\partial r_{\lambda'}}\frac{\partial^2\mu}{\partial r_\lambda\partial P} + g_s\frac{\partial^3\mu}{\partial r_{\lambda'}\partial r_\lambda\partial P} \\ + \frac{\partial^2 g_s}{\partial p_{\lambda'}\partial p_\lambda}\frac{\partial\mu}{\partial P} + \frac{\partial g_s}{\partial p_\lambda}\frac{\partial^2\mu}{\partial p_{\lambda'}\partial P} \\ + \frac{\partial g_s}{\partial p_{\lambda'}}\frac{\partial^2\mu}{\partial p_\lambda\partial P} + g_s\frac{\partial^3\mu}{\partial p_{\lambda'}\partial p_\lambda\partial P}\Bigg].\end{aligned} \tag{D.5}$$

This expression uses the exact SEO profile in every derivative. The safe profile is confined to the label transformation and does not enter the physical excess operator.

D.1.4 Continuity form

Because $d\bar{h}/dR$ is independent of P, the one-dimensional excess operator can be written as

$$\mathcal{Q}[f] = -\frac{\partial}{\partial P}J_P[f], \tag{D.6}$$

where

$$J_P[f] = \frac{\hbar}{8}\sum_{\lambda,\lambda'=1}^{2}\frac{d\bar{h}^{\lambda\lambda'}}{dR}\left[\frac{\partial^2 f}{\partial r_{\lambda'}\partial r_\lambda} + \frac{\partial^2 f}{\partial p_{\lambda'}\partial p_\lambda}\right]. \tag{D.7}$$

When a continuity-flow representation is selected, the additional bath velocity is

$$u_P = \frac{J_P[\widehat{\rho}_m]}{\widehat{\rho}_m}, \tag{D.8}$$

with a signed numerical floor applied only to the denominator.

D.2 Backward MInt map and higher monodromy tensors

D.2.1 Definitions

No second phase-space symbol is introduced for the backward point. For a post-step point $\boldsymbol{\mathcal{X}}^{n+1}$, the inverse MInt half-step defines the midpoint point already used in Chapter 5:

$$\boldsymbol{\mathcal{X}}^{n+1/2} = \Psi^{\mathrm{MInt}}_{-\Delta t/2}\left(\boldsymbol{\mathcal{X}}^{n+1}\right). \tag{D.9}$$

The first-, second-, and third-order derivative tensors of the inverse half-step map are

$$\mathsf{M}^{(1)} = D\Psi^{\mathrm{MInt}}_{-\Delta t/2}\left(\boldsymbol{\mathcal{X}}^{n+1}\right), \tag{D.10}$$

$$\mathsf{M}^{(2)} = D^2\Psi^{\mathrm{MInt}}_{-\Delta t/2}\left(\boldsymbol{\mathcal{X}}^{n+1}\right), \tag{D.11}$$

$$\mathsf{M}^{(3)} = D^3\Psi^{\mathrm{MInt}}_{-\Delta t/2}\left(\boldsymbol{\mathcal{X}}^{n+1}\right). \tag{D.12}$$

The implementation evaluates these tensors by forward-mode JAX automatic differentiation of the complete MInt map [87]. A central finite-difference fallback is retained when JAX is unavailable. No hand-coded derivative of the local eigenvectors is used in the production monodromy path.

D.2.2 Third-order chain rule

Let $f(\boldsymbol{\mathcal{X}})$ be a differentiable scalar density field, and let $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ be directions in the extended phase space. All derivatives of f below are evaluated at $\boldsymbol{\mathcal{X}}^{n+1/2}$. The first derivative of the pulled-back field is

$$D\left(f \circ \Psi\right)[\mathbf{v}_1] = Df\left[\mathsf{M}^{(1)}\mathbf{v}_1\right]. \tag{D.13}$$

The second derivative is

$$\begin{aligned} D^2\left(f \circ \Psi\right)[\mathbf{v}_1, \mathbf{v}_2] = D^2 f&\left[\mathsf{M}^{(1)}\mathbf{v}_1, \mathsf{M}^{(1)}\mathbf{v}_2\right] \\ &+ Df\left[\mathsf{M}^{(2)}[\mathbf{v}_1, \mathbf{v}_2]\right]. \end{aligned} \tag{D.14}$$

The complete third-order chain rule is

$$
\begin{aligned}
D^3 (f \circ \Psi) [\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3] = D^3 f &\left[\mathsf{M}^{(1)}\mathbf{v}_1, \mathsf{M}^{(1)}\mathbf{v}_2, \mathsf{M}^{(1)}\mathbf{v}_3\right] \\
&+ D^2 f \left[\mathsf{M}^{(2)}[\mathbf{v}_1, \mathbf{v}_2], \mathsf{M}^{(1)}\mathbf{v}_3\right] \\
&+ D^2 f \left[\mathsf{M}^{(2)}[\mathbf{v}_1, \mathbf{v}_3], \mathsf{M}^{(1)}\mathbf{v}_2\right] \\
&+ D^2 f \left[\mathsf{M}^{(1)}\mathbf{v}_1, \mathsf{M}^{(2)}[\mathbf{v}_2, \mathbf{v}_3]\right] \\
&+ D f \left[\mathsf{M}^{(3)}[\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3]\right].
\end{aligned}
\tag{D.15}
$$

For the first mapping contribution to the excess operator, select

$$
\mathbf{v}_1 = \mathbf{e}_P, \qquad \mathbf{v}_2 = \mathbf{e}_{r_\lambda}, \qquad \mathbf{v}_3 = \mathbf{e}_{r_{\lambda'}}, \tag{D.16}
$$

where $\mathbf{e}_P$ and $\mathbf{e}_{r_\lambda}$ are coordinate basis vectors in the extended phase space. For the second contribution, replace the last two directions by $\mathbf{e}_{p_\lambda}$ and $\mathbf{e}_{p_{\lambda'}}$.

D.2.3 Intrinsic and pulled-back operator paths

The excess operator has two distinct numerical evaluations.

1. The trajectory-level midpoint integrator evaluates the intrinsic operator directly at the current and midpoint support points:

   $$
   \mathcal{Q}\left[\widehat{\rho}_{m,n}\right](\boldsymbol{\mathcal{X}}_i^n), \qquad \mathcal{Q}\left[\widehat{\rho}_{m,n+1/2}\right]\left(\boldsymbol{\mathcal{X}}_i^{n+1/2}\right).
   $$

   No additional inverse half-step is applied inside these evaluations.

2. The Eulerian post-step interface first constructs $\boldsymbol{\mathcal{X}}^{n+1/2}$ from Eq. (D.9). Its intrinsic version differentiates the density with respect to the midpoint coordinates. The separate verification path differentiates the composed field $f \circ \Psi^{\mathrm{MInt}}_{-\Delta t/2}$ using Eq. (D.15).

The two operations coincide only when the inverse half-step map is the identity. At finite Δt, the monodromy terms in Eq. (D.15) distinguish derivatives with respect to the post-step coordinates from intrinsic derivatives at the midpoint point.

D.3 Discrete source matrices

For the product basis

$$
\Phi_j(\boldsymbol{\mathcal{X}}) = g_s(\mathbf{x}) k_j(\boldsymbol{\mathcal{X}}), \tag{D.17}
$$

define the physical-density source matrix exactly as in Chapter 5:

$$
\mathsf{L}^{(\rho)}_{ij} = \mathcal{Q}[\Phi_j]\big|_{\boldsymbol{\mathcal{X}}=\boldsymbol{\mathcal{X}}_i}. \tag{D.18}
$$

The collocated source value is

$$\mathcal{Q}[\widehat{\rho}_m]\,(\boldsymbol{\mathcal{X}}_i) = \sum_{j=1}^{N} \mathsf{L}^{(\rho)}_{ij} \zeta_j. \tag{D.19}$$

For the product surrogate,

$$\mathbf{u} = \mathsf{D}_g^{-1}\mathbf{y}, \qquad \mathsf{D}_g = \operatorname{diag}\left[g_{s,\mathrm{safe}}(\mathbf{x}_i)\right]. \tag{D.20}$$

Using $\boldsymbol{\zeta} = K_y^{-1}\mathbf{u}$, define

$$\mathsf{L} = \mathsf{D}_g^{-1}\mathsf{L}^{(\rho)}, \qquad \mathsf{A} = \mathsf{L}K_y^{-1}. \tag{D.21}$$

The transformed labels satisfy

$$\frac{d\mathbf{u}}{dt} = \mathsf{A}\mathbf{u}. \tag{D.22}$$

If the same frozen source system is written in physical labels, then

$$\frac{d\mathbf{y}}{dt} = \mathsf{L}^{(\rho)}K_y^{-1}\mathsf{D}_g^{-1}\mathbf{y}. \tag{D.23}$$

No additional symbol is introduced for this similar generator. For the direct-density GP, D_g^{-1} is absent and the coefficient solve uses $\mathbf{y}$ directly.

Appendix E

Numerical Propagation, Observables, and Diagnostics

This appendix first records the PBME baseline and the explicit midpoint scheme used for every result labelled MIDPOINT. Alternative integration branches are then documented separately. Their presence in the software does not imply that they were used in the production figures.

E.1 Production propagation schemes

The production comparison uses the MInt PBME baseline and the explicit midpoint correction-weight scheme. In the implementation, the word "weight" denotes a storage ratio; the physical mathematical variable remains the live density label y_i^n.

E.1.1 PBME baseline

The PBME scheme advances only the support:

$$\boldsymbol{\mathcal{X}}_i^{n+1} = \Psi_{\Delta t}^{\mathrm{MInt}}\left(\boldsymbol{\mathcal{X}}_i^n\right), \qquad y_i^{n+1} = y_i^n. \tag{E.1}$$

The GP is refitted on the new support geometry with unchanged physical labels.

E.1.2 Explicit midpoint correction-weight scheme

The command-line default uses the weight-based label integrator and the explicit midpoint rule. The word "weight" describes the internal storage; the mathematical update is written entirely in the physical labels y_i^n.

First, propagate the support to the midpoint:

$$\boldsymbol{\mathcal{X}}_i^{n+1/2} = \Psi_{\Delta t/2}^{\mathrm{MInt}}\left(\boldsymbol{\mathcal{X}}_i^n\right). \tag{E.2}$$

Evaluate the first source slope:

$$\mathcal{Q}_i^{(1)} = \mathcal{Q}\left[\widehat{\rho}_{m,n}\right](\boldsymbol{\mathcal{X}}_i^n). \tag{E.3}$$

The predicted midpoint label is

$$y_i^{n+1/2,*} = y_i^n + \frac{\Delta t}{2}\mathcal{Q}_i^{(1)}. \tag{E.4}$$

A temporary midpoint GP is fitted to

$$\left\{\boldsymbol{\mathcal{X}}_i^{n+1/2}, y_i^{n+1/2,*}\right\}_{i=1}^{N}.$$

The midpoint slope is

$$\mathcal{Q}_i^{(2)} = \mathcal{Q}\left[\widehat{\rho}_{m,n+1/2}\right]\left(\boldsymbol{\mathcal{X}}_i^{n+1/2}\right). \tag{E.5}$$

The physical labels are advanced by

$$y_i^{n+1} = y_i^n + \Delta t\,\mathcal{Q}_i^{(2)}. \tag{E.6}$$

Internally, the code obtains the stored correction-weight array by dividing the updated y_i^{n+1} by the fixed initial value y_i^0; this storage choice does not define a second density variable. The support then completes the second MInt half-step and the endpoint GP is refitted to $\{\boldsymbol{\mathcal{X}}_i^{n+1}, y_i^{n+1}\}$.

E.2 Alternative and inactive integration branches

The following schemes are retained for controlled comparisons or software compatibility. Unless a calculation explicitly identifies one of these branches, it should not be associated with the numerical results of this thesis.

E.2.1 Cayley correction-weight scheme

The optional Cayley rule introduces the local multiplicative rate

$$\sigma_i = \frac{\mathcal{Q}[\widehat{\rho}_m](\boldsymbol{\mathcal{X}}_i)}{\widehat{\rho}_m(\boldsymbol{\mathcal{X}}_i)}. \tag{E.7}$$

The equivalent physical-label update is

$$y_i^{\text{new}} = \frac{1 + \frac{\Delta t}{2}\sigma_i}{1 - \frac{\Delta t}{2}\sigma_i} y_i^{\text{old}}. \tag{E.8}$$

The midpoint refit and second source evaluation use the same staging as the explicit midpoint scheme.

E.2.2 Continuity-flow split

Let $f_{\text{flow}} \in [0,1]$ be the selected flow fraction. The support receives the additional bath-momentum displacement

$$\Delta P_i = f_{\text{flow}} \Delta t_{\text{leg}} u_{P,i}, \tag{E.9}$$

with u_P from Eq. (D.8). A numerical cap is applied to $|\Delta P_i|$ when requested.

The label rate along the modified support path is derived directly from Eq. (D.3). Since

$$\dot{P} = \dot{P}_{\text{PBME}} + f_{\text{flow}} u_P,$$

the total derivative is

$$\frac{d\rho}{dt} = \mathcal{Q}[\rho] + f_{\text{flow}} u_P \frac{\partial \rho}{\partial P}. \tag{E.10}$$

Thus the flow and label channels represent the same excess contribution without double counting. The value $f_{\text{flow}} = 0$ gives the pure label scheme. The current implementation restricts the added flow to the bath-momentum direction.

E.2.3 Crank–Nicolson source integrator

For the product representation, the experimental linear branch advances the transformed labels according to

$$\frac{d\mathbf{u}}{dt} = \mathsf{A}\mathbf{u}. \tag{E.11}$$

At frozen midpoint support and fixed hyperparameters, the Crank–Nicolson update is

$$\mathbf{u}^{n+1/2,+} = \left(I - \frac{\Delta t}{2}\mathsf{A}_{n+1/2}\right)^{-1} \left(I + \frac{\Delta t}{2}\mathsf{A}_{n+1/2}\right) \mathbf{u}^{n+1/2,-}. \tag{E.12}$$

The physical labels are then recovered from

$$\mathbf{y}^{n+1/2,+} = \mathsf{D}_g^{n+1/2}\mathbf{u}^{n+1/2,+}. \tag{E.13}$$

The implemented predictor–corrector variant first forms a half-step estimate with the initial generator, refits the midpoint GP, and then applies the full update using the midpoint generator. This branch is marked experimental and is not available for the density-difference representation. In the direct GP branch, the same algebra acts on $\mathbf{y}$ rather than $\mathbf{u}$.

E.2.4 Matrix-exponential source leg

The optional matrix-exponential branch uses the frozen source generator defined in Chapter 5. After the first MInt half-step,

$$\mathbf{u}^{n+1/2,-} = \left(\mathsf{D}_g^{n+1/2}\right)^{-1} \mathbf{y}^n. \tag{E.14}$$

With the midpoint support and hyperparameters frozen,

$$\mathbf{u}^{n+1/2,+} = \exp\left(\Delta t \mathsf{A}_{n+1/2}\right) \mathbf{u}^{n+1/2,-}, \tag{E.15}$$

and

$$\mathbf{y}^{n+1/2,+} = \mathsf{D}_g^{n+1/2} \mathbf{u}^{n+1/2,+}. \tag{E.16}$$

The midpoint support then completes the second MInt half-step. The exponential is exact for this frozen finite-dimensional source system, not for the continuum mapping-QCLE or for the complete split method.

E.2.5 Status of the excess-source clipping controls

The command-line interface retains `apply_q_clip`, `q_clip_frac`, and `q_clip_abs` for compatibility with earlier configurations. In the current midpoint implementation these values are recorded but no clipping is applied to the source update. They are therefore inactive interface controls, not active stabilization terms.

E.3 Physical observables and diagnostics

E.3.1 Mapping symbols

The physical population symbol for state λ is

$$c_{\lambda\lambda}(\mathbf{x}) = \frac{r_\lambda^2 + p_\lambda^2 - \hbar}{2\hbar}. \tag{E.17}$$

For the two-state implementation,

$$\operatorname{Re} c_{12}(\mathbf{x}) = \frac{r_1 r_2 + p_1 p_2}{2\hbar}, \tag{E.18}$$

$$\operatorname{Im} c_{12}(\mathbf{x}) = \frac{r_1 p_2 - r_2 p_1}{2\hbar}. \tag{E.19}$$

The subsystem trace symbol is

$$I_m(\mathbf{x}) = c_{11}(\mathbf{x}) + c_{22}(\mathbf{x}) = \frac{\mathbf{r}^2 + \mathbf{p}^2 - 2\hbar}{2\hbar}. \tag{E.20}$$

The cloud estimator for a population is

$$P_\lambda(t_n) \approx \sum_{i=1}^{N} c_i^n c_{\lambda\lambda}(\mathbf{x}_i^n). \tag{E.21}$$

The same measure is used for coherences, bath moments, trace, and physical energy. These cloud estimators are stored separately from analytic GP-integral estimates.

E.3.2 Signed-cloud diagnostics

For signed cloud contributions, use the notation

$$c_i^n = \begin{cases} \omega_i y_i^n, & \text{full-dimensional proposal,} \\ (y_i^n / y_i^0)/N, & \text{focused proposal.} \end{cases} \tag{E.22}$$

Then define

$$\mathrm{ESS}_{\mathrm{abs}} = \frac{\left(\sum_i |c_i^n|\right)^2}{\sum_i (c_i^n)^2}, \tag{E.23}$$

$$\mathrm{ESS}_{\pm} = \frac{\left(\sum_i c_i^n\right)^2}{\sum_i (c_i^n)^2}, \tag{E.24}$$

$$\chi_{\mathrm{cancel}} = \frac{|\sum_i c_i^n|}{\sum_i |c_i^n|}. \tag{E.25}$$

Small χ_{cancel} indicates that an observable is obtained by cancellation of large positive and negative contributions.

E.3.3 Source and regression diagnostics

For the collocated source values $\mathcal{Q}_i$, the calculation evaluates

$$\mathcal{Q}_{\mathrm{RMS}} = \left[\frac{1}{N}\sum_i \mathcal{Q}_i^2\right]^{1/2}, \tag{E.26}$$

$$\mathcal{Q}_{\max} = \max_i |\mathcal{Q}_i|, \tag{E.27}$$

$$r_{\mathrm{src}} = \max_i \frac{\Delta t |\mathcal{Q}_i|}{|y_i| + \epsilon}. \tag{E.28}$$

The regression diagnostics include the support-point RMS residual, LOO residuals, coefficient norms, kernel conditioning proxies, posterior uncertainty, and held-out operator-fidelity tests.

An optional ESS-resampling procedure replaces the carried labels by current surrogate predictions when the signed effective sample size falls below a selected threshold. This operation prevents uncontrolled variance growth but biases the cloud toward the current surrogate; it is therefore disabled by default and must be reported explicitly when used.

E.3.4 KDE comparison

The signed KDE used for diagnostic comparison is

$$\widehat{\rho}_{\mathrm{KDE}}(\boldsymbol{\mathcal{X}}) = \frac{1}{(2\pi)^{D/2}\prod_d h_d}\sum_{i=1}^{N} c_i \exp\left[-\frac{1}{2}\sum_d \frac{(\mathcal{X}_d - \mathcal{X}_{i,d})^2}{h_d^2}\right]. \tag{E.29}$$

The bandwidths use the multivariate Silverman scaling [88]

$$h_d = \left(\frac{4}{D+2} \right)^{1/(D+4)} \sigma_d N^{-1/(D+4)}. \tag{E.30}$$

The KDE does not participate in the QCLE propagation and imposes no positivity constraint.

Appendix F

Reference Calculations and Numerical Parameters

This appendix states the equations, domains, discretizations, and controlled approximations needed to interpret the independent TDSE and grid-QCLE reference calculations and the moving-cloud results. Its purpose is physical and numerical transparency: each approximation is associated with the quantity it can affect, and each reference is qualified by its boundary and refinement controls.

F.1 Reference TDSE and grid-QCLE calculations

The numerically resolved TDSE calculation propagates a two-component nuclear wavefunction on a one-dimensional spatial grid by a symmetric split-operator method [81, 89]. The potential half-steps use the local matrix exponential of the two-state diabatic Hamiltonian, and the kinetic full step is diagonal in the Fourier momentum representation. The uniform spatial grid is periodic as required by the discrete Fourier transform and is automatically enlarged so that the propagated wavepacket is negligible at the boundaries over the requested time interval; the implementation uses at least 4096 points. Populations and coherences are reported in the diabatic basis. A Wigner post-processing step constructs the matrix-valued phase-space fields used in the comparison plots.

The grid-QCLE solver uses cell-centred, uniform R and P grids without duplicating the periodic endpoint. Derivatives are Fourier pseudospectral: $k_R = 2\pi\ \mathrm{fftfreq}(N_R, \Delta R)$ and $k_P = 2\pi\ \mathrm{fftfreq}(N_P, \Delta P)$, with real-to-complex transforms used independently on each axis. No absorbing boundary is applied; the domain must be wide enough that the density does not reach a periodic boundary. Time integration uses classical fourth-order Runge–Kutta (RK4). The requested step is checked against separate advection, force, and quantum oscillation limits, using the RK4 imaginary-axis stability radius 2.828. The domains and resolved steps used in the comparisons are stated explicitly in Table F.1; none is inferred from a figure axis.

For the grid-QCLE reference, write

$$\hat{\rho}_W = \begin{pmatrix} \rho_W^{11} & \rho_{W,\mathrm{R}}^{12} + i\rho_{W,\mathrm{I}}^{12} \\ \rho_{W,\mathrm{R}}^{12} - i\rho_{W,\mathrm{I}}^{12} & \rho_W^{22} \end{pmatrix}.$$

The one-dimensional diabatic QCLE equations are

$$\partial_t \rho_W^{11} = -\frac{P}{M}\partial_R \rho_W^{11} + \frac{dV^{11}}{dR}\partial_P \rho_W^{11} + \frac{dV^{12}}{dR}\partial_P \rho_{W,\mathrm{R}}^{12} - \frac{2V^{12}}{\hbar}\rho_{W,\mathrm{I}}^{12}, \tag{F.1}$$

$$\partial_t \rho_W^{22} = -\frac{P}{M}\partial_R \rho_W^{22} + \frac{dV^{22}}{dR}\partial_P \rho_W^{22} + \frac{dV^{12}}{dR}\partial_P \rho_{W,\mathrm{R}}^{12} + \frac{2V^{12}}{\hbar}\rho_{W,\mathrm{I}}^{12}, \tag{F.2}$$

$$\begin{aligned}\partial_t \rho_{W,\mathrm{R}}^{12} = &-\frac{P}{M}\partial_R \rho_{W,\mathrm{R}}^{12} + \frac{dV_0}{dR}\partial_P \rho_{W,\mathrm{R}}^{12} + \frac{1}{2}\frac{dV^{12}}{dR}\partial_P \left(\rho_W^{11} + \rho_W^{22}\right) \\ &+ \frac{V^{11} - V^{22}}{\hbar}\rho_{W,\mathrm{I}}^{12},\end{aligned} \tag{F.3}$$

$$\begin{aligned}\partial_t \rho_{W,\mathrm{I}}^{12} = &-\frac{P}{M}\partial_R \rho_{W,\mathrm{I}}^{12} + \frac{dV_0}{dR}\partial_P \rho_{W,\mathrm{I}}^{12} \\ &+ \frac{1}{\hbar}\left[V^{12}\left(\rho_W^{11} - \rho_W^{22}\right) - \left(V^{11} - V^{22}\right)\rho_{W,\mathrm{R}}^{12}\right].\end{aligned} \tag{F.4}$$

The phase-space derivatives are Fourier-pseudospectral and the time integration uses explicit RK4 with an internally enforced stability limit.

Table F.1: Exact, separate interaction-window reference-refinement settings. Slash-separated entries list coarse/fine/finer values; no domain or nominal level is collapsed. Observable endpoints, both successive differences, guarded orders, and rejection reasons are reported in Tables 6.13 and 6.14.

Method	P_0	Mode	t_{final}	Δt ladder	Grid ladder	R domain ladder	P domain ladder	accepted edge mass	finest CFL ratio
TDSE	20	time	3000	0.2/0.1/0.05	8192/8192/8192	[-40,45.9895]/[-40,45.9895]/[-40,45.9895]	—	1.223884e-24	—
TDSE	20	grid	3000	0.05/0.05/0.05	2048/4096/8192	[-40,45.95801]/[-40,45.979]/[-40,45.9895]	—	6.164956e-25	—
TDSE	100	time	600	0.2/0.1/0.05	8192/8192/8192	[-40,45.9895]/[-40,45.9895]/[-40,45.9895]	—	4.825683e-26	—
TDSE	100	grid	600	0.05/0.05/0.05	8192/8192/8192	[-40,45.9895]/[-40,45.9895]/[-40,45.9895]	—	4.825683e-26	—
grid QCLE	20	time	3000	2/1/0.5	192×256/192×256/192×256	[-30,30]/[-30,30]/[-30,30]	[-35,35]/[-35,35]/[-35,35]	5.915682e-04	0.09195161
grid QCLE	20	grid	3000	0.5/0.5/0.5	48×64/96×128/192×256	[-30,30]/[-30,30]/[-30,30]	[-35,35]/[-35,35]/[-35,35]	5.915682e-04	0.09195161
grid QCLE	100	time	600	2/1/0.5	192×128/192×128/192×128	[-30,30]/[-30,30]/[-30,30]	[80,120]/[80,120]/[80,120]	2.152345e-07	0.1066453
grid QCLE	100	grid	600	0.5/0.5/0.5	48×32/96×64/192×128	[-30,30]/[-30,30]/[-30,30]	[80,120]/[80,120]/[80,120]	1.697013e-07	0.1066453

Table F.2: Numerical parameters used for the moving-cloud calculations. The regularization $\ell_2 = 0.05$ is used in the dynamical comparisons; the manufactured study additionally examines 10^{-6} and 0.01.

Purpose	Controlled values	Statistical interpretation
Time-step sensitivity	$N = 1000$, $\Delta t = 0.5, 0.25, 0.125$, seeds 11, 29, 47, 73	Three deterministic step sizes evaluated on four independent initial clouds
Cloud-size sensitivity	$N = 500, 1000, 2000$, $\Delta t = 0.25$, seeds 11, 29, 47	Independently sampled, non-nested clouds; no deterministic cloud-size order is inferred
Four-seed replication	$N = 1000$, $\Delta t = 0.25$, seeds 11, 29, 47, 73	Small-sample sensitivity intervals with three degrees of freedom
Manufactured operator	$N = 300, 600, 1200, 2400$; $\ell_2 = 10^{-6}, 0.01, 0.05$; seeds 123–125	Paired training and independent query clouds at each (N, ℓ_2)
Propagation interval	$t_{\text{final}} = 2t_c$	Includes approach to, passage through, and departure from the avoided crossing

The TDSE and grid-QCLE studies use $R_0 = -15$ and propagate to $t_{\text{final}} = 2t_c$, so every case traverses the avoided-crossing region. Time and grid refinement are separate: a time ladder holds the finest grid fixed, while a grid ladder holds the finest time step fixed. Both successive differences must exceed $\tau_{\text{noise}} = 10^{-12} + 10^{-12} \max_k |O_k|$, must decrease monotonically, and must give $0 < p \leq 6$; other rows are retained as roundoff- or saturation-limited, nonmonotone, or rapidly contracting but nonasymptotic. Expected temporal scaling is approximately second order for TDSE and fourth order for grid-QCLE RK4; spatial or phase-space-grid behaviour is assessed separately. The TDSE edge statistic is the maximum spatial-edge mass across the three levels. The grid-QCLE statistic is the larger finest-grid absolute physical-marginal edge fraction, with declared tolerance 10^{-3}; the sign-indefinite phase-space $|W|$ ringing diagnostic is not substituted for this boundary gate. For grid QCLE the dimensionless CFL ratio is $\Delta t/\Delta t_{\max}$, where $\Delta t_{\max}$ is the minimum of the RK4 imaginary-axis bound 2.828 divided by the spectral advection, force, and electronic-frequency scales; ratios below one pass this declared linear-stability check. TDSE is model-exact only within its displayed numerical controls; grid QCLE remains a numerical solution of the approximate QCLE.

F.2 Numerical parameters of the reported calculations

The dynamical comparisons use the one-dimensional two-state Tully model in atomic units with nuclear mass $M = 2000$, $\hbar = 1$, initial nuclear centre $R_0 = -15$, and initial momenta $P_{\text{init}} = 20$ and 100. The initial nuclear packet has $\sigma_R = 1$. PBME and MIDPOINT members of every paired comparison start from the same sampled cloud. Table F.2 collects the controlled variations used to separate deterministic time-step effects from stochastic cloud-to-cloud variation.

The moving-cloud density is represented by the product GP described in Chapter 5. Hyperparameters are optimized at the initial fit and then held fixed during each reported propagation, with $\ell_2 = 0.05$ in the dynamical studies. The source is applied by the tested MIDPOINT label update without an imposed SEO projection and without post-update conservation correction. These are defining features of the method under examination, not optional repairs introduced after observing the result.

F.3 Controlled approximations and interpretation

The product form imposes a separability approximation on the phase-space density. Ambient first through third derivatives are analytic derivatives of that fitted surrogate, but they are not exact derivatives of the unknown density away from the sampled manifold. The backward MInt half-step and its Jacobian enter the pulled-back source; higher footpoint derivatives omitted by the transported-profile approximation are not used to claim exact continuum dynamics. The KDE calculation is restricted to the identical-cloud reconstruction control and never supplies the QCLE source.

The TDSE reference solves the two-state quantum model within the stated finite spatial domain and split-operator discretization. The grid-QCLE reference solves the approximate quantum–classical equation using Fourier pseudospectral derivatives and RK4. Consequently agreement with TDSE and agreement with grid QCLE answer different physical questions. In both cases, time/grid refinement and boundary mass are examined before the reference is used in a PBME–MIDPOINT comparison.

Appendix G

Complete Per-Seed Numerical Evidence

This appendix retains the individual numerical replicates behind the compact tables in Chapter 6. Its purpose is scientific reconstructibility: means and confidence intervals remain useful summaries, but each conclusion about seed dispersion, regularization sensitivity, tail conditioning, or reference error can also be checked against the values from the independent seeds. The notation follows the estimator conventions of Section 6.1: P_{init} is nuclear momentum, ρ_{11}^{SN} and ρ_{22}^{SN} are signed-mass-normalized electronic populations, and raw quantities are never replaced by normalized display estimators in a conservation test.

G.1 Manufactured density, gradient, and excess action

The next three tables give every seed value at all four support sizes, all three regularizations, and both the training and independent query clouds. They include relative and absolute errors so that the small mean density error cannot conceal derivative or excess-action variation.

Table G.1: Manufactured-operator complete paired policy table: Density.

ℓ_2	N	Seed	Query	n_q	E_1	E_2	E_∞	MAE	RMSE	L^∞	floor used?	jitter	tries
0.01	1200	123	on_support	1200	0.009014162	0.009815349	0.009579275	8.186420e-05	1.418318e-04	8.381656e-04	no	0	1
0.01	1200	123	off_support	1000	0.00982301	0.01064346	0.01211077	9.740093e-05	1.714088e-04	0.00106018	no	0	1
0.01	1200	124	on_support	1200	0.006885515	0.007755661	0.008671584	6.138705e-05	1.056145e-04	6.656284e-04	no	0	1
0.01	1200	124	off_support	1000	0.0075248	0.008078198	0.01258177	7.097705e-05	1.174505e-04	0.001001714	no	0	1
0.01	1200	125	on_support	1200	0.004468482	0.00538544	0.008985748	4.216660e-05	7.817208e-05	6.708908e-04	no	0	1
0.01	1200	125	off_support	1000	0.005522433	0.006212368	0.01038027	5.362540e-05	9.544507e-05	7.900301e-04	no	0	1
0.01	2400	123	on_support	2400	0.007231469	0.007507373	0.009441734	6.854242e-05	1.107690e-04	8.337411e-04	no	0	1
0.01	2400	123	off_support	1000	0.007924089	0.008146642	0.007413902	6.979831e-05	1.076936e-04	6.080358e-04	no	0	1
0.01	2400	124	on_support	2400	0.004594798	0.005375094	0.009975039	4.271009e-05	7.785469e-05	8.333047e-04	no	0	1
0.01	2400	124	off_support	1000	0.005893327	0.007104605	0.02096923	5.576331e-05	1.091473e-04	0.001966247	no	0	1
0.01	2400	125	on_support	2400	0.009616188	0.009774071	0.009924131	9.165269e-05	1.456054e-04	8.725751e-04	no	0	1
0.01	2400	125	off_support	1000	0.01015035	0.0100307	0.01196987	9.071737e-05	1.421924e-04	8.556569e-04	no	0	1
0.01	300	123	on_support	300	0.00265337	0.00336907	0.00567316	2.322469e-05	4.475718e-05	2.759311e-04	no	0	1
0.01	300	123	off_support	1000	0.004833242	0.005420195	0.006267334	4.458566e-05	7.700215e-05	5.017507e-04	no	0	1
0.01	300	124	on_support	300	8.308844e-04	0.001165282	0.002461044	8.021247e-06	1.729559e-05	1.827424e-04	no	0	1
0.01	300	124	off_support	1000	0.008398002	0.01685498	0.04919432	7.630428e-05	2.340898e-04	0.00442341	no	0	1
0.01	300	125	on_support	300	1.908726e-04	2.341710e-04	2.631223e-04	1.932820e-06	3.636256e-06	1.987525e-05	no	0	1
0.01	300	125	off_support	1000	7.854081e-04	9.149795e-04	0.001645499	7.665786e-06	1.372944e-05	1.168916e-04	no	0	1
0.01	600	123	on_support	600	0.003734174	0.004975844	0.008743289	3.660073e-05	7.477071e-05	7.163581e-04	no	0	1
0.01	600	123	off_support	1000	0.005677611	0.008041849	0.02014455	5.417220e-05	1.162363e-04	0.001406656	no	0	1
0.01	600	124	on_support	600	0.01175659	0.0133328	0.01335845	1.072498e-04	1.885920e-04	9.870648e-04	no	0	1
0.01	600	124	off_support	1000	0.01355609	0.01526131	0.01467253	1.229195e-04	2.143085e-04	0.001218326	no	0	1
0.01	600	125	on_support	600	1.527614e-04	1.927147e-04	2.257010e-04	1.486939e-06	2.923655e-06	1.956015e-05	no	0	1
0.01	600	125	off_support	1000	0.00278207	0.004539234	0.01044349	2.642705e-05	6.626496e-05	8.294096e-04	no	0	1
0.05	1200	123	on_support	1200	0.008943139	0.009740394	0.009282458	8.121920e-05	1.407487e-04	8.121948e-04	no	0	1
0.05	1200	123	off_support	1000	0.009854253	0.01069589	0.01288285	9.771072e-05	1.722532e-04	0.001127767	no	0	1
0.05	1200	124	on_support	1200	0.006913316	0.007773885	0.008791995	6.163491e-05	1.058627e-04	6.748710e-04	no	0	1
0.05	1200	124	off_support	1000	0.007541942	0.008044298	0.01262961	7.113874e-05	1.169577e-04	0.001005523	no	0	1
0.05	1200	125	on_support	1200	0.004571642	0.005419907	0.00871959	4.314006e-05	7.867237e-05	6.510191e-04	no	0	1
0.05	1200	125	off_support	1000	0.005629439	0.006288444	0.01098364	5.466448e-05	9.661388e-05	8.359524e-04	no	0	1
0.05	2400	123	on_support	2400	0.007335374	0.007629057	0.009054397	6.952727e-05	1.125644e-04	7.995378e-04	no	0	1
0.05	2400	123	off_support	1000	0.008008034	0.008217393	0.006989033	7.053774e-05	1.086288e-04	5.731910e-04	no	0	1
0.05	2400	124	on_support	2400	0.004610731	0.005329478	0.009565732	4.285820e-05	7.719397e-05	7.991116e-04	no	0	1
0.05	2400	124	off_support	1000	0.005988653	0.008462112	0.0320238	5.666529e-05	1.300026e-04	0.003002814	no	0	1
0.05	2400	125	on_support	2400	0.00963106	0.009792824	0.01001909	9.179443e-05	1.458848e-04	8.809244e-04	no	0	1
0.05	2400	125	off_support	1000	0.01014542	0.01002593	0.01204717	9.067334e-05	1.421247e-04	8.611827e-04	no	0	1
0.05	300	123	on_support	300	0.002663867	0.003386343	0.005732291	2.331656e-05	4.498665e-05	2.788071e-04	no	0	1
0.05	300	123	off_support	1000	0.004807512	0.005382826	0.006099019	4.434831e-05	7.647127e-05	4.882757e-04	no	0	1
0.05	300	124	on_support	300	1.649389e-04	2.377362e-04	3.470750e-04	1.592298e-06	3.528578e-06	2.577171e-05	no	0	1
0.05	300	124	off_support	1000	0.00814153	0.01498683	0.03955623	7.397398e-05	2.081441e-04	0.003556781	no	0	1
0.05	300	125	on_support	300	2.143183e-04	2.801260e-04	3.605314e-04	2.170236e-06	4.349855e-06	2.723316e-05	no	0	1
0.05	300	125	off_support	1000	8.137621e-04	9.634440e-04	0.001631561	7.942528e-06	1.445666e-05	1.159015e-04	no	0	1
0.05	600	123	on_support	600	0.003738305	0.004954118	0.008509782	3.664122e-05	7.444423e-05	6.972263e-04	no	0	1
0.05	600	123	off_support	1000	0.005633914	0.007912708	0.01963205	5.375528e-05	1.143697e-04	0.001370869	no	0	1
0.05	600	124	on_support	600	0.01167381	0.01325484	0.01350935	1.064947e-04	1.874892e-04	9.982148e-04	no	0	1
0.05	600	124	off_support	1000	0.01365663	0.01531552	0.01470487	1.238311e-04	2.150698e-04	0.001221011	no	0	1
0.05	600	125	on_support	600	1.758445e-04	2.201651e-04	2.853613e-04	1.711624e-06	3.340103e-06	2.473054e-05	no	0	1
0.05	600	125	off_support	1000	0.003076129	0.00497573	0.01139755	2.922033e-05	7.263704e-05	9.051799e-04	no	0	1
1.000000e-06	1200	123	on_support	1200	0.008934996	0.009694503	0.009231523	8.114525e-05	1.400856e-04	8.077381e-04	no	0	1

ℓ_2	N	Seed	Query	n_q	E_1	E_2	E_∞	MAE	RMSE	L^∞	floor used?	jitter	tries
1.000000e-06	1200	123	off_support	1000	0.009827373	0.01062545	0.01280367	9.744419e-05	1.711188e-04	0.001120837	no	0	1
1.000000e-06	1200	124	on_support	1200	0.006903103	0.007837288	0.009240737	6.154386e-05	1.067261e-04	7.093163e-04	no	0	1
1.000000e-06	1200	124	off_support	1000	0.007603476	0.008275861	0.01301286	7.171916e-05	1.203244e-04	0.001036036	no	0	1
1.000000e-06	1200	125	on_support	1200	0.004540017	0.005377025	0.00881902	4.284164e-05	7.804992e-05	6.584427e-04	no	0	1
1.000000e-06	1200	125	off_support	1000	0.005552818	0.006258444	0.01092995	5.392045e-05	9.615297e-05	8.318661e-04	no	0	1
1.000000e-06	2400	123	on_support	2400	0.007283463	0.007560595	0.008790454	6.903524e-05	1.115542e-04	7.762305e-04	no	0	1
1.000000e-06	2400	123	off_support	1000	0.008002274	0.008224017	0.007244965	7.048701e-05	1.087164e-04	5.941807e-04	no	0	1
1.000000e-06	2400	124	on_support	2400	0.004617563	0.005409604	0.009857378	4.292170e-05	7.835455e-05	8.234755e-04	no	0	1
1.000000e-06	2400	124	off_support	1000	0.00609683	0.008741336	0.03313879	5.768888e-05	1.342922e-04	0.003107365	no	0	1
1.000000e-06	2400	125	on_support	2400	0.007576444	0.007545162	0.007635714	7.221172e-05	1.124011e-04	6.713670e-04	no	0	1
1.000000e-06	2400	125	off_support	1000	0.008308514	0.007921222	0.007154848	7.425624e-05	1.122890e-04	5.114588e-04	no	0	1
1.000000e-06	300	123	on_support	300	0.002658848	0.003376643	0.005700586	2.327263e-05	4.485779e-05	2.772650e-04	no	0	1
1.000000e-06	300	123	off_support	1000	0.004825317	0.005408517	0.006220588	4.451256e-05	7.683624e-05	4.980084e-04	no	0	1
1.000000e-06	300	124	on_support	300	1.635591e-04	2.323356e-04	3.187534e-04	1.578978e-06	3.448420e-06	2.366871e-05	no	0	1
1.000000e-06	300	124	off_support	1000	0.007833731	0.0144298	0.03715744	7.117731e-05	2.004078e-04	0.003341089	no	0	1
1.000000e-06	300	125	on_support	300	1.887309e-04	2.291136e-04	2.520941e-04	1.911132e-06	3.557725e-06	1.904222e-05	no	0	1
1.000000e-06	300	125	off_support	1000	8.680627e-04	0.001118876	0.002346343	8.472516e-06	1.678895e-05	1.666775e-04	no	0	1
1.000000e-06	600	123	on_support	600	0.003725047	0.004942137	0.008441224	3.651127e-05	7.426420e-05	6.916092e-04	no	0	1
1.000000e-06	600	123	off_support	1000	0.005636133	0.007921101	0.01964831	5.377645e-05	1.144910e-04	0.001372005	no	0	1
1.000000e-06	600	124	on_support	600	0.01178702	0.01341715	0.01370532	1.075274e-04	1.897851e-04	0.001012695	no	0	1
1.000000e-06	600	124	off_support	1000	0.0137646	0.01548151	0.01505052	1.248101e-04	2.174007e-04	0.001249712	no	0	1
1.000000e-06	600	125	on_support	600	1.756857e-04	2.198790e-04	2.855745e-04	1.710078e-06	3.335762e-06	2.474902e-05	no	0	1
1.000000e-06	600	125	off_support	1000	0.003087989	0.004989346	0.01140511	2.933299e-05	7.283582e-05	9.057805e-04	no	0	1

The on-support set contains all N training points; the independent off-support set contains 1000 points. Relative and absolute errors are printed. Production and manufactured regularizations are not conflated.

Table G.2: Manufactured-operator complete paired policy table: Gradient.

ℓ_2	N	Seed	Query	n_q	E_1	E_2	E_∞	MAE	RMSE	L^∞ floor used?	jitter	tries
0.01	1200	123	on_support	1200	0.01381068	0.01273613	0.0122272	1.407768e-04	2.485415e-04	0.002130649	no	0	1
0.01	1200	123	off_support	1000	0.01470273	0.01358334	0.01198748	1.506558e-04	2.663381e-04	0.002076504	no	0	1
0.01	1200	124	on_support	1200	0.01123255	0.0101648	0.01195415	1.165605e-04	2.045832e-04	0.002047084	no	0	1
0.01	1200	124	off_support	1000	0.0118276	0.01091929	0.01836483	1.275145e-04	2.233860e-04	0.003080516	no	0	1
0.01	1200	125	on_support	1200	0.007924313	0.007458699	0.01138812	8.630781e-05	1.557748e-04	0.001901368	no	0	1
0.01	1200	125	off_support	1000	0.008968954	0.008309177	0.01094293	9.594568e-05	1.700914e-04	0.00197109	no	0	1
0.01	2400	123	on_support	2400	0.01264406	0.0111974	0.01125544	1.309476e-04	2.207346e-04	0.002088465	no	0	1
0.01	2400	123	off_support	1000	0.01326216	0.01177186	0.01406341	1.332107e-04	2.210174e-04	0.002164787	no	0	1
0.01	2400	124	on_support	2400	0.01000832	0.009599094	0.01881739	1.048466e-04	1.904054e-04	0.003410278	no	0	1
0.01	2400	124	off_support	1000	0.01122594	0.01074639	0.01441494	1.140728e-04	2.080886e-04	0.002709288	no	0	1
0.01	2400	125	on_support	2400	0.01467655	0.01298208	0.01401147	1.534093e-04	2.566053e-04	0.00258684	no	0	1
0.01	2400	125	off_support	1000	0.01532929	0.01345802	0.0135072	1.553446e-04	2.611933e-04	0.002366947	no	0	1
0.01	300	123	on_support	300	0.006114691	0.005912716	0.006971883	5.972941e-05	1.102173e-04	0.001201622	no	0	1
0.01	300	123	off_support	1000	0.007080949	0.007364525	0.0229306	7.344907e-05	1.443469e-04	0.003663142	no	0	1
0.01	300	124	on_support	300	0.01201572	0.01748175	0.02723811	1.251458e-04	3.336215e-04	0.004088587	no	0	1
0.01	300	124	off_support	1000	0.01474146	0.02353975	0.07064009	1.538957e-04	4.670824e-04	0.01299518	no	0	1
0.01	300	125	on_support	300	0.001023041	0.001368993	0.004254468	1.131839e-05	2.840791e-05	7.471517e-04	no	0	1
0.01	300	125	off_support	1000	0.001420551	0.001794483	0.005640735	1.547731e-05	3.669910e-05	9.016302e-04	no	0	1
0.01	600	123	on_support	600	0.008205481	0.008974088	0.01530337	8.320891e-05	1.688016e-04	0.002577528	no	0	1
0.01	600	123	off_support	1000	0.008947166	0.009074606	0.0131523	9.504409e-05	1.852388e-04	0.002257817	no	0	1
0.01	600	124	on_support	600	0.01877588	0.01840655	0.01798034	1.944023e-04	3.560110e-04	0.002821162	no	0	1
0.01	600	124	off_support	1000	0.01926544	0.01815478	0.01864852	2.041025e-04	3.664082e-04	0.003013246	no	0	1
0.01	600	125	on_support	600	0.003845074	0.007032815	0.02653067	4.167845e-05	1.463209e-04	0.004890279	no	0	1
0.01	600	125	off_support	1000	0.005309437	0.00897138	0.03838834	5.543092e-05	1.758487e-04	0.006496764	no	0	1
0.05	1200	123	on_support	1200	0.01388032	0.01278612	0.01295094	1.414867e-04	2.495170e-04	0.002256765	no	0	1
0.05	1200	123	off_support	1000	0.01480934	0.01366713	0.01219024	1.517481e-04	2.679811e-04	0.002111628	no	0	1
0.05	1200	124	on_support	1200	0.01123848	0.01020291	0.01167045	1.166220e-04	2.053502e-04	0.001998502	no	0	1
0.05	1200	124	off_support	1000	0.01181162	0.01085343	0.0147539	1.273422e-04	2.220387e-04	0.002474818	no	0	1
0.05	1200	125	on_support	1200	0.008060593	0.007534976	0.01156534	8.779211e-05	1.573678e-04	0.001930958	no	0	1
0.05	1200	125	off_support	1000	0.009101206	0.008414441	0.01123765	9.736045e-05	1.722461e-04	0.002024176	no	0	1
0.05	2400	123	on_support	2400	0.01284314	0.01141505	0.01125313	1.330094e-04	2.250250e-04	0.002088038	no	0	1
0.05	2400	123	off_support	1000	0.01347736	0.01199165	0.01504708	1.353722e-04	2.251441e-04	0.002316205	no	0	1
0.05	2400	124	on_support	2400	0.009962957	0.009462773	0.01803145	1.043714e-04	1.877014e-04	0.003267841	no	0	1
0.05	2400	124	off_support	1000	0.01123927	0.01084602	0.02018133	1.142083e-04	2.100178e-04	0.00379308	no	0	1
0.05	2400	125	on_support	2400	0.01464575	0.01297376	0.01387009	1.530874e-04	2.564409e-04	0.002560739	no	0	1
0.05	2400	125	off_support	1000	0.01530645	0.01345217	0.01378637	1.551131e-04	2.610796e-04	0.002415867	no	0	1
0.05	300	123	on_support	300	0.00608848	0.005896822	0.006924985	5.947337e-05	1.099210e-04	0.001193538	no	0	1
0.05	300	123	off_support	1000	0.007031343	0.007278967	0.02220686	7.293451e-05	1.426700e-04	0.003547525	no	0	1
0.05	300	124	on_support	300	0.01101008	0.01626866	0.02762015	1.146720e-04	3.104708e-04	0.004145933	no	0	1
0.05	300	124	off_support	1000	0.0143715	0.02054293	0.0523268	1.500334e-04	4.076187e-04	0.009626205	no	0	1
0.05	300	125	on_support	300	0.001049353	0.001596514	0.005323761	1.160948e-05	3.312917e-05	9.349363e-04	no	0	1
0.05	300	125	off_support	1000	0.001435444	0.001663012	0.00500766	1.563957e-05	3.401037e-05	8.004378e-04	no	0	1
0.05	600	123	on_support	600	0.008121489	0.008837436	0.01483058	8.235717e-05	1.662312e-04	0.002497896	no	0	1
0.05	600	123	off_support	1000	0.00884429	0.008950677	0.01261426	9.395126e-05	1.827091e-04	0.002165453	no	0	1
0.05	600	124	on_support	600	0.01886662	0.01845761	0.0176037	1.953418e-04	3.569986e-04	0.002762065	no	0	1
0.05	600	124	off_support	1000	0.01951742	0.01847624	0.02906646	2.067720e-04	3.728960e-04	0.004696587	no	0	1
0.05	600	125	on_support	600	0.00416647	0.007467155	0.0267802	4.516221e-05	1.553575e-04	0.004936273	no	0	1
0.05	600	125	off_support	1000	0.005893704	0.01000346	0.04431314	6.153071e-05	1.960785e-04	0.007499465	no	0	1
1.000000e-06	1200	123	on_support	1200	0.01385778	0.01273317	0.01290138	1.412569e-04	2.484837e-04	0.002248129	no	0	1

ℓ_2	N	Seed	Query	n_q	E_1	E_2	E_∞	MAE	RMSE	L^∞	floor used?	jitter	tries
1.000000e-06	1200	123	off_support	1000	0.01477774	0.01360598	0.0121422	1.514243e-04	2.667820e-04	0.002103304	no	0	1
1.000000e-06	1200	124	on_support	1200	0.01136736	0.01033528	0.01225203	1.179594e-04	2.080143e-04	0.002098094	no	0	1
1.000000e-06	1200	124	off_support	1000	0.01193325	0.01127024	0.02445873	1.286534e-04	2.305656e-04	0.004102706	no	0	1
1.000000e-06	1200	125	on_support	1200	0.007957936	0.007457843	0.01172543	8.667401e-05	1.557569e-04	0.001957686	no	0	1
1.000000e-06	1200	125	off_support	1000	0.008952983	0.008314107	0.01164026	9.577484e-05	1.701923e-04	0.002096695	no	0	1
1.000000e-06	2400	123	on_support	2400	0.01286352	0.01138457	0.01122837	1.332205e-04	2.244242e-04	0.002083443	no	0	1
1.000000e-06	2400	123	off_support	1000	0.01355893	0.01202123	0.01665663	1.361916e-04	2.256995e-04	0.002563963	no	0	1
1.000000e-06	2400	124	on_support	2400	0.01006474	0.009639122	0.01870677	1.054377e-04	1.911994e-04	0.003390229	no	0	1
1.000000e-06	2400	124	off_support	1000	0.01142837	0.01109172	0.01899861	1.161299e-04	2.147754e-04	0.003570789	no	0	1
1.000000e-06	2400	125	on_support	2400	0.01235181	0.01060779	0.01082705	1.291095e-04	2.096748e-04	0.001998923	no	0	1
1.000000e-06	2400	125	off_support	1000	0.01287154	0.01089081	0.01167016	1.304381e-04	2.113688e-04	0.002045032	no	0	1
1.000000e-06	300	123	on_support	300	0.006107693	0.005909709	0.006959571	5.966106e-05	1.101613e-04	0.0011995	no	0	1
1.000000e-06	300	123	off_support	1000	0.007065454	0.007338592	0.02272475	7.328833e-05	1.438387e-04	0.003630258	no	0	1
1.000000e-06	300	124	on_support	300	0.01099517	0.01637672	0.02761677	1.145166e-04	3.125332e-04	0.004145426	no	0	1
1.000000e-06	300	124	off_support	1000	0.01394011	0.02003828	0.04965702	1.455299e-04	3.976052e-04	0.009135066	no	0	1
1.000000e-06	300	125	on_support	300	0.001064067	0.00147219	0.004889944	1.177228e-05	3.054934e-05	8.587512e-04	no	0	1
1.000000e-06	300	125	off_support	1000	0.001570009	0.002316406	0.009780247	1.710569e-05	4.737299e-05	0.001563301	no	0	1
1.000000e-06	600	123	on_support	600	0.008120539	0.008838295	0.01470354	8.234753e-05	1.662473e-04	0.002476498	no	0	1
1.000000e-06	600	123	off_support	1000	0.0088573	0.00896935	0.01264399	9.408947e-05	1.830902e-04	0.002170557	no	0	1
1.000000e-06	600	124	on_support	600	0.01904107	0.01864293	0.01807034	1.971479e-04	3.605830e-04	0.002835282	no	0	1
1.000000e-06	600	124	off_support	1000	0.01966909	0.01859996	0.02669076	2.083788e-04	3.753929e-04	0.00431272	no	0	1
1.000000e-06	600	125	on_support	600	0.004182154	0.007494829	0.02681367	4.533221e-05	1.559333e-04	0.004942443	no	0	1
1.000000e-06	600	125	off_support	1000	0.005921925	0.01004232	0.04444503	6.182534e-05	1.968402e-04	0.007521785	no	0	1

The on-support set contains all N training points; the independent off-support set contains 1000 points. Relative and absolute errors are printed. Production and manufactured regularizations are not conflated.

Table G.3: Manufactured-operator complete paired policy table: $Q[\rho]$.

ℓ_2	N	Seed	Query	n_q	E_1	E_2	E_∞	MAE	RMSE	L^∞	floor used?	jitter	tries
0.01	1200	123	on_support	1200	0.02495263	0.02293594	0.02343588	2.210497e-06	3.775668e-06	2.376403e-05	no	0	1
0.01	1200	123	off_support	1000	0.02527112	0.02552475	0.02812658	2.223905e-06	4.045188e-06	2.594301e-05	no	0	1
0.01	1200	124	on_support	1200	0.02384971	0.02382343	0.03058065	1.950988e-06	3.389745e-06	2.817423e-05	no	0	1
0.01	1200	124	off_support	1000	0.02377306	0.02340322	0.02595914	2.122700e-06	3.714951e-06	2.545415e-05	no	0	1
0.01	1200	125	on_support	1200	0.01767731	0.01750465	0.02806858	1.586907e-06	2.752048e-06	2.738945e-05	no	0	1
0.01	1200	125	off_support	1000	0.0194965	0.02102074	0.02669804	1.717250e-06	3.294729e-06	2.683047e-05	no	0	1
0.01	2400	123	on_support	2400	0.02572576	0.02418079	0.03154029	2.383925e-06	4.175206e-06	3.370155e-05	no	0	1
0.01	2400	123	off_support	1000	0.02465045	0.02355854	0.0307292	2.088560e-06	3.606000e-06	2.964522e-05	no	0	1
0.01	2400	124	on_support	2400	0.02277878	0.0203914	0.02178752	2.027536e-06	3.239126e-06	2.286872e-05	no	0	1
0.01	2400	124	off_support	1000	0.02190434	0.01877108	0.02594452	2.003904e-06	3.174270e-06	3.135960e-05	no	0	1
0.01	2400	125	on_support	2400	0.02314119	0.02148104	0.03038687	1.987548e-06	3.396954e-06	3.492927e-05	no	0	1
0.01	2400	125	off_support	1000	0.02278435	0.02221772	0.02920168	2.054149e-06	3.765528e-06	3.460748e-05	no	0	1
0.01	300	123	on_support	300	0.01332952	0.01290772	0.01330627	1.276266e-06	2.284761e-06	1.578742e-05	no	0	1
0.01	300	123	off_support	1000	0.01580796	0.01737402	0.03428965	1.406399e-06	2.773467e-06	3.418030e-05	no	0	1
0.01	300	124	on_support	300	0.03795065	0.05103838	0.07264642	4.186895e-06	9.948236e-06	7.110053e-05	no	0	1
0.01	300	124	off_support	1000	0.03990524	0.07617851	0.3025563	3.609563e-06	1.206463e-05	2.834295e-04	no	0	1
0.01	300	125	on_support	300	0.00343102	0.003587545	0.005178028	3.030703e-07	5.463187e-07	4.128140e-06	no	0	1
0.01	300	125	off_support	1000	0.003934398	0.004737009	0.008884326	3.421294e-07	7.537821e-07	8.676272e-06	no	0	1
0.01	600	123	on_support	600	0.0162425	0.01723954	0.02080866	1.597981e-06	3.150687e-06	2.318862e-05	no	0	1
0.01	600	123	off_support	1000	0.01869218	0.02033728	0.03539201	1.750126e-06	3.494492e-06	3.883303e-05	no	0	1
0.01	600	124	on_support	600	0.04056336	0.04229642	0.03666428	3.866622e-06	7.350711e-06	4.034005e-05	no	0	1
0.01	600	124	off_support	1000	0.04307655	0.04581997	0.04512252	3.872179e-06	7.834486e-06	5.205958e-05	no	0	1
0.01	600	125	on_support	600	0.0234627	0.03179386	0.07934014	2.192122e-06	5.140769e-06	6.570019e-05	no	0	1
0.01	600	125	off_support	1000	0.02377596	0.02429957	0.03914043	2.073783e-06	3.928177e-06	3.727809e-05	no	0	1
0.05	1200	123	on_support	1200	0.02524879	0.02282745	0.02232649	2.236733e-06	3.757809e-06	2.263911e-05	no	0	1
0.05	1200	123	off_support	1000	0.02574858	0.02571546	0.02642355	2.265923e-06	4.075412e-06	2.437219e-05	no	0	1
0.05	1200	124	on_support	1200	0.02391498	0.02427738	0.03381528	1.956327e-06	3.454335e-06	3.115432e-05	no	0	1
0.05	1200	124	off_support	1000	0.02376191	0.02372259	0.02645657	2.121704e-06	3.765646e-06	2.594191e-05	no	0	1
0.05	1200	125	on_support	1200	0.01840563	0.01851023	0.03187988	1.652289e-06	2.910144e-06	3.110853e-05	no	0	1
0.05	1200	125	off_support	1000	0.02001603	0.0220745	0.02867432	1.763010e-06	3.459891e-06	2.881655e-05	no	0	1
0.05	2400	123	on_support	2400	0.0260881	0.02441166	0.03113746	2.417502e-06	4.215070e-06	3.327112e-05	no	0	1
0.05	2400	123	off_support	1000	0.02539998	0.02411656	0.03109334	2.152066e-06	3.691412e-06	2.999651e-05	no	0	1
0.05	2400	124	on_support	2400	0.02243042	0.02001788	0.02101983	1.996529e-06	3.179793e-06	2.206293e-05	no	0	1
0.05	2400	124	off_support	1000	0.02163669	0.01846941	0.02473631	1.979418e-06	3.123257e-06	2.989922e-05	no	0	1
0.05	2400	125	on_support	2400	0.02305507	0.02141735	0.03044307	1.980152e-06	3.386882e-06	3.499387e-05	no	0	1
0.05	2400	125	off_support	1000	0.02274677	0.02227933	0.02963531	2.050761e-06	3.775970e-06	3.512139e-05	no	0	1
0.05	300	123	on_support	300	0.01323272	0.01283018	0.01322445	1.266998e-06	2.271035e-06	1.569034e-05	no	0	1
0.05	300	123	off_support	1000	0.01558751	0.01702482	0.03342239	1.386787e-06	2.717723e-06	3.331580e-05	no	0	1
0.05	300	124	on_support	300	0.03456274	0.04002157	0.06796313	3.813125e-06	7.800875e-06	6.651689e-05	no	0	1
0.05	300	124	off_support	1000	0.04255093	0.05487291	0.1542606	3.848874e-06	8.690396e-06	1.445087e-04	no	0	1
0.05	300	125	on_support	300	0.003575129	0.003953276	0.005340542	3.157998e-07	6.020130e-07	4.257703e-06	no	0	1
0.05	300	125	off_support	1000	0.004225826	0.004868325	0.00860964	3.674717e-07	7.746779e-07	8.408018e-06	no	0	1
0.05	600	123	on_support	600	0.01583588	0.0168725	0.02103765	1.557976e-06	3.083605e-06	2.344380e-05	no	0	1
0.05	600	123	off_support	1000	0.01828375	0.02022207	0.03681177	1.711886e-06	3.474697e-06	4.039082e-05	no	0	1
0.05	600	124	on_support	600	0.041708	0.04322571	0.03950195	3.975733e-06	7.512213e-06	4.346221e-05	no	0	1
0.05	600	124	off_support	1000	0.04437617	0.04651714	0.04398148	3.989003e-06	7.953692e-06	5.074311e-05	no	0	1
0.05	600	125	on_support	600	0.02350933	0.03185592	0.07694887	2.196478e-06	5.150803e-06	6.372002e-05	no	0	1
0.05	600	125	off_support	1000	0.02485847	0.0253088	0.03905059	2.168202e-06	4.091325e-06	3.719253e-05	no	0	1
1.000000e-06	1200	123	on_support	1200	0.02524368	0.02287573	0.02224125	2.236280e-06	3.765757e-06	2.255267e-05	no	0	1

ℓ_2	N	Seed	Query	n_q	E_1	E_2	E_∞	MAE	RMSE	L^∞	floor used?	jitter	tries
1.000000e-06	1200	123	off_support	1000	0.02566031	0.02565465	0.02640647	2.258154e-06	4.065774e-06	2.435643e-05	no	0	1
1.000000e-06	1200	124	on_support	1200	0.02445526	0.02457996	0.03131177	2.000524e-06	3.497388e-06	2.884782e-05	no	0	1
1.000000e-06	1200	124	off_support	1000	0.02434324	0.02390678	0.02622474	2.173610e-06	3.794885e-06	2.571458e-05	no	0	1
1.000000e-06	1200	125	on_support	1200	0.01782805	0.01783692	0.03088781	1.600439e-06	2.804287e-06	3.014047e-05	no	0	1
1.000000e-06	1200	125	off_support	1000	0.01951356	0.02124538	0.02746619	1.718752e-06	3.329938e-06	2.760243e-05	no	0	1
1.000000e-06	2400	123	on_support	2400	0.02641738	0.02456551	0.03053422	2.448015e-06	4.241634e-06	3.262654e-05	no	0	1
1.000000e-06	2400	123	off_support	1000	0.02585283	0.02432291	0.03083139	2.190434e-06	3.722998e-06	2.974381e-05	no	0	1
1.000000e-06	2400	124	on_support	2400	0.02317091	0.02085214	0.02148028	2.062441e-06	3.312313e-06	2.254623e-05	no	0	1
1.000000e-06	2400	124	off_support	1000	0.02232617	0.01914189	0.02453328	2.042495e-06	3.236975e-06	2.965381e-05	no	0	1
1.000000e-06	2400	125	on_support	2400	0.02280898	0.02056025	0.02462836	1.959015e-06	3.251343e-06	2.830994e-05	no	0	1
1.000000e-06	2400	125	off_support	1000	0.02192797	0.02031202	0.02577078	1.976941e-06	3.442544e-06	3.054146e-05	no	0	1
1.000000e-06	300	123	on_support	300	0.0133011	0.01289377	0.01329501	1.273546e-06	2.282292e-06	1.577406e-05	no	0	1
1.000000e-06	300	123	off_support	1000	0.01573514	0.01726174	0.03406999	1.399921e-06	2.755543e-06	3.396134e-05	no	0	1
1.000000e-06	300	124	on_support	300	0.03406622	0.03961109	0.07220197	3.758346e-06	7.720865e-06	7.066554e-05	no	0	1
1.000000e-06	300	124	off_support	1000	0.04158897	0.05324815	0.1446134	3.761862e-06	8.433077e-06	1.354713e-04	no	0	1
1.000000e-06	300	125	on_support	300	0.003704891	0.004049748	0.005704196	3.272620e-07	6.167038e-07	4.547623e-06	no	0	1
1.000000e-06	300	125	off_support	1000	0.004551804	0.00617556	0.01429375	3.958182e-07	9.826933e-07	1.395902e-05	no	0	1
1.000000e-06	600	123	on_support	600	0.01579231	0.01685731	0.02101144	1.553690e-06	3.080830e-06	2.341459e-05	no	0	1
1.000000e-06	600	123	off_support	1000	0.01829055	0.020326	0.03730445	1.712522e-06	3.492555e-06	4.093140e-05	no	0	1
1.000000e-06	600	124	on_support	600	0.04133859	0.04296094	0.03815824	3.940520e-06	7.466198e-06	4.198378e-05	no	0	1
1.000000e-06	600	124	off_support	1000	0.04406058	0.04650925	0.04511785	3.960634e-06	7.952343e-06	5.205419e-05	no	0	1
1.000000e-06	600	125	on_support	600	0.02349862	0.03186765	0.07749029	2.195478e-06	5.152700e-06	6.416836e-05	no	0	1
1.000000e-06	600	125	off_support	1000	0.02491152	0.02536426	0.03900783	2.172829e-06	4.100291e-06	3.715180e-05	no	0	1

The on-support set contains all N training points; the independent off-support set contains 1000 points. Relative and absolute errors are printed. Production and manufactured regularizations are not conflated.

G.2 Run-by-run time-step refinement

For each method, initial momentum, seed, and observable, the following table lists the three endpoint values, the two time-normalized differences, the independent-seed spread, the roundoff threshold, and the resulting identifiability verdict. Thus statements comparing step-size effects with seed variation do not rely on an unreported aggregate.

Table G.4: Run-by-run three-level time-step evidence with reconstructible endpoint values.

Method	P_{init}	Seed	Observable	h	$h/2$	$h/4$	$O_h(T)$	$O_{h/2}(T)$	$O_{h/4}(T)$	D_{12}	D_{23}	S_{seed}	τ_{noise}	p_{obs}	Verdict
PBME	20	11	ρ_{11}^{SN}	0.5	0.25	0.125	0.9560133	0.9560125	0.9560123	5.737737e-07	1.434436e-07	0.01123914	1.914096e-12	Not Computed	seed-variation reject
PBME	20	11	ρ_{22}^{SN}	0.5	0.25	0.125	0.04398666	0.04398752	0.04398774	5.737739e-07	1.434441e-07	0.01123914	1.202033e-12	Not Computed	seed-variation reject
PBME	20	11	trace	0.5	0.25	0.125	1	1	1	2.892425e-13	5.879888e-13	9.401662e-15	2.000000e-12	Not Computed	noise-floor reject
PBME	20	11	energy	0.5	0.25	0.125	0.1000455	0.1000455	0.1000455	4.831945e-09	1.207994e-09	8.363766e-05	1.100045e-12	Not Computed	seed-variation reject
PBME	20	11	R_mean	0.5	0.25	0.125	15.32609	15.32609	15.32609	1.236082e-06	3.090220e-07	0.02884268	9.823462e-12	Not Computed	seed-variation reject
PBME	20	11	P_mean	0.5	0.25	0.125	19.74085	19.74085	19.74084	2.885997e-06	7.215031e-07	0.05314185	2.121967e-11	Not Computed	seed-variation reject
PBME	20	11	R_var	0.5	0.25	0.125	2.40049	2.400492	2.400493	5.895229e-07	1.473914e-07	0.09507974	2.422371e-12	Not Computed	seed-variation reject
PBME	20	11	P_var	0.5	0.25	0.125	1.682751	1.682754	1.682755	3.241086e-06	8.102896e-07	0.09854947	2.783504e-12	Not Computed	seed-variation reject
PBME	20	29	ρ_{11}^{SN}	0.5	0.25	0.125	0.9623363	0.9623352	0.9623349	7.656746e-07	1.914187e-07	0.01151816	1.916744e-12	Not Computed	seed-variation reject
PBME	20	29	ρ_{22}^{SN}	0.5	0.25	0.125	0.03766369	0.03766484	0.03766513	7.656748e-07	1.914191e-07	0.01151816	1.203135e-12	Not Computed	seed-variation reject
PBME	20	29	trace	0.5	0.25	0.125	1	1	1	2.914842e-13	5.902366e-13	8.194463e-15	2.000000e-12	Not Computed	noise-floor reject
PBME	20	29	energy	0.5	0.25	0.125	0.1000009	0.1000009	0.1000009	5.058662e-09	1.264673e-09	1.068481e-04	1.100001e-12	Not Computed	seed-variation reject
PBME	20	29	R_mean	0.5	0.25	0.125	15.31321	15.31321	15.31321	1.594195e-06	3.985501e-07	0.0310464	9.829251e-12	Not Computed	seed-variation reject
PBME	20	29	P_mean	0.5	0.25	0.125	19.76514	19.76514	19.76514	3.818329e-06	9.545855e-07	0.03912341	2.123315e-11	Not Computed	seed-variation reject
PBME	20	29	R_var	0.5	0.25	0.125	2.483692	2.483694	2.483695	6.493605e-07	1.623497e-07	0.1111255	2.438778e-12	Not Computed	seed-variation reject
PBME	20	29	P_var	0.5	0.25	0.125	1.810827	1.810831	1.810832	3.421510e-06	8.554060e-07	0.1478631	2.865613e-12	Not Computed	seed-variation reject
PBME	20	47	ρ_{11}^{SN}	0.5	0.25	0.125	0.9749687	0.9749676	0.9749673	7.489555e-07	1.872392e-07	0.01197656	1.923031e-12	Not Computed	seed-variation reject
PBME	20	47	ρ_{22}^{SN}	0.5	0.25	0.125	0.02503132	0.02503245	0.02503273	7.489557e-07	1.872397e-07	0.01197656	1.200836e-12	Not Computed	seed-variation reject
PBME	20	47	trace	0.5	0.25	0.125	1	1	1	2.899230e-13	5.966728e-13	1.205172e-14	2.000000e-12	Not Computed	noise-floor reject
PBME	20	47	energy	0.5	0.25	0.125	0.10018	0.10018	0.10018	4.893639e-09	1.223418e-09	1.604910e-04	1.100180e-12	Not Computed	seed-variation reject
PBME	20	47	R_mean	0.5	0.25	0.125	15.35898	15.35898	15.35898	1.570382e-06	3.925975e-07	0.03474846	9.845722e-12	Not Computed	seed-variation reject
PBME	20	47	P_mean	0.5	0.25	0.125	19.85179	19.85178	19.85178	3.718001e-06	9.295051e-07	0.06335189	2.127512e-11	Not Computed	seed-variation reject
PBME	20	47	R_var	0.5	0.25	0.125	2.222558	2.222559	2.222559	3.297756e-07	8.245081e-08	0.1063266	2.307015e-12	Not Computed	seed-variation reject
PBME	20	47	P_var	0.5	0.25	0.125	1.620485	1.620488	1.620489	2.649810e-06	6.624781e-07	0.1076431	2.750642e-12	Not Computed	seed-variation reject
PBME	20	73	ρ_{11}^{SN}	0.5	0.25	0.125	0.96405	0.9640489	0.9640486	7.183384e-07	1.795847e-07	0.008792682	1.917252e-12	Not Computed	seed-variation reject
PBME	20	73	ρ_{22}^{SN}	0.5	0.25	0.125	0.03595001	0.03595109	0.03595135	7.183386e-07	1.795852e-07	0.008792682	1.203331e-12	Not Computed	seed-variation reject
PBME	20	73	trace	0.5	0.25	0.125	1	1	1	2.885387e-13	5.894481e-13	9.969494e-15	2.000000e-12	Not Computed	noise-floor reject
PBME	20	73	energy	0.5	0.25	0.125	0.1000153	0.1000153	0.1000153	4.941958e-09	1.235497e-09	9.702942e-05	1.100015e-12	Not Computed	seed-variation reject
PBME	20	73	R_mean	0.5	0.25	0.125	15.35666	15.35666	15.35665	1.511223e-06	3.778072e-07	0.0353145	9.829120e-12	Not Computed	seed-variation reject
PBME	20	73	P_mean	0.5	0.25	0.125	19.78235	19.78235	19.78235	3.567320e-06	8.918338e-07	0.03854744	2.123401e-11	Not Computed	seed-variation reject
PBME	20	73	R_var	0.5	0.25	0.125	2.126613	2.126614	2.126615	3.881799e-07	9.705314e-08	0.1078288	2.318511e-12	Not Computed	seed-variation reject
PBME	20	73	P_var	0.5	0.25	0.125	1.529694	1.529697	1.529698	2.670699e-06	6.676969e-07	0.135415	2.732006e-12	Not Computed	seed-variation reject
MIDPOINT	20	11	ρ_{11}^{SN}	0.5	0.25	0.125	1.066514	0.9877542	0.9832016	0.08972373	0.050708	0.2295475	1.971912e-12	Not Computed	seed-variation reject
MIDPOINT	20	11	ρ_{22}^{SN}	0.5	0.25	0.125	-0.06651382	0.01224584	0.01679843	0.08972373	0.050708	0.2295475	1.191940e-12	Not Computed	seed-variation reject
MIDPOINT	20	11	trace	0.5	0.25	0.125	1	1	1	4.666245e-13	1.317919e-12	6.557490e-13	2.000000e-12	Not Computed	noise-floor reject
MIDPOINT	20	11	energy	0.5	0.25	0.125	0.1014607	0.09429698	0.09801849	0.005230109	0.002733587	0.01698184	1.100177e-12	Not Computed	seed-variation reject
MIDPOINT	20	11	R_mean	0.5	0.25	0.125	15.819	15.23198	15.26093	0.2939675	0.1342496	2.148106	9.961444e-12	Not Computed	seed-variation reject
MIDPOINT	20	11	P_mean	0.5	0.25	0.125	20.45372	19.34415	19.65838	0.8908917	0.3437543	2.179107	2.141006e-11	Not Computed	seed-variation reject
MIDPOINT	20	11	R_var	0.5	0.25	0.125	44.82189	0.1936518	33.99422	14.20652	10.89001	24.43683	1.549145e-11	Not Computed	seed-variation reject
MIDPOINT	20	11	P_var	0.5	0.25	0.125	74.39819	-1.111782	54.20189	62.60563	44.03363	103.0828	5.975736e-11	Not Computed	seed-variation reject
MIDPOINT	20	29	ρ_{11}^{SN}	0.5	0.25	0.125	1.542495	0.8546812	0.5608961	0.5964768	0.2158285	0.3096014	2.272185e-12	Not Computed	seed-variation reject
MIDPOINT	20	29	ρ_{22}^{SN}	0.5	0.25	0.125	-0.542495	0.1453188	0.4391039	0.5964768	0.2158285	0.3096014	1.573282e-12	Not Computed	seed-variation reject
MIDPOINT	20	29	trace	0.5	0.25	0.125	1	1	1	5.179210e-13	1.438795e-12	1.145167e-12	2.000000e-12	Not Computed	noise-floor reject
MIDPOINT	20	29	energy	0.5	0.25	0.125	0.1327767	0.08575235	0.06000964	0.0351461	0.01682319	0.03067313	1.116282e-12	Not Computed	seed-variation reject
MIDPOINT	20	29	R_mean	0.5	0.25	0.125	19.45261	12.77719	8.162508	2.904486	2.248585	3.909657	1.107162e-11	Not Computed	seed-variation reject
MIDPOINT	20	29	P_mean	0.5	0.25	0.125	25.77563	17.8768	13.90546	5.251241	2.521823	3.875776	2.407294e-11	Not Computed	seed-variation reject
MIDPOINT	20	29	R_var	0.5	0.25	0.125	-6118.823	37.50319	-0.2730428	1952.901	12.70875	16.84964	1.942527e-09	Not Computed	seed-variation reject
MIDPOINT	20	29	P_var	0.5	0.25	0.125	-1.073411e+04	76.24654	23.93377	7280.647	35.90671	106.8695	7.228135e-09	Not Computed	seed-variation reject
MIDPOINT	20	47	ρ_{11}^{SN}	0.5	0.25	0.125	1.102969	0.9053043	0.782132	0.2882137	0.2712406	0.2447745	1.998661e-12	0.08756597	positive

Method	P_{init}	Seed	Observable	h	$h/2$	$h/4$	$O_h(T)$	$O_{h/2}(T)$	$O_{h/4}(T)$	D_{12}	D_{23}	S_{seed}	τ_{noise}	p_{obs}	Verdict
MIDPOINT	20	47	ρ_{22}^{SN}	0.5	0.25	0.125	-0.1029691	0.0946957	0.217868	0.2882137	0.2712406	0.2447745	1.378130e-12	0.08756597	positive
MIDPOINT	20	47	trace	0.5	0.25	0.125	1	1	1	9.002152e-13	1.278038e-12	8.659196e-13	2.000000e-12	Not Computed	noise-floor reject
MIDPOINT	20	47	energy	0.5	0.25	0.125	0.09557975	0.1180203	0.1187165	0.01557715	0.003456418	0.02622004	1.110889e-12	Not Computed	seed-variation reject
MIDPOINT	20	47	R_mean	0.5	0.25	0.125	15.94026	16.73255	17.17815	0.3844073	0.6547282	2.837444	1.058686e-11	Not Computed	seed-variation reject
MIDPOINT	20	47	P_mean	0.5	0.25	0.125	20.08246	21.31836	20.76889	1.218094	0.5841866	2.647474	2.199706e-11	Not Computed	seed-variation reject
MIDPOINT	20	47	R_var	0.5	0.25	0.125	33.33949	-66.19518	-84.40063	33.7071	5.95172	31.16055	2.852353e-11	Not Computed	seed-variation reject
MIDPOINT	20	47	P_var	0.5	0.25	0.125	50.53155	-94.13266	-129.2142	274.7254	177.4102	158.581	2.585982e-10	0.630901	positive
MIDPOINT	20	73	ρ_{11}^{SN}	0.5	0.25	0.125	1.006957	2.501948	0.94479	1.123698	1.150855	0.2151986	2.829080e-12	-0.03445137	zero/negative
MIDPOINT	20	73	ρ_{22}^{SN}	0.5	0.25	0.125	-0.006957091	-1.501948	0.05521001	1.123698	1.150855	0.2151986	2.113145e-12	-0.03445137	zero/negative
MIDPOINT	20	73	trace	0.5	0.25	0.125	1	1	1	3.913268e-12	2.686677e-12	6.636976e-13	2.000000e-12	0.542551	positive
MIDPOINT	20	73	energy	0.5	0.25	0.125	0.09627898	0.2072792	0.09805371	0.070917	0.06972971	0.01700668	1.154894e-12	0.02435808	positive
MIDPOINT	20	73	R_mean	0.5	0.25	0.125	15.25943	12.65081	15.03083	5.917192	5.784864	2.107136	9.803458e-12	0.03262986	positive
MIDPOINT	20	73	P_mean	0.5	0.25	0.125	19.60147	38.27287	19.46996	12.29472	12.37827	2.120072	3.037352e-11	-0.009771275	zero/negative
MIDPOINT	20	73	R_var	0.5	0.25	0.125	-37.46014	8.228257	-1.777438	12.53886	3.966624	17.02977	1.301533e-11	Not Computed	seed-variation reject
MIDPOINT	20	73	P_var	0.5	0.25	0.125	-61.97141	64.50552	-2.40664	136.1013	72.7522	89.76992	8.750683e-11	Not Computed	seed-variation reject
PBME	100	11	ρ_{11}^{SN}	0.5	0.25	0.125	0.1479964	0.1480061	0.1480085	6.853163e-06	1.713318e-06	0.01427847	1.697274e-12	Not Computed	seed-variation reject
PBME	100	11	ρ_{22}^{SN}	0.5	0.25	0.125	0.8520036	0.8519939	0.8519915	6.853163e-06	1.713317e-06	0.01427847	1.574092e-12	Not Computed	seed-variation reject
PBME	100	11	trace	0.5	0.25	0.125	1	1	1	5.547552e-14	1.183024e-13	2.203046e-15	2.000000e-12	Not Computed	noise-floor reject
PBME	100	11	energy	0.5	0.25	0.125	2.499976	2.499976	2.499976	5.995218e-08	1.498741e-08	4.217163e-04	3.499976e-12	Not Computed	seed-variation reject
PBME	100	11	R_mean	0.5	0.25	0.125	14.9137	14.9137	14.9137	4.075924e-07	1.018999e-07	0.03468144	9.634291e-12	Not Computed	seed-variation reject
PBME	100	11	P_mean	0.5	0.25	0.125	99.14157	99.14158	99.14158	6.905585e-06	1.726418e-06	0.016782	1.006587e-10	Not Computed	seed-variation reject
PBME	100	11	R_var	0.5	0.25	0.125	1.017735	1.017735	1.017736	2.138286e-07	5.345746e-08	0.05316163	2.005724e-12	Not Computed	seed-variation reject
PBME	100	11	P_var	0.5	0.25	0.125	0.4534352	0.4534464	0.4534491	7.683113e-06	1.920764e-06	0.01513736	1.389585e-12	Not Computed	seed-variation reject
PBME	100	29	ρ_{11}^{SN}	0.5	0.25	0.125	0.114679	0.1146881	0.1146904	6.565893e-06	1.641507e-06	0.02215294	1.694825e-12	Not Computed	seed-variation reject
PBME	100	29	ρ_{22}^{SN}	0.5	0.25	0.125	0.885321	0.8853119	0.8853096	6.565893e-06	1.641507e-06	0.02215294	1.593133e-12	Not Computed	seed-variation reject
PBME	100	29	trace	0.5	0.25	0.125	1	1	1	5.676560e-14	1.161225e-13	1.720199e-15	2.000000e-12	Not Computed	noise-floor reject
PBME	100	29	energy	0.5	0.25	0.125	2.499747	2.499747	2.499747	5.612978e-08	1.403183e-08	5.420477e-04	3.499747e-12	Not Computed	seed-variation reject
PBME	100	29	R_mean	0.5	0.25	0.125	14.8757	14.87571	14.87571	3.690112e-07	9.225516e-08	0.03255712	9.633030e-12	Not Computed	seed-variation reject
PBME	100	29	P_mean	0.5	0.25	0.125	99.10332	99.10333	99.10333	6.617685e-06	1.654449e-06	0.02662009	1.006413e-10	Not Computed	seed-variation reject
PBME	100	29	R_var	0.5	0.25	0.125	1.015867	1.015867	1.015868	1.602631e-07	4.006593e-08	0.0553631	2.008811e-12	Not Computed	seed-variation reject
PBME	100	29	P_var	0.5	0.25	0.125	0.4548972	0.4549077	0.4549104	7.412807e-06	1.853183e-06	0.0158835	1.396367e-12	Not Computed	seed-variation reject
PBME	100	47	ρ_{11}^{SN}	0.5	0.25	0.125	0.1421697	0.1421792	0.1421816	6.681729e-06	1.670459e-06	0.01285911	1.696563e-12	Not Computed	seed-variation reject
PBME	100	47	ρ_{22}^{SN}	0.5	0.25	0.125	0.8578303	0.8578208	0.8578184	6.681729e-06	1.670459e-06	0.01285911	1.578705e-12	Not Computed	seed-variation reject
PBME	100	47	trace	0.5	0.25	0.125	1	1	1	5.732040e-14	1.155373e-13	2.201188e-15	2.000000e-12	Not Computed	noise-floor reject
PBME	100	47	energy	0.5	0.25	0.125	2.500654	2.500654	2.500654	6.302828e-08	1.575639e-08	8.090533e-04	3.500654e-12	Not Computed	seed-variation reject
PBME	100	47	R_mean	0.5	0.25	0.125	14.8667	14.8667	14.8667	3.913304e-07	9.783438e-08	0.04407002	9.635628e-12	Not Computed	seed-variation reject
PBME	100	47	P_mean	0.5	0.25	0.125	99.14945	99.14946	99.14947	6.821278e-06	1.705340e-06	0.02355754	1.006714e-10	Not Computed	seed-variation reject
PBME	100	47	R_var	0.5	0.25	0.125	0.9232938	0.9232942	0.9232942	1.465423e-07	3.663628e-08	0.08070426	1.918052e-12	Not Computed	seed-variation reject
PBME	100	47	P_var	0.5	0.25	0.125	0.4338927	0.4339032	0.4339058	7.434943e-06	1.858719e-06	0.01494742	1.380701e-12	Not Computed	seed-variation reject
PBME	100	73	ρ_{11}^{SN}	0.5	0.25	0.125	0.1449208	0.1449305	0.144933	6.868726e-06	1.717209e-06	0.01280204	1.697336e-12	Not Computed	seed-variation reject
PBME	100	73	ρ_{22}^{SN}	0.5	0.25	0.125	0.8550792	0.8550695	0.855067	6.868726e-06	1.717209e-06	0.01280204	1.575887e-12	Not Computed	seed-variation reject
PBME	100	73	trace	0.5	0.25	0.125	1	1	1	5.608775e-14	1.148455e-13	2.978372e-15	2.000000e-12	Not Computed	noise-floor reject
PBME	100	73	energy	0.5	0.25	0.125	2.499827	2.499827	2.499827	6.027495e-08	1.506809e-08	4.870509e-04	3.499827e-12	Not Computed	seed-variation reject
PBME	100	73	R_mean	0.5	0.25	0.125	14.92102	14.92102	14.92102	4.072831e-07	1.018226e-07	0.04121457	9.633860e-12	Not Computed	seed-variation reject
PBME	100	73	P_mean	0.5	0.25	0.125	99.13556	99.13556	99.13557	6.933294e-06	1.733346e-06	0.01616718	1.006539e-10	Not Computed	seed-variation reject
PBME	100	73	R_var	0.5	0.25	0.125	0.9849752	0.9849757	0.9849758	2.017499e-07	5.043794e-08	0.04211461	1.978122e-12	Not Computed	seed-variation reject
PBME	100	73	P_var	0.5	0.25	0.125	0.4344405	0.4344509	0.4344534	7.380622e-06	1.845139e-06	0.01439101	1.381677e-12	Not Computed	seed-variation reject
MIDPOINT	100	11	ρ_{11}^{SN}	0.5	0.25	0.125	0.473597	0.4735177	0.4733697	14.16192	14.00279	13.98595	1.497159e-11	0.01630253	positive
MIDPOINT	100	11	ρ_{22}^{SN}	0.5	0.25	0.125	0.526403	0.5264823	0.5266303	14.16192	14.00279	13.98595	1.499124e-11	0.01630253	positive
MIDPOINT	100	11	trace	0.5	0.25	0.125	1	1	1	1.885706e-12	3.411351e-12	4.694354e-11	2.000000e-12	Not Computed	noise-floor reject
MIDPOINT	100	11	energy	0.5	0.25	0.125	2.413369	2.413347	2.413355	1.379421	1.368836	18.15773	3.875263e-12	Not Computed	seed-variation reject
MIDPOINT	100	11	R_mean	0.5	0.25	0.125	13.3778	13.37792	13.37812	35.0243	34.28368	341.3595	3.837367e-11	Not Computed	seed-variation reject

Method	P_{init}	Seed	Observable	h	$h/2$	$h/4$	$O_h(T)$	$O_{h/2}(T)$	$O_{h/4}(T)$	D_{12}	D_{23}	S_{seed}	τ_{noise}	p_{obs}	Verdict
MIDPOINT	100	11	P_mean	0.5	0.25	0.125	97.71719	97.71666	97.71667	41.51945	41.21954	345.2011	1.081727e-10	Not Computed	seed-variation reject
MIDPOINT	100	11	R_var	0.5	0.25	0.125	-7.744563e+30	-4.039919e+25	-2.069069e+25	2.267017e+30	5.769069e+24	4.655793e+45	2.267029e+18	Not Computed	seed-variation reject
MIDPOINT	100	11	P_var	0.5	0.25	0.125	-4.132086e+32	-2.155424e+27	-1.103881e+27	1.728612e+32	4.398829e+26	5.504872e+47	1.728621e+20	Not Computed	seed-variation reject
MIDPOINT	100	29	ρ_{11}^{SN}	0.5	0.25	0.125	-0.2926828	-0.3729309	-0.2880791	2.490533	4.253852	13.81807	5.231763e-12	Not Computed	seed-variation reject
MIDPOINT	100	29	ρ_{22}^{SN}	0.5	0.25	0.125	1.292683	1.372931	1.288079	2.490533	4.253852	13.81807	5.195442e-12	Not Computed	seed-variation reject
MIDPOINT	100	29	trace	0.5	0.25	0.125	1	1	1	2.587670e-13	7.089513e-13	2.715335e-11	2.000000e-12	Not Computed	noise-floor reject
MIDPOINT	100	29	energy	0.5	0.25	0.125	2.440822	2.563988	2.525908	0.430181	0.8896223	10.61481	3.679611e-12	Not Computed	seed-variation reject
MIDPOINT	100	29	R_mean	0.5	0.25	0.125	12.75439	11.45684	11.9006	2.814664	3.376706	197.4005	9.570691e-12	Not Computed	seed-variation reject
MIDPOINT	100	29	P_mean	0.5	0.25	0.125	97.49721	99.91029	99.22498	7.496233	17.23082	201.4809	1.026303e-10	Not Computed	seed-variation reject
MIDPOINT	100	29	R_var	0.5	0.25	0.125	-4.079150e+44	-4.633878e+43	-2.752764e+46	1.165235e+44	8.051140e+45	8.063352e+45	8.064395e+33	Not Computed	seed-variation reject
MIDPOINT	100	29	P_var	0.5	0.25	0.125	-2.383607e+46	-3.523996e+45	-1.913698e+48	1.192361e+46	9.518830e+47	9.533873e+47	9.536047e+35	Not Computed	seed-variation reject
MIDPOINT	100	47	ρ_{11}^{SN}	0.5	0.25	0.125	-1.568477	-1.479928	-1.500285	1.854444	1.487437	10.34699	2.846139e-12	Not Computed	seed-variation reject
MIDPOINT	100	47	ρ_{22}^{SN}	0.5	0.25	0.125	2.568477	2.479928	2.500285	1.854444	1.487437	10.34699	2.968964e-12	Not Computed	seed-variation reject
MIDPOINT	100	47	trace	0.5	0.25	0.125	1	1	1	1.514213e-13	4.659191e-13	2.712639e-11	2.000000e-12	Not Computed	noise-floor reject
MIDPOINT	100	47	energy	0.5	0.25	0.125	2.631183	2.620763	2.626583	0.128914	0.1313983	10.51904	3.551363e-12	Not Computed	seed-variation reject
MIDPOINT	100	47	R_mean	0.5	0.25	0.125	12.60575	12.73747	12.6782	2.128186	2.468764	197.2438	9.011635e-12	Not Computed	seed-variation reject
MIDPOINT	100	47	P_mean	0.5	0.25	0.125	100.0626	99.94141	100.0381	2.956775	2.898389	199.9079	1.011363e-10	Not Computed	seed-variation reject
MIDPOINT	100	47	R_var	0.5	0.25	0.125	-1.533659e+40	-4.487201e+42	-7.384480e+42	1.624195e+42	1.029708e+42	4.654550e+45	2.659376e+30	Not Computed	seed-variation reject
MIDPOINT	100	47	P_var	0.5	0.25	0.125	-9.663501e+41	-2.762485e+44	-4.597646e+44	3.292232e+44	2.100396e+44	5.503413e+47	5.403656e+32	Not Computed	seed-variation reject
MIDPOINT	100	73	ρ_{11}^{SN}	0.5	0.25	0.125	0.5849126	0.588819	0.5934345	4.460185	3.51947	10.076	4.755225e-12	Not Computed	seed-variation reject
MIDPOINT	100	73	ρ_{22}^{SN}	0.5	0.25	0.125	0.4150874	0.411181	0.4065655	4.460185	3.51947	10.076	4.731827e-12	Not Computed	seed-variation reject
MIDPOINT	100	73	trace	0.5	0.25	0.125	1	1	1	7.493326e-13	1.162229e-12	2.713450e-11	2.000000e-12	Not Computed	noise-floor reject
MIDPOINT	100	73	energy	0.5	0.25	0.125	2.517561	2.516863	2.513949	0.8780583	0.497507	10.522	3.665435e-12	Not Computed	seed-variation reject
MIDPOINT	100	73	R_mean	0.5	0.25	0.125	12.4996	12.5096	12.50658	17.84307	9.580874	197.3017	1.990051e-11	Not Computed	seed-variation reject
MIDPOINT	100	73	P_mean	0.5	0.25	0.125	99.93534	99.92516	99.87108	21.00832	10.59983	200.0288	1.035504e-10	Not Computed	seed-variation reject
MIDPOINT	100	73	R_var	0.5	0.25	0.125	-1.328630e+29	-1.349355e+30	-9.892515e+29	4.549185e+29	1.353487e+29	4.655793e+45	5.045855e+17	Not Computed	seed-variation reject
MIDPOINT	100	73	P_var	0.5	0.25	0.125	-8.492784e+30	-8.609717e+31	-6.308254e+31	5.309691e+31	1.583919e+31	5.504872e+47	5.890465e+19	Not Computed	seed-variation reject

N=1000; seeds 11,29,47,73. Differences are time-normalized L2 values on common saved times without extrapolation. The declared floor is tau_noise=1e-12+1e-12 max_k RMS(O_k); both differences must exceed it before an order is interpreted. For both stochastic moving-cloud methods, PBME and MIDPOINT, both differences must also exceed pooled independent-seed variation. Rejected orders remain visible.

G.3 Initial-label distribution and complete threshold sweep

Table G.5 gives the minimum and empirical distribution of $|y_i^0|$ for every paired momentum–seed case. Table G.6 then applies the single inclusion rule in Eq. (5.58) at every declared threshold, reporting both the affected-point fraction and the excluded absolute physical mass.

Table G.5: Per-seed distribution of the absolute initial labels $|y_i^0|$ shared by the paired PBME and MIDPOINT calculations.

P_{init}	Seed	N	Minimum	$q_{0.001}$	$q_{0.01}$	$q_{0.05}$	Median	$q_{0.95}$	$q_{0.99}$	$q_{0.999}$	Maximum
20	11	1000	2.517738e-06	3.698275e-06	3.529192e-05	1.501400e-04	0.001456908	0.002786677	0.00290662	0.002946748	0.002950469
20	29	1000	2.601808e-06	3.277947e-06	3.338633e-05	1.156339e-04	0.001472205	0.002799756	0.00291967	0.002947849	0.002949657
20	47	1000	1.108816e-06	4.885343e-06	2.479072e-05	1.791413e-04	0.001517404	0.002798699	0.002907885	0.002945863	0.00294957
20	73	1000	1.045298e-05	1.146347e-05	4.863888e-05	1.585822e-04	0.001518717	0.002776324	0.002919223	0.002949786	0.002952151
100	11	1000	2.517738e-06	3.698275e-06	3.529192e-05	1.501400e-04	0.001456908	0.002786677	0.00290662	0.002946748	0.002950469
100	29	1000	2.601808e-06	3.277947e-06	3.338633e-05	1.156339e-04	0.001472205	0.002799756	0.00291967	0.002947849	0.002949657
100	47	1000	1.108816e-06	4.885343e-06	2.479072e-05	1.791413e-04	0.001517404	0.002798699	0.002907885	0.002945863	0.00294957
100	73	1000	1.045298e-05	1.146347e-05	4.863888e-05	1.585822e-04	0.001518717	0.002776324	0.002919223	0.002949786	0.002952151

The inclusion set at threshold eta is M_eta={i: |y_i^0| > eta max_j |y_j^0|}. PBME and MIDPOINT use identical initial labels within each momentum/seed pair, verified by the stored label and cloud hashes.

Table G.6: Post-processing sensitivity to the $|y_i^0|$ inclusion threshold.

P_{init}	Method	Seed	η	Included	Excluded abs. mass	min $\lvert y_i^0\rvert$	max $\lvert y_i/y_i^0\rvert$	signed ESS	abs. ESS	N_{raw}	E_{raw}	$\Delta\rho_{11}^{\text{raw}}$	$\Delta\rho_{22}^{\text{raw}}$
20	PBME	11	0	1	0	2.517738e-06	1	1000	1000	1	0.1000455	0	0
20	PBME	11	1.000000e-14	1	0	2.517738e-06	1	1000	1000	1	0.1000455	0	0
20	PBME	11	1.000000e-12	1	0	2.517738e-06	1	1000	1000	1	0.1000455	0	0
20	PBME	11	1.000000e-10	1	0	2.517738e-06	1	1000	1000	1	0.1000455	0	0
20	PBME	11	1.000000e-08	1	0	2.517738e-06	1	1000	1000	1	0.1000455	0	0
20	PBME	11	1.000000e-06	1	0	2.517738e-06	1	1000	1000	1	0.1000455	0	0
20	PBME	11	1.000000e-05	1	0	2.517738e-06	1	1000	1000	1	0.1000455	0	0
20	PBME	11	1.000000e-04	1	0	2.517738e-06	1	1000	1000	1	0.1000455	0	0
20	PBME	11	0.001	0.999	1.000000e-03	3.699456e-06	1	999	999	0.999	0.09996119	-4.862702e-04	-5.137298e-04
20	PBME	11	0.01	0.99	0.01	3.540041e-05	1	990	990	0.99	0.09907715	-0.008534397	-0.001465603
20	MIDPOINT	11	0	1	0	2.517738e-06	333.0715	4.072953	45.97505	1.004406	0.09471243	0	0
20	MIDPOINT	11	1.000000e-14	1	0	2.517738e-06	333.0715	4.072953	45.97505	1.004406	0.09471243	0	0
20	MIDPOINT	11	1.000000e-12	1	0	2.517738e-06	333.0715	4.072953	45.97505	1.004406	0.09471243	0	0
20	MIDPOINT	11	1.000000e-10	1	0	2.517738e-06	333.0715	4.072953	45.97505	1.004406	0.09471243	0	0
20	MIDPOINT	11	1.000000e-08	1	0	2.517738e-06	333.0715	4.072953	45.97505	1.004406	0.09471243	0	0
20	MIDPOINT	11	1.000000e-06	1	0	2.517738e-06	333.0715	4.072953	45.97505	1.004406	0.09471243	0	0
20	MIDPOINT	11	1.000000e-05	1	0	2.517738e-06	333.0715	4.072953	45.97505	1.004406	0.09471243	0	0
20	MIDPOINT	11	1.000000e-04	1	0	2.517738e-06	333.0715	4.072953	45.97505	1.004406	0.09471243	0	0
20	MIDPOINT	11	0.001	0.999	1.000000e-03	3.699456e-06	333.0715	4.052153	45.90597	1.001824	0.09449482	-0.001255261	-0.001326153
20	MIDPOINT	11	0.01	0.99	0.01	3.540041e-05	333.0715	5.452548	44.00143	1.116344	0.1087534	0.1760884	-0.06415058
20	PBME	29	0	1	0	2.601808e-06	1	1000	1000	1	0.1000009	0	0
20	PBME	29	1.000000e-14	1	0	2.601808e-06	1	1000	1000	1	0.1000009	0	0
20	PBME	29	1.000000e-12	1	0	2.601808e-06	1	1000	1000	1	0.1000009	0	0
20	PBME	29	1.000000e-10	1	0	2.601808e-06	1	1000	1000	1	0.1000009	0	0
20	PBME	29	1.000000e-08	1	0	2.601808e-06	1	1000	1000	1	0.1000009	0	0
20	PBME	29	1.000000e-06	1	0	2.601808e-06	1	1000	1000	1	0.1000009	0	0
20	PBME	29	1.000000e-05	1	0	2.601808e-06	1	1000	1000	1	0.1000009	0	0
20	PBME	29	1.000000e-04	1	0	2.601808e-06	1	1000	1000	1	0.1000009	0	0
20	PBME	29	0.001	0.999	1.000000e-03	3.278624e-06	1	999	999	0.999	0.09988277	-0.001294076	2.940759e-04
20	PBME	29	0.01	0.99	0.01	3.343537e-05	1	990	990	0.99	0.09899846	-0.009276697	-7.233030e-04
20	MIDPOINT	29	0	1	0	2.601808e-06	175.6652	3.969117	92.63166	0.4393997	0.03767955	0	0
20	MIDPOINT	29	1.000000e-14	1	0	2.601808e-06	175.6652	3.969117	92.63166	0.4393997	0.03767955	0	0
20	MIDPOINT	29	1.000000e-12	1	0	2.601808e-06	175.6652	3.969117	92.63166	0.4393997	0.03767955	0	0
20	MIDPOINT	29	1.000000e-10	1	0	2.601808e-06	175.6652	3.969117	92.63166	0.4393997	0.03767955	0	0
20	MIDPOINT	29	1.000000e-08	1	0	2.601808e-06	175.6652	3.969117	92.63166	0.4393997	0.03767955	0	0
20	MIDPOINT	29	1.000000e-06	1	0	2.601808e-06	175.6652	3.969117	92.63166	0.4393997	0.03767955	0	0
20	MIDPOINT	29	1.000000e-05	1	0	2.601808e-06	175.6652	3.969117	92.63166	0.4393997	0.03767955	0	0
20	MIDPOINT	29	1.000000e-04	1	0	2.601808e-06	175.6652	3.969117	92.63166	0.4393997	0.03767955	0	0
20	MIDPOINT	29	0.001	0.999	1.000000e-03	3.278624e-06	43.279	21.2706	213.1542	0.6150648	0.0584367	0.227324	-0.05165888
20	MIDPOINT	29	0.01	0.99	0.01	3.343537e-05	43.279	23.79487	211.8329	0.628537	0.0596388	0.2267532	-0.03761586
20	PBME	47	0	1	0	1.108816e-06	1	1000	1000	1	0.10018	0	0
20	PBME	47	1.000000e-14	1	0	1.108816e-06	1	1000	1000	1	0.10018	0	0
20	PBME	47	1.000000e-12	1	0	1.108816e-06	1	1000	1000	1	0.10018	0	0
20	PBME	47	1.000000e-10	1	0	1.108816e-06	1	1000	1000	1	0.10018	0	0
20	PBME	47	1.000000e-08	1	0	1.108816e-06	1	1000	1000	1	0.10018	0	0
20	PBME	47	1.000000e-06	1	0	1.108816e-06	1	1000	1000	1	0.10018	0	0
20	PBME	47	1.000000e-05	1	0	1.108816e-06	1	1000	1000	1	0.10018	0	0
20	PBME	47	1.000000e-04	1	0	1.108816e-06	1	1000	1000	1	0.10018	0	0
20	PBME	47	0.001	0.999	1.000000e-03	4.889123e-06	1	999	999	0.999	0.1000611	-0.001450378	4.503780e-04

P_{init}	Method	Seed	η	Included	Excluded abs. mass	min $\lvert y_i^0 \rvert$	max $\lvert y_i/y_i^0 \rvert$	signed ESS	abs. ESS	N_{raw}	E_{raw}	$\Delta\rho_{11}^{\text{raw}}$	$\Delta\rho_{22}^{\text{raw}}$
20	PBME	47	0.01	0.989	0.011	2.978194e-05	1	989	989	0.989	0.09901861	-0.01198725	9.872457e-04
20	MIDPOINT	47	0	1	0	1.108816e-06	999.2251	0.6686498	32.34535	1.173566	0.1385047	0	0
20	MIDPOINT	47	1.000000e-14	1	0	1.108816e-06	999.2251	0.6686498	32.34535	1.173566	0.1385047	0	0
20	MIDPOINT	47	1.000000e-12	1	0	1.108816e-06	999.2251	0.6686498	32.34535	1.173566	0.1385047	0	0
20	MIDPOINT	47	1.000000e-10	1	0	1.108816e-06	999.2251	0.6686498	32.34535	1.173566	0.1385047	0	0
20	MIDPOINT	47	1.000000e-08	1	0	1.108816e-06	999.2251	0.6686498	32.34535	1.173566	0.1385047	0	0
20	MIDPOINT	47	1.000000e-06	1	0	1.108816e-06	999.2251	0.6686498	32.34535	1.173566	0.1385047	0	0
20	MIDPOINT	47	1.000000e-05	1	0	1.108816e-06	999.2251	0.6686498	32.34535	1.173566	0.1385047	0	0
20	MIDPOINT	47	1.000000e-04	1	0	1.108816e-06	999.2251	0.6686498	32.34535	1.173566	0.1385047	0	0
20	MIDPOINT	47	0.001	0.999	1.000000e-03	4.889123e-06	999.2251	1.194596	31.49862	1.521759	0.1799008	0.5050108	-0.1568182
20	MIDPOINT	47	0.01	0.989	0.011	2.978194e-05	999.2251	1.05035	31.28699	1.140712	0.1253721	-0.1378823	0.1050278
20	PBME	73	0	1	0	1.045298e-05	1	1000	1000	1	0.1000153	0	0
20	PBME	73	1.000000e-14	1	0	1.045298e-05	1	1000	1000	1	0.1000153	0	0
20	PBME	73	1.000000e-12	1	0	1.045298e-05	1	1000	1000	1	0.1000153	0	0
20	PBME	73	1.000000e-10	1	0	1.045298e-05	1	1000	1000	1	0.1000153	0	0
20	PBME	73	1.000000e-08	1	0	1.045298e-05	1	1000	1000	1	0.1000153	0	0
20	PBME	73	1.000000e-06	1	0	1.045298e-05	1	1000	1000	1	0.1000153	0	0
20	PBME	73	1.000000e-05	1	0	1.045298e-05	1	1000	1000	1	0.1000153	0	0
20	PBME	73	1.000000e-04	1	0	1.045298e-05	1	1000	1000	1	0.1000153	0	0
20	PBME	73	0.001	1	0	1.045298e-05	1	1000	1000	1	0.1000153	0	0
20	PBME	73	0.01	0.992	0.008	3.460859e-05	1	992	992	0.992	0.09926765	-0.00694802	-0.00105198
20	MIDPOINT	73	0	1	0	1.045298e-05	255.6513	0.02471431	73.47073	0.06211285	0.0128747	0	0
20	MIDPOINT	73	1.000000e-14	1	0	1.045298e-05	255.6513	0.02471431	73.47073	0.06211285	0.0128747	0	0
20	MIDPOINT	73	1.000000e-12	1	0	1.045298e-05	255.6513	0.02471431	73.47073	0.06211285	0.0128747	0	0
20	MIDPOINT	73	1.000000e-10	1	0	1.045298e-05	255.6513	0.02471431	73.47073	0.06211285	0.0128747	0	0
20	MIDPOINT	73	1.000000e-08	1	0	1.045298e-05	255.6513	0.02471431	73.47073	0.06211285	0.0128747	0	0
20	MIDPOINT	73	1.000000e-06	1	0	1.045298e-05	255.6513	0.02471431	73.47073	0.06211285	0.0128747	0	0
20	MIDPOINT	73	1.000000e-05	1	0	1.045298e-05	255.6513	0.02471431	73.47073	0.06211285	0.0128747	0	0
20	MIDPOINT	73	1.000000e-04	1	0	1.045298e-05	255.6513	0.02471431	73.47073	0.06211285	0.0128747	0	0
20	MIDPOINT	73	0.001	1	0	1.045298e-05	255.6513	0.02471431	73.47073	0.06211285	0.0128747	0	0
20	MIDPOINT	73	0.01	0.992	0.008	3.460859e-05	120.7644	3.706754	160.5606	0.4307875	0.04464812	0.3108087	0.05786592
100	PBME	11	0	1	0	2.517738e-06	1	1000	1000	1	2.499976	0	0
100	PBME	11	1.000000e-14	1	0	2.517738e-06	1	1000	1000	1	2.499976	0	0
100	PBME	11	1.000000e-12	1	0	2.517738e-06	1	1000	1000	1	2.499976	0	0
100	PBME	11	1.000000e-10	1	0	2.517738e-06	1	1000	1000	1	2.499976	0	0
100	PBME	11	1.000000e-08	1	0	2.517738e-06	1	1000	1000	1	2.499976	0	0
100	PBME	11	1.000000e-06	1	0	2.517738e-06	1	1000	1000	1	2.499976	0	0
100	PBME	11	1.000000e-05	1	0	2.517738e-06	1	1000	1000	1	2.499976	0	0
100	PBME	11	1.000000e-04	1	0	2.517738e-06	1	1000	1000	1	2.499976	0	0
100	PBME	11	0.001	0.999	1.000000e-03	3.699456e-06	1	999	999	0.999	2.497558	4.119373e-04	-0.001411937
100	PBME	11	0.01	0.99	0.01	3.540041e-05	1	990	990	0.99	2.475152	-8.737304e-04	-0.00912627
100	MIDPOINT	11	0	1	0	2.517738e-06	2.551741e+14	0.9612848	27.58835	-4.751139e+11	-1.146615e+12	0	0
100	MIDPOINT	11	1.000000e-14	1	0	2.517738e-06	2.551741e+14	0.9612848	27.58835	-4.751139e+11	-1.146615e+12	0	0
100	MIDPOINT	11	1.000000e-12	1	0	2.517738e-06	2.551741e+14	0.9612848	27.58835	-4.751139e+11	-1.146615e+12	0	0
100	MIDPOINT	11	1.000000e-10	1	0	2.517738e-06	2.551741e+14	0.9612848	27.58835	-4.751139e+11	-1.146615e+12	0	0
100	MIDPOINT	11	1.000000e-08	1	0	2.517738e-06	2.551741e+14	0.9612848	27.58835	-4.751139e+11	-1.146615e+12	0	0
100	MIDPOINT	11	1.000000e-06	1	0	2.517738e-06	2.551741e+14	0.9612848	27.58835	-4.751139e+11	-1.146615e+12	0	0
100	MIDPOINT	11	1.000000e-05	1	0	2.517738e-06	2.551741e+14	0.9612848	27.58835	-4.751139e+11	-1.146615e+12	0	0
100	MIDPOINT	11	1.000000e-04	1	0	2.517738e-06	2.551741e+14	0.9612848	27.58835	-4.751139e+11	-1.146615e+12	0	0
100	MIDPOINT	11	0.001	0.999	1.000000e-03	3.699456e-06	2.551741e+14	0.8807709	27.19023	-4.543648e+11	-1.096427e+12	-8.547367e+09	2.929645e+10
100	MIDPOINT	11	0.01	0.99	0.01	3.540041e-05	2.551741e+14	0.4454698	23.64206	-3.100984e+11	-7.355081e+11	-9.209522e+10	2.571108e+11
100	PBME	29	0	1	0	2.601808e-06	1	1000	1000	1	2.499747	0	0

P_{init}	Method	Seed	η	Included	Excluded abs. mass	min $\lvert y_i^0\rvert$	max $\lvert y_i/y_i^0\rvert$	signed ESS	abs. ESS	N_{raw}	E_{raw}	$\Delta\rho_{11}^{\mathrm{raw}}$	$\Delta\rho_{22}^{\mathrm{raw}}$
100	PBME	29	1.000000e-14	1	0	2.601808e-06	1	1000	1000	1	2.499747	0	0
100	PBME	29	1.000000e-12	1	0	2.601808e-06	1	1000	1000	1	2.499747	0	0
100	PBME	29	1.000000e-10	1	0	2.601808e-06	1	1000	1000	1	2.499747	0	0
100	PBME	29	1.000000e-08	1	0	2.601808e-06	1	1000	1000	1	2.499747	0	0
100	PBME	29	1.000000e-06	1	0	2.601808e-06	1	1000	1000	1	2.499747	0	0
100	PBME	29	1.000000e-05	1	0	2.601808e-06	1	1000	1000	1	2.499747	0	0
100	PBME	29	1.000000e-04	1	0	2.601808e-06	1	1000	1000	1	2.499747	0	0
100	PBME	29	0.001	0.999	1.000000e-03	3.278624e-06	1	999	999	0.999	2.497159	-5.187540e-04	-4.812460e-04
100	PBME	29	0.01	0.99	0.01	3.343537e-05	1	990	990	0.99	2.474747	-7.458807e-04	-0.009254119
100	MIDPOINT	29	0	1	0	2.601808e-06	7.610117e+23	0.4999376	16.41429	5.941656e+20	1.523433e+21	0	0
100	MIDPOINT	29	1.000000e-14	1	0	2.601808e-06	7.610117e+23	0.4999376	16.41429	5.941656e+20	1.523433e+21	0	0
100	MIDPOINT	29	1.000000e-12	1	0	2.601808e-06	7.610117e+23	0.4999376	16.41429	5.941656e+20	1.523433e+21	0	0
100	MIDPOINT	29	1.000000e-10	1	0	2.601808e-06	7.610117e+23	0.4999376	16.41429	5.941656e+20	1.523433e+21	0	0
100	MIDPOINT	29	1.000000e-08	1	0	2.601808e-06	7.610117e+23	0.4999376	16.41429	5.941656e+20	1.523433e+21	0	0
100	MIDPOINT	29	1.000000e-06	1	0	2.601808e-06	7.610117e+23	0.4999376	16.41429	5.941656e+20	1.523433e+21	0	0
100	MIDPOINT	29	1.000000e-05	1	0	2.601808e-06	7.610117e+23	0.4999376	16.41429	5.941656e+20	1.523433e+21	0	0
100	MIDPOINT	29	1.000000e-04	1	0	2.601808e-06	7.610117e+23	0.4999376	16.41429	5.941656e+20	1.523433e+21	0	0
100	MIDPOINT	29	0.001	0.999	1.000000e-03	3.278624e-06	7.610117e+23	0.3613749	15.72744	5.021200e+20	1.285239e+21	-4.774962e+19	-4.429600e+19
100	MIDPOINT	29	0.01	0.99	0.01	3.343537e-05	1.917684e+23	0.3589746	54.28124	-1.934841e+20	-4.733178e+20	2.017321e+20	-9.893818e+20
100	PBME	47	0	1	0	1.108816e-06	1	1000	1000	1	2.500654	0	0
100	PBME	47	1.000000e-14	1	0	1.108816e-06	1	1000	1000	1	2.500654	0	0
100	PBME	47	1.000000e-12	1	0	1.108816e-06	1	1000	1000	1	2.500654	0	0
100	PBME	47	1.000000e-10	1	0	1.108816e-06	1	1000	1000	1	2.500654	0	0
100	PBME	47	1.000000e-08	1	0	1.108816e-06	1	1000	1000	1	2.500654	0	0
100	PBME	47	1.000000e-06	1	0	1.108816e-06	1	1000	1000	1	2.500654	0	0
100	PBME	47	1.000000e-05	1	0	1.108816e-06	1	1000	1000	1	2.500654	0	0
100	PBME	47	1.000000e-04	1	0	1.108816e-06	1	1000	1000	1	2.500654	0	0
100	PBME	47	0.001	0.999	1.000000e-03	4.889123e-06	1	999	999	0.999	2.498062	-1.177156e-05	-9.882284e-04
100	PBME	47	0.01	0.989	0.011	2.978194e-05	1	989	989	0.989	2.472868	-0.001595268	-0.009404732
100	MIDPOINT	47	0	1	0	1.108816e-06	3.379605e+23	0.2035919	12.23083	-1.663047e+20	-4.358452e+20	0	0
100	MIDPOINT	47	1.000000e-14	1	0	1.108816e-06	3.379605e+23	0.2035919	12.23083	-1.663047e+20	-4.358452e+20	0	0
100	MIDPOINT	47	1.000000e-12	1	0	1.108816e-06	3.379605e+23	0.2035919	12.23083	-1.663047e+20	-4.358452e+20	0	0
100	MIDPOINT	47	1.000000e-10	1	0	1.108816e-06	3.379605e+23	0.2035919	12.23083	-1.663047e+20	-4.358452e+20	0	0
100	MIDPOINT	47	1.000000e-08	1	0	1.108816e-06	3.379605e+23	0.2035919	12.23083	-1.663047e+20	-4.358452e+20	0	0
100	MIDPOINT	47	1.000000e-06	1	0	1.108816e-06	3.379605e+23	0.2035919	12.23083	-1.663047e+20	-4.358452e+20	0	0
100	MIDPOINT	47	1.000000e-05	1	0	1.108816e-06	3.379605e+23	0.2035919	12.23083	-1.663047e+20	-4.358452e+20	0	0
100	MIDPOINT	47	1.000000e-04	1	0	1.108816e-06	3.379605e+23	0.2035919	12.23083	-1.663047e+20	-4.358452e+20	0	0
100	MIDPOINT	47	0.001	0.999	1.000000e-03	4.889123e-06	3.379605e+23	0.209684	12.18462	-1.687707e+20	-4.422352e+20	-2.903633e+16	-2.437008e+18
100	MIDPOINT	47	0.01	0.989	0.011	2.978194e-05	5.565055e+22	0.6477001	50.04449	9.648472e+19	2.443120e+20	-1.950072e+20	4.577966e+20
100	PBME	73	0	1	0	1.045298e-05	1	1000	1000	1	2.499827	0	0
100	PBME	73	1.000000e-14	1	0	1.045298e-05	1	1000	1000	1	2.499827	0	0
100	PBME	73	1.000000e-12	1	0	1.045298e-05	1	1000	1000	1	2.499827	0	0
100	PBME	73	1.000000e-10	1	0	1.045298e-05	1	1000	1000	1	2.499827	0	0
100	PBME	73	1.000000e-08	1	0	1.045298e-05	1	1000	1000	1	2.499827	0	0
100	PBME	73	1.000000e-06	1	0	1.045298e-05	1	1000	1000	1	2.499827	0	0
100	PBME	73	1.000000e-05	1	0	1.045298e-05	1	1000	1000	1	2.499827	0	0
100	PBME	73	1.000000e-04	1	0	1.045298e-05	1	1000	1000	1	2.499827	0	0
100	PBME	73	0.001	1	0	1.045298e-05	1	1000	1000	1	2.499827	0	0
100	PBME	73	0.01	0.992	0.008	3.460859e-05	1	992	992	0.992	2.480098	-0.002513259	-0.005486741
100	MIDPOINT	73	0	1	0	1.045298e-05	4.832604e+16	0.4782585	70.44788	-9.285806e+13	-2.337110e+14	0	0
100	MIDPOINT	73	1.000000e-14	1	0	1.045298e-05	4.832604e+16	0.4782585	70.44788	-9.285806e+13	-2.337110e+14	0	0
100	MIDPOINT	73	1.000000e-12	1	0	1.045298e-05	4.832604e+16	0.4782585	70.44788	-9.285806e+13	-2.337110e+14	0	0

P_{init}	Method	Seed	η	Included	Excluded abs. mass	min $\lvert y_i^0\rvert$	max $\lvert y_i/y_i^0\rvert$	signed ESS	abs. ESS	N_{raw}	E_{raw}	$\Delta\rho_{11}^{\mathrm{raw}}$	$\Delta\rho_{22}^{\mathrm{raw}}$
100	MIDPOINT	73	1.000000e-10	1	0	1.045298e-05	4.832604e+16	0.4782585	70.44788	-9.285806e+13	-2.337110e+14	0	0
100	MIDPOINT	73	1.000000e-08	1	0	1.045298e-05	4.832604e+16	0.4782585	70.44788	-9.285806e+13	-2.337110e+14	0	0
100	MIDPOINT	73	1.000000e-06	1	0	1.045298e-05	4.832604e+16	0.4782585	70.44788	-9.285806e+13	-2.337110e+14	0	0
100	MIDPOINT	73	1.000000e-05	1	0	1.045298e-05	4.832604e+16	0.4782585	70.44788	-9.285806e+13	-2.337110e+14	0	0
100	MIDPOINT	73	1.000000e-04	1	0	1.045298e-05	4.832604e+16	0.4782585	70.44788	-9.285806e+13	-2.337110e+14	0	0
100	MIDPOINT	73	0.001	1	0	1.045298e-05	4.832604e+16	0.4782585	70.44788	-9.285806e+13	-2.337110e+14	0	0
100	MIDPOINT	73	0.01	0.992	0.008	3.460859e-05	4.832604e+16	0.5116368	70.06995	-9.603127e+13	-2.414449e+14	-6.311465e+11	-2.542064e+12

The threshold sweep does not alter propagation. The inclusion set is M_eta={i: |y_i^0| > eta max_j |y_j^0|}; Included is the retained point fraction and Excluded abs. mass is the removed fraction of sum_i |omega_i y_i^0|. Raw quantities are not self-normalized.

G.4 Absolute reference errors for every seed

The field-error table reports absolute PBME and MIDPOINT relative L_1 and L_2 errors separately, before and after the explicitly identified unit-integral shape normalization. The observable table reports the absolute RMS and maximum time-series errors of both population estimators and then their paired MIDPOINT-minus-PBME differences. These tables are the individual values from which the paired intervals in Table 6.15 are calculated.

Table G.7: Absolute per-seed PBME and MIDPOINT field errors against the common physical references.

Reference	P_{init}	Seed	Field	PBME raw E_1	PBME raw E_2	MIDPOINT raw E_1	MIDPOINT raw E_2	PBME shape E_1	PBME shape E_2	MIDPOINT shape E_1	MIDPOINT shape E_2
QCLE	20	11	R–P density	0.7541632	0.5770252	1.069166	1.117634	0.7541631	0.577023	1.067571	1.114696
QCLE	20	29	R–P density	0.7567613	0.5829466	0.8523653	0.7018773	0.7567611	0.5829443	1.139695	1.036594
QCLE	20	47	R–P density	0.7373133	0.5466541	1.978803	2.931216	0.737313	0.5466517	1.789627	2.506695
QCLE	20	73	R–P density	0.7361042	0.5488264	1.106967	1.192537	0.736104	0.5488241	9.432692	14.94748
TDSE	20	11	nuclear R marginal	0.2450988	0.2084792	0.8531063	0.8577543	0.2450988	0.2084792	0.8512673	0.8549057
TDSE	20	29	nuclear R marginal	0.2544241	0.205672	0.7547542	0.7159306	0.2544241	0.205672	1.400726	1.261296
TDSE	20	47	nuclear R marginal	0.2255646	0.1795472	3.255473	2.873213	0.2255646	0.1795472	2.806255	2.452522
TDSE	20	73	nuclear R marginal	0.2086158	0.1697067	1.068677	1.231188	0.2086158	0.1697067	12.79815	12.44009
QCLE	100	11	R–P density	0.2878229	0.2908149	1.268491e+12	1.573286e+12	0.2878229	0.290815	3.188599	3.507764
QCLE	100	29	R–P density	0.3133281	0.3119393	1.408334e+21	2.299460e+21	0.313328	0.3119393	3.180984	4.058243
QCLE	100	47	R–P density	0.3217061	0.3186076	5.474794e+20	1.075108e+21	0.321706	0.3186075	3.804229	6.50817
QCLE	100	73	R–P density	0.334454	0.3342914	2.732187e+14	3.411279e+14	0.3344536	0.3342912	3.381936	3.798236
TDSE	100	11	nuclear R marginal	0.05512802	0.05769911	4.882285e+11	5.622169e+11	0.05512802	0.05769911	1.262088	1.181538
TDSE	100	29	nuclear R marginal	0.05190838	0.04884448	1.004255e+21	1.336119e+21	0.05190838	0.04884448	2.539953	2.539886
TDSE	100	47	nuclear R marginal	0.06263472	0.06170161	2.255690e+20	3.938376e+20	0.06263472	0.06170161	2.036377	2.502555
TDSE	100	73	nuclear R marginal	0.02955926	0.02732207	1.678693e+14	2.282472e+14	0.02955926	0.02732207	2.460424	2.57227

Raw errors retain each field integral. Shape errors divide each signed field by its own nonzero signed integral before comparison and therefore cannot be used as conservation evidence. E1 and E2 are relative L1 and L2 field errors.

Table G.8: Absolute per-seed population time-series errors and paired MIDPOINT-minus-PBME differences.

Reference	P_{init}	Seed	Observable	PBME E_{RMS}	MIDPOINT E_{RMS}	ΔE_{RMS}	PBME E_{∞}	MIDPOINT E_{∞}	ΔE_{∞}
QCLE	20	11	ρ_{11}^{SN}	0.01029402	0.06699834	0.05670432	0.035862	0.3753649	0.3395029
QCLE	20	11	ρ_{22}^{SN}	0.01029402	0.06699834	0.05670432	0.035862	0.3753649	0.3395029
QCLE	20	29	ρ_{11}^{SN}	0.01276918	0.1124632	0.09969404	0.03754402	0.5558304	0.5182863
QCLE	20	29	ρ_{22}^{SN}	0.01276918	0.1124632	0.09969404	0.03754402	0.5558304	0.5182863
QCLE	20	47	ρ_{11}^{SN}	0.01935254	0.2368523	0.2174998	0.03895302	3.480223	3.44127
QCLE	20	47	ρ_{22}^{SN}	0.01935254	0.2368523	0.2174998	0.03895302	3.480223	3.44127
QCLE	20	73	ρ_{11}^{SN}	0.01256275	1.149719	1.137156	0.02572772	19.25081	19.22509
QCLE	20	73	ρ_{22}^{SN}	0.01256275	1.149719	1.137156	0.02572772	19.25081	19.22509
TDSE	20	11	ρ_{11}^{SN}	0.00824413	0.06551272	0.05726859	0.03538828	0.375382	0.3399937
TDSE	20	11	ρ_{22}^{SN}	0.00824413	0.06551272	0.05726859	0.03538828	0.375382	0.3399937
TDSE	20	29	ρ_{11}^{SN}	0.01011709	0.1149561	0.104839	0.0329313	0.5558167	0.5228854
TDSE	20	29	ρ_{22}^{SN}	0.01011709	0.1149561	0.104839	0.0329313	0.5558167	0.5228854
TDSE	20	47	ρ_{11}^{SN}	0.0160436	0.2372376	0.221194	0.03273288	3.480207	3.447474
TDSE	20	47	ρ_{22}^{SN}	0.0160436	0.2372376	0.221194	0.03273288	3.480207	3.447474
TDSE	20	73	ρ_{11}^{SN}	0.009466471	1.147115	1.137649	0.02554337	19.25222	19.22668
TDSE	20	73	ρ_{22}^{SN}	0.009466471	1.147115	1.137649	0.02554337	19.25222	19.22668
QCLE	100	11	ρ_{11}^{SN}	0.006238786	197.3796	197.3733	0.01793169	9631.017	9630.999
QCLE	100	11	ρ_{22}^{SN}	0.006238786	197.3796	197.3733	0.01793169	9631.017	9630.999
QCLE	100	29	ρ_{11}^{SN}	0.01950467	1.698908	1.679404	0.03252643	48.38205	48.34952
QCLE	100	29	ρ_{22}^{SN}	0.01950467	1.698908	1.679404	0.03252643	48.38205	48.34952
QCLE	100	47	ρ_{11}^{SN}	0.00755677	3.523878	3.516321	0.02484807	152.5573	152.5324
QCLE	100	47	ρ_{22}^{SN}	0.00755677	3.523878	3.516321	0.02484807	152.5573	152.5324
QCLE	100	73	ρ_{11}^{SN}	0.004360389	3.090872	3.086512	0.01231975	115.4552	115.4429
QCLE	100	73	ρ_{22}^{SN}	0.004360389	3.090872	3.086512	0.01231975	115.4552	115.4429
TDSE	100	11	ρ_{11}^{SN}	0.006236986	197.3796	197.3733	0.01793051	9631.017	9630.999
TDSE	100	11	ρ_{22}^{SN}	0.006236986	197.3796	197.3733	0.01793051	9631.017	9630.999
TDSE	100	29	ρ_{11}^{SN}	0.01950896	1.698909	1.6794	0.03253133	48.38205	48.34952
TDSE	100	29	ρ_{22}^{SN}	0.01950896	1.698909	1.6794	0.03253133	48.38205	48.34952
TDSE	100	47	ρ_{11}^{SN}	0.007557963	3.523878	3.51632	0.02484644	152.5573	152.5324
TDSE	100	47	ρ_{22}^{SN}	0.007557963	3.523878	3.51632	0.02484644	152.5573	152.5324
TDSE	100	73	ρ_{11}^{SN}	0.004359862	3.090872	3.086512	0.01231704	115.4552	115.4429
TDSE	100	73	ρ_{22}^{SN}	0.004359862	3.090872	3.086512	0.01231704	115.4552	115.4429

PBME and MIDPOINT use the same initial cloud for every momentum/seed pair. Population curves are signed-mass-normalized estimators; raw normalization and trace are tested separately. Negative paired differences favor MIDPOINT for that seed and metric.

G.5 Versioned reproducibility release

The executable source is frozen in a clean release commit, and the complete numerical evidence is frozen in a deterministic archive with an embedded SHA-256 index. The LaTeX label of each generated numerical table and figure is also its unique artifact identifier in the table/figure manifest; that manifest records the generator and input digest for each retained artifact. This provides a versioned chain from the equations and parameters stated in the thesis to each displayed numerical result without placing implementation paths in the scientific narrative.

Public source repository: `https://github.com/sahandgit/gaussian_process_guided_mapping_quantumclassical_liouville_dynamics`.

Public release asset: `https://github.com/sahandgit/gaussian_process_guided_mapping_quantumclassical_liouville_dynamics/releases/download/thesis-final-2026-08-01-r2/frozen_numerical_evidence_payload.zip`.

Final release tag (identifies the final tagged document commit): `thesis-final-2026-08-01-r2`.

Archive filename: `frozen_numerical_evidence_payload.zip`.

Frozen source/evidence commit (retrievable at the repository above): `1f170df258cb43123af1a040a9973ca6996c804b`.

Frozen numerical-evidence archive SHA-256: `9752f3e16a305225ed30796d799d5a9795d09d8cdebd1ee74648054cf84af87e`.

Embedded checksum-index SHA-256: `e4749382e8ac55e1b20f129e2c80402b6ad112665bbd1a00be60ed6075b223b2`. The archive contains `final_reviewer_closure/README.md` (reproduction instructions), `environment.json` and `environment/pip_freeze.txt` (environment), `FINAL_RUN_MANIFEST.json` and the table/figure and per-run manifests, plus `PAYLOAD_SHA256SUMS.csv` for every archived file. The release also supplies `CLEAN_ROOM_VERIFICATION.json`, which records public download, checksum, extraction, manifest, and clean source-compilation checks. The public release asset and cited frozen source/evidence commit make this record independently retrievable and checksum-verifiable. The versioned GitHub release is not a DOI or an institutional persistent identifier.

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